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Fifty-Four Years of Atmospheric Bomb-Radiocarbon Decay Falsify the Central Assumptions of the IPCC Carbon-Cycle Framework

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Abstract

The IPCC carbon-cycle framework projects future CO_2 scenarios using the Bern impulse-response function, a multi-exponential parameterization. The atmospheric bomb radiocarbon record falsifies it. In the Hua et al. (2022) [55] global $\Delta^{14}C$ compilation, the post-1965 bomb signal decays as a single exponential with e-folding time $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr, residual scatter 0.539‰ and explained variance 99.998% in the 32-point bias-corrected fit. Every published Bern function diverges from the observed decay within the first year. The analysis rests on three independent empirical measurements: τ_{rel}; the fitted amplitude $A_{32} = 709.115 \pm 0.585$‰ at $t = 1965$; and the CO_2 molecular residence time $\tau_{\mathrm{res}} = 3.80 \pm 0.12$ yr from the Koutsoyiannis (2024a) [72] Refined Reservoir Routing fit to Mauna Loa and Barrow. Every other quantity is a first-principles consequence of these three and mass conservation, with no adjustable parameters. The cleaning is not curve fitting: the exponential form is established on the unaltered data (eight zonal fits, all $R^2 > 99.6$%) *before* any bias correction; plateau boundaries come from archival audit of Hua compilation events, confirmed by a first-principles smoothness criterion exceeded by factors of 6–14. A single exponential is mathematically incompatible with any sum of distinct exponentials at any parameter setting (Vandermonde and Cauchy-Schwarz proof). The operator-equivalence theorem extends this exclusion from the isotopic decay to the CO_2 mass impulse response, so no slow bulk-CO_2 mode escapes the bound. One first-order mass-balance ODE connects τ_{rel} and τ_{res} through the bridge equation, yielding $\phi_{\mathrm{slow}} = 0.203 \pm 0.006$, confirmed to 0.74σ by an independent two-box stratosphere-troposphere ODE ($\phi_{\mathrm{slow}}^{\mathrm{2box}} = 0.199 \pm 0.008$). The single-exponential shape requires the gross exchange rate constant while atmospheric CO_2 grew 31.6%; $F_{\mathrm{nat}}/M_{\mathrm{a}}$ is stable to ±0.4% year-to-year, and a Newey–West HAC regression of annual log-growth rates returns slope $\hat{\beta} = 0.980 \pm 0.042$, not rejecting $\beta = 1$ at 0.48σ; and $\delta^{13}C_{\mathrm{I}}$ is stationary at −13.2‰ over the 1978–2022 instrumental window [73] and across a 500-year sclerosponge and Law Dome extension; its 1977–1996 Francey trend is +2.94‰/decade at $p < 10^{-4}$ against a Hurst-Kolmogorov null, opposite to fossil dominance [41, 21]. Jungfraujoch measurements 2020–2024 [75] confirm the extrapolated exponential. The atmospheric CO_2 rise is dominated by non-fossil-fuel sources. The Bern parameterization and the long-conjectured "Suess effect" are both falsified by direct observation.

Keywords: carbon cycle; atmospheric mass balance; residence time; adjustment time; bomb-^{14}C; Bern model; Suess effect; biospheric expansion; isotopic ratio relaxation

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1. Introduction

1.1 An intuitive syllogism that was never derived from physical law

For decades the IPCC and most of the climate-science community have assumed, without empirical proof, a few seemingly intuitive things about our atmosphere, about how CO_2 behaves once it gets there, about what is driving the observed atmospheric CO_2 increase, and about how long human emissions persist. The reasoning runs as follows: nature was in balance before us; humans have added CO_2 by burning fossil fuels and changing land usage; therefore humans are obviously the cause of the rise; therefore nature must now be a net sink that absorbs what it can but cannot remove our emissions fast enough [17, 4, 14]; therefore the leftover "excess" atmospheric CO_2 from human activity persists for hundreds or thousands of years [60]. Familiar terms like "the Suess effect" [68], "residence time," "adjustment time," and "airborne fraction" are deployed throughout the climate-science literature and the popular press to support this story. Claims that 15–40% of an "emitted CO_2 pulse" remains in the atmosphere for over a thousand years [60] are repeated in textbooks, review papers, IPCC assessment reports [56, 57, 58, 59, 60], social-cost-of-carbon calculations, and net-zero policy documents.

The syllogism is intuitive but it is not derived from physical law. Each step is an assumption. The pre-industrial atmospheric balance is asserted as a state, not measured. The directional causality from human emissions to atmospheric increase is assumed, not derived from a conservation equation or empirical statistical analysis. The claim that nature is a net sink is itself not an established empirical fact but a derived accounting quantity: it follows from assuming that the entire human gross emission contributes to the atmospheric stock and identifying the remaining mass-balance residual as a natural net flux. The assumption is treated as definitional in the IPCC framework rather than as a physical conclusion. Nature is a far larger gross source (ocean outgassing, respiration, decomposition, volcanism) and a far larger gross sink (photosynthesis, ocean uptake, weathering) than human emissions, and the question of whether the human gross source or net imbalances among the non-fossil-fuel gross fluxes drives the observed atmospheric increase is empirical, not logical [7, 8, 47, 48, 49, 72, 20].

1.2 Undefined and inconsistently used terms: "airborne fraction," "adjustment time," "the Suess effect"

Several of the terms used to support the story do not have rigorous operational definitions in the documents that assert them. "Adjustment time" appears in the IPCC AR6 (2021) [60] Glossary (Annex VII, p. 2237) with no governing equation, no measurement procedure, and no derivation from physical law. "Airborne fraction" is defined in the same Glossary as the fraction of *cumulative* anthropogenic emissions remaining in the atmosphere (ABF_{L} in the notation of Koutsoyiannis (2024a) [72] and Cohler & Soon (2026a) [20]): the total anthropogenic mass still airborne divided by the total ever emitted, which is what "the fraction remaining" actually means. But the IPCC, and even the Global Carbon Budget (GCB), then consistently report "airborne fraction" as ∼44% using a totally different computation: the ratio of the *annual* atmospheric increase to the *annual* anthropogenic emission (ABF_{C}), a single-year rate ratio that says nothing about what fraction of emitted carbon *remains* in the air, and which contradictorily counts land-use change as part of "emissions." The two quantities are not interchangeable: ABF_{L} computed under mass-balance reservoir routing with residence time $\tau \approx 4$ yr is approximately 6% [72, 20], roughly an order of magnitude smaller than the $ABF_{\mathrm{C}} \approx 44\%$ value the IPCC reports. The IPCC defines the words one way in its glossary and then uses them to mean something totally different; and the difference is not a definitional subtlety. It is the difference between a measurement of atmospheric retention and a quantity that has no bearing on retention at all. Even taken on its own terms, the ∼44% value is not a stable property of the system: computed year by year from Global Carbon Budget [42] fossil and land-use emissions and Mauna Loa atmospheric growth over 1966–2019 [67, 46], ABF_{C} ranges

from 17.9% to 77.4%, a factor of 4.3×, with standard deviation = 12.85%, mean = 44.83%, and coefficient of variation = 28.7%. Smoothing with a 4-year trailing window chosen to match the residence time $\tau_{res} \approx 3.8$ yr [72, 20] reduces the standard deviation by a factor of 2.38× and leaves ABF_C still varying from 28.0% (1993, post-Pinatubo) to 53.9% (1980), a factor of 1.93×, with coefficient of variation 12.1%. The appearance of stability invoked to support "climate-carbon feedback has not yet emerged" arguments can be made only using smoothing windows of 15–30 yr, far longer than the physical residence time of the system; at such windows *any* noisy time series appears smooth, and the apparent constancy is a property of the smoothing operation rather than of the carbon cycle. The >2× collapse of ABF_C variance at smoothing windows comparable to τ_{res} is itself independent empirical confirmation of the short residence time, derived from a diagnostic that uses no bomb-^{14}C data and no isotopic information.

The same pattern of definitional drift afflicts "the Suess effect." The term originated with Suess (1955) [114], who observed that the $^{14}C/^{12}C$ ratio in pre-bomb tree rings had declined over the preceding fifty years. His proposed explanation, in his own words, was that "[t]he decrease can be attributed to the introduction of a certain amount of C^{14}-free CO_2 into the atmosphere by artificial coal and oil combustion and to the rate of isotopic exchange between atmospheric CO_2 and the bicarbonate dissolved in the oceans" [114]. The sentence is a two-mechanism conjecture, fossil-fuel combustion and oceanic exchange joined by "and," with no attribution between them and no quantitative commitment to either. "Can be attributed to" is the language of a hypothesis. "A certain amount" is deliberately unquantified. Suess offered an open question.

The subsequent carbon-cycle literature collapsed Suess's two-mechanism conjecture into a one-mechanism claim. The oceanic-exchange branch was dropped. The fossil-fuel branch was promoted from "can be attributed to" to "is caused by." The collapsed claim was named "the Suess effect" [68], as though Suess himself had identified fossil-fuel combustion as the sole mechanism. He had not. By the 2000s the institutional Suess effect, a one-mechanism fossil-fuel-only attribution, was treated as established background knowledge requiring no empirical verification, and it was consolidated in this form as the canonical reference for the field by Graven et al. (2020) [45].

The institutional Suess effect, so understood, is invoked throughout the carbon-cycle literature as a predicted regime change in the bomb-^{14}C decay as fossil-fuel dilution overtakes non-fossil-fuel-exchange decay. It has no quantitative criterion specifying when the transition occurs, what its functional form is, or how it should be detected in observational data. The term is deployed when needed to explain why the bomb-^{14}C record does not match the Bern impulse-response prediction, and withheld when the record's actual shape contradicts what a Suess transition would produce. The present paper applies the missing quantitative criterion (§9.2.5, Table 9). Fossil-fuel dilution is undetectable in the atmospheric ^{14}C record from 1965 through 2010. It emerges only after 2010, as a small tail correction once the bomb perturbation has decayed below ~8% of its peak: the Fossil Fuel Tail effect (Fig. 3(a), bottom). The dominant driver of the entire 1965–2019 decline is gross-flux exchange at $\tau_{res} = 3.8$ yr, directly measured from the bomb-^{14}C record. The $\delta^{13}C_I$ mass-balance analysis of Koutsoyiannis (2024b) [73] (see also Cohler & Soon (2026a) [20]) independently confirms the result: the net atmospheric carbon-input signature is constant at −13.2‰ from 1978 to 2022, with proxy extension back 500 years to the Little Ice Age. The institutional Suess effect is falsified. The Fossil Fuel Tail effect is what remains.

1.3 The Bern impulse-response function: a model of an ensemble mean of models

The IPCC's quantitative case for long atmospheric CO_2 persistence proceeds in three stages, none of which is ever tested against the one direct measurement of the atmospheric impulse response that exists. First, complex Earth-system models (the CMIP-class ESMs) and Earth-system models of intermediate complexity (EMICs) are run with an idealized CO_2 pulse; their decay curves are taken as the physical basis. Second, the multi-model *ensemble mean* of those runs is computed and an analytic Bern impulse-response function (IRF) of the form $R(t) = a_0 + \sum_i a_i\, e^{-t/\tau_i}$ is

fitted to that mean [65]. Third, this fitted IRF becomes the computational engine for every CO_2-equivalence metric the IPCC publishes: the Global Warming Potentials (GWP) and Absolute Global Warming Potentials (AGWP) at the standard horizons of 20, 100, and 500 years, with the AGWP integral evaluated out to 1000 years. The reduced-form function is therefore not merely a model; it is a model of the arithmetic mean of an ensemble of models. That ensemble mean is itself statistically illegitimate: the participating models are neither independent nor identically distributed (they share forcings, parameterizations, and structural assumptions, and are tuned to the same thermodynamically meaningless global-mean surface temperature), so averaging them and quoting a narrow spread invokes the Central Limit Theorem under conditions where none of its premises hold [20].

The Bern IRF parameters are unstable across assessments, which a function tracking a real physical process would not be. The permanent airborne fraction a_0 changes in every assessment: $a_0 = 0.1369$ in AR2, 0.152 in AR3, 0.217 in AR4, and 0.2173 in AR5/AR6, and the timescales shift with it, the longest mode running from $\tau_1 = 171.0$ to 394.4 yr (Table 7) [63, 64, 65]. AR2 carries five exponential modes; AR3 through AR6 carry three. The architecture is moreover physically impossible: the parallel-eigenvalue structure assigns each timescale an independent pathway from the atmosphere directly to a distinct ocean reservoir, whereas CO_2 enters the ocean only at the surface and all deeper transport is strictly serial [20]. Joos et al. (2013) [65] Section 2.1 states explicitly that the fitted coefficients "have no direct process-based meaning, but are fitting parameters." IPCC AR5 (2013) [59] Box 6.1 nevertheless conveys a mode-to-process identification through juxtaposition: it describes qualitative biogeochemical processes (land uptake, ocean invasion, carbonate reaction, silicate-weathering-era processes) alongside its citation of Joos et al. (2013) [65], in a form that invites the reader to associate the Bern IRF timescales with those processes. Box 6.1 does not itself literally assert the mapping, so the surreptitious structure permits a Position C retreat when challenged (§8.3.2); but Joos 2013 (the very source Box 6.1 cites) explicitly denies the association. Box 6.1 moreover refers to the [65] pulse-response output as "data," although [65] is a multi-model modeling study containing no empirical measurement: its curves are explicitly the multi-model mean with a ±2 standard-deviation ensemble spread, not data. The IPCC invokes "policy-relevant timescales" to motivate the reduced-form approach, but no quantitative definition of that phrase appears in the assessments, and the same machinery is then used to make claims of multi-millennial persistence (15–40% of a pulse airborne beyond 1000 years), far outside any horizon the phrase could plausibly bound. AR6 nominally moved its carbon-cycle assessment from the Bern IRF to CMIP6 ESM ensembles in Chapter 5 while retaining the AR5 Bern IRF in Chapter 7 for the GWP and AGWP metrics that policy actually uses, so the same invalid function remains operative.

1.4 Bomb-^{14}C: the only direct test of the atmospheric impulse response

At no stage is any of this compared to data. The bomb-^{14}C decay is the only direct empirical test of the atmospheric CO_2 impulse response that exists, and it falsifies the entire construction. The observed decay is a single exponential; the Bern IRF diverges from it within the first year and monotonically thereafter, at every AR2 through AR6 parameter choice, with or without a permanent fraction. By 2019 the AR2, AR3, AR4, and AR5/AR6 Bern functions predict that 33, 35, 42, and 48% respectively of the bomb-^{14}C pulse remains airborne, while the observed value is 4.3% (Table 7, Figure 2) [55, 72, 20]. The IPCC has never plotted its ESM or Bern IRF pulse decay curves against the bomb-^{14}C record in any assessment through AR6. Joos et al. (2013) [65], which supplies the Bern IRF fit used from AR5 forward, presents eight figures of multi-model pulse decays; none shows the ensemble against atmospheric $\Delta^{14}C$ observations. The Joos protocol uses bomb-^{14}C only for internal calibration: its Appendix A records that Bern-SAR "model parameters of the box-diffusion-type ocean model were determined such that the model reproduces the oceanic distribution of natural and bomb-produced radiocarbon," and that MESMO's "interior

ocean ventilation is well calibrated against natural radiocarbon on centennial timescale and against transient anthropogenic tracers on decadal timescales." The calibrated models are never plotted against the atmospheric bomb-^{14}C decay that constrains the same operator. When the omission was raised in the AR6 First Order Draft review by S. E. Schwartz (comment 25656), with the specific observation that observed IRFs "would be very different from those presented by Joos et al. (2013) [65]," the response was "Noted. We are exploring such an addition" [61]. No such addition appears in the published AR6 WG1 Chapter 5, which contains 33 figures, none of which is a CO_2 pulse impulse response. The one comparison that would expose the Bern architecture directly is the one that has never been drawn.

1.5 Why the ^{14}C tracer constrains the bulk-CO_2 response

The empirical falsification by bomb-^{14}C has been resisted in the institutional literature by a cluster of related fallacies that all assert, in different forms, that the bomb-^{14}C tracer behaves differently from atmospheric CO_2 and therefore cannot test claims about CO_2 persistence [17, 44, 45, 65]. We prove in Appendix B.2 the operator-equivalence theorem: any slow eigenmode present in the atmospheric impulse response for bulk CO_2 must appear in the bomb-^{14}C decay record at the same eigenvalue, because both are governed by the same first-order linear mass-balance operator acting on the same reservoir. A multi-exponential impulse response for bulk CO_2 requires a multi-exponential bomb-^{14}C decay. We show in this paper that the bomb-^{14}C decay is single-exponential; therefore the bulk-CO_2 impulse response is single-exponential. The tracer-is-not-the-bulk argument cannot escape this constraint without abandoning conservation of mass [111, 72].

Section 3 provides a comprehensive foreclosure of every form of the tracer-versus-bulk fallacy, demonstrating that the institutional position rests on a single conflation between the true statement that $^{14}CO_2$ is a trace species of negligible mass and the false statement that the mass-balance operator transports $^{14}CO_2$ and $^{12}CO_2$ at substantially different rates, and that no first-principles derivation, no published atmospheric measurement, and no coherent physical mechanism supports the asymmetry the institutional position requires.

1.6 Logical architecture of the τ_{rel} determination

Because the compilation-bias correction of the Hua et al. (2022) [55] record is prerequisite to the $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr authoritative recovery, and because superficial readings routinely conflate physics-grounded bias subtraction with parameter tuning, we summarize up front the complete logical chain (§9.1) and the station-by-station archival audit of the Hua compilation (§9.2) that underlies it. The procedure consists of two operations, neither of which is curve fitting: removal of physically-impossible compilation artifacts, and subtraction of measured constant biases on the stable-methodology plateaus between them. The single-exponential shape is established empirically *before* either operation, from the unedited 55-point global record R_{a} and seven unedited zonal subsets at $R^2 > 99\%$. Given that shape, the residuals r_{55} against the 55-point fit R_{55} are mathematically forced to consist of flat plateaus between documented Hua methodology-change events. The plateau *boundaries*, and equivalently the excluded artifact stretches, are determined by independent archival audit of those events (station exits such as Vermunt 1986, tree-ring-to-atmospheric weighting transitions, and retrospective data revisions such as the Wellington 1990–1993 and 1995–2005 revisions documented in Hua et al. (2022) [55] and Turnbull et al. (2017) [121]), identified *before* any residual analysis, and are then confirmed independently by a physics-derived smoothness criterion computed from independent atmospheric parameters with no reference to the residual data: the excluded stretches exceed the first-principles mass-balance bound by factors of 6 to 14, magnitudes no atmospheric physical process can produce.

The two routes, provenance and physics, converge on the same set of intervals. The plateau *levels* are arithmetic averages of residuals across those externally-bounded intervals, and therefore comprise

direct empirical measurements of Hua's compilation biases accumulated during all prior artifact regions, analogous to a calibrated instrument offset, not parameters fitted to optimize anything. The within-plateau flatness is then verified as a falsifiable prediction of the established exponential form: had the form been wrong or the boundaries misplaced, the residuals would not be flat; they are. No step fits a function to the data, adjusts a parameter to improve a fit, or uses the recovered τ_{rel} as an input; the recovered value is the *output* of bias subtraction at independently-determined locations. Readers who wish to evaluate the non-circular logical sequencing of the analysis should consult §9.1 and §9.2 before proceeding.

1.7 Scope, diversions set aside, and overview of this paper

A separate "what-aboutism" diversion in this literature invokes the observed declining atmospheric O_2/N_2 ratio as independent evidence for fossil-fuel attribution of the bulk CO_2 rise [70, 69, 6]. The O_2/N_2 argument is irrelevant to the present analysis. The bomb-^{14}C record and the conclusions derived from it have no dependence on the O_2/N_2 ratio or on its interpretation, and the question of whether the observed atmospheric oxygen decline is predominantly attributable to fossil combustion or to oceanic and biospheric processes does not enter this paper. Beyond that, the O_2/N_2 decline is itself slow on any policy-relevant timescale and on any timescale relevant to human beings, regardless of its attribution.

The IPCC anthropogenic-attribution framework, resting on the Bern impulse-response construction examined above, has been repeated across two or three generations of working scientists and tens of thousands of peer-reviewed papers, accumulating the rhetorical weight of established fact without acquiring the empirical or mathematical foundation that would justify it.

In this paper we subject the IPCC framework to direct empirical test using the only available observational record of atmospheric CO_2 impulse response: the bomb-produced ^{14}C pulse, measured at clean-air sites and in tree rings from the mid-1960s peak through 2019 and recently compiled by Hua et al. (2022) [55]. We derive from first principles, using only conservation of mass and independently measured atmospheric quantities, the mathematical relationship between the residence time of CO_2 molecules in the atmosphere (∼4 yr from gross-flux mass balance [115, 108, 37, 47, 48, 7, 8, 111, 72, 89, 122, 20]) and the bomb-^{14}C decay time (∼18 yr from the Hua data). The relationship has no adjustable parameters. The two timescales emerge from a single first-order linear ODE and are connected through the deep-mixed leak fraction $\bar{\phi}_{\mathrm{slow}} = 0.203 \pm 0.006$ predicted from independent atmospheric measurements, in agreement with the independent two-box value $\phi_{\mathrm{slow}}^{\mathrm{2box}} = 0.199 \pm 0.008$. The bomb-^{14}C data confirm a clean single-exponential atmospheric decay with characteristic time $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr. The IPCC multi-exponential impulse response, including the AR2–AR6 Bern parameterizations [56, 57, 58, 59, 60], deviates from the observed decay within the first year and grows monotonically worse: the AR3/AR4 and AR5/AR6 functions exceed the observed remaining fraction by a factor of ∼10× by the end of the 54-year observation window and by ∼300× near 120 years, an order of magnitude short of the millennial horizon on which the persistence claims rest. No multi-exponential impulse-response function, with or without a permanent fraction, is mathematically compatible with the observed single-exponential decay; we prove this structurally. The residence-time-vs.-adjustment-time distinction has no operational definition. The ≥ 1000-year persistence claim is mathematically incompatible with the bomb-^{14}C data. The "Suess-effect" framework, which predicts a regime change in the bomb-^{14}C decay as fossil-fuel dilution takes over from non-fossil-fuel-exchange decay [68, 44, 45, 55], fails because no such regime change is detectable.

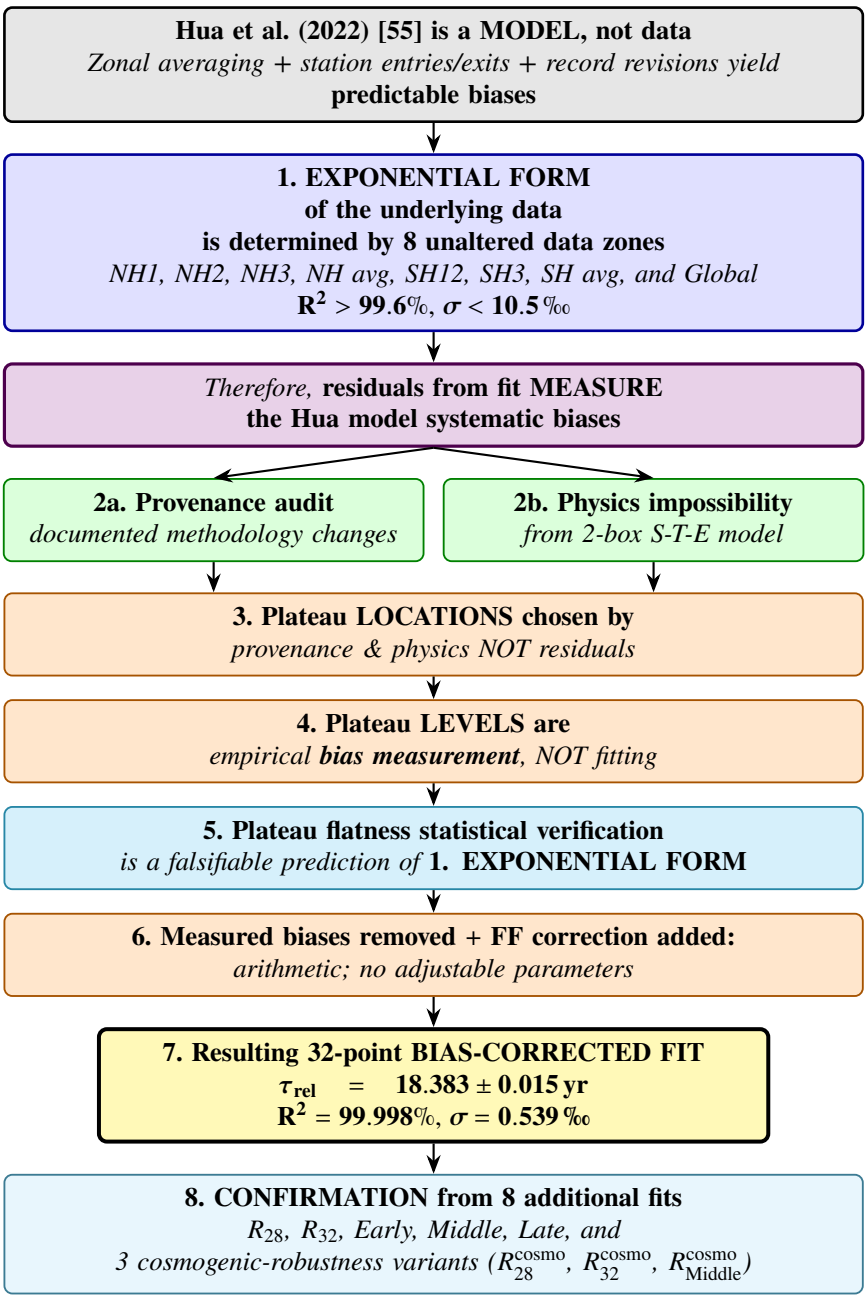


Figure 1: ***Falsifiable logical architecture of the Bomb-^{14}C τ_{rel} determination.*** *Hua et al. (2022) [55] presents a model built from averaging a changing station composition with retrospective record revisions; methodology changes predictably induce biases. Despite the biases, every one of the eight independent fits to the raw uncorrected data is clearly exponential (1). This makes the residuals themselves measurements of the biases, not free parameters. Plateau LOCATIONS come directly from provenance audit + confirmation by physical impossibility (2a, 2b), never from the residuals. Plateau LEVELS are arithmetic means (4). Within-plateau flatness (5) is a falsifiable consequence of the exponential form determined in Step 1: had the form been wrong or the plateau locations misplaced, the residuals would not be flat. They are. The bias-corrected fit (7) follows from bias subtraction (6) and is confirmed by consistency across eight additional fits (8): five primary (R_{28}, R_{32}, Early, Middle, Late) and three cosmogenic-robustness variants.* ***No step fits parameters to optimize; no step uses the output as input.*** *Full quantitative detail in §9.1 and §9.2.*

2. Review of previous works

The atmospheric residence time of CO_2, derived directly from observed gross fluxes through the atmospheric reservoir, has been consistently estimated in the peer-reviewed literature at approxi-

mately 3 to 5 yr [115, 108, 37, 47, 48, 7, 8, 111, 72, 89, 122, 20]. The *e*-folding decay time of the bomb-produced ^{14}C pulse, observed at multiple clean-air sites and in tree rings from peak in 1964 or 1965 through 2019, has been reported at values between 10 and 19 yr by various authors [94, 37, 47, 48, 72, 111], with the spread arising primarily from differences in fitting range, hemisphere, and data compilation. The numerical relationship between the short residence time of CO_2 in general and the longer bomb-^{14}C decay time has remained unresolved in the literature, with both sides of the long-standing controversy advancing positions that fall short of a quantitative derivation.

The short-residence-time literature has invoked the bomb-^{14}C record as empirical support for residence times of order 4–5 yr without a first-principles derivation of the relationship between the two measured timescales. Starr (1993) [112] treated the bomb-^{14}C decay history as "consistent with" a residence time of about 5 yr, alongside the seasonal photosynthesis swing, without distinguishing the two quantities. Essenhigh (2009) [37] similarly cited the bomb-^{14}C decay as one of several lines of evidence supporting short residence time. Berry (2019) [7] fit the bomb-^{14}C data to a single exponential, obtained 16.5 yr, and interpreted this as the residence time directly, without distinguishing it from the gross-flux mass-balance value of ∼4 yr. Stallinga (2023) [111] obtained 14.6 yr from older bomb-^{14}C data papers while separately citing 5 yr from gross-flux mass balance, proved that residence time and adjustment time are equal in any first-order linear reservoir, but did not fully resolve the empirical discrepancy. Koutsoyiannis (2024a) [72] obtained 17.2 yr from bomb-^{14}C and 3.68–3.91 yr from full Refined Reservoir Routing (RRR) fits to Mauna Loa and Barrow concentration data, and identified 17.2 yr as an upper bound on the ordinary CO_2 response time, conjecturing that the gap may be due to photosynthetic fractionation, stratospheric injection delay, and continued ^{14}C natural and human production. We address Koutsoyiannis's conjectures in Section 11.

The mainstream attribution literature asserts the opposite position: the bomb-^{14}C decay measures a quantity fundamentally distinct from bulk CO_2 residence time, but it does so without deriving an alternative value or providing an operational definition. Cawley (2011) [17], responding to Essenhigh, reframed the question in terms of net mass balance ("if natural sources exceed natural sinks, then nature must be a net source") without addressing the bomb-^{14}C argument directly or providing an alternative residence time derivation. Andrews (2020) [1] claimed the bomb-^{14}C analysis is mistaken and "confirms the prediction of a conventional model of the carbon cycle," but, as noted by Koutsoyiannis (2024a) [72], provided no calculation to support this assertion. IPCC AR5 (2013) [59] and IPCC AR6 (2021) [60] treat the residence-time–adjustment-time distinction as established despite the absence of any first-principles definition of "adjustment time": no governing equation, no operational measurement procedure, and no derivation from physical law are offered. [60] (Annex VII: Glossary, p. 2237) states:

> ***Turnover time (T)*** *(also called global atmospheric lifetime) is the ratio of the mass M of a* ***reservoir*** *(e.g., a gaseous compound in the* ***atmosphere****) and the total rate of removal S from the* ***reservoir****: $T = M/S$. For each removal process, separate turnover times can be defined. In soil carbon biology, this is referred to as mean residence time.*

but continues with "adjustment time" rhetorically as:

> *...****Response time or adjustment time*** *(T_a) is the time scale characterizing the decay of an instantaneous pulse input into the* ***reservoir****...In simple cases, where the global removal of the compound is directly proportional to the total mass of the reservoir, the adjustment time equals the* ***turnover time****: $T = T_a$.*

providing no operational formula. The glossary entry then continues with a remarkable passage that asserts, without derivation, the central claim of the IPCC carbon-cycle framework:

> *Carbon Dioxide (CO_2) is an extreme example. Its turnover time is only about 4 years because of the rapid exchange between the atmosphere and the ocean and terrestrial biota. However, a large part of that CO_2 is returned to the atmosphere within a few years. The adjustment time of CO_2 in the atmosphere is determined from the rates of removal of carbon by a range of processes with time scales from months to hundreds of thousands of years. As a result, 15 to 40% of an emitted CO_2 pulse will remain in the atmosphere longer than 1,000 years, 10 to 25% will remain about ten thousand years, and the rest will be removed over several hundred thousand years.*

No equation accompanies this passage. No physical law is cited. No mechanism is identified by which a well-mixed atmospheric gas with a measured turnover time of 4 years (a value the IPCC itself accepts) can retain 15–40% of a pulse for a thousand years. The word "extreme" is offered in place of a derivation, as though labeling the discrepancy excuses the absence of any explanation for it. The reader is asked to accept a four-order-of-magnitude gap between turnover time and persistence time on the authority of an adjective.

The numerical values used by the IPCC to support this assertion derive entirely from the multi-exponential IRF form originally introduced by Maier-Reimer and Hasselmann (1987) [80] and subsequently operationalized into the IPCC Bern parameterizations by Joos et al. [63, 64, 65] through various models or multi-model outputs. The parameters thus reflect the assumptions and outputs of the various underlying climate models, not *any* empirical data [20]. The bomb-^{14}C decay, which is the only direct empirical test of CO_2 atmospheric impulse-response structure available, shows a clean single exponential mathematically incompatible with any of the IPCC Bern IRF models [55, 72, 20].

To summarize the state of the literature: both sides of the residence-time-vs.-adjustment-time controversy have invoked the bomb-^{14}C empirical record without quantitatively deriving its relationship to gross-flux mass-balance residence time. The short-residence-time literature has treated the two quantities loosely as equivalent or as mutually supporting. The IPCC attribution literature has asserted them as distinct, without any operational definition or empirical demonstration. Stallinga (2023) [111] proves mathematically that residence time and adjustment time are equal in any first-order linear reservoir, but does not resolve the discrepancy between the turnover time of CO_2 molecules and the bomb-^{14}C e-folding time.

In this paper, we derive the relationship between the two timescales directly from atmospheric CO_2 mass balance, using only conservation of mass, the Mauna Loa CO_2 inventory record, IPCC gross non-fossil-fuel flux estimates, and the Hua et al. (2022) [55] global Δ^{14}C compilation. The result shows that the two timescales emerge from a single first-order linear ODE and are connected through the deep-mixed leak fraction and the mass-balance reconstruction of the gross inflow rate. The single-exponential shape of the observed decay is shown to be observational confirmation of proportional growth between gross non-fossil-fuel flux and atmospheric inventory, consistent with the temperature-driven biospheric expansion mechanism and inconsistent with anthropogenic CO_2 as the driver of atmospheric CO_2 growth [18, 20].

3. The Tracer-versus-Bulk Fallacy: A Comprehensive Foreclosure

The empirical falsification of the Bern impulse-response framework by the bomb-^{14}C record has, for decades, been deflected by an institutional response that asserts in various forms that the bomb-^{14}C tracer behaves differently from atmospheric CO_2 and therefore cannot test claims about CO_2 persistence [17, 44, 45, 65]. This section foreclosed that response at every level on which it has been mounted: chemical, physical, mathematical, and linguistic. The result, stated in advance: the institutional position rests on a single conflation between a true statement about isotopic abundance and a false statement about transport-operator asymmetry. No first-principles

derivation, no published atmospheric measurement, and no coherent physical mechanism supports the asymmetry the institutional position requires. The conflation is the sole foundation. Once named and refuted, the position has no remaining support.

3.1 Chemical identity: one species, one mixture

Atmospheric CO_2 is one molecular species, carbon dioxide, present as a mixture of three stable and one radioactive isotopologue. Natural abundance ratios are approximately 0.9889 for $^{12}CO_2$, 0.0111 for $^{13}CO_2$, and 1.2×10^{-12} for $^{14}CO_2$ [113]. There is no chemical referent "bulk CO_2" distinct from the mixture. The reservoir is the mixture. Mass-balance equations for atmospheric carbon are equations for the mixture as a whole, with the partial inventory of each isotopologue tracked by the same equations multiplied by the isotopic ratio of the relevant reservoir.

The institutional vocabulary that opposes "bulk CO_2" to "the ^{14}C tracer" as if they were separately-transported quantities is therefore not a description of two physical objects. It is a description of one physical object under two names, with the duplication smuggling in an asymmetry that the chemistry does not contain. Every subsequent argument in this section operates on the established fact that there is one species and one reservoir, and that the partial inventory of $^{14}CO_2$ is a property of the same reservoir whose total inventory is what climate science calls "atmospheric CO_2."

3.2 The air–sea gas-exchange operator

Gas exchange across the air–sea interface is governed by molecular diffusion through a thin boundary layer, with the volumetric flux per unit area set by the piston velocity k_w and the partial-pressure difference ΔpCO_2 between atmospheric CO_2 and the equilibrium CO_2 partial pressure of surface seawater [123, 116]:

$$F_{\text{gas}} = k_w \, \Delta pCO_2 \tag{1}$$

The piston velocity k_w depends on wind speed, surface roughness, and the Schmidt number of CO_2 in seawater. None of these depends on isotopic composition except through the small Schmidt-number dependence on molecular mass, which produces a kinetic isotope effect of approximately 0.7 ‰ between $^{14}CO_2$ and $^{12}CO_2$ [87, 40].

The operator transporting CO_2 across the interface is therefore identical for all isotopologues to within per-mil-scale fractionation corrections. There is no physical mechanism at the interface that produces an order-of-magnitude difference in transport rate between $^{14}CO_2$ and $^{12}CO_2$. Identical gas-phase mixing, identical molecular diffusion, identical solubility (to per-mil), identical chemistry on dissolution: the operator acts on the mixture as the mixture, transporting the partial inventory of each isotopologue at the same rate as it transports the total inventory.

3.3 The Revelle factor: what it constrains and what it does not

The Revelle factor, introduced by Revelle & Suess (1957) [99] and denoted R, is a *local intensive* property of a parcel of seawater. It is the dimensionless ratio

$$R(\mathbf{x}, t) = \left. \frac{\partial \ln pCO_2}{\partial \ln \text{DIC}} \right|_{\text{Alk}} \tag{2}$$

defined at a point $\mathbf{x}$ at instant t, where pCO_2 is the partial pressure of dissolved aqueous CO_2 in equilibrium with that parcel, DIC is its dissolved inorganic carbon, and Alk its total alkalinity. The factor is derived from the carbonate equilibrium constants K_1 and K_2 evaluated at the local temperature, salinity, DIC, and alkalinity. Reported values of approximately 9–15 refer to measurements at a small number of specific oceanic locations [116]. Like any intensive property of a non-equilibrium spatially-extended system, $R(\mathbf{x}, t)$ has no aggregate global value [38, 23], and its measured behavior at any one location places no constraint on the carbon-uptake dynamics of the global ocean as a whole.

Three facts about $R(\mathbf{x}, t)$ are decisive for the present foreclosure.

First, $R(\mathbf{x}, t)$ is a local equilibrium chemical property of a seawater parcel. It is intensive. It does not aggregate over the global ocean by any operation grounded in physics, for the same reasons that any intensive property fails to aggregate across a non-equilibrium spatially-extended system [38, 23]. The published values reported in the literature are point measurements at specific oceanic locations [116]. The institutional carbon-cycle framework extrapolates these point values to global ocean uptake behavior. This extrapolation has no measured global counterpart, no derivation from global ocean dynamics, and no physical justification beyond assertion. It is the same local-to-global fallacy that disqualifies global mean surface temperature as a physical quantity [38], applied to a different intensive property. The fact that R has been measured at a small number of points does not constitute evidence that any global Revelle effect has ever operated, is now operating, or will ever operate. No such measurement exists.

Second, even at a local point, $R(\mathbf{x}, t)$ describes a relationship between equilibrium concentrations. It is a thermodynamic-equilibrium chemical property at that point. It does not describe a rate process. It does not appear in Eq. 1 or in any other formulation of the gas-exchange operator. It cannot, by its mathematical structure, produce a difference in the rate at which CO_2 molecules cross the air–sea interface at that point, because the rate at which molecules cross the interface is governed by k_w and ΔpCO_2, neither of which is a function of R.

Third, the carbonate equilibrium constants K_1 and K_2 depend on the electronic structure of the C–O bond and the protonation states of HCO_3^- and CO_3^{2-}. None of these structural quantities depends on whether the central carbon atom is ^{12}C, ^{13}C, or ^{14}C, to within the per-mil precision of measurable carbonate equilibrium isotope effects [87]. The local Revelle factor for $^{14}CO_2$ and for $^{12}CO_2$ therefore differs at the per-mil level, not at the order-of-magnitude level that the institutional asymmetry between bomb-^{14}C decay (18 yr) and the Bern IRF (centuries and millenia) would require.

The Revelle factor, accurately characterized as a local intensive chemical-equilibrium property of seawater parcels at specific points where it has been measured, cannot produce differential transport rates between isotopologues. It does not describe transport. It does not discriminate between isotopologues. It has no measured global referent. The institutional invocation of "the Revelle factor" as a mechanism distinguishing tracer dynamics from bulk dynamics at the global atmospheric scale is therefore unsupported on three independent grounds: it conflates a local intensive property with a global aggregate that does not exist, it conflates an equilibrium chemical property with a transport rate, and it conflates a property that does not discriminate between isotopologues with one that would have to.

3.4 The two statements: negligible mass versus differential operator

The institutional position has been sustained by a conflation between two statements that look similar in casual writing but are physically distinct. State them precisely.

Statement A. *$^{14}CO_2$ is a trace species, present at approximately 10^{-12} of total atmospheric CO_2. A bomb-^{14}C perturbation does not measurably alter the bulk thermodynamic or chemical state of the atmosphere or ocean carbonate system.*

Statement B. *The mass-balance operator transporting CO_2 between atmosphere, ocean, and biosphere acts on $^{14}CO_2$ at one rate and on $^{12}CO_2$ at a substantially different rate, producing different eigenvalues for the two isotopologues in the linearized atmospheric impulse response.*

Statement A is true. It is the standard observation about isotopic tracers, and it follows immediately from the natural abundance ratio. Statement A is consistent with operator equivalence: a trace species rides on the dominant species at the same operator eigenvalues, because the operator acts on the mixture as the mixture. Statement A does not assert differential transport. It asserts negligible

perturbation to bulk thermodynamic state.

Statement B is the claim the institutional position requires in order to reconcile fast bomb-^{14}C decay (τ_{rel} = 18.38 yr) with the slow Bern IRF (multi-exponential decay with permanent fraction). Statement B has no first-principles physical derivation. No paper in the institutional literature provides one. The fact that no such derivation exists is verifiable by direct inspection of the lineage of foundational papers [3, 99, 95, 63, 65]; none contains an equation of the form "$^{14}CO_2$ transport velocity equals X, $^{12}CO_2$ transport velocity equals Y, $X \neq Y$ because of physical mechanism M."

The institutional defense of Statement B reduces, in every instance the present authors have located, to one of the following:

1. Restatement of Statement A as if it implied Statement B (the load-bearing conflation).
2. Invocation of the Revelle factor as a buffering mechanism, without specifying how the Revelle factor produces differential transport (refuted in §3.3).
3. Citation to a model implementation in which different exchange velocities are coded for the two isotopologues, without specifying what physical mechanism the coded asymmetry represents (addressed in §3.5).

The conflation in route 1 is the fallacy this section is designed to foreclose. The Revelle invocation in route 2 has been refuted. The model citation in route 3 is examined next.

3.5 Model implementations: hand-coded asymmetry without physical derivation

The Bern lineage of carbon-cycle box models [95, 63, 65] implements the atmospheric CO_2 reservoir as a single state variable for total atmospheric carbon, with additional state variables for isotopic tracers when isotopic observations are to be simulated. In the isotopically resolved configurations, the flux equations for $^{14}CO_2$ are constructed by multiplying the $^{12}CO_2$ fluxes by appropriate isotopic ratios and applying fractionation factors. This construction, taken at face value, encodes operator equivalence: the same operator transports both isotopologues, and the isotopic ratio of each reservoir evolves under the same first-order linear dynamics that govern total carbon.

The departure from operator equivalence enters through model choices that are external to the air–sea exchange physics. Specifically, in the institutional configurations, the rate of net uptake of ^{12}C-DIC perturbations by the global surface ocean is modulated by a parameterization labeled "Revelle buffering," while the gross exchange of the isotopic label is not so modulated. The justification offered is that Revelle buffering applies to net DIC perturbations but not to label exchange. This justification is incorrect on two grounds. First, the Revelle factor is a local intensive property whose extrapolation to a global ocean buffering parameter has no empirical or physical basis (§3.3); the "buffering" invoked in the model is a parameterized global quantity that has no measured global counterpart. Second, even at the local points where R has been measured, it describes the equilibrium relationship between DIC and pCO_2, not the rate of gross-flux exchange across the interface (§3.3). The institutional asymmetry between ^{14}C exchange and ^{12}C uptake in the model is therefore justified by a parameterization that conflates local intensive chemistry with global transport dynamics at every step.

The operational result of the institutional configuration is that the model produces fast bomb-^{14}C decay through gross flux exchange while producing slow ^{12}C bulk persistence through Revelle-buffered net uptake. The two timescales emerge from the same model not because the model encodes a derived physical asymmetry but because the model encodes a parametric asymmetry in the rate equations for the two isotopologues. The parametric asymmetry is justified by Revelle-versus-tracer language whose physical content has been refuted in §3.3 and §3.4.

This is the second form of the load-bearing fallacy. The first form is the conflation between Statement A and Statement B. The second form is the model implementation that requires Statement B and justifies it by language that supports only Statement A. Neither form has a physical derivation. Both have been sustained by repetition rather than by derivation in a citation network that all points back to the same parametric choices originating in Revelle & Suess (1957) [99] and Oeschger et al. (1975) [95] and propagated through [63, 65] into the IPCC assessment reports [59, 60].

The internal contradiction of the institutional position is visible in the canonical institutional textbook itself. Sarmiento and Gruber (2006) [104], Panel 1.2.1, opens with the unqualified statement that "radiocarbon follows the same biological and mixing pathways as the stable carbon isotopes ^{12}C and ^{13}C," with the only stated difference being radioactive decay at $\lambda = 1/8267\ \mathrm{yr}^{-1}$, a per-mil-scale effect on the 54-year bomb-^{14}C decay history from Hua et al. (2022) [55]. The panel then derives the volumetric exchange rate between surface and deep ocean from the surface-to-deep radiocarbon disequilibrium, a calculation that depends on the opening statement being true and that is meaningless if it is false. The institutional textbook therefore commits explicitly to operator equivalence in the precise form proven in Appendix B.2: $^{14}CO_2$ and $^{12}CO_2$ are transported by the same operator, with the ^{14}C radioactive decay constant entering only as a per-mil-scale correction. The Bern carbon-cycle-model framework, which requires the operator to act differently on the two isotopologues at the order-of-magnitude level, contradicts the textbook canon by which the same research community calibrates its ocean exchange rates.

The contradiction is internal to the institutional position and has stood unresolved in print for four decades. The multi-exponential ocean carbon impulse response was introduced by Maier-Reimer and Hasselmann (1987) [80], propagated through the Bern lineage [63, 65] into the IPCC assessment reports [56, 57, 58, 59, 60], and remains the foundation of every downstream carbon-cycle metric currently in use. Throughout this period, the operator-equivalence principle that the framework violates has been the unstated foundation of the parallel ocean-mixing-rate literature, made explicit in the canonical textbook [104] two decades ago. The late Jorge Sarmiento (1946–2026) served as Director of Princeton's Atmospheric and Ocean Sciences Program (1980–1990, 2006–2015) and founding Director of the Princeton/GFDL Cooperative Institute for Climate Science (2003) and the Cooperative Institute for Modeling the Earth System (2018), establishing Princeton/GFDL as a principal center of institutional ocean carbon modeling for over four decades. Nicolas Gruber is Full Professor of Environmental Physics and Chair of the Center for Climate Systems Modeling at ETH Zürich, where his group contributes to the Global Carbon Project's annual ocean carbon assessments [42]. Gruber completed his PhD at the University of Bern in 1997, within the institutional lineage that produced the Bern impulse-response framework, and his postdoctoral work at Princeton in collaboration with Sarmiento. Their published work is the principal observational and modeling foundation cited by the IPCC WGI carbon-cycle chapters [59, 60, 14] that propagate the Bern IRF framework. The textbook those chapters cite states explicitly that radiocarbon and bulk carbon *follow the same pathways*. The framework those chapters propagate requires that the two isotopologues be transported on timescales differing by orders of magnitude, which is only possible if the pathways are not the same. The contradiction has stood for four decades, sustained by senior figures who have produced neither a derivation of the asymmetry the framework requires nor an atmospheric measurement that would support it. Resolving it would require either retracting Panel 1.2.1 of Sarmiento and Gruber (2006) [104], which would remove the institutional method for calibrating ocean exchange rates, or retracting the Bern IRF framework, which would remove the foundation of every downstream IPCC carbon-cycle metric. Neither retraction has been performed. The institutional position is therefore maintained by sustained non-acknowledgment of a contradiction that has been implicit in the institutional carbon-cycle framework for four decades and explicit in its own canonical textbook for two. The mechanism by which a research community holds two contradictory positions simultaneously, each defended within its own discursive context, with the contradiction sustained by vocabulary

that keeps the contexts separate, is the Orwellian doublethink that Orwell (1949) [96] identified as the characteristic failure mode of institutional discourse under sustained political and economic pressure. The tracer-versus-bulk fallacy is doublethink in the precise sense: the principle that radiocarbon follows the same pathways as bulk carbon is taught in the textbook used to calibrate ocean exchange rates, while the framework that requires the opposite is propagated in the UN IPCC assessment reports used to set international climate policy, and neither the textbook nor the framework is permitted to acknowledge the other.

3.6 Operator equivalence as the closing mathematical constraint

The mathematical foundation of operator equivalence is proven in Appendix B.2. The statement, applied to the present question: any first-order linear mass-balance system on atmospheric CO_2 admits a single eigenvalue spectrum, and that spectrum applies to every isotopologue identically up to per-mil fractionation corrections. A multi-exponential impulse response for total atmospheric CO_2 requires a multi-exponential bomb-^{14}C decay with the same exponents. A single-exponential bomb-^{14}C decay forbids any slow eigenmode in the total atmospheric CO_2 impulse response that does not appear in the bomb-^{14}C record at the same eigenvalue.

The bomb-^{14}C record, analyzed in Section 8 of this paper, is single-exponential at $\tau_{\text{rel}} = 18.383 \pm 0.015$ yr across 52 years of data, with $R^2 = 99.998\%$ and residual standard deviation $\sigma = 0.539‰$. The Bern IRF, with its multi-exponential decay and permanent fraction, is therefore forbidden by the bomb-^{14}C record via operator equivalence. The institutional asymmetry that would permit the Bern IRF to coexist with the bomb-^{14}C record requires Statement B, which has been refuted at every level on which it has been articulated.

The mathematical constraint is therefore closed. Mass conservation, applied to a mixture of isotopologues transported by a single operator, forbids the asymmetry the institutional position requires.

3.7 Foreclosure of remaining escape routes

The following enumerates the institutional escape routes that have appeared, or might appear, in defense of the tracer-versus-bulk asymmetry, and refutes each. The list is not claimed to be exhaustive of all conceivable future arguments, but it covers every route the present authors have located in the published literature [17, 44, 45, 65, 77, 50].

Escape 1: Bomb-^{14}C measures gross flux, Bern IRF describes net flux. Refutation: Gross flux determines the operator eigenvalue. Net flux is the steady-state response under continuous forcing. The eigenvalue is what the bomb-^{14}C decay reports, because the bomb pulse was an impulse perturbation and the decay is the relaxation toward the unperturbed steady state. Different forcing scenarios produce different steady states, not different eigenvalues. The Bern IRF encodes eigenvalues; bomb-^{14}C measures the eigenvalue. They are the same kind of object and must agree.

Escape 2: Different reservoirs have different timescales; ^{14}C samples them differently because of its initial distribution. Refutation: Operator equivalence holds eigenvalue-by-eigenvalue. Every eigenmode present in the total-CO_2 dynamics must appear in the $^{14}CO_2$ dynamics at the same eigenvalue, regardless of initial conditions. Initial conditions determine the amplitude with which each eigenmode is excited; they do not create or remove eigenmodes. A slow eigenmode absent from the $^{14}CO_2$ record cannot be present in the total-CO_2 dynamics.

Escape 3: Carbonate chemistry buffers ^{12}C-DIC perturbations but does not buffer ^{14}C-DIC perturbations because the latter are negligible. Refutation: The Revelle factor is a local intensive equilibrium property of seawater parcels at specific points where it has been measured (§3.3). It does not aggregate to a global ocean buffering parameter, has no measured global counterpart, and has never been demonstrated to operate at the global scale, in 4.5 billion years of Earth's carbon

cycle, by any direct measurement. Its institutional invocation as a global buffering mechanism is a local-to-global conflation of the same form that disqualifies global mean surface temperature [38]. Even granting the local Revelle factor at the points where it has been measured, it governs the equilibrium partition between dissolved aqueous CO_2 and total DIC at that point. It does not govern the rate of air–sea exchange (§3.3). A small $^{14}CO_2$ perturbation rides on the same gas-exchange operator at the same eigenvalue as a large $^{12}CO_2$ perturbation would. The difference in perturbation amplitude does not produce a difference in eigenvalue, because the atmospheric carbon reservoir responds linearly to perturbations on the timescales relevant to the bomb-^{14}C record. This linearity is not assumed in the present paper; it is empirically demonstrated by the bomb-^{14}C decay itself, which is single-exponential at $R^2 = 99.998\%$ across 52 years and an amplitude range of a factor of 13 (Figure 3(b)). A nonlinear reservoir would produce amplitude-dependent decay rates and curvature in the log-transformed decay; neither is observed. The institutional invocation of Revelle-driven nonlinearity in surface ocean uptake is therefore not merely physically misapplied (§3.3) but empirically falsified by the single-exponential bomb-^{14}C record on which the operator-equivalence constraint is anchored. The consensus claim that the reservoir is nonlinear is itself circular: it is asserted by appeal to local Revelle chemistry, which is then extrapolated to the global reservoir without empirical support, while the direct empirical test of the global reservoir's linearity (the bomb-^{14}C decay) is either dismissed via the tracer-versus-bulk fallacy this section has refuted, or ignored.

Escape 4: Biosphere fractionation produces differential ^{14}C behavior. Refutation: Biospheric and oceanic fractionation produce per-mil isotopic offsets, characterized by Farquhar et al. (1989) [40] for C3 photosynthesis and Mook (1986) [87] for inorganic aqueous chemistry. These are small corrections to the isotopic ratio at exchange. They do not produce order-of-magnitude eigenvalue differences in the transport operator, because they enter as small multiplicative corrections to the flux ratio of ^{14}C to ^{12}C at each interface, not as separate transport equations with separate eigenvalues.

Escape 5: The institutional models reproduce both the atmospheric CO_2 rise and the bomb-^{14}C decline, therefore the asymmetry is empirically validated. Refutation: The models are tuned to fit both observations under the institutional assumption that the atmospheric CO_2 rise is fossil-driven and net-uptake-limited. The fit is then cited as confirmation of the assumption that produced it. This is the circularity that the bomb-^{14}C analysis exposes. The bomb-^{14}C record, properly understood as a measurement of the operator eigenvalue, falsifies the assumption that the atmospheric CO_2 rise is net-uptake-limited at the multi-exponential Bern timescales, because the operator eigenvalue is $\tau_{\text{rel}} = 18.38$ yr and admits no slower mode. Joint fitting of two observations under a falsified assumption does not validate the assumption.

Escape 6: $^{14}CO_2$ is consumed by radioactive decay; this introduces a sink that does not act on $^{12}CO_2$. Refutation: The half-life of ^{14}C is 5,700 years [71]. The corresponding decay constant is $\lambda_{14} \equiv \ln 2/5{,}700\ \text{yr}^{-1} = 1.22 \times 10^{-4}\ \text{yr}^{-1}$. On the ∼50-year timescale of the bomb-^{14}C record, radioactive decay removes approximately 0.6% of the ^{14}C inventory. This correction is below the precision of the bomb-^{14}C measurement and is routinely included in the data reduction [113]. It is a per-mil-scale effect, not an order-of-magnitude effect, and cannot account for the discrepancy between $\tau_{\text{rel}} = 18.38$ yr and the multi-exponential Bern IRF.

Escape 7: The cosmogenic ^{14}C source produces additional $^{14}CO_2$ during the period under analysis, distorting the apparent decay. Refutation: The steady-state cosmogenic ^{14}C production rate of approximately 7.5 kg/yr [74, 82] is implicit in the pre-bomb atmospheric ^{14}C baseline on which the Hua compilation [55] is anchored, and is therefore already absorbed into the fitted exponential as a constant baseline term (§9.3.2). No separate correction for steady-state cosmogenic input is required. The primary fits reported in this paper (§9.3.3) apply no cosmogenic correction of any kind. A separate diagnostic correction of −1‰ at the two years 2009 and 2010 is examined in §9.3.2 as a robustness check, testing whether a physically plausible cosmogenic excess of the

magnitude expected for the deep cycle-24 solar minimum (the deepest of the modern era) brings the Middle window into even closer agreement with the two independent Early and Late windows. The diagnostic correction is a rough estimate, not a fitted parameter and not a first-principles calculation; it is presented transparently to demonstrate that the falsification of the Bern IRF does not depend on it. The primary 32-point $\tau_{32} = 18.383 \pm 0.015$ yr fit and its single-exponential falsification of the Bern IRF stand without any cosmogenic correction. The diagnostic correction is included only to show that a plausible cosmogenic perturbation, if anything, tightens the agreement across the three temporally disjoint windows rather than weakening it.

No escape route survives. The institutional position has no remaining argumentative move that does not reduce to one of the routes refuted above.

3.8 Conclusion of the foreclosure

The tracer-versus-bulk fallacy is a single load-bearing conflation between two distinct statements: the true statement that $^{14}CO_2$ is a trace species of negligible mass (Statement A), and the false statement that the mass-balance operator transports $^{14}CO_2$ and $^{12}CO_2$ at substantially different rates (Statement B). Statement A is consistent with operator equivalence and is supported by the chemistry. Statement B is required by the institutional position, is implemented in the Bern lineage of carbon-cycle models, and is supported by no first-principles physical derivation, no published atmospheric measurement, and no coherent physical mechanism not refuted above.

The empirical foreclosure was performed decades ago by the bomb tests, which injected a ^{14}C pulse into the atmosphere and allowed the eigenvalue of the atmospheric mass-balance operator to be measured directly through the relaxation of the isotopic label. The measurement gives the isotopic relaxation time $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr (19), single-exponential, no tail, no permanent fraction. This is the relaxation timescale of the label, not the average dynamical atmospheric turnover time. The two are distinct physical quantities linked by the bridge equation (23):

$$\tau_{\mathrm{rel}} = \bar{\mathcal{D}}\,\frac{\tau_{\mathrm{res}}}{\bar{\phi}_{\mathrm{slow}}}$$

where $\tau_{\mathrm{res}} = 3.80 \pm 0.12$ yr (18) is the average dynamic turnover time determined from the mass-balance ODE (3) (RRR fit to Mauna Loa and Barrow concentration records, 1958–2023; §6.1), $\bar{\phi}_{\mathrm{slow}}$ is the deep-mixed leak fraction (21), and $\bar{\mathcal{D}} = 0.9827 \pm 0.0005$ is the average atmospheric disequilibrium factor averaged over 1965–2019 (22). The label relaxation is slower than the turnover because the fast-recycled fraction of each turnover returns carbon at the contemporary atmospheric isotopic ratio, so only the deep-mixed leak fraction $\bar{\phi}_{\mathrm{slow}}$ of each turnover permanently removes the label; this stretches the relaxation time by $\bar{\mathcal{D}}/\bar{\phi}_{\mathrm{slow}}$ relative to the turnover time. By operator equivalence (Appendix B.2), the eigenvalue spectrum measured by the bomb-^{14}C relaxation is also the eigenvalue spectrum of the bulk atmospheric CO_2 mass impulse response, because both are governed by the same first-order linear mass-balance operator acting on the same reservoir. The Bern IRF, which encodes a multi-exponential decay with eigenvalues of $O(10^2)$ yr and a permanent fraction a_0 – effectively $O(\infty)$ – requires eigenvalues that do not appear in the measured bomb-^{14}C spectrum at any amplitude consistent with the data. It is therefore falsified by direct measurement, with the falsification mediated by the operator-equivalence theorem proven in Appendix B.2 and the bridge equation derived in §6.1.

The institutional Bern IRF framework, introduced by Maier-Reimer and Hasselmann (1987) [80] and propagated through the IPCC assessment cycles from IPCC SAR (1996) [56] through IPCC AR6 (2021) [60], has rested on the tracer-versus-bulk fallacy for approximately four decades. The falsifying data, in the form of atmospheric bomb-^{14}C measurements, has been accumulating in the literature since the late 1950s; the present analysis is enabled by the current authoritative compilation [55], whose precision and zonal resolution allow the single-exponential structure to be demonstrated at $R^2 = 99.998\%$ across eight independent fits. The falsification has not been

performed in the institutional literature because the institutional vocabulary obscures the operator-equivalence constraint and because the institutional modeling framework parameterizes around it. Once the vocabulary is clarified and precisely stated, the operator-equivalence theorem proved, the bridge equation derived from the mass-balance ODE, and the cleaned bomb-^{14}C record fit to its physically required single-exponential form, the falsification follows directly. This paper performs that falsification in the open. The fallacy is named, the escape routes are foreclosed, and the institutional position has no remaining physical, mathematical, or logical support.

The positive physics of the atmospheric CO_2 reservoir, developed in Section 4 and following, proceeds on the established understanding that the mass-balance operator acts identically on all isotopologues to within per-mil fractionation corrections, that the bomb-^{14}C record measures the eigenvalue of that operator, and that the eigenvalue so measured governs the relaxation of all atmospheric CO_2 perturbations regardless of isotopic composition.

4. First-Principles Physics of the Atmospheric CO2 Reservoir: Turnover, Impulse Response, and Isotopic Disequilibrium

Having foreclosed in Section 3 the tracer-versus-bulk fallacy and established that the mass-balance operator acts identically on all isotopologues to within per-mil fractionation corrections, we now develop the first-principles physics of the atmospheric CO_2 reservoir. The reservoir admits two different observational probes that answer fundamentally different physical questions. Conflating them is the central methodological error of the IPCC Bern IRF framework [65, 60], and untangling them is the prerequisite for any meaningful discussion of CO_2 residence and adjustment timescales.

4.1 The kinematic probe: instantaneous turnover

The atmosphere holds an inventory $M_a(t)$ of CO_2, exchanged with surface reservoirs by gross inflow $F_{in}(t)$ (terrestrial respiration, fire, ocean outgassing, volcanism, fossil and land-use emissions) and gross outflow $F_{out}(t)$ (photosynthetic uptake, ocean dissolution, weathering). Mass conservation gives

$$\frac{dM_a}{dt} = F_{in}(t) - F_{out}(t) \tag{3}$$

The atmospheric record establishes that $F_{in} > F_{out}$ throughout the post-1958 instrumental window: the Mauna Loa concentration has risen monotonically from 315 ppm in 1958 to 412 ppm in 2019, with $dM_a/dt \approx 5$ GtC/yr in 2019 [76]. The system is therefore in *sustained disequilibrium*, not steady state. This distinction matters because the disequilibrium is small in relative terms: $F_{in} \approx 235$ GtC/yr and $F_{out} \approx 230$ GtC/yr in 2019, with their difference (5 GtC/yr) ~2.3% of either gross flow. The atmosphere is being flushed by counterflowing fluxes nearly two orders of magnitude larger than the net accumulation rate.

The instantaneous residence time of CO_2 *currently in the atmospheric reservoir* is, following standard hydrological convention [10, 36], the ratio of inventory to gross *outflow*:

$$\tau_{out}(t) \equiv \frac{M_a(t)}{F_{out}(t)} \tag{4}$$

since F_{out} is the rate at which the existing inventory is removed by photosynthetic uptake, ocean dissolution, and weathering. For a reservoir in steady state, $F_{in} = F_{out}$ and $M_a/F_{in} = M_a/F_{out}$, but the atmospheric CO_2 reservoir has been in sustained disequilibrium with $dM_a/dt > 0$ throughout the post-1750 industrial period. We therefore use $F_{out}(t) = F_{in}(t) - dM_a/dt$, with dM_a/dt obtained directly from the Mauna Loa CO_2 growth rate. Equation (4) is a snapshot quantity: it depends only on the ratio of inventory to flow at a single instant and assumes nothing about reservoir coupling, return fractions, or perturbation dynamics. It does not require steady state; the calculation is identical whether F_{in} equals, exceeds, or is less than F_{out}, because τ_{out} is a property of forward flow

alone. Similarly, we define κ_{out}, κ_{in}

$$\kappa_{\text{out}}(t) \equiv \frac{F_{\text{out}}(t)}{M_{\text{a}}(t)} \quad \kappa_{\text{in}}(t) \equiv \frac{F_{\text{in}}(t)}{M_{\text{a}}(t)} \tag{5}$$

and the disequilibrium ratio ($\mathcal{D}$)

$$\mathcal{D}(t) \equiv \frac{F_{\text{out}}(t)}{F_{\text{in}}(t)} = \frac{\kappa_{\text{out}}(t)}{\kappa_{\text{in}}(t)} \tag{6}$$

Much of the carbon-cycle literature also refers to τ_{out} as the *atmospheric residence time* of CO_2. It is synonymous with the hydrological "turnover time" $T = M/S$ used by the IPCC AR6 glossary. We define the term τ_{res} as the average of Mauna Loa and Barrow, Alaska multi-decadal Refined Reservoir Routing measurement (Koutsoyiannis (2024a) [72]), and we define the term *ratio relaxation time* τ_{rel} as the isotopic perturbation decay timescale derived in Section 6. We show that "adjustment time," as defined by the IPCC, represents something entirely different, corresponding to nothing in the physical world.

The kinematic probe answers a single question: *how long does a CO_2 molecule remain in the atmosphere before some surface flux removes it?* The answer, $\tau_{\text{res}} = 3.8$ yr, already falsifies any claim that CO_2 persists in the atmosphere for centuries. The IPCC concedes this kinematic turnover time as "approximately 4 years" but asserts, without empirical support, that returning carbon is preferentially trapped in slow reservoirs (deep ocean, recalcitrant soils), so that *gross turnover* is fast while *net adjustment* is slow. The latter "adjustment time" is a purely model-based construct with no counterpart in the observable carbon cycle. We dissect this history of the IPCC Bern IRF framework and prove its mathematical incompatibility with the observed single exponential in §8.3.

The two dynamical probes developed below measure two distinct real physical timescales of the atmospheric reservoir: the bomb-^{14}C impulse-response natural experiment yields the ratio relaxation time $\tau_{\text{rel}} = 18.383$ yr, and Koutsoyiannis's Refined Reservoir Routing fit yields the residence time $\tau_{\text{RRR}} = 3.8$ yr (virtually identical to the kinematic $\langle\tau_{\text{out}}\rangle = 3.70$). We adopt τ_{RRR}, recovered dynamically from the integrated multi-decadal response, as the definition of our τ_{res}. Both τ_{res} and τ_{rel} are short, both are stable across methods, and they are connected to one another through the exact bridge equation derived in Section 6.1. The IPCC "adjustment time" corresponds to nothing measurable in either probe.

4.2 Two dynamical probes: bomb-^{14}C impulse response and Refined Reservoir Routing

A test of perturbation dynamics in any well-mixed reservoir requires a labeled subset whose redistribution can be tracked. This is a standard solved problem in residence-time theory, developed independently in chemical reactor engineering [28], pharmacokinetics [62], and hydrology [10]: a tracer pulse injected into a well-mixed reservoir decays at a rate that directly reveals the reservoir's residence time, regardless of whether the total mass is in equilibrium. The atmosphere is precisely such a reservoir, and three carbon isotopes are candidates for the tracer that labels three different subsets of atmospheric CO_2. However, they differ dramatically as dynamical probes: ^{12}C comprises ~99% of atmospheric carbon, so it *is* the bulk and therefore cannot be used as a tracer; δ^{13}C fractionates thermodynamically at every phase boundary (air–sea, air–leaf), producing source-dependent offsets that conflate transport with chemistry; whereas Δ^{14}C is dominated by radioactive decay alone and received a historically unique, massive, precisely documented impulse from US and USSR atmospheric nuclear testing (1952–1963) that nearly doubled the atmospheric ^{14}C inventory [55], with the global composite peaking at ~720‰ above the pre-bomb baseline of −27‰ (~923‰ in NH zone 1), yielding a signal-to-noise ratio of $O(10^2$–$10^3)$. By comparison, dye-injection experiments in chemical reactors typically achieve SNR $O(10$–$10^2)$ [28], fluorescent

groundwater tracers reach $O(10^2)$ at best, and isotope-dilution bolus experiments in pharmacokinetics operate at SNR $O(10–50)$ [62]. Levin and Hesshaimer called ^{14}C "a unique tracer of global carbon cycle dynamics" [78]; Koutsoyiannis described the bomb-^{14}C injection as "an accidental real-world experiment" producing "a real-world situation close to an ideal" for estimating the impulse response function [72]. In quantitative terms, the bomb-^{14}C decay surpasses, by one to two orders of magnitude in SNR, any tracer experiment reported in the reactor-engineering, pharmacokinetic, or hydrological literature, and constitutes the highest-fidelity realization of the classical tracer experiment ever applied to a geophysical reservoir.

4.2.1 Nature's bomb-^{14}C tracer experiment

Atmospheric thermonuclear weapons testing in the late 1950s and early 1960s drove global atmospheric $\Delta^{14}C$ from approximately −27‰ in 1950 to a peak of approximately +693‰ by 1965 [55, 35], nearly doubling atmospheric ^{14}C from ~878 kg to ~1580 kg. Throughout this paper, we work with the bomb excess above the actual pre-bomb atmospheric baseline, obtained by adding 27‰ to every $\Delta^{14}C$ value in the Hua et al. (2022) [55] compilation.[1] Thus the bomb-excess anomaly peaks at $R_a(1965) = 693 + 27 = 720$‰ and decays to $R_a(2019) = 4 + 27 = 31$‰.

The surface ocean and terrestrial biosphere remained near their pre-bomb $\Delta^{14}C$ values at the moment of the spike, because their large reservoirs had not yet exchanged substantial CO_2 with the labeled atmosphere. The bomb spike therefore created a sudden and large isotopic disequilibrium between the atmosphere and its surface exchange partners. This disequilibrium constituted a large-amplitude natural impulse tracer experiment. The signal does not depend on small fractionation offsets. It is a near-100% relative labeling of one reservoir, applied as a sharp transient pulse that peaked in 1964–1965 and has decayed continuously as a single exponential ever since. The decay, observed continuously in direct measurements of ^{14}C in air samples and tree rings from all over the world [55], tracks the true mixing rate of atmospheric CO_2 with surface reservoirs, with the labeled fraction serving as an unambiguous quantitative tracer, and thus constitutes a direct empirical realization of the impulse-response experiment that the IPCC Bern IRF formalism purports to describe. The observed atmospheric $\Delta^{14}C$ has decayed as a near-perfect single exponential (R^2 = 99.998% on the cleaned record; Section 9) with an 18.383 ± 0.015 yr e-folding timescale. A single exponential is the unambiguous signature of a linear reservoir with one characteristic timescale; the structural consequences for the IPCC Bern impulse-response architecture are developed in §8.3 and §8.4.

4.2.2 Refined Reservoir Routing

Koutsoyiannis's RRR framework [72] models the atmosphere as a single reservoir whose mass balance obeys the continuity equation (adopting his original notation)

$$\frac{dS(t)}{\mathrm{d}t} = I(t) - \frac{S(t)}{W(t)} \tag{7}$$

where $S(t)$ is atmospheric CO_2 storage, $I(t)$ is total inflow (natural + anthropogenic), and $W(t)$ is the instantaneous residence time of CO_2. This is solved over the full observational period (monthly steps from the 1960s onward) and fitted jointly to the Mauna Loa and Barrow CO_2 records. The resulting characteristic residence time $\tau_{RRR} = 3.8 \pm 0.12$ yr is a dynamical system

[1] The quantity $\Delta^{14}C$ is defined exactly as $\Delta^{14}C = (A_{SN}/A_{abs} - 1) \times 1000$‰, where A_{abs} is 95% of the ^{14}C activity of NBS oxalic acid I (HOxI, prepared from 1955 sugar beets), normalized to AD 1950 and $\delta^{13}C = -25$‰ [113]. This laboratory standard fixes the zero point by convention; it is not a snapshot of the atmosphere on any specific day in 1950. The Hua et al. (2022) [55] global compilation shows $\Delta^{14}C \approx -24$ to −26‰ in 1950–1953 while several regional values reach −27‰ because the actual atmospheric $^{14}C/^{12}C$ ratio at those times was slightly lower than the fixed laboratory reference. Real atmospheric measurements around the reference epoch therefore sit slightly below zero, and the +27‰ offset removes this convention to expose the true bomb-excess decay.

property. It characterizes the integrated response and decay behavior across decades of data, not an instantaneous snapshot ratio of stock to flux at any single time. The RRR model reproduces observed concentrations with explained variance >99.9% while remaining stable to within a few percent across 1965–2010. Koutsoyiannis notes correctly that the bomb-^{14}C data provide an independent upper bound on the characteristic residence time. He calculates Δ^{14}C perturbation decay time of 17.2 years which is slower than the true molecular turnover time due to back-flux from fast reservoirs.

4.3 $\delta^{13}C$ of the bulk atmosphere is dynamically uninformative

The inter-reservoir δ^{13}C offsets are small and have been approximately stationary on millennial timescales. The atmosphere–biosphere fractionation is approximately −18‰ to −20‰ for C3 photosynthesis [87, 40], and the atmosphere–ocean equilibrium offset is approximately −8‰ once kinetic and equilibrium components are combined [87]. Pre-industrial values were $\delta^{13}\mathrm{C} \approx -6.5$‰ in the atmosphere [39, 102], +1 to +2‰ in surface-ocean DIC, and approximately −25‰ in the bulk terrestrial biosphere [87], with atmospheric δ^{13}C nearly stationary across the past millennium [102]. The atmospheric and biospheric reservoirs were therefore in approximate *isotopic* quasi-equilibrium for at least 1000 years, even though they exchange large gross fluxes annually and were never in *mass* equilibrium.

The dynamical implication is direct. Because the inter-reservoir isotopic offsets are governed by fractionation physics and have been stationary for a millennium, the slow post-1750 atmospheric δ^{13}C decline of approximately 2‰ [102] is a small perturbation on an already near-equilibrated isotopic system. Such a perturbation carries essentially no information about gross redistribution timescales between atmosphere, biosphere, and surface ocean: any isotopic signal injected into the atmosphere is rapidly smeared across the coupled reservoirs on the residence-time scale (3.8 yr), after which the system relaxes back toward its stationary fractionation offset. The bulk δ^{13}C decline therefore cannot adjudicate between fast and slow redistribution dynamics. This is precisely why the bomb-^{14}C probe, with its three-orders-of-magnitude advantage in SNR, is indispensable.

A separate line of evidence directly falsifies the conventional attribution of the δ^{13}C decline to fossil-fuel input: the so-called "Suess effect," as institutionalized in the modern literature by the Graven et al. (2020) [45] Grand Challenges review. Koutsoyiannis (2024b) [73] extracts the δ^{13}C signature of the *net atmospheric input* ($\delta^{13}\mathrm{C_I}$) from the joint time series of [CO_2] and δ^{13}C using standard Keeling-plot intercept analysis, and finds it constant at $\delta^{13}\mathrm{C_I} \approx -13.2$‰ across the 44-year instrumental period from 1978 to 2022 and over the 500-year proxy record extending back to the Little Ice Age. The constancy of this value directly falsifies the fossil-source hypothesis: were fossil-fuel combustion ($\delta^{13}\mathrm{C} \approx -28$‰) the dominant marginal source, $\delta^{13}\mathrm{C_I}$ would have shifted progressively toward more negative values as emissions grew fivefold over the instrumental period. Instead, the invariant −13.2‰ intercept rules out fossil-fuel combustion as a detectable source and points unambiguously to a natural input dominated by terrestrial respiration and oceanic outgassing [20].

Cohler & Soon (2026c) [22] formalize this constraint using the mass partition equation $\delta^{13}\mathrm{C_I} = \mathrm{f}\,\delta^{13}\mathrm{C_F} + (1-\mathrm{f})\,\delta^{13}\mathrm{C_N}$ with the standard fossil signature $\delta^{13}\mathrm{C_F} \approx -28$‰ and the physically admissible range $-13.2‰ \le \delta^{13}\mathrm{C_N} \le -8$‰. The 44-year Scripps four-site intercept $\delta^{13}\mathrm{C_I} = -13.2 \pm 0.2$‰ forces the fossil fraction f into the interval $[0, 0.26]$ across the instrumental window alone. Independent Keeling regression of the Law Dome ice-core record of Koutsoyiannis (2024b) [73] yields the same intercept. The five-century sclerosponge constancy of $\delta^{13}\mathrm{C_I}$ across a roughly 10^4-fold rise in fossil emissions then collapses the admissible band to $f \approx 0$. The fossil fraction of the net atmospheric CO_2 addition is bounded above at approximately 1.3%, indistinguishable from zero within the ±0.2‰ precision of the intercept estimate. The result does not rely on any climate model: it follows from conservation of mass, the Keeling relation, and standard isotopic signature values used throughout the carbon-cycle literature.

We return to this result in Section 5.4 as an independent confirmation that the fast-recycled fraction $\phi_{\text{fast}}(t)$ is time-invariant.

A third, complementary isotopic falsification comes from the *sign* and *significance* of the recent-window trend in $\delta^{13}C_I$. Cohler & Soon (2026b) [21] apply the Koutsoyiannis (2024b) [73] Equation 13 fixed-$\Delta[CO_2]$ extraction to the Francey et al. (1999) [41] Law Dome ice-core, firn, and Cape Grim Air Archive record, and test the 1977–1996 recent-window trend, spanning the fastest sustained rise in global fossil-fuel emissions on the record. Fossil dominance requires this trend to be negative and large on the scale set by the fossil-to-non-fossil-fuel signature gap of approximately −15‰. The observed trend is +2.94‰/decade against a 95% Hurst-Kolmogorov noise envelope of [−1.26, +1.24]‰/decade at $H = 0.921$, with $p < 10^{-4}$: statistically significant, opposite in sign to the fossil-dominance prediction, and larger in magnitude than twice the upper bound of the noise envelope in the wrong direction. The result is robust across data treatments: the Rubino et al. (2013) [102] Law Dome revision extended with the three excluded 1994–1996 Cape Grim Air Archive years, on the same CSIRO GASLAB measurement chain, gives +1.81‰/decade at $p < 10^{-4}$. The Rubino-alone treatment fails significance only through the 1993 endpoint truncation that removes precisely those three years. The positive sign is consistent with a growing ocean-outgassing contribution to the net atmospheric input under surface warming, in which the ocean fraction of the enhanced non-fossil-fuel flux grows relative to the biosphere fraction because ocean pCO_2 responds instantly via van't Hoff sensitivity $\partial \ln pCO_2/\partial T \approx 0.0423\ K^{-1}$ while biosphere respiration is buffered by thermal inertia and biological turnover on decadal timescales [21].

4.4 Why both probes are required and how they converge

The kinematic probe (§4.1) establishes that gross turnover is fast at multiple instants in time. Snapshots agreeing to within a few percent over 54 years (Section 10) constitute strong evidence against any framework that requires gross turnover to be slow, but they do not by themselves rule out an IPCC "adjustment time" loophole in which fast gross turnover masks a slow underlying adjustment process. The radiocarbon data closes this loophole by directly measuring the perturbation-decay timescale. Because the bomb-^{14}C natural experiment labels the atmosphere with a tracer that the surface reservoirs did not initially share, the subsequent decay of atmospheric $\Delta^{14}C$ is the response of the actual coupled system to an actual large-amplitude isotopic perturbation, precisely the quantity that the IPCC claims persists for longer than 1000 yr for 15–40% of emitted CO_2.

The two kinds of probes, kinematic and dynamical, converge on the same conclusion from independent epistemic foundations. The kinematic probe makes no model assumptions whatsoever: it is the ratio of two directly measured quantities (§10.1). The dynamical kind comprises two distinct probes, one based on the RRR mass-balance linear-reservoir framework fit to the $[CO_2]$ record, the other on the bomb-^{14}C tracer experiment that nature itself performed. The four numerical estimates that emerge from these probes are summarized in Table 1.

The factor of ∼4.8 separating the residence-time and relaxation-time estimates is not a discrepancy between methods. It is the physically expected consequence of the bomb-^{14}C label probing isotopic re-equilibration of the labeled atmospheric reservoir with the initially unlabeled surface ocean and biosphere, a process that proceeds on the relaxation timescale rather than the residence timescale. Both timescales are outputs of the same mass-balance equation, as we now derive.

Table 1: Four estimates of CO_2 atmospheric timescales. The two IPCC-based estimates agree with the RRR residence-time estimate to within 0.4σ (4-decades), 0.1σ (2010–2019, AR6), despite resting on entirely independent foundations. τ_{rel} is the e-folding time of the bomb-^{14}C isotopic perturbation decay, measured directly from the cleaned 32-point fit to the Hua et al. (2022) [55] compilation (§9.3); it differs from τ_{res} by an empirical factor of $\tau_{\mathrm{rel}}/\tau_{\mathrm{res}} = \bar{\mathcal{D}}/\bar{\phi}_{\mathrm{slow}} = 4.84 \pm 0.16$, related to $\bar{\phi}_{\mathrm{slow}}$ and the average disequilibrium factor $\mathcal{D}(t)$ through the bridge equation (23). All four values are short, stable across methods, and incompatible with the century-to-millennial modes of the IPCC framework by two to three orders of magnitude.

Quantity	Value (yr)	Source
τ_{out} (AR6)	3.83	Kinematic; 2010-2019 (§10.1, Table 11)
$\langle\tau_{\mathrm{out}}\rangle$ (AR3–AR6)	3.70 ± 0.21	Kinematic; 4-decade snapshot average (§10.1, Table 11)
$\tau_{\mathrm{res}} = \tau_{\mathrm{RRR}}$	3.80 ± 0.12	Dynamic; RRR fit, MLO+BRW [72]
τ_{rel}	18.383 ± 0.015	Empirical; 32-point joint fit Hua et al. (2022) [55] (§9.3)

5. Mass-Balance Framework

5.1 The atmospheric carbon and ^{14}C balances

Let $M_{\mathrm{a}}(t)$ denote atmospheric carbon inventory in GtC, $R_{\mathrm{a}}(t)$ the atmospheric Δ^{14}C ratio in per mille, $F_{\mathrm{in}}(t)$ and $F_{\mathrm{out}}(t)$ the gross fluxes of carbon into and out of the atmospheric reservoir in GtC/yr, and $R_{\mathrm{in}}(t)$ the flux-weighted average ratio of incoming carbon. Conservation of total atmospheric carbon gives (3) reproduced here for reference:

$$\frac{\mathrm{d}M_{\mathrm{a}}}{\mathrm{d}t} = F_{\mathrm{in}}(t) - F_{\mathrm{out}}(t) \tag{8}$$

Conservation of ^{14}C in the atmosphere, with $N_{\mathrm{a}}(t) = M_{\mathrm{a}}(t)R_{\mathrm{a}}(t)$ the atmospheric ^{14}C content, gives

$$\frac{d(M_{\mathrm{a}}R_{\mathrm{a}})}{\mathrm{d}t} = F_{\mathrm{in}}(t)R_{\mathrm{in}}(t) - F_{\mathrm{out}}(t)R_{\mathrm{a}}(t) \tag{9}$$

where the well-mixed assumption for the atmosphere implies that outgoing flux carries the atmospheric ratio R_{a}. Radioactive decay of ^{14}C is negligible on the timescales of interest: with a half-life of 5,700 yr [71], the decay over 54 yr amounts to $1 - e^{-(54 \ln 2/5{,}700)} = 0.65\%$, well below the precision of any flux estimate used here.

Expanding the left-hand side of (9) and using (8) to eliminate F_{out}:

$$M_{\mathrm{a}}\frac{\mathrm{d}R_{\mathrm{a}}}{\mathrm{d}t} + R_{\mathrm{a}}\frac{\mathrm{d}M_{\mathrm{a}}}{\mathrm{d}t} = F_{\mathrm{in}}R_{\mathrm{in}} - \left(F_{\mathrm{in}} - \frac{\mathrm{d}M_{\mathrm{a}}}{\mathrm{d}t}\right)R_{\mathrm{a}} \tag{10}$$

The $R_{\mathrm{a}}(\mathrm{d}M_{\mathrm{a}}/\mathrm{d}t)$ terms cancel, so defining gross inflow rate $\kappa_{\mathrm{in}}(t) \equiv F_{\mathrm{in}}(t)/M_{\mathrm{a}}(t)$ yields

$$\frac{\mathrm{d}R_{\mathrm{a}}}{\mathrm{d}t} = \kappa_{\mathrm{in}}(t)\,[R_{\mathrm{in}}(t) - R_{\mathrm{a}}(t)] \tag{11}$$

This is exact under conservation of mass and the well-mixed atmosphere assumption. The atmospheric ratio relaxes toward R_{in} at the instantaneous rate κ_{in}.

5.2 Two-pool decomposition of the incoming ratio

The CO_2 entering the atmosphere comes from two complementary sources that sum to 1, distinguished by where the carbon was before it entered:

- The *fast-recycled* fraction $\phi_{\mathrm{fast}}(t)$ has been in atmospheric exchange on timescales of days to months and returns at the contemporary atmospheric ratio R_{a}. This includes surface-ocean outgassing of recently absorbed CO_2, plant respiration, soil respiration of recent litter, decay of recent biomass, and wildfires.

- The *deep-mixed* (slow) fraction $\phi_{\text{slow}}(t)$ returns at an effective baseline ratio R_{b}, justified below to be ≈ 0‰ on the Hua+27 bomb-excess scale after fossil-fuel carbon is separately accounted for. This includes deep-ocean upwelling of water isolated from the post-bomb atmosphere and refractory soil organic matter respiration; fossil-fuel combustion (which carries no ^{14}C and contributes a strongly negative Δ^{14}C signature) is handled separately as an exact mass-balance correction (Section 9.2.5).

Why $R_{\text{b}} \approx 0$. The deep-mixed pool $\phi_{\text{slow}}(t)$ is defined by isolation from the atmosphere since before the bomb tests. Each of its constituent sources carries, by construction, the pre-bomb baseline ratio: deep-ocean water that was last in atmospheric contact decades to millennia ago carries the pre-bomb atmospheric Δ^{14}C of its sequestration epoch; refractory soil organic matter mineralized decades to centuries ago carries the same; and biogenic respiration of recalcitrant carbon pools likewise returns at near-baseline. On the Hua+27 bomb-excess scale used throughout this paper, where zero is set to the actual pre-bomb atmospheric value, all three sit within a few permil of zero, against a peak signal of 720‰.

The single exception is fossil-fuel carbon, which is ^{14}C-dead and sits at -973‰ on the Hua+27 scale. Fossil carbon is therefore not absorbed into the $R_{\text{b}} \approx 0$ idealization. It is handled separately as a first-principles isotopic mass-balance correction (Section 9.2.5), in which the directly measured annual E_{FOS} and M_{a} fix the dilution rate exactly, with no free parameter. With fossil carbon removed in this way, the remaining deep-mixed sources all carry the pre-bomb baseline within an effective $|R_{\text{b}}| \lesssim 20$‰, i.e. within $\sim$3% of the initial bomb excess.

Section 15.4.1 demonstrates that the framework is robust to this idealization: the exact ODE with arbitrary constant R_{b} yields a single exponential of $(R_{\text{a}} - R_{\text{b}})$ with the same τ_{rel}, and admitting effective R_{b} in the range -5 to -20‰ shifts the inferred ϕ_{fast} by at most 1%, well inside the 1–2% precision floor set by the $\delta^{13}C_{\text{I}}$ constraint (§5.4). The $R_{\text{b}} = 0$ choice is therefore a notational convenience adopted for clarity throughout §§5–6.1, not a load-bearing assumption.

The composite isotopic ratio of all incoming CO_2 is therefore the flux-weighted average of the two pools:

$$R_{\text{in}}(t) = \phi_{\text{fast}}(t)\, R_{\text{a}}(t) + \phi_{\text{slow}}(t)\, R_{\text{b}} \tag{12}$$

Because $R_{\text{b}} \approx 0$ on the Hua+27 bomb-excess scale, this reduces to

$$R_{\text{in}}(t) = \phi_{\text{fast}}(t)\, R_{\text{a}}(t) \tag{13}$$

Substituting into (11):

$$\frac{\mathrm{d}R_{\text{a}}}{\mathrm{d}t} = -\phi_{\text{slow}}(t)\kappa_{\text{in}}(t)R_{\text{a}}(t) = -\kappa(t)R_{\text{a}}(t) \tag{14}$$

with the instantaneous decay rate

$$\kappa(t) \equiv \phi_{\text{slow}}(t)\kappa_{\text{in}}(t) \tag{15}$$

the solution is

$$R_{\text{a}}(t) = R_{\text{a}}(t_0)\, \exp\left(-\int_{t_0}^{t} \kappa(s)\, ds\right) \tag{16}$$

5.3 Surface ocean mixed-layer dynamics

The fast-cycled oceanic component of $\phi_{\text{fast}}(t)$ is the surface mixed layer. Gas exchange of CO_2 occurs only at the air–sea interface. However, wind-driven turbulence, waves, and convection homogenize the upper ocean to a well-defined depth h (typically 50–200 m) on timescales of hours to days. The mixed-layer depth h is rigorously defined as the depth at which the density increase

from the surface reaches a threshold ($\Delta\sigma_\theta = 0.03$ kg m^{-3} or $\Delta T = 0.2$°C). The equilibration timescale between the atmosphere and the mixed layer is

$$\tau = \frac{h}{k_w}$$

where k_w (piston velocity) is the effective transfer velocity (5–20 cm h^{-1}). Thus τ ranges from ∼10 days to ∼4 months. The top ∼1 mm (aqueous boundary layer) is continuously flushed by bulk mixed-layer water, so the concentration (and Δ^{14}C ratio) at the interface is always identical to the bulk mixed-layer value. The mixed layer in turn equilibrates with the atmosphere on the timescale $\tau = h/k_w$ above (hours to a few months), so outgassing returns carbon at the bulk mixed-layer ratio, which lags the atmosphere by that short equilibration time. On the $\tau_{\text{rel}} = 18.4$ yr decay timescale this lag is a constant factor on the exponential (§5.4), so the fast oceanic channel returns carbon at effectively the contemporary atmospheric ratio $R_{\text{a}}(t)$, as required for the fast-recycled fraction $\phi_{\text{fast}}(t)$.

5.4 Physical meaning and time-invariance of $\phi_{\text{fast}}(t)$

The fast-recycled fraction $\phi_{\text{fast}}(t)$ is the portion of total gross inflow F_{in} that originates from the surface ocean mixed layer and the active terrestrial biosphere (recent photosynthate and shallow soil carbon). These reservoirs exchange with the atmosphere on hours-to-months timescales, short compared with $\tau_{\text{rel}} = 18.4$ yr, and therefore return carbon at a ratio $R_{\text{fast}}(t) = R_{\text{a}}(t - \delta)$ that lags the atmosphere only by that short equilibration time δ. During the bomb spike the fast pool sits below the rapidly rising atmosphere, and during the subsequent decline slightly above it; but once the atmosphere is in its single-exponential decay a short constant lag against an exponential is a constant multiplicative factor, $R_{\text{fast}}(t) = e^{\delta/\tau_{\text{rel}}} R_{\text{a}}(t)$ with $e^{\delta/\tau_{\text{rel}}} \approx 1$. This factor renormalizes the inferred $\phi_{\text{slow}}(t)$ by at most a percent and leaves τ_{rel} and the single-exponential shape exactly unchanged; the non-exponential transient of 1965–1966, where the lag is not a constant factor, is excluded a priori from the fit (§C.8). To the precision of this paper the fast channel therefore returns carbon at effectively the contemporary atmospheric ratio $R_{\text{a}}(t)$. The complementary deep-mixed fraction $\phi_{\text{slow}}(t)$ originates from reservoirs isolated from the atmosphere since before the bomb tests (deep ocean below the thermocline, refractory soil carbon, fossil fuels) and returns at the pre-bomb baseline $R_{\text{b}} \approx 0$.

From the exact solution (16), a pure single-exponential decay (observed in Hua et al. (2022) [55] with residuals at the few-per-mille level over 54 years) requires the integrand $\kappa(t) = \phi_{\text{slow}}(t)\,\kappa_{\text{in}}(t)$ to be effectively constant. This is a constraint on the *product* $\phi_{\text{slow}}\,\kappa_{\text{in}}$, not on the individual factors. The 32-point cleaned recovery (§9.3) constrains $\kappa_{\text{rel}} \equiv 1/\tau_{\text{rel}}$ to within ∼1% over the 52-years 1967–2019: any larger drift would produce systematic curvature in the residuals exceeding the $\sigma_{32} = 0.539$‰ noise floor. Mass-balance analysis (Section 10) shows that $\kappa_{\text{in}}(t)$ has a small but statistically real secular trend of +1.3% over 1961–2023, so constancy of κ_{rel} permits a compensating drift in $\phi_{\text{slow}}(t)$ of the same magnitude. The implied $\phi_{\text{fast}}(t)$ rise (from 0.7952 to 0.7983 over 58 years, an absolute shift of 0.31%) is far below the $\delta^{13}C_{\text{I}}$ detection threshold discussed next.

The constancy of $\delta^{13}C_{\text{I}}$ at −13.2‰ across the 40-45-year instrumental period (at Barrow, La Jolla, Mauna Loa, and the South Pole), and the 500-year proxy record [73] constrains $\phi_{\text{fast}}(t)$ to be constant to within ∼1–2%, consistent with both exact constancy and the small permitted drift above. The fast-recycled fraction $\phi_{\text{fast}}(t)$ and the deep-mixed fraction $\phi_{\text{slow}}(t)$ represent a physical partitioning of CO_2 molecules between two return channels. This partitioning is isotope-independent: the same molecule that returns quickly on the Δ^{14}C scale also returns quickly on the δ^{13}C scale, because the transport processes (surface-ocean re-emission, leaf respiration of recent photosynthate) are the same. The two isotope systems observe the same physical partition through different isotopic lenses. If $\phi_{\text{fast}}(t)$ were varying substantially over time, the relative contributions

of the two return channels to the net atmospheric input would change, shifting the flux-weighted $\delta^{13}C$ of that input. The invariance of $\delta^{13}C_I$ for 500 years requires that any drift in $\phi_{fast}(t)$ be small ($\lesssim$1–2%), consistent with the 0.31% permitted by the bomb-^{14}C product constraint. The three independent empirical inputs (τ_{rel} from bomb-^{14}C linearity, τ_{res} from the Koutsoyiannis RRR fit, and $\delta^{13}C_I$ constancy from the Keeling intercept) are therefore mutually consistent at the precision floor of each, and the product $\phi_{slow}(t)\,\kappa_{in}(t)$ is effectively constant.

6. The Three Measurements and the Timescales They Determine

6.1 What is measured, and what follows from the physics

This paper rests on three independent empirical measurements. Every other quantity in the analysis is a consequence of these three and the first-order linear-reservoir mathematics derived in Section 5.

Two of the three are fitted parameters of the single-exponential solution $R_a(t) = A\,e^{(1965-t)/\tau_{rel}}$ to the bomb-^{14}C decay record:

1. the time constant $\tau_{rel} = 18.383 \pm 0.015$ yr, taken from the 32-point bias-corrected fit τ_{32} (Section 9.3, Table 10); and
2. the amplitude $A = 709.115 \pm 0.585$‰ at $t = 1965$, taken from the same fit as A_{32}. A is the value of the fitted exponential at $t = 1965$, not the observed 1965 datum, which is a separate measurement with its own uncertainty.

The single-exponential functional form is not assumed. It is established empirically by single-exponential fits to the unaltered Hua et al. (2022) [55] data, every zonal series returning $R^2 > 99.6\%$ with $\sigma < 10.5$‰ before any bias correction (Section 8.1, Table 6). Mass balance then identifies what that shape requires: the solution (16) of the ODE (14), $R_a(t) \propto \exp(-\int \kappa\,dt)$, is single-exponential only if the product $\kappa(t) = \phi_{slow}(t)\,\kappa_{in}(t)$ is constant, which Sections 5.4 and 10 establish ($\phi_{slow}(t)$ and $\kappa_{in}(t)$ each varying by $\lesssim 1\%$ over the record with compensating drifts, below the $\delta^{13}C_I$ detection threshold). On this established form, the 32-point bias-corrected recovery yields a residual of 0.539‰ against the fit amplitude $A_{32} = 709.115$‰ (0.076%), below the reported measurement uncertainty of every one of the 32 points.

The third measurement, made independently of the bomb decay and from a different dataset, is the CO_2 molecular residence time:

3. $\tau_{res} = 3.80 \pm 0.12$ yr, from the Koutsoyiannis Refined Reservoir Routing fit [72] to the Mauna Loa and Barrow [CO_2] records.

The remainder of this subsection establishes τ_{res}, shows that the gross inflow rate $\bar{\kappa}_{in}$ is reconstructed from it rather than measured separately, and states how the two measured timescales are joined.

The CO_2 molecular residence time can be estimated two ways: *kinematically*, as the ratio of atmospheric inventory to gross outflow, and *dynamically*, from a full linear-reservoir fit to the observed concentration record. We define τ_{res} by the dynamical fit and use the kinematic estimate as an independent corroboration.

The kinematic estimate is the inverse of the rate at which CO_2 molecules leave the atmosphere as a fraction of current inventory, $\tau_{out}(t) = M_a(t)/F_{out}(t)$ (4). Evaluated on the directly measured inventory and the gross-flux budgets published in IPCC AR3 through AR6, the four decadal values average to $\langle \tau_{out}^{IPCC} \rangle = 3.70 \pm 0.21$ yr (Section 10.1).

The dynamical estimate is the characteristic residence time of a linear-reservoir fit to the concentration time series. Koutsoyiannis (2024a) [72], applying Refined Reservoir Routing to the Mauna

Loa and Barrow records (1958–2023), obtains a linear reservoir ($b = 1$) with explained variance exceeding 99.7% on storage. The annual-mean residence time, formed as the geometric mean of the seasonal extremes $W_m = \sqrt{W_{min}\, W_{max}}$, is $W_m = 3.68$ yr at Mauna Loa and 3.91 yr at Barrow, with split-sample validation (calibration 1958–2002, prediction 2003–2023) confirming robustness (Koutsoyiannis (2024a) [72], Table 4).

Because this paper analyzes the global Hua et al. (2022) [55] $\Delta^{14}C$ series, the residence time must itself be a global value. The leak fraction $\phi_{slow}(t)$ and the gross outflow $F_{out}(t)$ both vary regionally with the local mix of ocean, bare land, and active vegetation, so a single-station residence time would be physically inconsistent with the globally averaged isotopic record.

Koutsoyiannis reports the two stations separately and forms no global value. We construct one as the equal-weight mean of the two fitted stations, with half the inter-station difference as its uncertainty:

$$\langle W_m \rangle_{MLO,BRW} = \frac{3.68 + 3.91}{2} = 3.80 \pm 0.12 \text{ yr} \tag{17}$$

The two stations bracket the meridional range of atmospheric turnover. Mauna Loa (19.5°N) samples subtropical free-tropospheric air and Barrow (71°N) samples Arctic air; the fitted residence time is shorter at the lower-latitude station (3.68 yr) and longer in the Arctic (3.91 yr), as expected, since gross atmosphere-to-surface exchange, whose largest component is terrestrial gross primary production, is concentrated in the productive low- and mid-latitude biosphere and falls off sharply at high northern latitudes (Beer et al. (2010) [5]). A flux-weighted global mean, which weights the high-exchange low-latitude regions most heavily, therefore lies below the equal-weight average. Adopting the equal-weight value is conservative: it errs toward the longer residence time, giving smaller gross non-fossil-fuel fluxes F_{out} and F_{nat} and hence a larger anthropogenic share of gross inflow than a flux-weighted value would, so the choice does not understate the anthropogenic contribution our results bound. The equal-weight value is independently corroborated as a global figure by the four-decade IPCC flux budget, 3.70 ± 0.21 yr (above), with which it agrees to within half the joint standard deviation. We therefore take

$$\tau_{res} \equiv \langle W_m \rangle_{MLO,BRW} \tag{18}$$

as the global residence time used throughout.

The ratio relaxation time τ_{rel} is the measured e-folding time of the Hua et al. $\Delta^{14}C$ global series plus 27 ‰. It is measured directly using an unweighted non-linear least-squares single-exponential fit to the post-peak record (Appendix A). After correction for the documented compilation bias in the Hua et al. (2022) [55] series (Section 9), the 32-point fit R_{32} yields

$$\tau_{rel} \equiv 18.383 \pm 0.015 \text{ yr} \tag{19}$$

with $R^2 = 99.998\%$ and $\sigma_{32} = 0.539‰$ over the 1967–2019 window. The 28-point fit R_{28} (1967–2015) yields $\tau_{28} = 18.382 \pm 0.016$ yr, agreeing with the 32-point value to within 0.001 yr (8.8 hr; Table 10).

The $\Delta^{14}C$ anomaly relaxes toward equilibrium with the connected fast-exchanging pool at the rate $\kappa_{rel} \equiv 1/\tau_{rel}$. The relationship between instantaneous gross inflow and outflow rates is derived from the mass balance ODE (3) as

$$F_{in} = F_{out} + dM_a/dt \implies \frac{F_{in}}{M_a} = \frac{F_{out}}{M_a} + \frac{1}{M_a} dM_a/dt \implies$$
$$\kappa_{in} = \kappa_{out} + g(t) \text{ where } g(t) = d \ln M_a/dt$$

substituting κ_{res} (18) for the unavailable κ_{out}, over 1965–2019 yields

$$\bar{\kappa}_{in} \equiv 0.2632 + \bar{g} = 0.2678 \pm 0.0010 \text{ yr}^{-1} \tag{20}$$

from (15) we have $\phi_{\text{slow}}(t) = \kappa(t)/\kappa_{\text{in}}(t)$, but from bias-corrected fit R_{32} we know $\kappa(t) = \kappa_{\text{rel}}$ within 0.08‰. Therefore,

$$\bar{\phi}_{\text{slow}} = \kappa_{\text{rel}}/\bar{\kappa}_{\text{in}} = 0.203 \pm 0.006 \tag{21}$$

Again substituting κ_{res} for the unavailable κ_{out}, the average disequilibrium factor $\bar{\mathcal{D}}$ (6) can be rewritten as

$$\bar{\mathcal{D}} \equiv \frac{\kappa_{\text{res}}}{\bar{\kappa}_{\text{in}}} = \frac{1}{\bar{\kappa}_{\text{in}}\,\tau_{\text{res}}} = 0.9827 \pm 0.0005 \tag{22}$$

From (21), substitute $\kappa_{\text{rel}}/\bar{\phi}_{\text{slow}}$ for $\bar{\kappa}_{\text{in}}$ yielding $\bar{\mathcal{D}} = \bar{\phi}_{\text{slow}}/(\kappa_{\text{rel}}\tau_{\text{res}})$ and the *bridge equation*. Equivalently, from (22) we have $\bar{\mathcal{D}}\,\tau_{\text{res}} = 1/\bar{\kappa}_{\text{in}}$, so the direct form $\bar{\phi}_{\text{slow}} = 1/(\tau_{\text{rel}}\,\bar{\kappa}_{\text{in}})$ of (21) and the bridge form below are algebraically identical:

$$\tau_{\text{rel}} = \bar{\mathcal{D}}\,\frac{\tau_{\text{res}}}{\bar{\phi}_{\text{slow}}} \qquad \Longleftrightarrow \qquad \bar{\phi}_{\text{slow}} = \bar{\mathcal{D}}\,\frac{\tau_{\text{res}}}{\tau_{\text{rel}}} \tag{23}$$

The isotopic relaxation time τ_{rel} is the molecular turnover time τ_{res} stretched by the reciprocal of the deep-mixed leak fraction $\bar{\phi}_{\text{slow}}$ times the disequilibrium factor $\bar{\mathcal{D}}$, which is ~1.7% less than unity (Section 10).

The value of τ_{res} sets the numerical magnitudes alone, not the structural results. The relaxation rate $\kappa_{\text{rel}} = \bar{\phi}_{\text{slow}}\,\bar{\kappa}_{\text{in}} = 1/\tau_{\text{rel}}$ is fixed by the independently measured τ_{rel} (19), so changing τ_{res} rescales ϕ_{slow} and κ_{in} in inverse proportion while leaving their product unchanged, and with it the single-exponential relaxation, the exclusion of the Bern multi-exponential partition, and the proportionality $F_{\text{nat}} \propto M_{\text{a}}$. Only the partition of $1/\tau_{\text{rel}}$ between ϕ_{slow} and κ_{in} and the absolute flux magnitudes depend on the choice, and the ±0.12 yr inter-station spread is propagated into every such quantity, where it is the dominant uncertainty (Table 2).

The deep-mixed leak fraction ϕ_{slow}, the fast-recycled fraction ϕ_{fast}, the disequilibrium identity, the fossil-fuel correction, and the operator-equivalence exclusion of the IPCC Bern modes are all consequences of these three measurements and the single-exponential mathematics. No adjustable parameter enters at any stage.

The dynamical value (3.80 ± 0.12 yr) and the kinematic four-decade average (3.70 ± 0.21 yr) are defined differently but agree to within half the joint standard deviation. That agreement is itself confirmation that the linear-reservoir model holds in the actual data.

6.2 The deep-mixed leak fraction

τ_{res} and τ_{rel} emerge from the same mass-balance equation (11) yet measure different physical quantities: τ_{res} is the gross-flux molecular turnover time, while τ_{rel} is the timescale on which the atmospheric Δ^{14}C ratio approaches equilibrium with the connected fast-exchanging pool, which returns CO_2 at effectively the contemporary atmospheric ratio (the short equilibration lag is a constant factor on the exponential, §5.4). *There is no inconsistency between them.* The bridge equation (23) relates the two through the deep-mixed leak fraction ϕ_{slow}: the slow isotopic relaxation is the fast molecular turnover seen through the fraction ϕ_{slow} of each cycle that draws on the deep-mixed pool. Both timescales are fixed empirically and independently, τ_{res} from the Koutsoyiannis Refined Reservoir Routing fit (18) and τ_{rel} from the bomb-^{14}C decay (19), with the gross inflow rate $\bar{\kappa}_{\text{in}} = 0.2678\ \text{yr}^{-1}$ reconstructed from τ_{res} by mass balance (20) (equivalently $1/\bar{\kappa}_{\text{in}} = 3.73$ yr).

Expressing the bridge equation (23) through the reconstructed inflow rate (20) gives the partition in the form used for computation:

$$\phi_{\text{slow}} = \frac{1}{\tau_{\text{rel}}\,\bar{\kappa}_{\text{in}}} \qquad \phi_{\text{fast}} = 1 - \frac{1}{\tau_{\text{rel}}\,\bar{\kappa}_{\text{in}}} \tag{24}$$

Equation (24) is exact: it uses the instantaneous mass-balance reconstruction of κ_{in} and makes no steady-state assumption. With $\tau_{\mathrm{rel}} = 18.383$ yr from the single-exponential fit, any long-mode component of the form posited by the Bern impulse-response partition is excluded at the $\sigma_{32} = 0.539‰$ level (Appendix B.1); the atmosphere–surface system is first-order with a single τ_{res}. The disequilibrium then satisfies the exact identity

$$\frac{F_{\mathrm{in}} - F_{\mathrm{out}}}{F_{\mathrm{out}}} = \frac{\mathrm{d}M_{\mathrm{a}}/\mathrm{d}t}{F_{\mathrm{out}}} = \tau_{\mathrm{res}} \, \frac{1}{M_{\mathrm{a}}} \frac{\mathrm{d}M_{\mathrm{a}}}{\mathrm{d}t} \tag{25}$$

i.e. the relative disequilibrium equals the residence time multiplied by the relative growth rate of M_{a}, with no steady-state assumption.

Using Mauna Loa annual growth rates and annual mean concentrations, (25) gives a disequilibrium that rises on 10-year rolling means from 1.30% in 1965–1974 to 2.37% in 2016–2025 (Figure 4), tracking the acceleration of $\mathrm{d}\ln M_{\mathrm{a}}/\mathrm{d}t$ scaled by the fixed τ_{res}. Because $\delta^{13}\mathrm{C_I}$ is constant at $-13.2‰$ across the entire 1978–2022 instrumental record (Section 4.3), the source ensemble feeding F_{in} has not changed in isotopic composition; the accelerating source driving the rising disequilibrium is therefore biospheric, not anthropogenic.

We compute ϕ_{fast} throughout from the exact expression (24), using the dynamically fixed τ_{res} and the reconstructed $\bar{\kappa}_{\mathrm{in}}$ (20) (central value 0.2678 yr^{-1}, Section 10), and carry the full ∼2.3–2.4% disequilibrium rather than approximating it away; no result in this work relies on $F_{\mathrm{in}} = F_{\mathrm{out}}$. Propagating the uncertainties in τ_{rel} and $\bar{\kappa}_{\mathrm{in}}$ through (24), with $\sigma_{\phi_{\mathrm{slow}}}/\phi_{\mathrm{slow}} = [(\sigma_\tau/\tau_{\mathrm{rel}})^2 + (\sigma_\kappa/\bar{\kappa}_{\mathrm{in}})^2]^{1/2}$ where $\sigma_\kappa \approx \sigma_{\tau_{\mathrm{res}}}/\tau_{\mathrm{res}}^2$, gives the central values and uncertainties of all derived quantities collected in Table 2. The dominant uncertainty is τ_{res} (the Mauna Loa–Barrow inter-station spread); the formal fit error in τ_{rel} is negligible by comparison.

Table 2: Derived quantities from the two empirically measured timescales $\tau_{\mathrm{res}} = 3.80 \pm 0.12$ yr and $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr. Uncertainties propagated via first-order error analysis; the dominant contribution in every case is $\sigma_{\tau_{\mathrm{res}}}$. Central values are used for narrative inline calculations throughout this paper.

Quantity	Definition	Value
Turnover rate	$\kappa_{\mathrm{res}} = 1/\tau_{\mathrm{res}}$	0.263 ± 0.008 yr^{-1}
Gross inflow rate	$\bar{\kappa}_{\mathrm{in}}$	0.2678 ± 0.0010 yr^{-1}
Deep-mixed leak fraction	$\phi_{\mathrm{slow}} = 1/(\tau_{\mathrm{rel}}\,\bar{\kappa}_{\mathrm{in}})$	0.203 ± 0.006
Fast-recycled fraction	$\phi_{\mathrm{fast}} = 1 - 1/(\tau_{\mathrm{rel}}\,\bar{\kappa}_{\mathrm{in}})$	0.797 ± 0.006
Timescale ratio $\tau_{\mathrm{rel}}/\tau_{\mathrm{res}}$	empirical	4.84 ± 0.16

The empirical timescale ratio $\tau_{\mathrm{rel}}/\tau_{\mathrm{res}} = 4.84 \pm 0.16$ and the exact bridge value $1/\phi_{\mathrm{slow}} = 4.92 \pm 0.15$ differ by the ∼2% atmospheric disequilibrium factor $F_{\mathrm{in}}/F_{\mathrm{out}}$, confirming internal consistency. Both are determined to ±3% with no adjustable parameters and no assumptions beyond mass conservation and the two-pool decomposition. In the remainder of this paper central values are used for narrative calculations; results sensitive to the uncertainty range (e.g. the predicted leading-order τ_{rel} and the anthropogenic share of the deep-mixed leak) are reported with propagated uncertainty.

Table 3: Symbols, units, values, and status. "Measured" marks quantities obtained directly from data; "fitted" marks parameters extracted from the bomb-^{14}C single-exponential at three bias removal / correction levels; "anchor" marks the two committed empirical values (τ_{res} and τ_{rel}) on which the bridge equation and every quantity derived from it rest; "derived" marks quantities determined from the anchor and input values through the first-order-reservoir mathematics of Section 5; "input" marks externally sourced values or datasets supplied to the framework; "defined" marks notational identities; "physical constant" marks invariants of nature.

Symbol	Description	Value (units)	Status
Bomb-^{14}C related			
$R_a(t)$	Atmospheric Δ^{14}C, Hua+27 bomb-excess scale	‰	input
$R_a^*(t)$	Adjusted atmospheric Δ^{14}C: original Hua+27 values with plateau-level removals (1979–1984, 1992–1994, 1998–2010), emerge-shift corrections (2011–2016, 2016–2019), and FF correction C_{FF} added for $t \geq 2011$ (§ 9.2.5)	‰	derived
$R_{in}(t)$	Flux-weighted input Δ^{14}C, $\phi_{fast}(t)\, R_a(t) + \phi_{slow}(t)\, R_b$	‰	derived
R_b	Deep-mixed baseline ratio on Hua+27 scale; τ_{rel} exactly invariant for any constant R_b (§ 15.4.1)	0‰	defined
$R_{55}, A_{55}, \tau_{55}$	Fit to $R_a(t) : A_{55} \exp((1965 - t)/\tau_{55})$ full 55-point fit to uncorrected data	τ_{55} = 18.389±0.139 yr, A_{55} = 708.847 ± 3.565 ‰	fitted
$R_{28}, A_{28}, \tau_{28}$	Fit to $R_a^*(t) : A_{28} \exp((1965 - t)/\tau_{28})$ 28-point fit to bias-corrected data	τ_{28} = 18.382±0.016 yr, A_{28} = 709.138 ± 0.631‰	fitted
$R_{32}, A_{32}, \tau_{32}$	Fit to $R_a^*(t) : A_{32} \exp((1965 - t)/\tau_{32})$ 32-point fit to bias-corrected data (and 2011–2019 FF-corrected tail)	τ_{32} = 18.383±0.015 yr, A_{32} = 709.115 ± 0.585 ‰	fitted
$r_n(t)$	Residual of the Hua compilation against the n-point single-exponential fit at year t: $r_n(t) \equiv R_a(t) - A_n \exp[(1965 - t)/\tau_n]$, where $n \in \{55, 28, 32\}$ identifies the fit. Used in the smoothness test of Appendix C as r_{55}.	‰	derived
τ_{rel}	Our defined relaxation time $\tau_{rel} \equiv \tau_{32}$	18.383 ± 0.015 yr	anchor
Independent empirical results			
W_m	RRR mean residence time at MLO/BRW	3.68 yr, 3.91 yr	input
τ_{res}	$\langle W_m \rangle_{MLO,BRW}$	3.80 ± 0.12 yr	anchor
$\delta^{13}C_I$	Net atmospheric input δ^{13}C (Keeling intercept)	−13.2 ± 0.3‰	input
Defined rates			
τ_{out}	Kinematic instantaneous residence time, M_a/F_{out}	—	defined
κ_{out}	Kinematic instantaneous outflow rate, $F_{out}/M_a = 1/\tau_{out}$	—	defined
κ_{in}	Kinematic instantaneous inflow rate, F_{in}/M_a	—	defined
Derived rates			
$\langle \tau_{out}^{IPCC} \rangle$	IPCC 40-yr average residence time, M_a/F_{out}	3.70 ± 0.21 yr	derived
κ_{res}	Inverse of dynamical residence time, $1/\tau_{res}$	0.263 ± 0.008 yr^{-1}	derived
κ_{rel}	Constant relaxation rate, $\equiv 1/\tau_{rel}$	0.05440 ± 0.00004 yr^{-1}	derived
$\bar{\kappa}_{in}$	Gross inflow turnover, $\langle 1/\tau_{res} + \mathrm{d} \ln M_a/\mathrm{d}t \rangle_{1965–2019}$ (20)	0.2678 ± 0.0010 yr^{-1}	derived
$\kappa(t)$	Instantaneous isotopic decay rate, $\phi_{slow}(t)\, \kappa_{in}(t)$; empirically $\kappa(t) \approx \kappa_{rel}$ (§ 5.4)	~0.0544 yr^{-1}	derived
Reservoir partition (bridge equation)			
$\phi_{fast}(t)$	Fast-recycled inflow fraction; empirically constant (§ 5.4)	0.797 ± 0.006	derived
$\phi_{slow}(t)$	Deep-mixed inflow fraction, $1 - \phi_{fast}(t)$; empirically constant (§ 5.4)	0.203 ± 0.006	derived

Continued on next page

Table 3 continued from previous page

Symbol	Description	Value (units)	Status
Reservoir partition (two-box stratosphere–troposphere model)			
$\phi_{\text{fast}}^{\text{2box}}$	Fast-recycled return fraction in the two-box model (§ 7.3)	0.801 ± 0.008	derived
$\phi_{\text{slow}}^{\text{2box}}$	Deep-mixed return fraction, $1 - \phi_{\text{fast}}^{\text{2box}}$ (§ 7.3)	0.199 ± 0.008	derived
Atmospheric state and gross fluxes			
$M_{\text{a}}(t)$	Atmospheric carbon inventory (Mauna Loa)	GtC	input
$\mathrm{d}M_{\text{a}}/\mathrm{d}t$	Centered 5-point OLS growth rate of M_{a}	GtC yr^{-1}	derived
$E_{\text{FOS}}(t)$	Fossil-fuel emissions (GCB)	GtC yr^{-1}	input
$E_{\text{LUC}}(t)$	Land-use-change emissions (GCB)	GtC yr^{-1}	input
$F_{\text{anth}}(t)$	Anthropogenic atmospheric inflow, $E_{\text{FOS}} + E_{\text{LUC}}$ (GCB)	GtC yr^{-1}	input
$F_{\text{in}}(t)$	Gross atmospheric inflow	GtC yr^{-1}	derived
$F_{\text{out}}(t)$	Gross atmospheric outflow	GtC yr^{-1}	derived
$F_{\text{nat}}(t)$	Natural atmospheric inflow, $F_{\text{in}} - F_{\text{anth}}$	GtC yr^{-1}	derived
Fossil-fuel correction (§9.2.5)			
$D_{\text{FF}}(t)$	Instantaneous dilution coefficient, $1000\, E_{\text{FOS}}/M_{\text{a}}$	$‰\ \text{yr}^{-1}$	derived
$L_{\text{FF}}(t)$	Level-dependent dilution rate, $D_{\text{FF}}(0.973 + R_{\text{a}}/1000)$	$‰\ \text{yr}^{-1}$	derived
$\bar{D}_{\text{FF}}(t)$	Average FF dilution through year $t-2$ implicit in the 55-point fit R_{55}	$‰\ \text{yr}^{-1}$	derived
$D_{\text{excess}}(t)$	Per-year excess dilution, $L_{\text{FF}} - \bar{D}_{\text{FF}}$	$‰\ \text{yr}^{-1}$	derived
$C_{\text{FF}}(t)$	Fossil-fuel correction in $R_{\text{a}}^{*}(t)$ for t = 2011–2019; sum of D_{excess} from 2011 to t	‰	derived
Operator-equivalence appendix (§B.2)			
$\mathbf{A}$	Linear exchange operator on coupled reservoirs	yr^{-1}	defined
λ_j	Eigenvalue of $\mathbf{A}$, mode j	yr^{-1}	defined
$\mathbf{v}_j, \mathbf{w}_j$	Right and left eigenvectors of $\mathbf{A}$		defined
λ_{14}	^{14}C radioactive decay constant, $\ln 2/5{,}700$	$1.22 \times 10^{-4}\ \text{yr}^{-1}$	constant
$\mathbf{e}_{\text{a}}$	Atmospheric unit vector in reservoir state space		defined
$\delta\mathbf{M}, \delta\mathbf{X}$	Mass and ^{14}C-content anomaly vectors	GtC; GtC·‰	input

7. Two-reservoir stratosphere-troposphere model: cross-consistency with the bridge equation and derivation of the smoothness criterion

The bridge equation framework of §6.1 derives the relationship between the residence time τ_{res} and the relaxation time τ_{rel} from a turnover-flux mass-balance argument applied to the atmosphere as a whole. In this section we present a complementary derivation from a different mathematical entry point: the explicit coupled mass-balance ODEs governing ^{14}C exchange between the stratospheric and tropospheric reservoirs. The two-reservoir analysis serves three purposes. First, it provides an independent cross-validation of the bridge-equation ϕ partition through different mathematics and different physical inputs: the two-box framework derives $\phi_{\text{slow}}^{\text{2box}}$ from coupled-ODE eigenvalue analysis with the gross stratosphere–troposphere mass exchange F_{BDC} as an independent atmospheric input, whereas the bridge equation derives ϕ_{slow} from the turnover-flux algebra with τ_{res} and τ_{rel} as inputs. The two frameworks share only the empirical anchors τ_{res} and τ_{rel} but combine them through entirely different mathematical structures; agreement of the resulting ϕ_{slow} values to within 0.46 joint standard deviation provides genuine cross-validation of the bridge equation against an independent dynamical derivation. Second, the slow-mode quasi-equilibrium analysis yields the smoothness criterion used in Appendix C to distinguish smooth atmospheric physics from compilation artifacts in the Hua et al. (2022) [55] residual series. Third, the two-box analysis confirms that the linear-reservoir behavior established by the Koutsoyiannis RRR test on the whole-atmosphere CO_2 record is dynamically consistent with the explicit stratosphere–troposphere

reservoir structure.

7.1 Motivation: a minimal global-mean reduction

The Global Radiocarbon Exploration model (GRACE) of Hesshaimer (1997) [51] and Naegler & Levin (2006) [90] uses 22 atmospheric boxes (6 troposphere, 16 stratosphere) to resolve the latitudinal and vertical structure of bomb-^{14}C deposition and transport. The proliferation of boxes serves spatial resolution of the HASL observations [117], not different physics: each box obeys identical mass-balance and tracer-conservation equations, differing only in geometric volume and exchange coefficients with neighbors.

For our analysis the input is the globally averaged Hua et al. Δ^{14}C compilation, in which spatial structure has already been integrated away. The relevant physics for the global mean then reduces to a two-reservoir system: the global stratospheric pool S and the global tropospheric pool T, coupled by the global gross Brewer–Dobson mass exchange and drained to ocean and biosphere through the troposphere. This is the minimum model that exhibits two distinct timescales (S–T equilibration and combined-pool decay) necessary to explain the empirical decay record. We derive it explicitly below.

7.2 Reservoir masses and gross fluxes

Stratospheric and tropospheric carbon masses. The dry-atmosphere mass split between troposphere and stratosphere is approximately constant at 78.5%/21.5% over the analysis window, with the WMO thermal tropopause at a global mean of ∼12 km [118, 52, 24, 25, 29, 27]. With atmospheric carbon $M_{\mathrm{a}}(t) = [\mathrm{CO_2}](t) \times 2.124\,\mathrm{GtC/ppm}$ and applying a small correction for the ∼2 ppm depletion of stratospheric CO_2 relative to surface measurements [52], the 1965 reservoir masses are

$$M_{\mathrm{S}}(1965) = 144\ \mathrm{GtC} \qquad M_{\mathrm{T}}(1965) = 534\ \mathrm{GtC} \tag{26}$$

Both quantities scale with the atmospheric CO_2 growth over the record at the same rate.

Gross Brewer–Dobson mass flux from first principles. The total dry-air mass above the 100 hPa surface follows from hydrostatic balance:

$$M_{\mathrm{strat}}^{\mathrm{air}} = \frac{P\,A_{\mathrm{Earth}}}{g} = \frac{10^4\ \mathrm{Pa} \times 5.10 \times 10^{14}\ \mathrm{m^2}}{9.81\ \mathrm{m/s^2}} = 5.2 \times 10^{17}\ \mathrm{kg} \tag{27}$$

The mean age of stratospheric air, measured from balloon-borne in-situ samples of SF_6 and CO_2 at northern mid-latitudes (32–51°N) above 24 km altitude, is $\Gamma \approx 5.0$ yr, with no statistically significant change observed over the 1975–2016 record [33, 34]. This mid-latitude upper-altitude mean age constitutes an upper bound on the mass-weighted bulk stratospheric residence time τ_{strat}, since lower-altitude and tropical air has shorter transit time from the tropopause. A direct calculation of the bulk turnover time from Holton et al. (1995) [53] gives $\tau_{\mathrm{strat}} \approx 2$ yr at the 100 hPa surface. Taking the bulk-averaged value $\tau_{\mathrm{strat}} = 2.0$ yr with a range 1.5–3.0 yr that brackets both first-principles and balloon-derived constraints:

$$F_{\mathrm{air}} = \frac{M_{\mathrm{strat}}^{\mathrm{air}}}{\tau_{\mathrm{strat}}} = \frac{5.2 \times 10^{17}\ \mathrm{kg}}{2.0\ \mathrm{yr}} = (2.6 \pm 0.9) \times 10^{17}\ \mathrm{kg\ yr^{-1}} \tag{28}$$

This value is independently confirmed by the downward-control dynamical calculation of Holton et al. (1995) [53], which gives $F_{\mathrm{air}} = 2.7 \times 10^{17}\ \mathrm{kg\ yr^{-1}}$ at the 100 hPa surface from observed zonal-mean geopotential heights, in agreement with the balloon-derived value to within 4%.

Gross carbon flux. The gross carbon flux corresponding to F_{air} is obtained from the air-mass flux, the CO_2 mole fraction, and the ratio of carbon to dry-air molar masses (12 g/mol and 29 g/mol respectively). Each kilogram of dry air contains $1/29$ mol of air, of which a fraction equal to the CO_2 mole fraction is CO_2 (one mole of carbon per mole of CO_2). The mass of carbon per kilogram of air is therefore $(12/29) \times [CO_2] \times 10^{-6}$, where $[CO_2]$ is atmospheric concentration in ppm. With the 1965 Mauna Loa annual mean $[CO_2] = 320.04$,

$$F_{\text{BDC}}(1965) = F_{\text{air}} \times \frac{12}{29} \times \chi_{CO_2}(1965) = 34.4 \pm 11.5 \text{ GtC yr}^{-1} \tag{29}$$

where the ±33% uncertainty propagates from the τ_{strat} range above.

Effective gross atmospheric outflow to ocean and biosphere. The Koutsoyiannis Refined Reservoir Routing test (§4.2.2) applied to the Mauna Loa and Barrow CO_2 records measures the gross molecular turnover time $\tau_{\text{res}} = 3.80 \pm 0.12$ yr for the atmosphere as a whole. The corresponding gross outflow is

$$F_{\text{out}}(1965) = \frac{M_{\text{a}}}{\tau_{\text{res}}} = \frac{678 \text{ GtC}}{3.80 \text{ yr}} = 178 \pm 6 \text{ GtC yr}^{-1} \tag{30}$$

in 1965. This is the total gross outflow from the atmosphere to ocean and biosphere combined; the troposphere is the conduit.

7.3 Coupled mass-balance ODEs

Mass conservation for ^{14}C in each reservoir, with the stratosphere exchanging only with the troposphere via the Brewer–Dobson circulation [92, 13], and the troposphere exchanging with both the stratosphere and the Earth surface sinks:

$$\frac{d(M_{\text{S}}\, R_{\text{S}})}{\text{d}t} = F_{\text{BDC}}\,(R_{\text{T}} - R_{\text{S}}) \tag{31}$$

$$\frac{d(M_{\text{T}}\, R_{\text{T}})}{\text{d}t} = F_{\text{BDC}}\,(R_{\text{S}} - R_{\text{T}}) - F_{\text{out}}(1965)\,(R_{\text{T}} - R_{\text{E,eff}}) \tag{32}$$

where $R_{\text{E,eff}}$ is the effective return signature of carbon flowing back from ocean and biosphere into the troposphere. We write this as $R_{\text{E,eff}} = \phi^{\text{2box}}_{\text{fast}}\, R_{\text{T}}$: a fraction $\phi^{\text{2box}}_{\text{fast}}$ of returning carbon comes back at the contemporary atmospheric ratio (the fast-recycled fraction), while the complement $\phi^{\text{2box}}_{\text{slow}} = 1 - \phi^{\text{2box}}_{\text{fast}}$ comes back at the pre-bomb baseline (0‰ on the +27 scale). The net effective drainage rate of ^{14}C from the troposphere is then $F_{\text{out}}(1965)\,(1 - \phi^{\text{2box}}_{\text{fast}})\, R_{\text{T}}$, identical in physical content to the deep-mixed leak fraction of §6.2.

Expanding the product derivatives and dividing through by the reservoir masses:

$$\frac{\text{d}R_{\text{S}}}{\text{d}t} = \frac{F_{\text{BDC}}}{M_{\text{S}}}\,(R_{\text{T}} - R_{\text{S}}) - R_{\text{S}}\,\frac{\text{d}\ln M_{\text{S}}}{\text{d}t} \tag{33}$$

$$\frac{\text{d}R_{\text{T}}}{\text{d}t} = \frac{F_{\text{BDC}}}{M_{\text{T}}}\,(R_{\text{S}} - R_{\text{T}}) - \frac{F_{\text{out}}(1965)\,(1 - \phi^{\text{2box}}_{\text{fast}})}{M_{\text{T}}}\,R_{\text{T}} - R_{\text{T}}\,\frac{\text{d}\ln M_{\text{T}}}{\text{d}t} \tag{34}$$

Quantifying the $\text{d}\ln M/\text{d}t$ term. The atmospheric carbon mass grows at $\text{d}\ln M_{\text{a}}/\text{d}t \approx 0.0047\,\text{yr}^{-1}$ averaged over 1965–2019 (NOAA Mauna Loa). Since both reservoirs share this growth, the $\text{d}\ln M/\text{d}t$ term acts as a uniform additive shift on both eigenvalues. The term ranges from 1.7 to 0.2‰/yr against dominant terms of ∼ 80 and ∼ 3‰/yr at the bomb-spike and modern periods respectively, a 2–7% correction to dominant dynamics. It is structurally important: it accounts for the fact that the atmospheric reservoir into which ^{14}C is decaying is itself growing, and the decay rate observed against a growing denominator differs from the underlying physical drainage rate.

7.4 Eigenvalue analysis

Define the rate constants

$$k_S = \frac{F_{BDC}}{M_S} \qquad k_T = \frac{F_{BDC}}{M_T} \qquad k_E = \frac{F_{out}(1965)\,(1-\phi_{fast}^{2box})}{M_T} \qquad k_{dM} = \frac{d\ln M_a}{dt} \tag{35}$$

With numerical values at 1965:

$$k_S = 0.239 \pm 0.080\ \mathrm{yr}^{-1} \qquad k_T = 0.0644 \pm 0.0215\ \mathrm{yr}^{-1} \qquad k_{dM} = 0.0047\ \mathrm{yr}^{-1} \tag{36}$$

where the uncertainties on k_S and k_T propagate from F_{BDC}. The linear system written in matrix form is

$$\frac{d}{dt}\begin{pmatrix} R_S \\ R_T \end{pmatrix} = (A - k_{dM}\,I)\begin{pmatrix} R_S \\ R_T \end{pmatrix} \quad \text{where} \quad A = \begin{pmatrix} -k_S & k_S \\ k_T & -(k_T + k_E) \end{pmatrix} \tag{37}$$

The $-k_{dM}\,I$ term shifts both eigenvalues of A uniformly by $-k_{dM}$.

The eigenvalues of A are the roots of

$$\lambda^2 + (k_S + k_T + k_E)\,\lambda + k_S\,k_E = 0 \tag{38}$$

giving

$$\lambda_{\pm}^{(A)} = \frac{-(k_S + k_T + k_E) \pm \sqrt{(k_S + k_T + k_E)^2 - 4\,k_S\,k_E}}{2} \tag{39}$$

The observed eigenvalues are shifted by $-k_{dM}$:

$$\lambda_{\pm}^{obs} = \lambda_{\pm}^{(A)} - k_{dM} \tag{40}$$

The slow observed eigenvalue corresponds to the empirical τ_{rel}, and the fast eigenvalue gives the S–T equilibration timescale τ_{fast}:

$$\tau_{slow} = -\frac{1}{\lambda_+^{obs}} = \tau_{rel} \qquad \tau_{fast} = -\frac{1}{\lambda_-^{obs}} \tag{41}$$

7.5 Calibration to the observed τ_{rel}

The independent parameters M_S, M_T, F_{BDC}, and k_{dM} are sourced from atmospheric measurements (§7.2). The remaining parameter ϕ_{fast}^{2box} (equivalently k_E via the relation $k_E = (F_{out}(1965)/M_T)\,(1-\phi_{fast}^{2box})$) is calibrated by requiring the observed slow eigenvalue to match the empirical τ_{rel} from the 55-point uncorrected fit:

$$\tau_{rel} = 18.389 \pm 0.139\ \mathrm{yr} \quad \Rightarrow \quad \lambda_+^{obs} = -\frac{1}{18.389}\ \mathrm{yr}^{-1} \tag{42}$$

We use the 55-point uncleaned value here rather than the cleaned 32-point value to avoid circularity: the cleaning is itself justified by the smoothness criterion derived from this model (Appendix C), so the calibration must use a value computed without that cleaning. The choice of 55-point versus 32-point τ_{rel} shifts the calibrated k_E by less than 0.1% (Table 10) and is therefore operationally irrelevant.

Solving for k_E algebraically:

$$k_E = -\frac{L(L + k_S + k_T)}{L + k_S} \qquad \text{where} \quad L = \lambda_+^{(A)} = -\frac{1}{\tau_{rel}} + k_{dM} \tag{43}$$

Numerical result: $k_E = 0.0666 \pm 0.0019\ \mathrm{yr}^{-1}$, equivalent to $F_{OE,eff} = k_E\,M_T = 35.6 \pm 1.0\,\mathrm{GtC/yr}$ of effective net ^{14}C drainage from the troposphere.

The complete two-box parameter set with the eigenvalue outputs is collected in Table 4. The corresponding fast eigenvalue gives $\tau_{\text{fast}} = 3.08 \pm 1.04$ yr (the uncertainty propagates from the ±33% uncertainty in F_{BDC}, which dominates), and the slow eigenvector ratio fixes $\phi_{\text{fast}}^{\text{2box}} = 0.801 \pm 0.008$.

The fast eigenvalue τ_{fast} characterizes the stratosphere–troposphere equilibration timescale, which is set by the Brewer–Dobson circulation. The value $\tau_{\text{fast}} = 3.08$ yr falls between the bulk-averaged stratospheric turnover time of ~ 2 yr inferred from Holton et al. (1995) [53] and the mid-latitude upper-stratospheric mean age of ~ 5 yr from Engel et al. (2009) [33] and Engel et al. (2017) [34], exactly as expected for a mass-weighted bulk S–T equilibration time. The two-box framework thus passes an internal consistency check against independent stratospheric turnover measurements.

Table 4: Two-box model parameters at 1965 and the eigenvalue outputs, with uncertainties. Only τ_{rel} is used to calibrate the model; all other quantities are independently sourced or emerge from the eigenvalue analysis. Reported uncertainties propagate F_{BDC}, τ_{rel}, and τ_{res} uncertainties in quadrature, treated as independent.

Quantity	Value	Source
Independently sourced inputs:		
$M_{\text{S}}(1965)$	144 GtC	Hesshaimer & Levin (2000) [52] air-mass split
$M_{\text{T}}(1965)$	534 GtC	Hesshaimer & Levin (2000) [52] air-mass split
$M_{\text{strat}}^{\text{air}}$	5.2×10^{17} kg	first principles ($P\,A_{\text{Earth}}/g$)
τ_{strat}	2.0 (1.5–3.0) yr	Holton et al. (1995) [53]; Engel et al. (2009, 2017) [33, 34]
$F_{\text{BDC}}(1965)$	34.4 ± 11.5 GtC yr^{-1}	derived from $M_{\text{strat}}^{\text{air}}/\tau_{\text{strat}}$
τ_{res}	3.80 ± 0.12 yr	Koutsoyiannis (2024a) [72] RRR test
$F_{\text{out}}(1965)$	178 ± 6 GtC yr^{-1}	$M_{\text{a}}/\tau_{\text{res}}$
$\mathrm{d}\ln M_{\text{a}}/\mathrm{d}t$	0.00463 yr^{-1}	Mauna Loa 1965–2019 average
Calibration target:		
τ_{rel}	18.389 ± 0.139 yr	55-point uncorrected fit (Table 10)
Derived rate constants:		
$k_{\text{S}} = F_{\text{BDC}}/M_{\text{S}}$	0.239 ± 0.080 yr^{-1}	derived
$k_{\text{T}} = F_{\text{BDC}}/M_{\text{T}}$	0.0644 ± 0.0215 yr^{-1}	derived
k_{E}	0.0666 ± 0.0019 yr^{-1}	calibrated to τ_{rel}
$F_{\text{OE,eff}}$	35.6 ± 1.0 GtC yr^{-1}	$k_{\text{E}}\,M_{\text{T}}$
Eigenvalue outputs:		
τ_{fast}	3.08 ± 1.04 yr	from eigenvalue analysis
τ_{slow}	18.39 yr	by construction (calibration)
$\phi_{\text{fast}}^{\text{2box}}$	0.801 ± 0.008	from slow eigenvector ratio
$\phi_{\text{slow}}^{\text{2box}}$	0.199 ± 0.008	$1 - \phi_{\text{fast}}^{\text{2box}}$

7.6 Cross-consistency with the bridge-equation framework

The bridge-equation framework of Equation (23) derives the relationship between τ_{res}, τ_{rel}, and the deep-mixed leak fraction from the turnover-flux mass balance of the whole atmosphere, with τ_{res} measured dynamically by the Koutsoyiannis RRR test on Mauna Loa and Barrow CO_2 records (§4.2.2) and τ_{rel} measured by single-exponential fitting to the post-peak Δ^{14}C record. The two-box derivation above starts from a different mathematical entry point (coupled S–T reservoir ODEs with explicit gross fluxes) and sources M_{S}, M_{T}, F_{BDC}, and k_{dM} from independent atmospheric measurements.

Inputs and outputs of the two-box analysis. The two-box model takes the independently sourced inputs M_{S}, M_{T}, F_{BDC}, k_{dM}, $F_{\text{out}}(1965) = M_{\text{a}}/\tau_{\text{res}}$, and the calibration target τ_{rel}, and produces three mathematical outputs from the eigenvalue analysis: the two eigenvalues τ_{fast}, τ_{slow}, and the slow-mode eigenvector ratio $R_{\text{S}}/R_{\text{T}}$. By construction, $\tau_{\text{slow}} = \tau_{\text{rel}}$; this is the calibration target. The

remaining outputs, τ_{fast} and the eigenvector ratio (equivalently $\phi_{\text{fast}}^{\text{2box}}$), follow from the calibrated k_{E} together with the independently sourced k_{S} and k_{T}.

Cross-validation of ϕ. The bridge-equation algebra computes the deep-mixed leak fraction from $\phi_{\text{fast}} = 1 - 1/(\tau_{\text{rel}}\,\bar{\kappa}_{\text{in}})$ Eq. (23), where $\bar{\kappa}_{\text{in}} = \langle 1/\tau_{\text{res}} + \mathrm{d}\ln M_{\text{a}}/\mathrm{d}t\rangle_{1965\text{–}2019}$ is the gross atmospheric inflow rate. The two-box algebra computes ϕ_{fast} from $\phi_{\text{fast}} = 1 - k_{\text{E}}\,M_{\text{T}}/F_{\text{out}}(1965)$, where k_{E} is determined through the coupled-ODE eigenvalue calibration. The two frameworks share only the empirical anchors τ_{res} and τ_{rel} but combine them through entirely different mathematical structures, and the two-box analysis additionally takes F_{BDC} as an independent input derived from atmospheric mass and balloon-borne measurements of stratospheric mean age. That the two computations yield the same ϕ partition is therefore a meaningful cross-validation: it verifies that the explicit coupled-reservoir dynamics is consistent with the whole-atmosphere turnover-flux mass-balance framework, and that the first-principles F_{BDC} value is dynamically compatible with the observed bomb-^{14}C decay.

Joint uncertainty in the comparison. The bridge equation gives $\phi_{\text{slow}}^{\text{bridge}} = 0.203 \pm 0.006$ and $\phi_{\text{fast}}^{\text{bridge}} = 0.797 \pm 0.006$, with uncertainty dominated by $\tau_{\text{res}} = 3.80 \pm 0.12$ yr through $\bar{\kappa}_{\text{in}}$. The two-box analysis gives $\phi_{\text{slow}}^{\text{2box}} = 0.199 \pm 0.008$ and $\phi_{\text{fast}}^{\text{2box}} = 0.801 \pm 0.008$, with uncertainty dominated by $F_{\text{BDC}} = 34.4 \pm 11.5$ GtC/yr. For the direct comparison $\Delta\phi = \phi^{\text{2box}} - \phi^{\text{bridge}}$, naive quadrature combination of the individual uncertainties would give $\sigma^{\text{naive}} = 0.010$. However, τ_{rel} and τ_{res} enter both frameworks: τ_{rel} appears explicitly in the bridge formula and is the calibration target for the two-box k_{E}; τ_{res} appears in $\bar{\kappa}_{\text{in}}$ for the bridge and in $F_{\text{out}}(1965)$ for the two-box. Sensitivity analysis of the difference $\Delta\phi(\tau_{\text{rel}}, \tau_{\text{res}}, F_{\text{BDC}})$ shows that the τ_{rel} and τ_{res} contributions cancel between the two frameworks to better than 5% of their individual contributions, leaving F_{BDC} as the only material contributor to the joint uncertainty in $\Delta\phi$:

$$\sigma_{\Delta\phi} = \left|\frac{\partial\Delta\phi}{\partial F_{\text{BDC}}}\right| \sigma_{F_{\text{BDC}}} = 0.0054 \tag{44}$$

The observed difference $\Delta\phi_{\text{slow}} = -0.004$ then corresponds to $\Delta\phi_{\text{slow}}/\sigma_{\Delta\phi} = 0.74$. The two independent frameworks agree on the ϕ partition to within 0.75σ of joint uncertainty, with F_{BDC} as the dominant remaining source of comparison uncertainty.

Direct sensitivity of $\phi_{\text{slow}}^{\text{2box}}$ to the τ_{rel} calibration target, and its cancellation in $\Delta\phi$. The claim of τ_{rel}-cancellation in $\sigma_{\Delta\phi}$ above is quantitative. Perturbing τ_{rel} at fixed k_{S}, k_{T}, k_{dM} and recomputing k_{E} from the slow-mode eigenvalue condition $\lambda_{\text{slow}} = -1/\tau_{\text{rel}}$ gives

$$\frac{\partial\phi_{\text{slow}}^{\text{2box}}}{\partial\tau_{\text{rel}}} = \frac{M_{\text{T}}}{F_{\text{out}}(1965)}\,\frac{\partial k_{\text{E}}}{\partial\tau_{\text{rel}}} = -0.0127\,\text{yr}^{-1} \tag{45}$$

The bridge-equation partial is $\partial\phi_{\text{slow}}^{\text{bridge}}/\partial\tau_{\text{rel}} = -\phi_{\text{slow}}/\tau_{\text{rel}} = -0.0110\,\text{yr}^{-1}$. Both frameworks therefore respond to τ_{rel} in the same direction with nearly the same magnitude, and in the difference the two contributions cancel to

$$\frac{\partial\Delta\phi}{\partial\tau_{\text{rel}}} = -0.0127 - (-0.0110) = -0.00166\,\text{yr}^{-1} \tag{46}$$

about an order of magnitude below either individual sensitivity. Propagating through the τ_{rel} uncertainty envelope gives direct contributions to $\sigma_{\Delta\phi}$ of $\pm 2.5\times10^{-5}$ at the cleaned-fit uncertainty $\sigma_{\tau_{\text{rel}}} = 0.015$ yr (Table 2) and $\pm 2.3\times10^{-4}$ at the much larger uncleaned 55-point uncertainty $\sigma_{\tau_{55}} = 0.139$ yr (Table 10). Against the F_{BDC}-dominated $\sigma_{\Delta\phi} = 0.0054$, the τ_{rel} contribution is ∼0.5% at the cleaned uncertainty and ∼4% at the uncleaned uncertainty, confirming the "better than

5%" cancellation claimed above. The $\Delta\phi$ comparison is therefore controlled by F_{BDC} across the full τ_{rel} uncertainty envelope, and the two-box ϕ partition is a cross-validation of the bridge-equation framework, not a restatement of its calibration input.

7.7 Slow-mode quasi-equilibrium and the smoothness criterion

The slow eigenmode of the two-box system has eigenvalue $\lambda_+^{\mathrm{obs}} = -1/\tau_{\mathrm{rel}}$. Its eigenvector satisfies

$$(A - \lambda_+^{(A)}\, I)\, \mathbf{v}_{\mathrm{slow}} = 0 \tag{47}$$

Table 5: Quantitative comparison between the bridge-equation framework (23) and the two-reservoir eigenvalue analysis. The bridge-equation column reproduces Table 2. The two-box column gives the values computed in the two-box analysis with uncertainties propagated from $F_{\mathrm{BDC}} = 34.4 \pm 11.5$ GtC/yr and the calibrated τ_{rel}. The τ_{rel} row is the calibration target by construction; the ϕ rows are independent cross-validations between two mathematically distinct frameworks. The $\Delta\phi/\sigma_{\Delta\phi}$ column uses correlation-aware joint uncertainty (see text): τ_{rel} and τ_{res} appear in both frameworks and their contributions to $\sigma_{\Delta\phi}$ largely cancel, leaving F_{BDC} as the dominant comparison uncertainty.

Quantity	Bridge equation	Two-box model	$\lvert\Delta\rvert/\sigma_\Delta$
τ_{rel}	18.389 ± 0.139 yr	18.389 yr	(calibration target)
ϕ_{slow}	0.203 ± 0.006	0.199 ± 0.008	0.74
ϕ_{fast}	0.797 ± 0.006	0.801 ± 0.008	0.74

The first row gives

$$(-k_{\mathrm{S}} - \lambda_+^{(A)})\, v_{\mathrm{S}} + k_{\mathrm{S}}\, v_{\mathrm{T}} = 0 \quad \Rightarrow \quad \frac{v_{\mathrm{S}}}{v_{\mathrm{T}}} = \frac{k_{\mathrm{S}}}{k_{\mathrm{S}} + \lambda_+^{(A)}} \tag{48}$$

Numerically: $v_{\mathrm{S}}/v_{\mathrm{T}} = 1.263 \pm 0.135$.

Quasi-equilibrium stratospheric–tropospheric gap. In the slow mode, after the fast S–T equilibration has settled (formally $\sim 3\,\tau_{\mathrm{fast}} \approx 9$ yr post-test-ban), R_{S} tracks R_{T} with the constant ratio

$$R_{\mathrm{S}}(t) = (1.263 \pm 0.135)\, R_{\mathrm{T}}(t) \qquad \text{equivalently} \quad R_{\mathrm{S}}(t) - R_{\mathrm{T}}(t) = (0.263 \pm 0.135)\, R_{\mathrm{T}}(t) \tag{49}$$

This ratio is a prediction of the model. Cross-checking against Hesshaimer & Levin (2000) [52] Table 2 for 1978 with $R_{\mathrm{T}} \approx 350$‰: model predicts $R_{\mathrm{S}} \approx 442 \pm 47$‰, which falls within the range of HASL lower-stratospheric observations for that year.

STE contribution to $\mathrm{d}R_{\mathrm{T}}/\mathrm{d}t$. The stratospheric source contribution to the tropospheric rate of change, in the slow mode, is

$$\left(\frac{\mathrm{d}R_{\mathrm{T}}}{\mathrm{d}t}\right)_{\mathrm{STE}} = k_{\mathrm{T}}\,(R_{\mathrm{S}} - R_{\mathrm{T}}) = 0.0644 \times 0.263\, R_{\mathrm{T}}(t) = (0.0169 \pm 0.0018)\, R_{\mathrm{T}}(t)\ [\text{‰/yr}] \tag{50}$$

This is a smooth, monotonically decreasing function of time as R_{T} decays. It has no year-to-year noise term. STE delivers high-Δ^{14}C carbon to the troposphere at a rate determined entirely by the quasi-equilibrium gap, which itself decays smoothly with the slow mode.

Second derivative and the smoothness criterion. Differentiating equation (50) once more in time:

$$\left(\frac{\mathrm{d}^2 R_{\mathrm{T}}}{\mathrm{d}t^2}\right)_{\mathrm{STE}} = 0.0169\, \frac{\mathrm{d}R_{\mathrm{T}}}{\mathrm{d}t} = -\frac{0.0169}{\tau_{\mathrm{rel}}}\, R_{\mathrm{T}}(t) = -(0.000920 \pm 0.000099)\, R_{\mathrm{T}}(t)\ \left[\text{‰/yr}^2\right] \tag{51}$$

This is the maximum smoothness deviation that STE physics can produce in the second derivative of tropospheric Δ^{14}C. Since the single-exponential fit's contribution to d^2R_T/dt^2 is also smooth, the second difference of the residual against the fit must satisfy the same bound:

$$\left|\frac{d^2 r_{55}}{dt^2}\right| \leq 0.000920\, R_T(t)\ \left[‰/\mathrm{yr}^2\right] \tag{52}$$

This is the *smoothness criterion* used in Appendix C to test individual years of the Hua et al. (2022) [55] compilation. Any year-by-year second difference of the residuals exceeding this bound by a substantial factor cannot be smooth STE physics; it must be either a documented ocean–biosphere perturbation event or a compilation artifact. The discrete implementation of equation (52) with the relevant buffer factors and noise floors is given in §C.4 and §C.5.

7.8 Summary

The two-reservoir model derived here uses independently sourced atmospheric inputs: reservoir masses M_S, M_T from the standard atmospheric mass split, gross stratospheric mass flux F_{BDC} from first-principles atmospheric mass combined with balloon-borne measurements of stratospheric mean age, atmospheric growth rate k_{dM} from the NOAA Mauna Loa record, and gross outflow $F_{out}(1965)$ from τ_{res} measured by the Koutsoyiannis RRR test on CO_2 records. One free parameter k_E (equivalently ϕ_{fast}) is calibrated against the empirical $\tau_{rel} = 18.389 \pm 0.139$ yr.

The two-box deep-mixed leak fraction $\phi_{slow}^{2box} = 0.199 \pm 0.008$ agrees with the bridge-equation value $\phi_{slow}^{bridge} = 0.203 \pm 0.006$ with $\Delta\phi_{slow} = -0.004$ at 0.74σ of the correlation-aware joint uncertainty $\sigma_{\Delta\phi} = 0.0054$ (dominated by F_{BDC}, since τ_{rel} and τ_{res} enter both frameworks and their contributions to $\Delta\phi$ cancel). The two frameworks share only the empirical anchors τ_{res} and τ_{rel} but combine them through entirely different mathematical structures, and the two-box analysis additionally constrains F_{BDC} from independent atmospheric mass and balloon mean-age data. This agreement therefore constitutes genuine cross-validation: it confirms that the turnover-flux mass-balance framework of §6 is dynamically consistent with the explicit coupled-reservoir ODE structure, and that the first-principles F_{BDC} value is compatible with the observed bomb-^{14}C decay.

The fast eigenvalue $\tau_{fast} = 3.08 \pm 1.04$ yr falls between the bulk-averaged stratospheric turnover time ~ 2 yr [53] and the mid-latitude upper-stratospheric mean age ~ 5 yr [33, 34], exactly as expected for a mass-weighted bulk S–T equilibration time. The two-box framework thus passes an internal consistency check against independent stratospheric turnover measurements.

The slow-mode quasi-equilibrium analysis yields the smoothness criterion $|d^2r_{55}/dt^2| \leq 0.000920\, R_T(t)$, which provides the first-principles physical bound used in Appendix C to identify compilation artifacts in the Hua et al. (2022) [55] record. The bound is derived entirely from coupled mass conservation in the two reservoirs and uses no information from the compilation itself.

8. Bomb-^{14}C Decay: Empirical Determination of τ_{rel} and Falsification of the IPCC Bern IRFs

All single-exponential fits in this paper of data from Hua et al. (2022) [55] are obtained by *unweighted* nonlinear least squares on the central values. This choice is dictated by the error structure of the Hua compilation and the multiplicative physics of Δ^{14}C decay rather than by convention; the full justification, including why inverse-variance ($1/\sigma^2$) weighting would be actively misleading for a multi-station composite product, is given in Appendix A.

8.1 Zonal single-exponential fits

The Hua et al. (2022) [55] zonal data (Tables S1a–S1c) show a pre-bomb baseline of $\Delta^{14}C \approx -27‰$ in 1950–1953 across all zones. To work with the bomb excess above this baseline, we add a constant

+27‰ offset to all values before fitting the single-exponential model, where n is the number of points in the fit:

$$R_n(t) = R_n(t_0)\ \exp\left((t_0 - t)/\tau\right) \tag{53}$$

The single-exponential bomb-decay model describes the slow eigenmode of the coupled stratosphere-troposphere-earth system derived in §7. The slow eigenmode is the physical regime of the atmosphere only after fast-mode stratosphere-troposphere equilibration has decayed enough that its contribution to $\mathrm{d}R_\mathrm{T}/\mathrm{d}t$ is small compared to the slow-mode contribution. US and USSR atmospheric nuclear testing ended in late 1962, with most bomb ^{14}C injected directly into the stratosphere; French and Chinese atmospheric tests continued through 1976 (§9.2.2). Each tropospheric zone reached its own peak when the rate of stratosphere-to-troposphere input from STE equaled the slow-mode loss rate, the moment when $\mathrm{d}R_\mathrm{T}/\mathrm{d}t = 0$. By two years after each peak, the fast-mode contribution has decayed by a factor of $\exp(-2/\tau_\mathrm{fast}) = 0.59$ while the slow-mode contribution remains at $\exp(-2/\tau_\mathrm{slow}) = 0.90$. The first two post-peak years are therefore in the fast-mode-dominated regime; the slow-mode regime begins approximately at peak + 2 years.

Two fits are presented for each series: a full fit from the boreal-summer peak year of that series through 2019, and a slow-mode fit that excludes the first two post-peak years. The physical criterion is uniform; the slow-mode fit is the recommended one for each series only where the data establish that fast-mode contamination is empirically detectable. See §8.1.1 for the metric-by-metric comparison that determines the choice.

A single-exponential model (53) explains more than 99.6% of the variance in every zonal series and > 99.8% in the global average fit, with $\sigma < 8$‰. Every series decays as a single exponential. No series requires a second exponential, a permanent fraction, or a regime change. Because each fit starts at its own series' peak, the global timing of the bomb pulse is implicit in t_0 and does not affect τ; the systematic increase in τ from north to south is therefore not a mixing-delay artifact but a real gradient in the local atmospheric relaxation timescale.

8.1.1 Where fast-mode contamination is empirically detectable

The two-box analysis of §7 predicts that fast-mode contamination is largest in the Northern Hemisphere near the high-latitude stratospheric injection region where most atmospheric nuclear testing occurred. SH zones receive their bomb ^{14}C through interhemispheric transport that smooths and delays the input, producing a weaker fast-mode transient [12, 125]. Table 6 presents both the full fit and the slow-mode (first-two-years-excluded) fit for every series, with the recommended fit indicated in boldface. Inspection of the table shows that only three series exhibit empirically detectable fast-mode contamination. For these three series the slow-mode fit improves R^2, reduces σ, and tightens the relative uncertainties on both τ and $R_n(t_0)$ simultaneously, while the recovered τ shifts at the $\sim 2\sigma$ level or above:

NH zone 1 ($\Delta\tau = +1.026$ yr, 3.04σ separation), NH zone 2 ($\Delta\tau = +0.382$ yr, 1.80σ separation), and NH avg ($\Delta\tau = +0.328$ yr, 1.56σ separation). All three series sit in the high-latitude or mid-latitude Northern Hemisphere where bomb-test injection was strongest. For NH zone 1 specifically, the full fit's $\sigma = 20.028$‰ is dominated by fast-mode residuals in 1964–1965; excluding these two years reduces σ to 9.965‰, a factor of two improvement, and improves R^2 from 99.134% to 99.701%. The recovered τ shifts from 16.722 yr (biased low by fast-mode contamination) to 17.748 yr (slow eigenmode).

Table 6: Single-exponential fits to the Hua et al. (2022) [55] boreal-summer annual zonal $\Delta^{14}C$ means, with +27‰ offset applied. Each full fit runs from the series' peak year t_0^{peak} to 2019; each slow-mode fit runs from $t_0^{\text{peak}} + 2$ to 2019, excluding the first two post-peak years where the system is still in the fast-mode regime. $R_n(t_0)$ is the fitted amplitude at the starting year; $\sigma = \sqrt{\text{SSR}/\text{n}}$; parameter errors use the $n - 2$ convention. Boldface indicates the recommended fit for each series. The slow-mode fit is preferred only where the data establish that fast-mode contamination is empirically detectable (NH zones 1, 2, and NH avg); elsewhere the full fit is preferred because the fast-mode contamination is below detection and the full fit provides better leverage on τ via the high-amplitude early data points.

Series	Fit	t_0	n	$R_n(t_0)$ (‰)	τ (yr)	R^2 (%)	σ (‰)
NH zone 1	full	1964	56	823.314 ± 9.502	16.722 ± 0.287	99.134	20.028
NH zone 1	**slow-mode**	**1966**	**54**	**694.574 ± 4.632**	**17.748 ± 0.177**	**99.701**	**9.965**
NH zone 2	full	1964	56	782.246 ± 4.613	17.590 ± 0.155	99.766	9.918
NH zone 2	**slow-mode**	**1966**	**54**	**685.344 ± 3.676**	**17.972 ± 0.145**	**99.805**	**7.946**
NH zone 3	**full**	**1965**	**55**	**712.072 ± 4.541**	**18.254 ± 0.175**	**99.722**	**9.890**
NH zone 3	slow-mode	1967	53	631.503 ± 4.312	18.479 ± 0.191	99.680	9.404
NH avg	full	1964	56	774.343 ± 4.413	17.763 ± 0.151	99.781	9.524
NH avg	**slow-mode**	**1966**	**54**	**680.912 ± 3.691**	**18.091 ± 0.147**	**99.800**	**7.999**
SH zone 3	**full**	**1965**	**55**	**698.767 ± 4.766**	**18.724 ± 0.192**	**99.673**	**10.481**
SH zone 3	slow-mode	1967	53	635.415 ± 4.553	18.467 ± 0.200	99.641	9.928
SH zone 12	**full**	**1965**	**55**	**688.787 ± 3.746**	**18.903 ± 0.155**	**99.790**	**8.268**
SH zone 12	slow-mode	1967	53	620.970 ± 3.794	18.854 ± 0.174	99.735	8.336
SH avg	**full**	**1965**	**55**	**690.283 ± 3.843**	**18.876 ± 0.158**	**99.780**	**8.478**
SH avg	slow-mode	1967	53	623.123 ± 3.873	18.795 ± 0.177	99.727	8.500
Global avg	**full**	**1965**	**55**	**708.847 ± 3.565**	**18.389 ± 0.139**	**99.825**	**7.787**
Global avg	slow-mode	1967	53	631.426 ± 3.487	18.537 ± 0.155	99.788	7.613

The remaining five series (NH zone 3, SH zone 3, SH zone 12, SH avg, Global avg) show no material change from the two-year exclusion. For these series the τ difference between full and slow-mode fits is below 1σ, and the relative uncertainty on τ is lower in the full fit than in the slow-mode fit due to the longer time baseline and the leverage provided by the high-amplitude early data points. For the global average specifically, the full fit has τ relative uncertainty of 0.756% vs. 0.836% in the slow-mode fit, and $R^2 = 99.825\%$ vs. 99.788%, while σ differs by only 0.174‰ in a direction (lower in slow-mode) that is mechanically due to removing two high-noise early points rather than improving fit quality.

The pattern of where the slow-mode fit is empirically required, and where it is not, is itself a quantitative confirmation of the two-box prediction. Fast-mode contamination is detectable in the NH high-latitude and mid-latitude zones nearest the stratospheric injection region. It is below detection in the NH low-latitude zone (NH zone 3, equatorward of the principal test sites), in all Southern Hemisphere zones, and in the hemispherically-averaged global record where NH and SH contributions partially cancel. The physical criterion (peak + 2 years for slow-mode onset) is uniform across all series; the empirical pattern of where it produces a material change reveals the spatial structure of the fast-mode transient predicted by the two-box theory.

8.2 The hemispheric τ_{rel} asymmetry is geography, not transport delay

The boreal-summer hemispheric averages of τ_{rel} from single-exponential fits to the Hua et al. (2022) [55] compilation (Table 6) are $\tau_{\text{NH avg}} = 18.091$ yr and $\tau_{\text{SH avg}} = 18.879$ yr, and the global average is $\tau_{55} = 18.389 \pm 0.139$ yr (±0.76%). The hemispheric asymmetry $\tau_{\text{SH}} - \tau_{\text{NH}} = 0.788$ yr is quantitative and consistently directed: the Southern Hemisphere, with much more ocean surface

area, exhibits slower bomb-^{14}C decay. We show here that this hemispheric asymmetry is a parameter-free prediction of the first-principles two-pool mass-balance framework of §6 from one bomb-^{14}C anchor (the global τ_{55}) and one independent physical constraint, with no zonal or hemispheric bomb-^{14}C data entering the prediction; the hemispheric τ asymmetry is unrelated to the delay time of ^{14}C arrival from north to south.

Two-pool decomposition of ϕ_{slow}. The bridge relation derived in Eq. (23) reads

$$\tau_{\text{rel}} = \frac{1}{\phi_{\text{slow}}\,\bar{\kappa}_{\text{in}}} \tag{54}$$

where $\bar{\kappa}_{\text{in}} = \langle 1/\tau_{\text{res}} + \mathrm{d}\ln M_{\text{a}}/\mathrm{d}t\rangle_{1965\text{–}2019} = 0.2678\,\text{yr}^{-1}$ is the gross atmospheric throughput rate, derived from independently measured CO_2 quantities (§10, stable to ±0.4% across the full record), and ϕ_{slow} is the slow-leak fraction of gross atmospheric inflow.

The fast-recycling fraction has two parallel area-scaled contributions: an oceanic component proportional to the local ocean-area fraction o and a biospheric component proportional to the local land-area fraction $1 - o$. With α, β, and γ denoting the per-unit-area rate coefficients for oceanic fast exchange, biospheric fast exchange, and slow leak respectively, mass balance gives

$$\phi_{\text{slow}}(o) = \frac{\gamma}{\alpha\, o + \beta(1-o) + \gamma} = \frac{1}{a\, o + b(1-o) + 1} \tag{55}$$

where $a \equiv \alpha/\gamma$ and $b \equiv \beta/\gamma$ are dimensionless rate ratios. The functional form is fixed by the two-pool decomposition; the two parameters (a, b) require external constraints.

Fit selection for hemispheric and global τ. The hemispheric and global τ values used here are the bold fits from Table 6: with the first two post-peak years excluded for NH avg (1965-66; fit runs 1967-2019), and the full record (1965-2019) for SH avg and global. The NH avg exclusion is justified by the fast-eigenmode decay timescale. The last large atmospheric detonation occurred in December 1962, so by 1967 the fast eigenmode had been decaying for 4 years. With $\tau_{\text{fast}} = 3.08$ yr from the two-box eigenvalue analysis (Table 4), the fast-mode amplitude in 1967 is $\exp(-4/3.08) = 0.273$ of its December 1962 value, meaning 73% of the fast-mode decay has already occurred. This is 1.88 fast-mode half-lives ($\tau_{\text{fast}} \ln 2 = 2.13$ yr) past the last detonation. The fast component continues to decay rapidly across the remainder of the fit window: by 1974 it is below 5% of its December 1962 amplitude, and is essentially zero for the last forty years of the fitted record. The residual fast-mode contribution to τ_{NH} from 1967-2019 is therefore well below the formal fit uncertainty. For τ_{SH} and global τ_{55}, the fast-eigenmode contribution is below detection from the start of the record because the bomb-^{14}C source was overwhelmingly Northern Hemisphere stratospheric injection; the full-record fit provides better leverage on τ via the high-amplitude early data points. In each case the fit represents the slow-eigenmode-dominated dynamics that the two-pool framework models.

Single bomb-^{14}C anchor: the global τ. Earth's overall ocean-area fraction from the NOAA NGDC ETOPO1 integration [32] is $o_{\text{global}} = 0.708$. With $\tau_{\text{global}} = 18.389$ yr and $\bar{\kappa}_{\text{in}} = 0.2678\,\text{yr}^{-1}$, the global slow-leak fraction is

$$\phi_{\text{slow}}^{\text{global}} = \frac{1}{18.389 \times 0.2678} = 0.20306 \tag{56}$$

and Eq. (55) evaluated at $o = 0.708$ yields the single calibration equation

$$0.708\, a + 0.292\, b = 1/\phi_{\text{slow}}^{\text{global}} - 1 = 3.9247 \tag{57}$$

Independent physical constraint on b/a. The ratio b/a is fixed independently from IPCC AR6 (2021) [60] (Figure 5.12) gross-flux estimates, with one correction for the radiocarbon-relevant fast pool. The global air-sea gross exchange is approximately 80 GtC/yr over the ocean surface ($0.708 \times 510 \times 10^6$ km^2), giving a gross flux density of 0.222 ktC yr^{-1}km^{-2}. The gross air-biosphere flux from Gross Primary Production (GPP) is approximately 120 GtC/yr over the land surface, but most GPP is returned to the atmosphere on sub-annual timescales by autotrophic respiration and rapid decomposition and so cannot contribute to fast cycling at the bomb-^{14}C residence timescale $\tau_{\text{res}} \approx 3.8$ yr. The radiocarbon-relevant fraction of GPP is the carbon entering the active biospheric pool with turnover comparable to τ_{res}: leaves, fine roots, fine litter, and labile soil organic matter. From compartmental analysis of terrestrial carbon cycling [120, 109] this fast biospheric pool exchanges with the atmosphere at approximately 25 GtC/yr globally, or 0.169 ktC yr^{-1}km^{-2} of land surface. The corresponding ratio is

$$\frac{b}{a} = \frac{0.169}{0.222} = 0.76 \tag{58}$$

independent of any bomb-^{14}C data

Calibration and parameter values. Combining Eq. (57) with $b = 0.76\,a$ from Eq. (58) gives

$$a = \frac{3.9247}{0.708 + 0.292 \times 0.76} = 4.2205 \tag{59}$$

$$b = 0.76 \times a = 3.2076 \tag{60}$$

The two-pool model is then specified completely:

$$\tau_{\text{rel}}(o) = \frac{1.0129\,o + 4.2076}{0.2678}\ \text{yr} \tag{61}$$

Hemispheric ocean fractions from independent geography. The standard published hemispheric ocean fractions from the NOAA NGDC ETOPO1 1-arc-minute global relief integration [32] are

$$o_{\text{NH}} = 0.607 \tag{62}$$

$$o_{\text{SH}} = 0.809 \tag{63}$$

with global ocean fraction $o_{\text{global}} = 0.708$. These are geographic constants derived from Earth's coastlines and topography, independent of any carbon-cycle data.

Hemispheric-average τ_{rel} predictions. Evaluating Eq. (61) at the hemispheric ocean fractions yields

$$\tau_{\text{rel}}^{\text{NH, pred}} = \frac{1.0129 \times 0.607 + 4.2076}{0.2678} = 18.008\,\text{yr} \tag{64}$$

$$\tau_{\text{rel}}^{\text{SH, pred}} = \frac{1.0129 \times 0.809 + 4.2076}{0.2678} = 18.772\,\text{yr} \tag{65}$$

The observed hemispheric averages from the single-exponential fits to the boreal-summer data (Table 6) are $\tau_{\text{rel}}^{\text{NH, obs}} = 18.091$ yr and $\tau_{\text{rel}}^{\text{SH, obs}} = 18.879$ yr. The agreement is

$$\tau_{\text{rel}}^{\text{NH, pred}} - \tau_{\text{rel}}^{\text{NH, obs}} = -0.083\,\text{yr}\ (-0.46\%) \tag{66}$$

$$\tau_{\text{rel}}^{\text{SH, pred}} - \tau_{\text{rel}}^{\text{SH, obs}} = -0.107\,\text{yr}\ (-0.57\%) \tag{67}$$

Both predictions agree with observation to within 0.57%, comfortably inside the ±0.76% formal uncertainty of the global τ_{55} anchor itself. The predicted hemispheric asymmetry

$$\tau_{\mathrm{rel}}^{\mathrm{SH,\,pred}} - \tau_{\mathrm{rel}}^{\mathrm{NH,\,pred}} = 0.764\,\mathrm{yr} \tag{68}$$

matches the observed asymmetry $\tau_{\mathrm{rel}}^{\mathrm{SH,\,obs}} - \tau_{\mathrm{rel}}^{\mathrm{NH,\,obs}} = 0.788\,\mathrm{yr}$ to within 0.024 yr (3.0% of the asymmetry).

Non-circularity. The prediction uses exactly one bomb-^{14}C measurement (the global τ_{55}) and one external physical constraint (the b/a ratio from IPCC AR6 gross fluxes corrected for the radiocarbon-relevant biospheric pool). No hemispheric or zonal bomb-^{14}C value enters the calibration. The two hemispheric ocean fractions $o_{\mathrm{NH}} = 0.607$ and $o_{\mathrm{SH}} = 0.809$ are geographic constants independent of any carbon-cycle data. The predicted hemispheric τ values and their asymmetry are therefore genuine predictions of the two-pool framework.

Summary. The two-pool mass-balance framework, with parameters fixed by one bomb-^{14}C anchor (the global τ_{55}) and one independent flux-ratio constraint, predicts the boreal-summer hemispheric averages $\tau_{\mathrm{rel}}^{\mathrm{NH}}$ and $\tau_{\mathrm{rel}}^{\mathrm{SH}}$ to within 0.57% of observation, and the total north-south τ difference to within 3%. The agreement is well inside the global anchor's own ±0.76% formal uncertainty, and the prediction uses no hemispheric bomb-^{14}C data. The N-S asymmetry in atmospheric ^{14}C decay is therefore a direct quantitative consequence of Earth's hemispheric ocean-area asymmetry under the mass-balance bridge relation of Eq. (23), requiring no appeal to interhemispheric transport delay or any other dynamical mechanism beyond first-principles area-scaled fast cycling.

8.3 Direct falsification of the IPCC Bern IRFs

Figure 2 compares the observed global Δ^{14}C decay following the bomb tests with the IPCC Bern impulse-response functions used in AR2 through AR6. The Bern IRFs form the CO_2 component of every emissions metric reported in AR6 (Chapter 7 and Table 7.15), including GWP-20, GWP-100, GWP-500, GTP-50, and GTP-100. The four Bern models predict that 33–48% of a CO_2 pulse remains in the atmosphere after 54 years. In contrast, the bomb-^{14}C observations compiled by Hua et al. (2022) [55] show only ∼4.3% remaining at the same time. The discrepancy is already visible within the first two years and reaches a factor of ∼10× at 54 years. At 100 and 200 years, the Bern IRFs deviate from the observed single-exponential decay by factors of ∼90× and ∼18,500×, respectively. Because the identical long-tailed Bern IRF is integrated out to 500 years to produce the official GWP-500 values in Table 7.15, the mismatch directly falsifies the CO_2 response function that underpins IPCC metrics, carbon budgets, and net-zero accounting on all policy-relevant timescales.

8.3.1 The IPCC Bern parallel-serial fallacy and parameter instability

The multi-exponential form that defines the Bern impulse-response function was introduced by Maier-Reimer and Hasselmann (1987) [80] on the structurally false premise that deep ocean reservoirs communicate *directly* with the atmosphere through parallel sink channels. This is the parallel-serial fallacy described by Cohler & Soon (2026a) [20]. In physical reality, the atmosphere exchanges CO_2 only with the surface mixed layer; every path to the thermocline, intermediate water, or deep ocean is strictly serial through that surface, not parallel from the atmosphere. The Bern form therefore embeds multiple parallel physical pathways that do not exist in nature.

The Joos et al. papers [63, 64, 65] adopted the multi-exponential form [80] unchanged and tuned its coefficients and timescales to the pulse-response outputs of successively more complex ocean general circulation models whose own ocean compartmentalizations repeat the same parallel-sink

assumption and which omit ultimate geological sinks. The AR5/AR6 version is fitted to a multi-model ensemble mean, making it a parameterization of an ensemble of parameterized models [20]. The resulting Bern IRFs have never matched any real-world carbon-cycle observable.

8.3.2 The "policy-relevant" escape hatch and its collapse into self-contradiction

Confronted with the bomb-^{14}C falsification, the institutional defense of the Bern IRF proceeds by retreat: the Bern IRF is said to be a reduced-form parameterization for policy horizons, not a physical model whose long-tail behavior can be falsified by atmospheric measurement, while the multi-millennial persistence claim is attributed to other models and parameterizations elsewhere in the assessment. The documentary record of AR5, AR6, and the papers they cite does not support a consistent form of this defense. Three positions are maintained simultaneously, and at most two can be true.

Position A: Multi-millennial persistence is a physical claim of the IPCC. IPCC AR6 (2021) [60] Chapter 5 (Canadell et al., p. 713) states that "CO_2 has multiple residence times in the atmosphere—from one year to many thousands of years." IPCC AR5 (2013) [59] Box 6.1 (p. 472), titled "Multiple Residence Times for an Excess of Carbon Dioxide Emitted in the Atmosphere," describes CO_2 cycling in qualitative bands: "every few years" for atmosphere–surface exchange, "decades to centuries" for land vegetation and the upper ocean, "centuries to millennia" for deeper soils and the deep sea, and "up to millions of years" for geological reservoirs including deep-sea carbonate sediments. The same box states that removal of anthropogenic CO_2 "will take a few hundred thousand years (*high confidence*)." The box cites Joos et al. (2013) [65] as its source. Box 6.1 does not itself explicitly identify the individual Bern IRF coefficients a_0, a_1, a_2, a_3 or their decay timescales τ_i with the qualitative processes it describes. It conveys the identification by juxtaposition only, placing the Joos 2013 citation alongside qualitative biogeochemical process descriptions in a form that invites the reader to make the mode-to-process association. But Joos 2013, the source Box 6.1 cites, explicitly denies the association: Joos et al. (2013) [65], Section 2.1, states that the fitted coefficients "have no direct process-based meaning, but are fitting parameters." The physical interpretation of the Bern IRF modes that Box 6.1 conveys by juxtaposition is therefore not merely unasserted; it is directly contradicted by the cited source. The specific numerical timescales $\sim 10^4$ yr (carbonate compensation) and $\sim 10^5$ yr (silicate weathering) do not appear in Box 6.1 itself; they come from the downstream literature, in particular Archer et al. (2009) [2], as an input assumption of that paper (see Position A source trace below). The multi-millennial persistence of atmospheric CO_2 is nevertheless propagated as a physical fact through the AR6 glossary and the public communications of NASA, NOAA, and the EPA.

Position B: AR6 uses the Bern IRF, specifically the Joos et al. (2013) [65] parameter set, to compute every CO_2-equivalence metric published. IPCC AR6 (2021) [60] Chapter 7 (Forster et al., §7.6, p. 1013) states explicitly: "Although there has been greater understanding since AR5 of the carbon cycle responses to CO_2 emissions (Sections 5.4 and 5.5), *there has been no new quantification of the response of the carbon cycle to an instantaneous pulse of CO_2 emission since Joos et al. (2013).*" The same chapter's Table 7.15 publishes GWP-20, GWP-100, **GWP-500**, GTP-50, GTP-100, CGTP-50, and CGTP-100 for CO_2, CH_4, N_2O, HFC-32, HFC-134a, CFC-11, and PFC-14. Every metric in that table uses the Joos 2013 IRF as the CO_2 denominator. The GWP-500 integration for CO_2 extends the Joos 2013 IRF out to 500 years. These numbers feed national emissions-accounting frameworks, Nationally Determined Contributions under the Paris Agreement, the social cost of carbon, and net-zero policy.

Position C: The Bern IRF is a reduced-form parameterization whose coefficients have no physical meaning and whose validity does not extend beyond 1000 years. Joos et al. (2013) [65], Section 2.1, states that the fitted coefficients "have no direct process-based meaning, but are fitting parameters." The same paper, Section 3, states: "the fits only apply for the period from 0 to 1000 yr," and the authors explicitly "do not provide results for the multi-millennial CO_2 evolution." IPCC AR5 (2013) [59] Chapter 6, Box 6.1 nevertheless *implies* such a mapping through juxtaposition of qualitative process descriptions with its Joos et al. (2013) [65] citation, without literally asserting any correspondence between the Bern IRF coefficients a_i or timescales τ_i and particular biogeochemical processes (Position A), and AR6 Chapter 7 relies on the same Joos 2013 fit for every published emissions metric (Position B).

The trilemma. The three positions are mutually incompatible. At most two may be held:

- If Position A (multi-millennial persistence is physical) and Position B (AR6 uses the Joos 2013 IRF for all metrics) are both true, then Position C is false: the Joos 2013 IRF is being used to produce physical claims beyond its stated 1000-year validity, and the coefficients are being read literally as physical processes despite their authors' disclaimer.
- If Position B and Position C are both true, then Position A is false: the metrics in Table 7.15 are reduced-form policy tools, and the IPCC has no empirical basis for the multi-millennial persistence claim in Chapter 5.
- If Position A and Position C are both true, then Position B is false: the physical claim in Chapter 5 rests on something other than the Joos 2013 IRF, and the AR6 metrics are decoupled from the physical account of atmospheric CO_2 response. No such alternative IRF has been published in AR6.

Where the multi-millennial claim actually comes from. When the trilemma is pressed and Position B+C is defended, Position A is relocated to a second body of literature: Archer et al. (2009) [2], an Annual Review intercomparison of reduced-complexity Earth System Models of Intermediate Complexity (EMICs) and box models. The full model list in Archer et al. (2009) [2] is CLIMBER-2, CC_SED, GENIE8, GENIE16, GEOCYC, LTCM, MESMO, MPI-UW, and UVIC2.8. No comprehensive ESM appears. The model runs are limited to 10^4 yr: Archer et al. (2009) [2] states that "the silicate weathering time constant is not resolved by the duration of these model experiments." The 10^5-year persistence figure cited by AR5 Box 6.1 and AR6 Chapter 5 is not a simulation output of Archer et al. (2009) [2]. It is an *input assumption*: Archer et al. (2009) [2] writes, verbatim, "If we assume however that 10% of the fossil fuel remains in the atmosphere until it is neutralized by silicate weathering on a time scale of 100,000 years"—and then uses that assumed persistence to derive the mean atmospheric lifetime of 12–14 kyr. The multi-millennial persistence claim traced to its source is therefore (i) an input assumption of 10% residual at 10^5 yr, (ii) propagated through EMICs and box models whose duration cannot resolve the assumed timescale, (iii) presented in AR6 as a physical attribute of atmospheric CO_2.

The collapse. The "policy-relevant" defense dissolves the moment it is examined:

- If the Bern IRF is a reduced-form policy tool whose long-tail behavior cannot be read as a physical claim (Position C), then GWP-500 in Table 7.15 has no physical warrant, and every national emissions-accounting framework built on it rests on a parameterization the authors themselves disclaim.
- If multi-millennial persistence is a physical fact (Position A), then it must be tested against the one direct atmospheric measurement of impulse-response behavior that exists: the bomb-^{14}C decay. That test has been performed in this paper and the physical claim is falsified: the

single-exponential decay at $\tau_{\text{rel}} = 18.383 \pm 0.015$ yr leaves residual atmospheric signal of 4×10^{-3} at $t = 100$ yr and 2×10^{-5} at $t = 200$ yr, both below any detection threshold and both bounded structurally by the operator-equivalence theorem and the 32-point residual at one part in ∼1316 (§8.4).

- If the multi-millennial persistence claim rests on Archer et al. (2009) [2], then it rests on an explicit input assumption whose timescale the underlying EMIC runs cannot resolve, and the claim is not a measurement of the atmospheric response but a restatement of the assumption fed into the models.

There is no fourth position. The institutional literature oscillates between Positions A, B, and C depending on which is being challenged. The present analysis closes all three exits at once: the single-exponential bomb-^{14}C decay directly falsifies the physical content of Position A; the operator-equivalence theorem closes the Bern-architecture escape of Position B at any parameter setting; and the input-assumption structure of Archer et al. (2009) [2] exposed in its own text closes the "it comes from other models" retreat of Position C to an assumption, not an observation.

8.3.3 Terminology of the Bern modes

In this paper, *permanent fraction* refers strictly to a_0 (with $\tau_0 = \infty$ by construction), which is the sole origin of the IPCC's multi-millennial persistence claim within the Bern IRF itself. The *finite modes* are a_1, a_2, a_3 (and a_4, a_5 in AR2) with the timescales listed in Table 7. The longest finite mode a_1 has drifted across assessments from $\tau_1 = 371.6$ yr in AR2 to $\tau_1 = 394.4$ yr in AR5/AR6, passing through $\tau_1 = 171.0$ yr in AR3 and $\tau_1 = 172.9$ yr in AR4; it is therefore a *multi-century* mode in all four assessments, not a millennial one. Its residual weight at $t = 1000$ yr under AR5/AR6 is $a_1\, e^{-1000/394.4} \approx 0.224 \times 0.0796 \approx 0.0178$, i.e. 1.78% of the pulse. Together with $a_0 \approx 0.2173$, this gives the AR5/AR6 Bern IRF a 1000-yr residual of approximately 0.235: the IPCC's multi-millennial persistence claim quantitatively. The multi-millennial persistence is therefore *carried entirely by* a_0; the finite modes have decayed to under 2% of the pulse at $t = 1000$ yr even for the longest one.

8.3.4 Mathematical incompatibility with any single exponential

This incompatibility is not empirical but structural. The multi-exponential architecture, introduced by Maier-Reimer and Hasselmann (1987) [80] and adopted by Joos et al. (1996) [63] and all subsequent IPCC assessments, expresses the impulse response as a constant permanent fraction plus a sum of decaying exponentials with widely separated timescales (a positive permanent fraction a_0 in every assessment, from 0.1369 in AR2 to 0.2173 in AR5/AR6). A single exponential is the response of a reservoir with one characteristic timescale. These two forms are not approximations of one another that happen to differ in fitted parameters; they are categorically distinct.

See Appendix B.1 for the proof of why this is true. Geometrically, a sum of exponentials with distinct time constants curves upward on a semi-log plot, whereas a single exponential is exactly a straight line. A curve cannot be simultaneously bent and straight, so the two forms necessarily diverge (see Figure 2). The discrepancy is in the shape itself and no parameter adjustment can remove it. Every IPCC Bern IRF curves upward away from the observed linear decay, immediately and irreversibly. The IPCC Bern architecture is falsified not by a poor fit to the data but by the wrong *shape that no parameter adjustment can correct*.

To match the accuracy of our 32-point parameter recovery, this structural result also becomes a quantitative bound for any proposed empirically equivalent multi-exponential. Each multi-exponential component must be small enough to keep the cleaned residuals within $\sigma_{32} = 0.539‰$ against the fit amplitude $A_{32} = 709.115‰$, i.e. below one part in ∼1316 over the 52-yr recovery window. The AR5/6 IPCC Bern IRF permanent fraction exceeds this bound by a factor of ∼285.

Two facts must be kept apart. The first is absolute: a single exponential is never equal to a sum of exponentials with different decay rates, with or without a constant term. They are different kinds of function. They match only in in degenerate cases: all rates forced equal, or all terms but one set to zero. The proof is in Appendix B.1.

The second fact is the one that matters here. The IPCC Bern IRF functions are emphatically not such degenerate cases. Each carries a positive permanent fraction ($a_0 = 0.1369$ in AR2, up to 0.2173 in AR5/AR6) and all four (AR2, AR3, AR4, AR5/AR6) have several well-separated timescales by design. That is precisely what produces the multi-thousand-year persistence that IPCC's carbon budgets rely on. So they fall under the structural result, and the bomb-^{14}C record falsifies them directly: every IPCC Bern curve departs from the observed single exponential within the first few years of the 1965–2019 record and diverges from then on: an immediate, large, and growing failure inside the data, on precisely the timescales that the IPCC calls "policy-relevant."

8.4 Why bomb-^{14}C constrains the CO_2 mass IRF

A formal objection sometimes raised against using the bomb-^{14}C decay to constrain the atmospheric CO_2 mass impulse-response function is that the isotopic ratio and the total inventory respond to different perturbations of the coupled reservoir system, so that a slow mass mode could in principle exist while the bomb-^{14}C decay remains a clean single exponential. This objection fails on two independent grounds, one structural and one empirical.

The structural ground is the operator-equivalence theorem proved in Appendix B.2. Under linear exchange between reservoirs, the atmospheric mass anomaly $\delta\mathbf{M}$ and the atmospheric ^{14}C-content anomaly $\delta\mathbf{X}$ evolve under operators that share eigenvectors (modulo the known $-\lambda_{14}$ shift, negligible over 54 years). For any eigenmode j of the exchange operator $\mathbf{A}$, the atmospheric expression of that mode is governed by the same product $(\mathbf{w}_j^T \mathbf{e}_a)\, v_{j,a}$ in both responses, because both impulses are atmospheric at $t = 0$. Any mode with nonzero atmospheric expression in the Bern mass IRF must therefore appear with the same fractional amplitude in the atmospheric Δ^{14}C decay. The cleaned 32-point recovery ($\sigma_{32} = 0.539‰$ against the fit amplitude $A_{32} = 709.115‰$) bounds any such mode at one part in ~1316 (~0.076%); the AR5/AR6 Bern permanent fraction $a_0 \approx 0.22$ exceeds this bound by a factor of ~285.

The IPCC defends the Bern multi-mode architecture through a rhetorical structure that implies, without literally asserting, a mapping of each exponential coefficient to a distinct physical process acting on a distinct horizon. IPCC AR5 (2013) [59] Box 6.1 (p. 472) describes qualitative biogeochemical processes alongside a citation of Joos et al. (2013) [65] without explicitly identifying the Bern coefficients a_i or timescales τ_i with any of them; IPCC AR6 (2021) [60] Chapter 5 (Canadell et al., p. 713) carries this implied mapping forward with the statement that "CO_2 has multiple residence times in the atmosphere—from one year to many thousands of years"; and IPCC AR6 (2021) [60] Chapter 7 (Forster et al., §7.6, Table 7.15 and Supplementary Table 7.SM.7) applies the resulting IRF to compute GWP-20, GWP-100, GWP-500, GTP-50, GTP-100, and CGTP. The physical interpretation this structure conveys is, however, directly contradicted by the source Box 6.1 cites: Joos et al. (2013) [65] Section 2.1 states that the fitted coefficients "have no direct process-based meaning, but are fitting parameters." See §8.3.2 for the full trilemma. The implicit rhetorical position is that the permanent fraction a_0 (with $\tau_0 = \infty$ by construction) and the multi-century mode a_1 (with $\tau_1 = 394.4$ yr in AR5/AR6, down from $\tau_1 = 371.6$ yr in AR2 and $\tau_1 = 171.0$ yr in AR3, 172.9 yr in AR4; Table 7) govern only the long horizons, leaving the shorter policy horizons unaffected by whatever "physical processes" those slow modes are claimed to represent.

The arithmetic refutes the position. A permanent fraction is permanent from $t = 0$: at $t = 1$ yr, $a_0 = 0.2173$ contributes 21.73% of the AR5/AR6 impulse response, exactly the fraction it contributes at $t = 54$ yr and at $t = 1000$ yr. The finite modes (a_1, a_2, a_3) with AR5/AR6 timescales

of 394.4, 36.54, and 4.304 yr also contribute at every t, decaying only by their own exponential factors, so all four coefficients affect the response at every timescale down to a few years. Evaluated at $t = 1, 10, 54, 100, 200$ yr, the AR5/AR6 Bern function returns 0.94, 0.68, 0.48, 0.37, 0.35; the single-exponential fit to the bomb-^{14}C record returns 0.95, 0.58, 0.053, 4×10^{-3}, 2×10^{-5} at the same times. The Bern-to-observed ratios are 0.99, 1.17, 9.0, 86, ~18,800. The failure is not confined to any horizon, policy-relevant or otherwise. It begins in the first year, exceeds a factor of ten within the observation window, and diverges to $\sim 10^{23}$ at 10^3 yr, on which timescale the AR5/AR6 permanent fraction alone still asserts 0.235 airborne.

The empirical ground is that the linearity premise on which the operator-equivalence theorem rests is not assumed; it is directly tested in Section 10 over the same M_a range where the Bern IRF claims its purported slow modes operate. A clean single-exponential decay of R_a requires the product $\kappa = \phi_{slow}\,\kappa_{in}$ to be constant to within ~1% over the fit window (§5.4). The $\delta^{13}C_I$ constancy independently bounds ϕ_{fast} to within 1–2% (§5.4). Together these force κ_{in} to be constant across the observation window. The independent test is the Koutsoyiannis $b = 1$ linear-reservoir RRR fit [72], which achieves $R^2 > 99.7\%$ on storage with a stable τ_{res} across split-sample validation; under that stable τ_{res}, the mass-balance reconstruction $\kappa_{in}(t) = 1/\tau_{res} + \mathrm{d}\ln M_a/\mathrm{d}t$. Averaged over 1965 to 2019 yields $\bar{\kappa}_{in} = 0.2678\ \mathrm{yr}^{-1} \pm 0.4\%$ (Table 12), across which M_a rises by 31.6%.

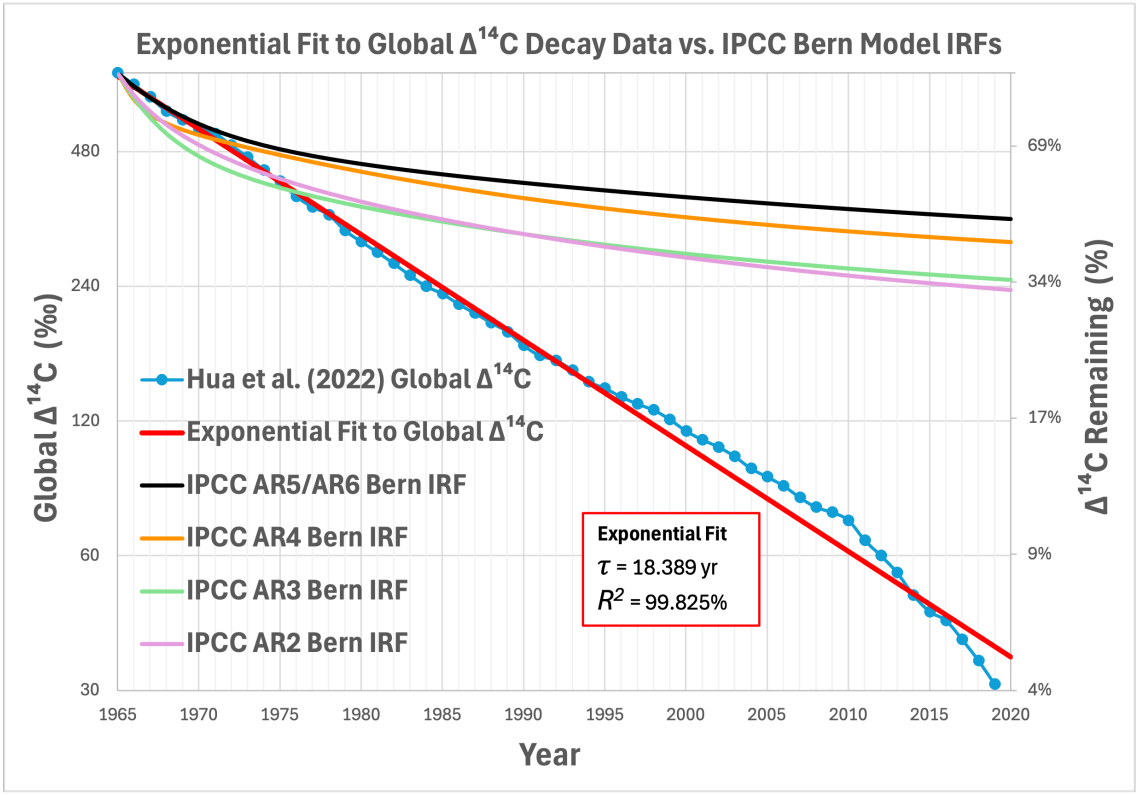


Figure 2: Observed Hua et al. (2022) [55] global $\Delta^{14}C$ bomb excess (blue) decay compared to IPCC Bern impulse-response functions from AR2, AR3, AR4, and AR5/AR6 (purple, green, orange, black) and a single-exponential non-linear least squares fit (red). On a semi-log plot, a single-exponential displays as a straight line while all multi-exponentials with positive coefficients display as upward curves. All IPCC Bern IRFs deviate from observation throughout the record with monotonically growing discrepancy after the initial crossover point for each, which occurs between years 2 and 11. By 2019, observed $\Delta^{14}C$ is 4.3% of its peak value vs. 33, 35, 42, and 48% respectively for AR2, AR3, AR4, and AR5/6.

Table 7: IPCC Bern IRF parameters for the functions shown in Figure 2. Every Bern IRF has the form $R(t) = a_0 + \sum_{i=1}^{n} a_i\, e^{-t/\tau_i}$, normalized so that $R(0) = \sum_{i=0}^{n} a_i = 1$, where a_0 is the permanent (non-decaying) fraction and each τ_i is in years. The e-folding time $t_{1/e}$ (in years) is the time at which $R(t) = 1/e$. AR2 uses five exponential modes ($n = 5$); AR3, AR4, and AR5/AR6 each use three ($n = 3$). Each amplitude a_i is paired with its timescale τ_i in the column immediately to its right; entries "–" are unused in that parameterization. Parameters are listed in reverse chronological order by the IPCC assessments in which they are used [63, 64, 65, 56, 57, 58, 59, 60].

IPCC Report	$t_{1/e}$	a_0	a_1	τ_1	a_2	τ_2	a_3	τ_3	a_4	τ_4	a_5	τ_5
AR5/AR6 (2013, 2021)	165	0.2173	0.2240	394.4	0.2824	36.54	0.2763	4.304	–	–	–	–
AR4 (2007)	95.6	0.217	0.259	172.9	0.338	18.51	0.186	1.186	–	–	–	–
AR3 (2001)	45.6	0.152	0.253	171.0	0.279	18.0	0.316	2.57	–	–	–	–
AR2 (1995)	41.2	0.1369	0.1298	371.6	0.1938	55.7	0.2502	17.01	0.2086	4.16	0.0807	1.33

This is the decisive test. Any nonlinear ocean or biospheric mechanism strong enough to generate Bern-scale slow atmospheric mass modes would necessarily produce a non-constant κ_{in} as M_{a} grows: gross fluxes scaling sub-linearly (or super-linearly) with atmospheric inventory is precisely the signature of the saturating or buffering sinks invoked to justify multi-millennial persistence. No such curvature is present. F_{in} tracks M_{a} to ±0.4% over 58 years and a 31.6% inventory rise; F_{nat} tracks M_{a} to 0.9% over the same period (Figure 4). The two transient departures (1991 Pinatubo, 2005–2016 PDO cool phase) are at the few-tenths-of-a-percent level and resolve to documented surface-energy-balance perturbations.

The linearity premise of the operator-equivalence theorem is therefore not contingent; it is verified by independent measurement across the full industrial-era inventory range that the Bern IRF purports to describe. Any objection that posits a slow mass mode hidden from the bomb-^{14}C observable must specify a nonlinear mechanism that simultaneously (a) appears in $\delta M_{\text{a}}(t)$ under emission pulses, (b) has zero expression in the eigenstructure of $\mathbf{A}$ acting on $\delta\mathbf{X}$, and (c) produces no curvature in $F_{\text{in}}(M_{\text{a}})$ across a 31.6% rise in M_{a}. No such mechanism has been identified, proposed, or observed.

Consequences. The theorem proves that an emission pulse of CO_2 and the bomb-^{14}C pulse are seen by the atmosphere through the same lens. Both start in the atmosphere; both propagate through the same web of physical reservoirs at the same rates; the only difference is that one carries extra mass and the other carries extra ^{14}C label. Whatever modes of response the atmosphere–ocean–biosphere system actually has, those modes are properties of the coupled reservoir network itself, not of which kind of perturbation excites them. Any purported slow mode that would let a CO_2 pulse linger in the atmosphere for centuries must therefore also let a ^{14}C pulse linger in the atmosphere for centuries, with the same fractional weight.

This forecloses the standard defense of the IPCC framework against the bomb-^{14}C evidence. That defense holds that the bomb-^{14}C decay measures only how isotopic labels redistribute among reservoirs, not how excess CO_2 mass eventually disappears, so the 18-year Δ^{14}C decay can coexist with the IPCC's claimed century-to-millennial CO_2 persistence. The theorem proves this is not possible under linear exchange: any purported slow mode that would retain 15 to 40% of an emitted CO_2 pulse for more than 1000 years must also retain 15 to 40% of the bomb-^{14}C pulse on the same timescale. The bomb-^{14}C record shows the opposite. The pulse decayed cleanly to ~4% of its peak in 54 years, with the cleaned-recovery residuals bounded at 0.075% of the original signal. There is no room in the data for any such purported slow component. The Bern permanent fraction a_0 in particular would show up as an essentially constant offset over the observation window (it decays only through the 5,700-year ^{14}C radioactive half-life); no such offset exists.

The IPCC has never produced empirical evidence for the existence of any of these purported slow

modes. The Bern impulse-response function is fitted to the outputs of multi-compartment ocean general circulation models whose own internal structure presupposes the parallel-topology coupling that the serial physics of the actual carbon cycle rules out ([20], and §8.3.1). The architecture is a parameterization of an ensemble of parameterizations, not a fit to any atmospheric observable. The single empirical test available, the bomb-^{14}C decay, falsifies it directly. The purported slow modes are not modes that the data fail to detect at the current precision; they are modes for which the only direct atmospheric impulse-response experiment ever performed places a structural upper bound two to three orders of magnitude below the values the framework requires.

For climate science, the consequence is that the multi-millennial atmospheric persistence claim, the foundation of the IPCC's cumulative-emissions framework, of remaining carbon budgets, and of the entire policy edifice built on the idea that today's emissions commit the climate for centuries, rests on modes that have no empirical referent and that the bomb-^{14}C record directly excludes. The Bern IRF and every quantitative claim derived from it are falsified not by a poor fit to the data but by a structural incompatibility between their architecture and what the atmosphere actually did when it was given an impulse to respond to.

Application to the four IPCC Bern IRFs. The four Bern parameterizations carried by AR2/SAR, AR3, AR4, and AR5/AR6 share the structural form

$$g(t) = a_0 + \sum_{i=1}^{n} a_i \, e^{-t/\tau_i} \tag{69}$$

with longest modes τ_1 = 371.6, 171.0, 172.9, and 394.4 yr, respectively (Table 7; Cohler & Soon (2026a) [20]). The multi-millennial atmospheric persistence claim in the AR6 glossary derives from the permanent fraction a_0, which takes values of 0.1369 (AR2), 0.152 (AR3), 0.217 (AR4), and 0.2173 (AR5/AR6). Under the stated linear exchange-operator premises, the bomb-^{14}C amplitude bound structurally excludes the Bern IRF architecture as an atmospheric impulse-response description, including both its finite-τ_i modes and its permanent fraction a_0. The cleaned 32-point recovery bounds the atmospheric expression of any Bern mode at one part in ∼1316 of the fit amplitude (the ratio of A_{32} = 709.115‰ to the σ_{32} = 0.539‰ residual), i.e., below 0.076% (§8.3). The four a_0 amplitudes (0.1369, 0.152, 0.217, 0.2173) exceed this bound by factors of ∼180 to ∼285. The amplitude bound is tight for the permanent fraction a_0 and for slow finite modes with $\tau_i \gg 52$ yr; intermediate finite modes, which the amplitude bound alone constrains only approximately, are excluded independently by the Vandermonde argument (Appendix B.1) and by the multi-window fit consistency of §9.3. Figure 2 shows the direct comparison: all four parameterizations curve upward away from the observed linear decay throughout the 1965–2019 record. The empirical exclusion is structural and present at all timescales within the observation window.

On the absence of a global buffer factor. A specific version of the objection above invokes the Revelle factor as the carbonate-chemistry mechanism through which deep-ocean saturation produces slow atmospheric mass modes invisible to bomb-^{14}C. The Revelle factor $\beta = (\partial \ln pCO_2 / \partial \ln \mathrm{DIC})_{\mathrm{T,S,A_T}}$ is, however, a local intensive thermodynamic ratio defined point-by-point on a seawater parcel at specified temperature, salinity, alkalinity, and DIC. It varies across the ocean surface from ∼8 in warm tropical waters to ∼15 in cold high-latitude waters and varies further with depth, season, and biological state. No integration procedure produces a single global β that appears as a parameter in any atmosphere–ocean conservation equation; spatial averaging of an intensive ratio over a heterogeneous flux field yields a statistic of that field, not a derivative of any physical quantity. The categorical operational status of the Revelle factor as a local-only intensive property is the same as that of local surface temperature: a measurable field-valued quantity whose area-average is not itself an operationally defined physical parameter [38, 18, 23].

The empirical test of carbonate-chemistry nonlinearity in the actual heterogeneous ocean is therefore not whether some constructed global β varies with M_a, but whether the integrated air–sea exchange flux, taken over the actual surface ocean with its actual local $\beta(x, y, z, t)$ field and local exchange coefficients, produces a non-constant κ_{in} as atmospheric inventory grows. Table 12 and Figure 4 answer that question directly: κ_{in} is constant to ±0.4% over a 31.6% rise in M_a spanning the full industrial-era CO_2 trajectory. Whatever the actual local Revelle field is doing, its integrated effect on the gross exchange flux is linear in M_a to within measurement precision.

8.5 Miller et al. (2025): A contemporary restatement of the falsified framework

Miller et al. (2025) [86] present the most recent high-profile institutional restatement of the framework this paper falsifies. They simulate atmospheric $\Delta^{14}CO_2$ for 2000–2012 in the TM5 3D chemical-transport model, using monthly gridded input fluxes for fossil-fuel combustion, net ecosystem exchange, biomass burning, net ocean exchange, terrestrial and oceanic isotopic disequilibrium return, cosmogenic production, nuclear-reactor emissions, and methane oxidation. They compare the simulation to approximately 5,000 atmospheric flask measurements from the NOAA/INSTAAR, Heidelberg, UC Irvine, and SIO networks.

The framework commits explicitly to everything our analysis falsifies. Their introduction states that "the observed decline... is being driven dominantly by the addition of fossil CO_2," invoking [44] and [77] in support. They report that if fossil CO_2 alone controlled the trend, the simulated decline would be more than twice as steep as observed. To match observations they introduce large compensating return fluxes from ocean and biosphere, invoke Revelle-buffered surface-ocean uptake, and carry the multi-compartment box-model architecture of the Bern lineage forward through TM5. All of this is excluded by the operator-equivalence theorem (Appendix B.2) at the amplitude bound $\sigma_{32} = 0.539$ ‰ against the fit amplitude $A_{32} = 709.115$ ‰.

The site-level trends Miller et al. (2025) [86] tabulate span −2.87 to −5.40 ‰/yr across 29 surface sites for 2000–2012 (their Table 2). The single-exponential fit of §9.3 reproduces this range from $\tau_{rel} = 18.383 \pm 0.015$ yr and the first-principles fossil-fuel correction of §9.2.5. No 3D transport model. No Revelle buffering. No compensating disequilibrium return fluxes. One first-order linear ODE on a well-mixed reservoir with one characteristic timescale.

Agreement with observations obtained by adjusting input flux amplitudes until the model matches is not a test of the framework; it is enforcement of it. Miller et al. (2025) [86] acknowledge openly that fossil CO_2 input alone produces a simulated decline more than twice as steep as observed (their p. 2). They close the gap by inventing compensating ocean and biosphere disequilibrium return fluxes whose amplitudes are set by the very gap they must close, and by applying posterior flux scalings of 0.7 and 0.8 to their $^{14}CO_2$ mole-fraction components (their §2.3). No independent measurement of either return flux at the required global amplitude is invoked. The framework is falsified on three independent grounds.

Mathematical. The multi-mode Bern-lineage architecture Miller et al. (2025) [86] carry through TM5 cannot produce a single exponential at any parameter setting (Appendix B.1). The bomb-^{14}C record is single-exponential at $R^2 = 99.998\%$ across 52 years. By the operator-equivalence theorem (Appendix B.2), the atmospheric amplitude of any posited slow mode in the bulk CO_2 impulse response is bounded at one part in ∼1316 of the fit amplitude $A_{32} = 709.115$ ‰. The AR5/AR6 Bern permanent fraction exceeds this bound by a factor of ∼285.

Methodological. The Revelle factor they invoke to parameterize the ocean disequilibrium return flux is a local intensive thermodynamic property of seawater parcels at the sparse surface points where it has been measured (§3.3). Extrapolating it to a global rate limit on bulk ocean CO_2 uptake conflates a local intensive property with a global aggregate that has no measured counterpart. The fingerprints a global Revelle-buffered uptake at the required amplitude would leave in the ocean are absent: no uniform monotonic surface-ocean DIC rise, no sustained air-sea pCO_2 disequilibrium at

the required magnitude, no uniform acidification, no uniform rise of carbonate saturation horizons.

Empirical. The geological record directly falsifies the multi-millennial persistence premise the framework requires. Every Phanerozoic high-CO_2 episode was fully drawn down. The Paleocene–Eocene Thermal Maximum excursion drew down in ∼100–200 kyr. The Permian–Triassic excursion drew down. If Revelle-buffered multi-millennial persistence were a real physical constraint on bulk ocean CO_2 uptake, Earth would have hit carbonate saturation during one of these episodes and remained there. It did not.

A tuned model reproducing observations it was tuned against is not evidence for the architecture it embeds. The single-exponential bomb-^{14}C decay is the direct test, and the architecture fails it.

8.6 Hua et al.'s unsupported interpretive claims

Hua et al. (2022) [55] (p. 738) attaches two causal interpretations to the decay, citing [95], [31], and [78] for the first, and citing [77] and [43] for the second:

> *Decreases in atmospheric $\Delta^{14}C$ from the mid-1960s to mid-1980s are mainly due to rapid exchange between the atmosphere and the* ***biosphere and oceans*** *. . . while combustion of* ***fossil fuels*** *free of ^{14}C is the main causal factor for the $\Delta^{14}C$ decline since the late 1980s and early 1990s. . .* (emphasis added)

Neither claim is accompanied by a quantitative calculation. No mass-balance estimate of the fossil-fuel dilution rate is provided. No comparison of that rate to the observed Δ^{14}C decline rate is performed. No test for a slope change between the pre-1988 and post-1988 portions of the record is reported. No model fit demonstrating that a two-regime interpretation is statistically preferred over a single exponential is presented.

Moreover, the claims are directly contradicted by the data in Table 6. If rapid exchange dominated the early decay and fossil-fuel dilution dominated the late decay, the semi-log plot would show a visible slope change around the mid-1980s. It does not: the single exponential fits the entire 55-year record at $R^2_{55} = 99.825\%$ (Figure 2). The residual structure that is present ($\sigma_{55} = 7.787‰$) is entirely accounted for by documented compilation artifacts (Section 9); once these are removed, the cleaned 32-point recovery achieves $R^2_{32} = 99.998\%$ with $\sigma_{32} = 0.539‰$ and no remaining structure. The fossil-fuel dilution rate, computed from first-principles mass balance in Section 9.2.5, grows from ∼5‰/yr in 1965 to ∼11‰/yr by 2010. In every year it is well below 2σ of the original fit, and detectable only after 2010 when the bomb excess has decayed to ∼28‰.

The Appendix of Hua et al. (2022) [55] (p. 745) provides further insight into the consequences of adopting an interpretive framework without testing it. For the practical purpose of extrapolating the Δ^{14}C record beyond 2019 for radiocarbon dating, the authors fit a single exponential to the compiled monthly data, implicitly accepting that the post-peak decay is exponential. They report trying multiple start dates "from the 1980s onwards" and finding that restricting the window to 1993–2019 yielded the "best fit or highest R^2 value." No R^2 is reported for this or any other fit, and no residual analysis is performed.

The restriction to 1993–2019 is consistent with the authors' two-regime interpretation: if the early decay is driven by fast cycling and the late decay by fossil-fuel dilution, then for forward extrapolation one would naturally fit only the "current regime." But the reasoning is circular. The truncation presupposes the regime change that should have been tested, and the result contradicts the authors' own stated criterion. A single-exponential fit to the full 1965–2019 record yields $R^2 = 99.8\%$; the truncated 1993–2019 fit yields $R^2 = 98.3\%$. The full record is the actual best fit by the very metric Hua et al. invoke. The truncated fit achieves a lower absolute standard error ($\sigma = 4.9‰$ versus 7.9‰) only because the late-record data points are smaller in magnitude, not because the fit captures a larger fraction of the variance. The truncation also biases τ upward from

18.39 to 19.11 yr, because the 1993–2019 window is disproportionately affected by the plateau B compilation offset (+9.5‰; Section 9.2.4) and the fossil-fuel tail departure (Section 9.2.5).

The unquantified fossil-fuel attribution that Hua et al. (2022) [55] inherit from Levin et al. (2010) [77] and Graven et al. (2012) [43] has been propagated into 21st-century projections by Graven (2015) [44] and consolidated into the modern institutional "Suess effect" framework by Graven et al. (2020) [45], the AGU Grand Challenges review that has since served as the canonical reference for fossil-fuel attribution of $\Delta^{14}C$ and $\delta^{13}C$ trends. Graven (2015) [44] uses the Oeschger et al. (1975) [95] one-dimensional box-diffusion ocean model with one-box atmospheric and biospheric reservoirs, driven by IPCC AR5 Representative Concentration Pathways, to project that fossil-fuel emissions under RCP8.5 will drive atmospheric $\Delta^{14}CO_2$ to −250‰ by 2100, an apparent radiocarbon age of ∼2000 yr. The architecture is a multi-compartment diffusion model of the same family as the IPCC Bern parameterizations; the operator-equivalence theorem of §8.4 applies to it identically, and the bomb-^{14}C amplitude bound excludes its slow-mode atmospheric expression on the same structural grounds that exclude the Bern a_0 permanent fraction. Graven et al. (2020) [45] retains this same architectural family for both ^{14}C and ^{13}C trajectories under the SSP scenarios and therefore inherits the same structural exclusion.

Section 9 demonstrates that the pre-1993 deviations that motivated the truncation are entirely explained by documented station transitions, weighting shifts, and retrospective data revisions within the Hua compilation itself, not by any transition from one physical mechanism to another. When these compilation artifacts are identified and removed, the cleaned 32-point record spanning 52 years yields $R^2_{32} = 99.998\%$ with structureless residuals of $\sigma_{32} = 0.539‰$ (Table 10), confirming that the recovered single exponential describes the entire record.

9. Complete Reconciliation of the Bomb-^{14}C Decay Record

Before proceeding with the residual diagnostic, the station-by-station archival audit, and the construction of the cleaned 32-point fit, we lay out in §9.1 the complete logical chain governing every cleaning step, and in §9.2 the documented Hua compilation methodology changes that determine the plateau boundaries. The cleaning is bias subtraction, not curve fitting: the single-exponential shape is established empirically *before* any cleaning from the unedited 55-point global record and seven unedited zonal subsets at $R^2 > 99\%$; plateau boundaries are documented station exits, tree-ring-to-atmospheric weighting transitions, and retrospective data revisions identified by independent archival audit *before* any residual analysis; plateau levels are arithmetic averages of residuals across those externally-bounded intervals, which are direct empirical measurements of Hua's cumulative compilation bias, analogous to a calibrated instrument offset, not fitted parameters; and artifact-stretch exclusions follow from a physics-derived smoothness criterion computed from independent atmospheric parameters with no reference to the residual data. No step fits a function to the data, adjusts a parameter to improve a fit, or uses the cleaned τ_{rel} as an input. Readers should consult both subsections before assessing the cleaning analysis.

9.1 Logical chain of our methodology

Step 1: The shape of bomb-^{14}C decay is empirical, not assumed. The single-exponential shape of the post-peak atmospheric $\Delta^{14}C$ decay is established by direct observation *before* any cleaning. The 55-point uncorrected global Hua et al. (2022) [55] record fits a single exponential at $R^2 = 99.825\%$ with residual standard deviation $\sigma_{55} = 7.787‰$ against the fit amplitude $A_{55} = 708.847‰$, a 1.1% error (Table 6). Each of the seven unedited zonal and hemispheric series individually fits a single exponential at $R^2 > 99\%$, with values ranging from 99.134% (NH zone 1) to 99.827% (SH average), and residual σ between 7.298 and 20.028‰ on initial amplitudes of 670 to 823‰ (§8.1). The single-exponential shape emerges from the raw data of every series without any selection, exclusion, or correction. The systematic north-to-south increase in τ from

16.722 yr (NH zone 1) to 19.271 yr (SH average) is itself a physical gradient consistent with the hemispheric difference in ocean area and fast-cycling exchange volume under the bridge relation $\tau_{\mathrm{rel}} = \tau_{\mathrm{res}}/\phi_{\mathrm{slow}}$ (§8.1, Eq. (23); no series requires a second exponential, a permanent fraction, or a regime change to achieve $R^2 > 99\%$.

The same single-exponential shape is further confirmed by an independent within-record test on the cleaned 32-point set. Cutting the cleaned record into three non-overlapping windows (Early: 1967, 1979–1984, 1992–1994; 10 points spanning 27 years and a 4.4× amplitude decay from $R_{\mathrm{a}} = 636$ to 146‰; Middle: 1998–2010, 13 contiguous Plateau B points; Late: 2011–2019, 9 fossil-fuel-corrected tail points) and fitting the single-exponential model independently in each, the recovered τ values are 18.379 ± 0.021, 18.395 ± 0.180 (cosmogenic-corrected), and 18.428 ± 0.322 yr respectively (Table 10, §9.3.1). The three windows are temporally disjoint, use different data subsets, and are fit independently with no shared parameters; their agreement to within 0.05 yr on an 18.4 yr timescale is the falsifying observable for any multi-regime interpretation. A genuine transition from non-fossil-fuel-exchange-driven decay to fossil-fuel-driven decay in the late 1980s or early 1990s [55, 77, 43, 44, 45] would necessarily produce window-to-window drift in the recovered τ; none is present.

By contrast, every published IPCC Bern impulse-response function (AR2, AR3, AR4, and AR5/AR6) deviates from the same data visibly and monotonically grows further wrong thereafter (Figure 2; §8.3.4). The mathematical incompatibility between any sum of exponentials with distinct decay rates and a single exponential is structural and proved in Appendix B.1; the published Bern parameters make the divergence immediate and obvious. The shape is therefore not assumed by us and is not consistent with the framework that would prefer multi-mode behavior. Eleven independent single-exponential recoveries (seven zonal and hemispheric subsets of the raw data, the global 55-point fit, and three independent non-overlapping windows of the cleaned 32-point record) converge on a single τ to within the latitude-gradient range expected from independently measured atmospheric physics. The empirical foundation is not a fitted shape, but an observed one.

Step 2: Plateau levels are biases by mathematical necessity. Given that the underlying physics produces a single exponential (Step 1), and given that Hua et al. (2022) [55] is a model-based compilation that documents discrete methodology changes (station exits, station entries, weighting transitions, retrospective revisions) at specific years, the residual series $r_{55}(t) = R_{\mathrm{a}}(t) - R_{55}(t)$ has a mathematically forced structure. Between methodology-change points, Hua's compilation bias is constant (the methodology is unchanged), so $r_{55}(t)$ must be constant up to measurement noise. At methodology-change points, the bias steps to a new constant value. The residual series therefore necessarily consists of flat plateaus separated by step transitions. The plateau *levels* are measurements of Hua's cumulative compilation bias, not free parameters; they are determined uniquely by the residual values within each stable-methodology period. The plateau *boundaries* are the documented methodology-change points, independently identified by the archival audit of §9.2. No optimization is performed. No constant is fit to anything. The plateau means are arithmetic averages of residuals across externally-defined intervals.

Step 3: Artifact stretches are excluded by physics, not by fit quality. The intervals between plateaus (1968–1978, 1985–1991, 1995–1997) correspond to periods of methodology change in Hua et al. (2022) [55], during which the bias is not constant. The smoothness criterion derived from the coupled two-box stratosphere–troposphere ODEs in §7, $|d^2 r_{55}/\mathrm{d}t^2| \leq 0.000920\, R_{\mathrm{T}}(t)$, is exceeded within these intervals by factors of 6 to 14, with sign-flipping curvature inconsistent with any smooth coupled-reservoir physics. The smoothness threshold is computed from independently sourced atmospheric parameters (M_{S}, M_{T}, F_{BDC}, $\mathrm{d}\ln M_{\mathrm{a}}/\mathrm{d}t$) with no reference to the residual data.

The exclusion of these stretches is therefore grounded in physics, not in their failure to fit a curve, and is independently confirmed by the archival audit of §9.2.

Step 4: Endpoint and FF-tail handling are physics-driven. The a priori exclusion of 1965–1966 (§C.8) follows from the fast-mode equilibration timescale $3\,\tau_{\text{fast}} \approx 9$ yr derived in §7 from atmospheric parameters independent of the Hua data. The C_{FF} correction applied to 2011–2019 (§9.2.5) is computed from directly measured $E_{\text{FOS}}(t)$ [42] and $M_{\text{a}}(t)$ [76] with no adjustable parameters. The two emergence-offset locations (2010→2011 and 2015→2016) are predicted independently: one from when C_{FF} crosses the noise envelope, the other from the documented station-network composition change in Hua et al. (2022) [55]. Neither is chosen to match a feature of the residual data. The offset magnitudes are constrained by physical continuity of the smooth bomb-decay across discrete compilation steps, and τ_{rel} is insensitive to these magnitudes at the 0.16% level (§C.11.4).

Step 5: The bias-corrected fit is a measurement, not a target. After Steps 1–4, the retained 32 points are the data with Hua's documented compilation biases subtracted. The single-exponential fit to these 32 points, the authoritative result reported throughout this paper, yields $\tau_{\text{rel}} = 18.383 \pm 0.015$ yr, $R^2 = 99.998\%$, $\sigma_{32} = 0.539‰$ against the fit amplitude $A_{32} = 709.115‰$. This fit is the *output* of the cleaning procedure, not its target. The cleaning procedure never references τ_{32}. The robustness of τ_{32} to the cleaning choices is documented in Table 10: across all five fits (55-point uncorrected, 28-point corrected, 28-point cosmo-corrected, 32-point corrected, 32-point cosmo-corrected) the central τ value spans only 18.374 to 18.389 yr (0.08% range). The cleaning sharpens the residual standard deviation by a factor of 14.4 ($\sigma_{55} = 7.787‰ \rightarrow \sigma_{32} = 0.539‰$) without moving the central τ value outside its own 1-σ uncertainty. As an independent corroboration, applying the diagnostic −1‰ cosmogenic-minimum correction at 2009 and 2010 (§9.3.2), a value within the integer-display precision of Hua et al. (2022) [55] and justified by the documented galactic cosmic-ray enhancement during the cycle-24 deep solar minimum but not derived from an independent solar-modulation reconstruction at sub-permil precision, shifts τ by only 0.008 yr to 18.375 ± 0.013 yr and improves σ by ∼12% to 0.472‰. The cosmogenic correction is reported as a diagnostic and a robustness check rather than as part of the authoritative recovery; it changes no conclusion of this paper.

Mathematical definition of the adjusted series and corrections. To render every numerical result fully traceable, we now introduce precise notation for the adjusted series. All corrections are constant functions of time over the specified domains except the fossil-fuel term $C_{\text{FF}}(t)$, whose explicit first-principles form is already given in §9.2.5.

Define the signed correction constants together with the years on which each applies using only the unadjusted residual series r_{55}, the first-principles fossil-fuel correction C_{FF} (2011–2019) and the first-principles cosmogenic correction (2009–2010):

$$
\begin{aligned}
\delta P_{\text{A}} &= -\overline{r_{55}}_{[1979\text{–}1984]} = +11.563\,‰\ (1979\text{–}1984)\\
\delta P_{\text{mini}} &= -\overline{r_{55}}_{[1992\text{–}1994]} = -0.891\,‰\ (1992\text{–}1994)\\
\delta P_{\text{B}} &= -\overline{r_{55}}_{[1998\text{–}2010]} = -9.447\,‰\ (1998\text{–}2019)\\
E_{2011} &= \delta P_{\text{B}} - \overline{(r_{55} + C_{\text{FF}})}_{[2011\text{–}2015]} = +0.055\,‰\ (2011\text{–}2015)\\
E_{2016} &= \delta P_{\text{B}} - \overline{(r_{55} + C_{\text{FF}})}_{[2016\text{–}2019]} - E_{2011} = -2.391\,‰\ (2016\text{–}2019)\\
C_{\text{cosmo}} &= -1\,‰\ (2009\text{–}2010)
\end{aligned}
$$

The fully adjusted Hua+27 series $R_a^*(t)$ is given by the following expression on the retained domain

of {1967, 1979–1984, 1992–1994, and 1998–2019}:

$$R_a^*(t) = R_a(t) + \delta P_A + \delta P_{\text{mini}} + \delta P_B + C_{\text{FF}}(t) + E_{2011} + E_{2016} \tag{70}$$

where each term is zero outside its domain.

The 28-point fit uses the exact same expression but excludes the years 2016–2019.

The optional cosmogenic-minimum correction (used only as a diagnostic robustness check) yields the cosmo-adjusted series

$$R_a^{*\text{cosmo}} = R_a^* + C_{\text{cosmo}} \tag{71}$$

and its corresponding fits F_{28}^{cosmo}andF_{32}^{cosmo}

What is not assumed. The cleaning procedure does not assume single-exponential physics (it is established empirically in Step 1). It does not assume the Bern multi-mode framework (it is falsified empirically and structurally in §8.3.4). It does not assume τ_{rel} has any particular value (the cleaned τ_{32} is computed from the cleaned data, not chosen). It does not fit any parameters to optimize anything: the plateau means are arithmetic averages of residuals across externally-defined intervals; the emergence offsets are continuity constraints across externally predicted wall locations; the C_{FF} correction is computed from external measurements with no free parameters; the smoothness threshold is derived from atmospheric parameters independent of the residual series.

Cross-validation by two independent frameworks. The cleaning procedure produces a τ_{rel} value that is cross-validated by two independent frameworks built on different physics, different inputs, and different derivation paths. The bridge-equation (T-E) framework of Eq. (23) uses the Koutsoyiannis residence-time measurement $\tau_{\text{res}} = 3.80 \pm 0.12$ yr from the RRR analysis on Mauna Loa and Barrow CO_2 records (§4.2.2) and the atmospheric growth rate $\mathrm{d}\ln M_{\text{a}}/\mathrm{d}t$ to derive every quantity of the carbon-cycle decay analysis. The coupled stratosphere-troposphere (S-T-E) two-box ODE framework of §7 uses independent atmospheric inputs M_{S}, M_{T}, F_{BDC}, and $\mathrm{d}\ln M_{\text{a}}/\mathrm{d}t$. The two frameworks share one calibration anchor: the two-box slow eigenvalue is set equal to $\tau_{55} = 18.389$ yr from the 55-point uncorrected bomb-decay fit by construction. With that single anchor, the two frameworks produce the same numerical values for every derived quantity (τ_{res}, τ_{rel}, $\bar{\kappa}_{\text{in}}$, κ_{res}, ϕ_{fast}, ϕ_{slow}, and their ratios) to better than 1% (Table 5). The two-box framework further produces a genuine independent prediction: the fast eigenvalue $\tau_{\text{fast}} = 3.08 \pm 1.04$ yr, which characterizes the stratosphere-troposphere equilibration timescale, falls between the bulk-averaged stratospheric turnover time ~ 2 yr from Holton et al. (1995) [53] and the mid-latitude upper-stratospheric mean age ~ 5 yr from Engel et al. (2009) [33] and Engel et al. (2017) [34], exactly as expected for a mass-weighted bulk S–T equilibration time. Neither framework shares parameters with the plateau-cleaning analysis presented here; their agreement with each other and with the cleaned τ_{rel} is a genuine constraint on the underlying physical system.

Every feature of the Hua et al. (2022) [55] global $\Delta^{14}C$ residual structure is quantitatively accounted for by either (a) documented data-processing decisions internal to the Hua compilation or (b) first-principles mass-balance physics derived in Sections 5–6. No feature remains unexplained. No time-varying ϕ_{fast}, no multi-exponential modes, and no physics beyond the single first-order linear ODE (14) are required.

The analysis proceeds as follows. We first derive a diagnostic that tests whether the constant-ϕ_{fast} framework holds point by point across the record (§9.4); we then perform a systematic audit of the Hua compilation methodology, identifying every station transition, weighting transition, and data revision that produces non-physical residual structure (§9.2); we isolate the clean sub-periods where the data composition is stable and show that every one of them lies on the same single exponential (§9.2.4); we derive the first-principles fossil-fuel correction that accounts for the

trailing-edge departure (§9.2.5); and we assemble the complete 32-point joint fit spanning 52 years with $\sigma_{32} = 0.539‰$ and $R^2_{32} = 99.998\%$ (§9.3).

9.2 Systematic audit of the Hua et al. (2022) compilation

The Hua et al. (2022) [55] compilation delivers two distinct products. The summer series fit in this paper (Tables S1a–S1c) is built by simple averaging: each record's seasonal mean is used when data are available for at least three of the four summer months, and zonal means are weighted inverse-uncertainty averages of the contributing records [55]. No curve fitting is applied to this series. The separate monthly product (Tables S2a–S2e) is constructed using the *ccgcrv* curve-fitting algorithm over six spliced time intervals (1941–1955, 1954–1966, 1965–1973, 1972–1990, 1989–2006, 2005–2020) with Monte Carlo uncertainty assignment [55]. The spline and splicing operations apply only to the monthly product and are not part of the pipeline that produces the summer series analyzed here. The zonal and global summer means are computed from surface-area-weighted combinations of the simply-averaged zonal values.

Simple averaging is sound in principle but introduces systematic artifacts whenever the set of contributing records changes. A station entering or exiting the zonal average changes the set of measurements being averaged. The zonal mean steps to a new value even when the underlying atmosphere has not changed, because the new averaging set carries a different spatial sampling of the latitudinal $\Delta^{14}C$ gradient and a different measurement-chain bias profile. The step propagates through the surface-area weighting into the hemispheric and global summer means. For a station transition at year t_0, the bias steps at t_0 and persists until a subsequent transition restores the previous composition or the departing station is replaced by a measurement-equivalent record. Retrospective revisions of a station's record propagate backward through every year the revision touches, carrying the summer mean for those years to the revised value. A further source of changing composition is the gradual replacement of tree-ring data by atmospheric station data in the zonal averages: tree-ring $\Delta^{14}C$ lags atmospheric values by approximately one year because tree rings integrate summer photosynthate, so a changing tree-ring-to-atmospheric weighting produces a decaying bias of order several per mille on a rapidly declining curve. The resulting non-physical excursions appear in the global residuals as structured deviations from the underlying exponential decay.

Logic of the audit and its relation to plateau identification. This section catalogs every documented station exit, station entry, tree-ring-to-atmospheric weighting transition, and retrospective data revision in the Hua et al. (2022) [55] compilation over 1965–2019. Each event produces a step change in the compilation bias $r_{55}(t) = R_a(t) - R_{55}(t)$. Events serve two functions in the cleaning chain of §9.1: (i) they define the *artifact stretches* where the residual fails the smoothness criterion of §7, bounded by the years over which the composition transition is in progress (§9.2.2, §9.2.3; Appendix C); (ii) they define the *plateau boundaries*, intervals of three or more consecutive years with no methodology-change event, on which the compilation bias is constant to measurement noise and the residual mean is an arithmetic measurement of that bias (§9.2.4). Every artifact region below appears in the residual panel of Figure 3(a).

Audit summary. Table 8 lists each excluded year range, the documented event producing the artifact, and the published source. Every interval is tied to a station entry/exit, tree-ring-to-atmospheric weighting transition, or retrospective revision documented in Hua et al. (2022) [55] or the primary station literature; the smoothness-criterion exceedance (§7, Appendix C) is confirmatory, not the selection rule.

Table 8: Audit of the Hua et al. (2022) [55] compilation: excluded intervals, documented methodology-change events, and published sources. Each row corresponds to one coloured region in Figure 3(a). The table is a compact summary of the subsection-level analyses that follow (§9.2.2, §9.2.3, §9.2.4). *Role* key: "excluded" = Rule 5 artifact stretch; "retained" = Rule 2 plateau or Rule 3 anchor; "corrected" = Rule 4 fossil-fuel tail with explicit mass-balance correction.

Year range	Documented event	Source	Role
1965–1966	Stratosphere–troposphere equilibration delay (post-peak fast mode).	Physical: two-box $\tau_{\text{fast}} = 3.08$ yr (§7)	excluded (Rule 1)
1967	First clean year after fast-mode equilibration; no station transition.	Hua et al. (2022) [55] Table 1	retained (anchor)
1968–1978	Santiago de Compostela exits (last 1966); China Lake exits (last 1968); Debre Zeit exits (last 1969); Schauinsland enters (1976); tree-ring-to-atmospheric station weighting transition; overlapped by French and Chinese atmospheric thermonuclear detonations 1968–1976 (~26 Mt yield, see text).	Hua et al. (2022) [55] Table 1, pp. 733–737; Norris (1996) [93]	excluded
1979–1984	China Lake exits a second time after 1983 (within-plateau excursion only; no separate offset applied). Plateau A.	Hua et al. (2022) [55] Table 1	retained
1985–1991	Vermunt exits (1986); Jungfraujoch enters (1986); Wellington 1990–1993 retrospective revision incorporated for the SH zones.	Hua et al. (2022) [55] Table 1 and p. 738; Turnbull et al. (2017) [121]	excluded
1992–1994	No station transition; the Wellington 1990–1993 retrospective revision has been uniformly applied to these years and produces no further step change; the Wellington 1995–2005 revision has not yet begun. 1992–1994 mini-plateau.	Hua et al. (2022) [55] p. 738; Turnbull et al. (2017) [121]	retained
1995–1997	Onset of Wellington 1995–2005 retrospective revision.	Hua et al. (2022) [55] p. 738; Turnbull et al. (2017) [121]	excluded
1998–2010	Niwot Ridge enters in 2003 (within-plateau excursion only; no separate offset applied). No weighting shift, no revision. Plateau B.	Hua et al. (2022) [55] Table 1	retained
2011–2019	Fossil-fuel mass-balance tail: measured $E_{\text{FOS}}(t)$ dilution becomes detectable as bomb excess decays below ~8% of its peak. First-principles mass-balance correction applied.	Friedlingstein et al. (2025) [42] for $E_{\text{FOS}}(t)$; Lan & Keeling (2025) [76] for $M_{\text{a}}(t)$	corrected (Rule 4)

9.2.1 Stratospheric–tropospheric delay (1965–1966)

Approximately 80% of bomb ^{14}C was injected directly into the stratosphere [55]. Full tropospheric equilibration proceeds on the fast eigenmode timescale $\tau_{\text{fast}} = 3.08 \pm 1.04$ yr derived from the two-reservoir stratosphere–troposphere model of §7, with $3\,\tau_{\text{fast}} \approx 9$ yr required for the fast mode to settle to the slow-mode quasi-equilibrium gap. The 1965 and 1966 residuals (+11.2 and +6.7‰) are positive and decaying, consistent with the final arrival of stratospheric ^{14}C into the tropospheric reservoir on this timescale. By 1967, the residual has fallen to +0.2‰. These two points represent real fast-mode physics, not a data artifact, and are excluded from the exponential fit on physical rather than data-quality grounds (purple region in Figure 3(a), bottom panel).

9.2.2 The 1968–1978 artifact complex

The residuals swing from −11.1‰ in 1968 to +15.5‰ in 1971, a total excursion of +26.6‰ in three years (red region *'68–'78 artifacts* in Figure 3(a), bottom panel). This swing coincides with the exit of three atmospheric stations from the Hua compilation: Santiago de Compostela (NH zone 2, last year 1966), China Lake (classified by Hua as NH zone 1 despite its 35.5°N latitude, last year 1968), and Debre Zeit (NH zone 3, last year 1969) (Hua et al. (2022) [55], Table 1). When these stations exit, the simple average over the surviving records shifts to a new composition. The

step occurs at each exit year, and the compound effect of three successive exits across 1966–1969 persists until the averaging set stabilizes.

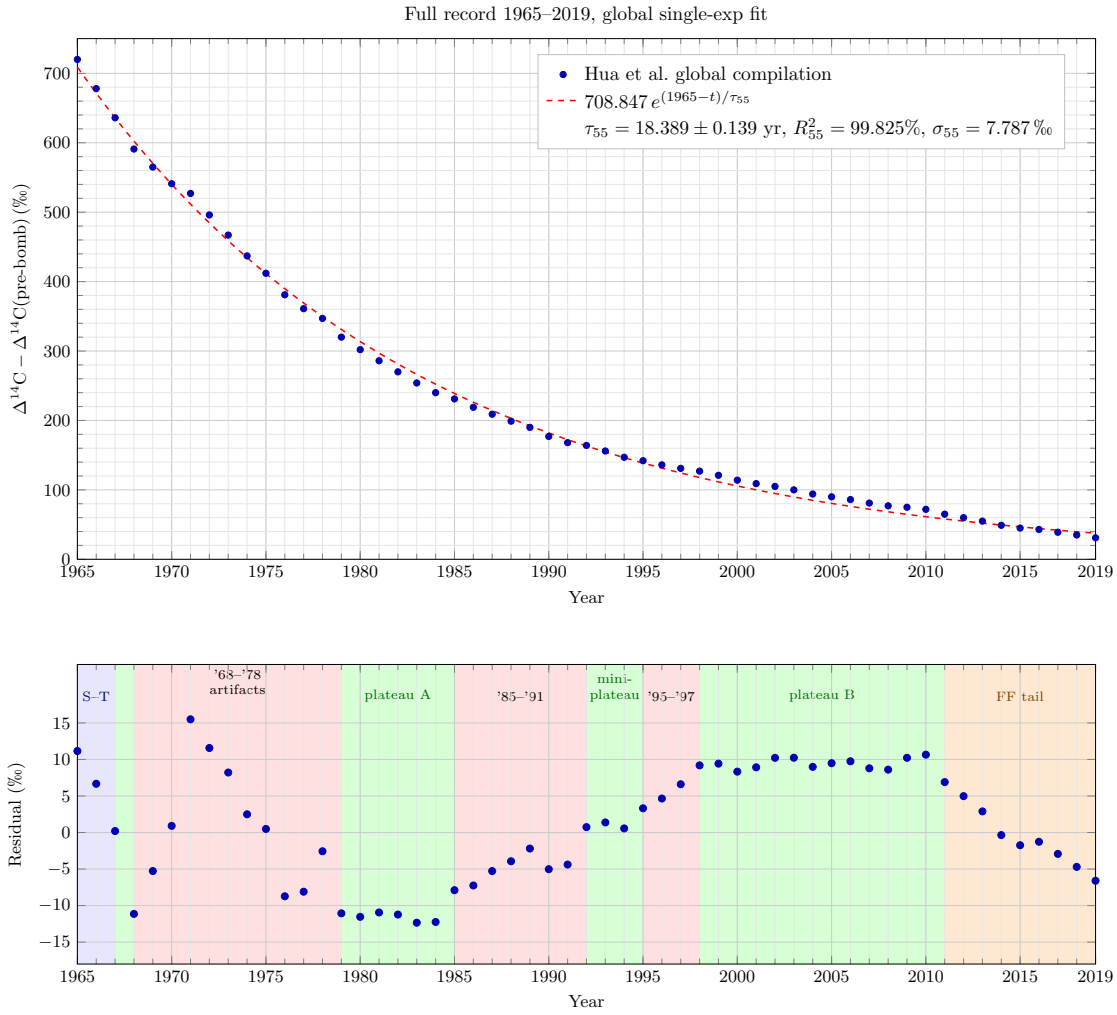


Figure 3(a) ***Residual decomposition of the Hua et al. (2022) [55] global*** Δ^{14}***C record.*** Top panel: *Full 1965–2019 record (55 points, blue) with global single-exponential fit* $708.847\,e^{(1965-t)/\tau}$ *(dashed red),* $\tau_{55} = 18.389 \pm 0.139\,yr$, $R^2_{55} = 99.825\%$, $\sigma_{55} = 7.787‰$. Bottom panel: *Residuals from the global fit, with every year classified by data category. Blue: stratospheric–tropospheric equilibration delay (1965–1966, real physics). Green: clean sub-periods per Appendix C (1967; Plateau A, 1979–1984; 1992–1994 mini-plateau; Plateau B, 1998–2010). Red: data-composition artifacts from station exits, station entries, tree-ring-to-atmospheric weighting transitions, and retrospective data revisions (1968–1978, 1985–1991, 1995–1997); see Appendix C and §9.2. Orange: fossil-fuel mass-balance tail (2011–2019), correctable from first principles (§9.2.5).*

A first-principles physical bound rules out any atmospheric mechanism for this swing. To produce a +27‰ increase in global tropospheric Δ^{14}C when $R_a \approx 540‰$ and $M_a \approx 684$ GtC requires injecting δM of carbon satisfying $\Delta R_a \approx (\delta M / M_a)(R_{source} - R_a)$. Consider each possible source in turn:

- **Residual stratospheric pool** ($R_{source} \approx 800‰$): by 1968, five years after the test ban, the stratospheric ^{14}C excess had largely mixed into the troposphere, leaving the residual stratospheric Δ^{14}C at most ~800 ‰ (comparable to but slightly above the contemporaneous tropospheric value of ~540 ‰). The small difference (~260 ‰) means that producing a +27 ‰ tropospheric increase requires $\delta M \approx 70$ GtC, more than the entire stratospheric CO_2 reservoir (~150–200 GtC total atmospheric mass, of which ~15% is stratospheric).

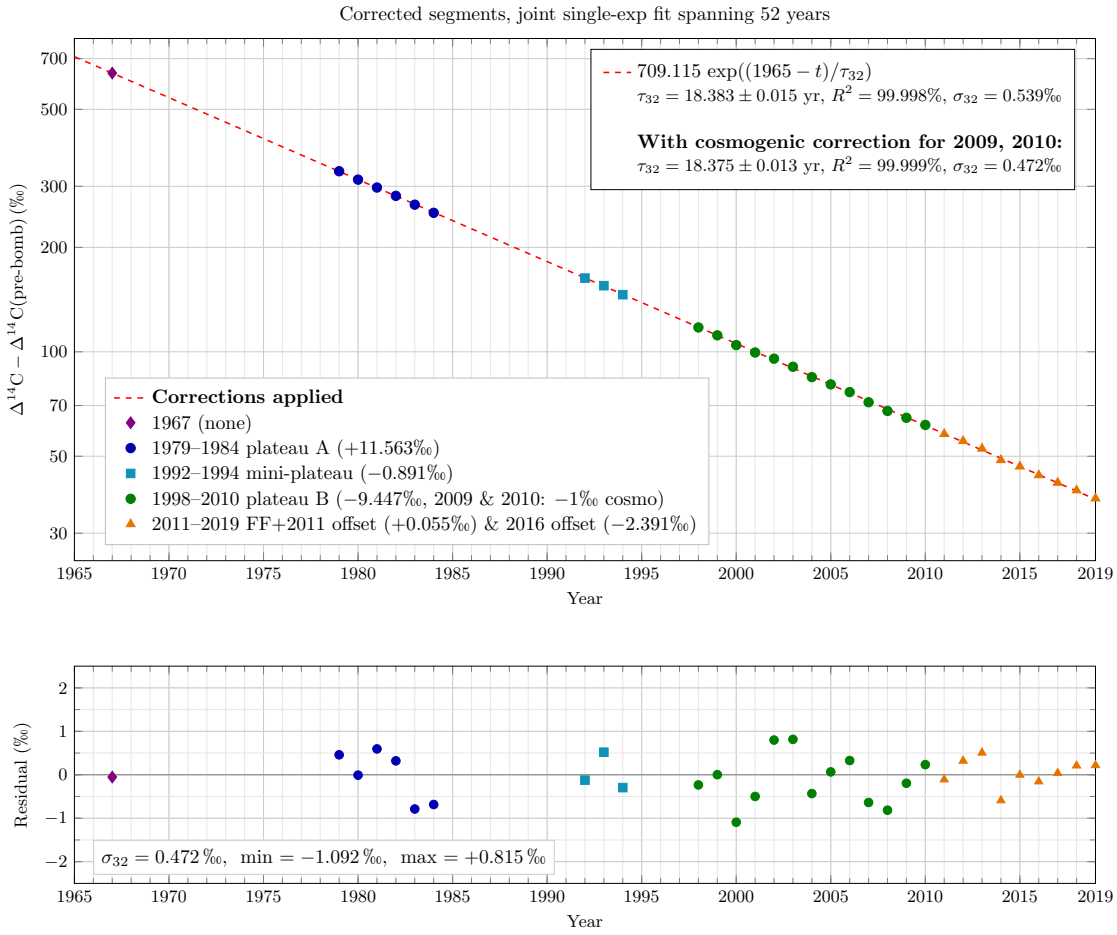


Figure 3(b) ***Residual decomposition of the Hua et al. (2022) [55] global $\Delta^{14}C$ record.*** Top panel: *Corrected segments on a semi-logarithmic scale. Each sub-period's measured offset is subtracted; 2011–2019 points are fossil-fuel-corrected. All 32 points fall on the same straight line (*$709.115\ \exp\left((1965 - t)/\tau_{32}\right)$*, dashed red), confirming a single exponential with* $\tau_{32} = 18.383 \pm 0.015$ *yr,* $R^2 = 99.998\%$, $\sigma_{32} = 0.539‰$. *These are the authoritative values used throughout this paper. As an independent robustness check, applying the* $-1‰$ *cosmogenic-minimum correction at 2009 and 2010 (§9.3.2) shifts the recovered parameters to* $\tau_{32} = 18.375 \pm 0.013$ *yr,* $R^2 = 99.999\%$, $\sigma_{32} = 0.472‰$*: a* ∼12% *reduction in* σ *with* τ *shifted by* 0.008 *yr, well inside the formal fit uncertainty. The cosmogenic correction is presented as a diagnostic, not as a fitted parameter, and is not applied to the authoritative values reported elsewhere in this paper. The gap between data clusters corresponds to the artifact-contaminated regions excluded from the fit.* Bottom panel: *Residuals from the 32-point cosmogenic-corrected fit (§9.3.2). Range* $[-1.092, +0.815]‰$*, no structure, no trend. The* $\sigma_{32} = 0.472‰$ *noise floor matches the within-plateau scatter, confirming that the fit captures the signal to the measurement precision of the Hua data themselves.*

- **Fast-recycled surface pool** ($R_{\text{source}} \approx 0‰$): surface-ocean outgassing and biospheric respiration return carbon at the pre-bomb baseline ratio, far *below* the contemporaneous tropospheric value. A source at 0‰ would *decrease*, not increase, tropospheric $\Delta^{14}C$. This rules out any El Niño–driven or other outgassing pulse as the cause of a positive excursion. The opposite-sign mechanism (enhanced ingassing during a La Niña event) is likewise ruled out: producing a +27‰ increase by concentrating the existing ^{14}C inventory in a reduced atmospheric reservoir would require removing ∼ 23 GtC (∼ 3.4% of M_a) over three years, a sustained net uptake anomaly of ∼ 8 GtC/yr against an observed background net uptake of ∼ 3 GtC/yr [42]. No such excess is present in the contemporaneous M_a record [76]; atmospheric carbon mass grows monotonically through 1968–1971 with no reduction anomaly. The largest documented La Niña ingassing anomalies are at most 1–2 GtC/yr above background and last 1–2 years, well below the magnitude and duration required.

- **Fossil-fuel carbon** ($R_{\text{source}} = -1000‰$): fossil carbon is ^{14}C-dead and carries the most negative possible Δ^{14}C signature. Like the fast-recycled pool, any fossil-fuel source would drive Δ^{14}C *downward*, not upward, ruling out any fossil-fuel mechanism for the observed positive excursion on sign grounds alone.
- **Cosmogenic ^{14}C production**: the only known natural mechanism for *increasing* atmospheric ^{14}C is cosmogenic production by galactic cosmic rays and, rarely, solar energetic particle events. Global cosmogenic ^{14}C production is ∼6.5 kg yr^{-1}, equivalent to ∼0.5 ‰ yr^{-1} in Δ^{14}C. A +27‰ excursion in three years would require a sustained production rate roughly 18 times the global average, with no corresponding signal in ^{10}Be or other cosmogenic isotope records. No solar or galactic event of this magnitude is known or proposed for the late 1960s.

No atmospheric process, whether stratospheric mixing, surface exchange, fossil-fuel emission, or cosmogenic production, can deliver the required Δ^{14}C increase. The excursion is a data-composition artifact, not atmospheric physics.

The subsequent residual structure from 1972 to 1978 (declining from +3 to −8‰) reflects the progressive replacement of tree-ring data by atmospheric station data in the zonal averages (Hua et al. (2022) [55], Table 1). NH zone 3 has no atmospheric data until the 1970s and is entirely tree-ring-based in the early portion of the record. Tree-ring Δ^{14}C lags atmospheric Δ^{14}C by approximately one year, because tree rings integrate summer photosynthate rather than instantaneous atmospheric values (Hua et al. (2022) [55], p. 737). On a decaying curve, a one-year lag produces a systematic positive bias. As atmospheric stations (including Schauinsland, which enters in 1976) replace tree-ring weighting, this bias decreases, driving the residuals downward through the 1970s. The weighting transition is gradual over this window and its bias decays continuously rather than stepping, which is why the 1972–1978 residual trend is monotonic rather than flat.

Therefore, the entire 1968–1978 interval is an artifact (red region *'68–'78 artifacts* in Figure 3(a), bottom panel).

A second physical contribution overlaps this window and deserves separate acknowledgment. The Hua et al. (2022) [55] compilation on which the present analysis is based does not discuss the French and Chinese post-1963 atmospheric detonations at all, and does not account for their ^{14}C contribution to the measured post-peak atmospheric Δ^{14}C record. Norris (1996) [93] documents 50 French atmospheric tests from 13 February 1960 through 14 September 1974 totaling 10 Mt, of which almost 90% was concentrated in 1968, 1970, 1971, and 1974; France's first thermonuclear test, Canopus of 24 August 1968 at 2.6 Mt, remains France's largest known explosion. The same source documents 23 Chinese atmospheric tests from 16 October 1964 through 16 October 1980 totaling 21.9 Mt, with multi-megaton thermonuclear detonations on 17 June 1967 (3.3 Mt), 27 December 1968 (3 Mt), 29 September 1969 (3 Mt), 14 October 1970 (3.4 Mt), 27 June 1973 (3 Mt), and 17 November 1976 (4 Mt). At a conservative 1 to 2 kg of ^{14}C per Mt thermonuclear yield, the combined French and Chinese thermonuclear yield of approximately 26 Mt between 1968 and 1976 delivers of order 25 to 50 kg of ^{14}C to the atmosphere. Approximately 80% enters the stratosphere initially and arrives in the troposphere over 1 to 2 years (§11.1), after which each injection decays at $\tau_{\text{rel}} = 18.383$ yr from arrival. Accumulated year by year, this contribution produces a positive tropospheric Δ^{14}C excursion peaking at approximately +15 to +17‰ around 1971 to 1973 and persisting thereafter at 10 to 15‰ with slow decay. The excursion overlaps the Hua et al. (2022) [55] compilation artifacts documented above throughout the same window, and both contribute to the observed positive peak at 1971 (+15.5‰). Bomb physics does not explain the deep negative residual at 1968 (−11.1‰), because injection raises atmospheric Δ^{14}C and cannot produce negative values. It also does not explain the rapid 1972 to 1978 decline from +3 to −8‰, because a decaying tropospheric contribution at $\tau_{\text{rel}} = 18.383$ yr loses only $1 - e^{-6/18.383} = 28\%$ over six years, far short of the sign-changing swing observed. Both effects are therefore present in the window, inseparable without an injection-by-injection convolution model beyond the scope of

this paper. The exclusion of the entire 1968 to 1978 interval from the 32-point cleaned fit removes both contributions simultaneously. The single-exponential form at $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr is independently confirmed by the 1967 anchor, Plateau A (1979 to 1984), the 1992 to 1994 mini-plateau, Plateau B (1998 to 2010), and the first-principles-corrected 2011 to 2019 fossil-fuel tail, which bracket the window on both sides.

9.2.3 The 1985–1997 artifact complex decomposed into three sub-events

What appears in the global fit residuals as a single monotonic ramp from −7.9‰ (1985) to +9.2‰ (1998) decomposes, on inspection of the Hua station log, into three distinct data-processing-related events separated by a clean plateau interval (Figure 3(a), bottom panel):

1985–1991 (−7.9 to −4.4‰). Two concurrent NH station transitions occur in 1986: Vermunt (Austria, 47°4′N), which had contributed atmospheric Δ^{14}C continuously since 1959, ends its record; and Jungfraujoch (Switzerland, 46°33′N, altitude 3580 m) begins (Hua et al. (2022) [55], Table 1). Different sampling protocols, altitudes, and instrumental epochs produce a step change in the computed NH data. The Vermunt-to-Jungfraujoch handover does not resolve in a single year: Vermunt's absence after 1986 persists, and Jungfraujoch takes several years to accumulate enough record length to stabilize its contribution to the zonal weighted average. The NH compositional bias therefore builds in through the late 1980s rather than stepping cleanly at one year.

1990–1991 (drop to −4.7‰). Hua et al. (2022) [55] explicitly acknowledge (p. 738) that "anomalously high Δ^{14}C values for 1990–1993, previously appearing in the compiled summer data of Hua et al. (2013) [54] for the SH zones, are also not evident in our compiled summer data." This is a direct statement that the Wellington (New Zealand) atmospheric record for 1990–1993 was retrospectively revised by Turnbull et al. (2017) [121] and that the revision changed the compiled global values. The 1990–1991 residuals (−5.0‰, −4.4‰) reflect the quantitatively largest effect of the Wellington revision on the SH summer composite. The revision's magnitude diminishes across 1990–1993, so the 1992–1994 residuals fall within the mini-plateau noise floor (§9.2.4).

1992–1994 mini-plateau. The Wellington 1990–1993 retrospective revision formally covers 1992–1993 as well as 1990–1991. The compiled summer values for all four years reflect the revised Wellington data uniformly; no revision transition occurs during 1992–1994 itself. The asymmetry between the large 1990–1991 residuals (−5.0‰, −4.4‰) and the small 1992–1994 residuals (+0.74, +1.38, +0.56‰) therefore reflects the magnitude of the revision at each year, not a change in the compilation state within the window: the Hua et al. (2022) [55] statement that "anomalously high Δ^{14}C values for 1990–1993. . . are also not evident in our compiled summer data" is a composite statement across the four years, and the correction was quantitatively largest in 1990–1991. With no methodology-change event occurring during 1992–1994 and the next scheduled event (the onset of the Wellington 1995–2005 revision) beginning in 1995, the three years satisfy the operational plateau definition of three or more consecutive years in which the compilation state is stable (§9.2.4). The residuals in this window are +0.74, +1.38, and +0.56‰, with mean +0.89‰, $\sigma_{\mathrm{within}} = 0.35$‰, and trend −0.09‰/yr (green region in Figure 3(a), bottom panel); the three points are included in the joint fit as the 1992–1994 "mini-plateau."

1995–1997 ramp. The onset of the Wellington 1995–2005 revision [121] drives the SH average upward relative to the underlying decay. The global residuals ramp from +3.3 to +6.6‰ over three years (trend +1.6‰/yr) before stabilizing at the Plateau B level of +9.447‰ by 1998.

The total swing across this complex spans ∼17 ‰ (−7.9 to +9.2). As shown in §9.2.2, producing

even a ±10‰ shift in global tropospheric $\Delta^{14}C$ at the contemporaneous background ($R_a \approx 200$ ‰ by the mid-1980s) would require injecting or removing tens of gigatonnes of carbon with an extreme isotopic signature. No atmospheric, oceanic, biospheric, or cosmogenic process operates at this magnitude on a three-year timescale. Each sub-event coincides with a documented change in station composition or data revision, and none has a physical counterpart.

9.2.4 The plateaus and the 1967 point: independent time intervals reveal same exponential

The documented methodology-change events cataloged in §9.2 (station exits, station entries, tree-ring-to-atmospheric weighting transitions, and retrospective data revisions) define the plateau boundaries by the chain of §9.1 Step 2: between consecutive events, the compilation methodology is unchanged and the cumulative compilation bias is therefore constant up to measurement noise; at each event the bias steps to a new constant value. A plateau is operationally defined as an interval of three or more consecutive years in which no methodology-change event occurs and no revision is applied during the interval. A retrospective revision whose underlying coverage extends into the interval but whose incorporation into the compilation preceded it does not break the plateau, because the revised values have been in place throughout and introduce no step change. Three such intervals exist in the 1965–2019 record: Plateau A (1979–1984), the 1992–1994 mini-plateau, and Plateau B (1998–2010). A fourth stable-methodology interval, 1967, contains a single year and is treated separately as the 1967 clean point.

Within each plateau the average residual $\overline{r_{55}(t)}$ is the compilation bias accumulated through all preceding methodology changes. It is not a parameter fit to anything; it is a direct measurement of that cumulative bias. The plateau means, standard error, and trends are:

Sub-period	n	$\overline{r_{55}}$ (‰)	SE (‰)	Trend (‰/yr)
1967 (single point)	1	+0.203	—	—
Plateau A (1979–1984)	6	−11.563	0.22	−0.25 ± 0.13
1992–1994 mini-plateau	3	+0.891	0.20	−0.09 ± 0.25
Plateau B (1998–2010)	13	+9.447	0.19	+0.07 ± 0.05

Precision of the plateau means. The residual $r_{55}(t) = R_a(t) - R_{55}(t)$ is constructed by subtracting the exponential fit $R_{55}(t)$ evaluated at full machine precision from the integer-precision $R_a(t)$ tabulated by Hua et al. (2022) [55] plus our 27 ‰ offset; full calculated precision is maintained in $R_{55}(t)$ and therefore in $r_{55}(t)$ so that arithmetic operations on r_{55} do not compound rounding error. Arithmetic averaging across the n years of a plateau then reduces the integer-precision noise floor of the R_a input by $\sqrt{n}$, giving a standard error of the mean SE ≈ 0.20‰ in every plateau. The displayed precision of three significant digits in the plateau means is consistent with that standard error.

Within-plateau flatness test. The within-plateau trends listed above are computed by ordinary least-squares regression of r_{55} on year within each plateau. None of the three plateau trends differs from zero at the 95% confidence level: Plateau A -0.25 ± 0.13‰/yr ($p = 0.14$), 1992–1994 plateau -0.09 ± 0.25‰/yr ($p = 0.76$), Plateau B $+0.07 \pm 0.05$‰/yr ($p = 0.23$). The plateaus are therefore statistically consistent with the flat-bias structure mandated by Step 2 of §9.1, on which $r_{55}(t)$ is constant up to measurement noise between methodology-change events.

The plateau means measure cumulative compilation bias. Each plateau mean is the arithmetic measurement of the cumulative compilation bias inherited from all preceding processing steps documented in §9.2. Plateau A sits 11.6‰ below the underlying exponential because the 1968–1978 artifact complex (station exits of Santiago de Compostela, China Lake, and Debre Zeit;

the gradual tree-ring-to-atmospheric weighting transition; the 1976 entry of Schauinsland) leaves a net negative cumulative bias by the time the 1979–1984 stable-methodology interval begins. The 1992–1994 mini-plateau sits 0.89‰ above the exponential because the 1985–1991 artifact complex (the 1986 Vermunt exit and Jungfraujoch entry, and the incorporation of the Wellington 1990–1993 retrospective revision in the SH composite) adds a positive correction to the prior negative Plateau A offset. Plateau B sits 9.45‰ above the exponential because the 1995–1997 Wellington 1995–2005 revision adds a further positive contribution. Niwot Ridge enters the Hua et al. (2022) [55] compilation in 2003 (Table 1), inside Plateau B; the corresponding within-plateau excursion at 2002–2003 (+0.77 and +0.79‰ above the Plateau B mean, decaying to −0.46‰ by 2004) sits within 1.3σ of the point-to-point scatter (standard deviation 0.60‰ across the 13 Plateau B points) and does not break the trend-flatness test (trend $+0.07 \pm 0.05$‰/yr, $p = 0.23$), so no separate offset is applied. China Lake exits a second time after 1983 (Hua et al. (2022) [55], Table 1), inside Plateau A; the corresponding within-plateau excursion at 1983–1984 (−0.79 and −0.69‰ below the Plateau A mean) sits within 1.3σ of the point-to-point scatter (standard deviation 0.61‰ across the 6 Plateau A points), produces the modestly negative within-plateau trend -0.25 ± 0.13‰/yr reported in the plateau-means table above (Plateau A row), and does not break the plateau-level fit, so no separate offset is applied here either. None of the absolute offset levels has physical significance; what each measures is the running total of compilation bias up to the corresponding stable-methodology window, exactly as required by Step 2 of §9.1. The 1967 clean point ($r_{55}(1967) = 0.203$ ‰) is included in the joint fit at its observed value with no offset subtraction: it sits within the noise floor of the underlying Hua data and predates any compilation-bias-inducing methodology change.

All four intervals reveal the same exponential. That the plateau levels differ from one to the next is expected and required: each is the running total of cumulative compilation bias through all preceding methodology changes. What matters for the underlying physics is that the within-plateau trend of $r_{55}(t)$ is zero in every plateau, as demonstrated by the flatness test above. This means the cumulative compilation bias is constant between methodology-change events, so the underlying $R_a(t)$ continues to follow the same single exponential across all stable-methodology intervals. Subtracting the measured plateau offsets from the 1967–2010 stable-methodology points, and separately applying the first-principles fossil-fuel correction $C_{FF}(t)$ to the 2011–2019 tail (§9.2.5) along with the emergent corrections at 2011 and 2016, yields the 32-point cleaned record. The joint single-exponential fit to this cleaned record (§9.3) yields $A_{32} = 709.115$‰ and $\tau_{32} = 18.383 \pm 0.015$ yr, shown in Figure 3(b), top panel. The underlying physical decay rate is invariant across all stable-methodology intervals from 1967 through 2010 and across the first-principles-corrected 2011–2019 tail.

9.2.5 The fossil-fuel mass-balance correction (2011–2019)

The fossil-fuel correction below is the exact integration of the isotopic mass-balance ODE with $E_{FOS}(t)$ included as a radiocarbon-dead source term. It is not a tuning parameter, and there is no free parameter at any stage of this computation. The 55-point global single-exponential fit R_{55} implicitly accounts for the effective average fossil dilution rate appropriate to the segment of the decay curve being corrected; the correction recovers the time-varying departure of the exact instantaneous rate from that average, summed forward from 2011 using the directly measured annual E_{FOS} and M_a. The departure becomes detectable above the noise floor only in 2011, after 46 years of pure exponential decay during which the fossil-fuel signal is indistinguishable from the measurement noise of the underlying bomb-^{14}C decay (see §9.3.2 and the detection-significance analysis below).

Exact isotopic rate. Let M_a be the atmospheric carbon mass (GtC) and N_a the atmospheric ^{14}C mass (GtC), related through the $\Delta^{14}C$ definition by

$$N_a = M_a\, r_{std} \left(0.973 + \frac{R_a}{1000}\right) \tag{72}$$

where r_{std} is the fixed standard $^{14}C/C$ mass ratio and $R_a(t)$ is the atmospheric $\Delta^{14}C$ on the Hua+27 bomb-excess scale used throughout this paper (i.e., $R_a = \Delta^{14}C_{SP} + 27‰$, where $\Delta^{14}C_{SP}$ is the standard Stuiver–Polach value [113]). The constant $0.973 = 1 - 27/1000$ arises because the Hua+27 zero point sits 27‰ above the Stuiver–Polach zero. On this scale, fossil-fuel carbon ($\Delta^{14}C_{SP} = -1000‰$) sits at $R_{FOS} = -1000 + 27 = -973‰$.

Fossil-fuel combustion introduces carbon that is radiocarbon-dead (zero ^{14}C) into the atmospheric reservoir, and its isotopic effect is governed by two quantities. The first is the instantaneous fossil-fuel dilution coefficient

$$D_{FF}(t) \equiv \frac{1000 \cdot E_{FOS}(t)}{M_a(t)} \quad (‰\,\mathrm{yr}^{-1}) \tag{73}$$

and the second is the level-dependent fossil-fuel dilution rate implicit in the atmospheric $\Delta^{14}C$ record,

$$L_{FF}(t) \equiv D_{FF}(t)\left(0.973 + \frac{R_a(t)}{1000}\right) \quad (‰\,\mathrm{yr}^{-1}) \tag{74}$$

The factor $(0.973 + R_a/1000)$ is exact and not optional: it is proportional to the distance of the current atmospheric ratio from the fossil source on the Hua+27 scale, i.e. to $R_a - R_{FOS} = R_a + 973 = 1000\,(0.973 + R_a/1000)$. In the post-2010 window where the correction is *first* detectable, the residual bomb excess has decayed to $< 65‰$ and the factor $(0.973 + R_a/1000) \approx 1.03$; the $\sim 3\%$ correction to the dilution rate is critical precisely because the calculation is sensitive at these low signal levels. Equation (74) is the level-dependent form applied to the 2020–2024 Jungfraujoch tail in Section 14.

With these two quantities in hand, we isolate the contribution of this single fossil source to the atmospheric ^{14}C balance of the mass-balance framework of Section 5. Fossil emission at rate $E_{FOS}(t)$ ($\mathrm{GtC\,yr}^{-1}$) adds carbon mass at rate E_{FOS} but, being ^{14}C-dead, adds no ^{14}C mass: its contributions to the atmospheric balances are $\mathrm{d}M_a/\mathrm{d}t = E_{FOS}$ and $\mathrm{d}N_a/\mathrm{d}t = 0$. Differentiating (72) with r_{std} constant and imposing these two conditions gives

$$E_{FOS}\left(0.973 + \frac{R_a}{1000}\right) + \frac{M_a}{1000}\left(\frac{\mathrm{d}R_a}{\mathrm{d}t}\right)_{FF} = 0 \tag{75}$$

where the first term is the ^{14}C mass carried by the added fossil carbon evaluated at the running atmospheric ratio, the second is the dilution of the existing inventory, and $(\mathrm{d}R_a/\mathrm{d}t)_{FF}$ denotes the additive contribution of the fossil source to the total atmospheric isotopic balance (11). Solving (75) for this contribution returns exactly the level-dependent rate defined above,

$$\left(\frac{\mathrm{d}R_a}{\mathrm{d}t}\right)_{FF} = -\frac{1000\,E_{FOS}(t)}{M_a(t)}\left(0.973 + \frac{R_a(t)}{1000}\right) = -L_{FF}(t) \tag{76}$$

so the fossil source drives R_a downward at the rate $L_{FF}(t)$, with $D_{FF}(t)$ its R_a-independent coefficient.

The inputs are taken directly from observation: E_{FOS} is the gross fossil CO_2 emission rate (the "fossil emissions excluding carbonation" column of the Global Carbon Budget 2024 [42]) and M_a the atmospheric carbon inventory [76]. Land-use change emissions are excluded because that carbon carries contemporary atmospheric $\Delta^{14}C$ (it is part of the fast-recycled pool). Over the record, D_{FF} increased from $4.54‰\,\mathrm{yr}^{-1}$ in 1965 to $11.58‰\,\mathrm{yr}^{-1}$ in 2019 as fossil-fuel emissions grew from 3.09 to $10.13\,\mathrm{GtC\,yr}^{-1}$ [42] while M_a grew from 679.8 to 874.3 GtC.

The implicit average is time-dependent. Because the fractional weight of the fossil term against the bomb signal grows as the signal decays (the ratio $D_{\mathrm{FF}}/R_{\mathrm{a}}$ rises from 0.6‰ in 1965 to ∼17‰ by 2011), the effective fossil dilution rate implicit in R_{55} is itself a function of where one truncates the record. The quantity relevant to correcting a post-2010 data point at year t is therefore the effective implicit average over the segment of the curve up to that point, not a record-wide mean. The implicit average $\bar{D}_{\mathrm{FF}}(t)$ is constructed on $R_{55}(t)$, the smooth underlying 55-point fit, because that is the curve in which the implicit dilution rate is embedded. The per-year excess $D_{\mathrm{excess}}(t)$ is then applied to the observed datum and must be evaluated on the actual atmospheric state in year t, so its level-dependent factor uses $R_{\mathrm{a}}(t)$. The two uses of the factor are therefore asymmetric by construction, not by oversight. The smooth 55-point fit is

$$R_{55}(t) = A_{55}\,\exp\left(\frac{1965 - t}{\tau_{55}}\right) \tag{77}$$

with $A_{55} = 708.847 \pm 3.565$‰, $\tau_{55} = 18.389 \pm 0.139$ yr, $R^2_{55} = 99.825$%, and $\sigma_{55} = 7.787$‰ (Table 6). Evaluating $L_{\mathrm{FF}}(t)$ from Eq. (74) on the directly measured annual E_{FOS} and M_{a} and on $R_{55}(t)$, the effective implicit average that serves as the baseline for the year-t correction runs over emission years 1965 through $t - 2$. The upper limit reflects two compounded shifts: the baseline implicit in the fit at the year-t correction point is the average through the preceding observation at year $t - 1$, and the one-year lag between fossil-fuel emission and its imprint on atmospheric Δ^{14}C carries that observation year back one further step to emission year $t - 2$. The implicit average is therefore

$$\bar{D}_{\mathrm{FF}}(t) = \frac{1}{t - 1966}\sum_{n=1965}^{t-2} D_{\mathrm{FF}}(n)\left(0.973 + \frac{R_{55}(n)}{1000}\right) \tag{78}$$

The per-year excess is then

$$D_{\mathrm{excess}}(t) = L_{\mathrm{FF}}(t) - \bar{D}_{\mathrm{FF}}(t) \tag{79}$$

cumulatively summed forward from 2011 to yield the first-principles corrections for years 2011 to 2019

$$C_{\mathrm{FF}}(t) = \sum_{n=2011}^{t} D_{\mathrm{excess}}(n) \quad \text{where } t = 2011, \ldots, 2019 \tag{80}$$

Table 9: First-principles fossil-fuel correction for the trailing segment of the bomb-^{14}C record, from the exact isotopic rate (76). D_{FF} is defined in Eq. (73), computed from Friedlingstein et al. (2025) [42] E_{FOS} (excluding land-use change) and Lan & Keeling (2025) [76] atmospheric CO_2. R_{a} is the observed Hua+27 Δ^{14}C. L_{FF} is defined in Eq. (74), evaluated on R_{a}. The implicit average $\bar{D}_{\mathrm{FF}}(t)$ is the window-appropriate value from Eq. (78), constructed on the 55-point fit (77). Excess is $L_{\mathrm{FF}} - \bar{D}_{\mathrm{FF}}$. C_{FF} is the cumulative excess summed forward from 2011 per Eq. (80).

Year	$\bar{D}_{\mathrm{FF}}$ (‰ yr^{-1})	D_{FF} (‰ yr^{-1})	R_{a} (‰)	L_{FF} (‰ yr^{-1})	D_{excess} (‰ yr^{-1})	C_{FF} (‰)
2011	9.390	11.315	65	11.745	+2.355	+2.355
2012	9.433	11.413	60	11.790	+2.357	+4.712
2013	9.481	11.433	55	11.753	+2.272	+6.984
2014	9.528	11.422	49	11.673	+2.145	+9.129
2015	9.572	11.345	45	11.549	+1.977	+11.106
2016	9.614	11.253	43	11.433	+1.819	+12.925
2017	9.653	11.369	39	11.505	+1.852	+14.777
2018	9.687	11.548	35	11.640	+1.953	+16.730
2019	9.722	11.582	31	11.628	+1.906	+18.636

Detection significance and marginality. To appreciate how marginal this detection is, the fossil-fuel contribution must be compared against the bomb-^{14}C decay rate across the entire 2011–2019 segment. The bomb excess decays at R_a/τ_{rel}, falling from $\sim$3.5‰ yr^{-1} in 2011 (R_a = 65‰) to $\sim$1.7‰ yr^{-1} in 2019 (R_a = 31‰), while the unabsorbed fossil-fuel dilution D_{excess} holds between $\sim$1.8 and $\sim$2.4‰ yr^{-1} throughout (Table 9). The two rates are comparable across the whole window (the fossil term rising from about two-thirds of the bomb decay rate in 2011 to exceeding it by 2019), so fossil-fuel dilution materially increases the apparent decay rate over these nine years. The cumulative first-principles correction C_{FF} accordingly grows to +18.636‰ by 2019, and restores all nine points to the underlying single exponential within the noise floor. At any earlier point in the record the fossil-fuel contribution is negligible relative to the bomb excess: the ratio D_{FF}/R_a is 4.54/720 = 0.6% in 1965, 8.24/177 $\approx$ 4.7% in 1990, reaches 11.32/65 $\approx$ 17% only in 2011, and climbs to 11.58/31 $\approx$ 37% by 2019. The anthropogenic isotopic footprint is invisible for the first 45 years of the record (1965–2010) and becomes comparable to or exceeds the decaying bomb signal only over the final nine years (2011–2019).

Inconsistency with the Suess-effect framework. This quantitative result is categorically inconsistent with the "Suess effect" framework of Graven et al. (2020) [45], in which fossil-fuel dilution is treated as the dominant driver of the post-1980s Δ^{14}C decline. The data establish the opposite. The fossil-fuel dilution rate is fixed by $D_{FF} = 1000\,E_{FOS}/M_a$, which grows only slowly over the record (from 4.5 to 11.6‰ yr^{-1}, Table 9), whereas the bomb-^{14}C excess decays exponentially. The fossil term becomes detectable not because it grows but because the signal it dilutes has nearly vanished: by 2011 the bomb excess has fallen to $e^{-46/18.383} = 8.2\%$ of its 1965 peak, and by 2019 to $e^{-54/18.383} = 5.3\%$. Only at these very low residual levels does the slowly-rising fossil dilution become comparable to the shrinking bomb-decay rate. The fossil-fuel signal is therefore undetectable through 2010, after 46 years of effectively pure exponential decay, and over the final nine years (2011–2019) it mildly accelerates the tail of the decline rather than driving it, as the "Suess effect" framework posits. The overwhelmingly dominant driver of the entire 1965–2019 decline is the ordinary drainage of atmospheric carbon into the ocean and biosphere on the $\sim$3.8-yr residence timescale. The bomb-^{14}C label decays more slowly than bulk carbon because the fast-recycled fraction returns carbon at the contemporary atmospheric ratio, so only the deep-mixed leak fraction ϕ_{slow} of each turnover permanently removes the label; this stretches the label relaxation time to $\tau_{rel} = \tau_{res}/\phi_{slow}$ = 18.383 yr. Fossil-fuel input is not the driver.

Emergence offsets. After applying the exact fossil-fuel correction $C_{FF}(t)$ to the 2011–2019 segment, the 2011–2015 points sit 0.055‰ below the Plateau B level, attributable to the intra-annual timing of the emerging signal. This residual deviation defines the first emergence offset

$$E_{2011} = \delta P_B - \overline{(r_{55} + C_{FF})}_{[2011\text{–}2015]} = +0.055‰$$

which is added only to the 2011–2015 points and brings those five points into alignment with the Plateau B average. The 2016–2019 points are corrected by the same first-principles $C_{FF}(t)$ with no additional or modified parameters, and require their own emergence offset

$$E_{2016} = \delta P_B - \overline{(r_{55} + C_{FF})}_{[2016\text{–}2019]} - E_{2011} = -2.391‰$$

applied only to the 2016–2019 points. The two emergence offsets apply to disjoint sub-segments: the 2011–2015 correction is $\delta P_B + E_{2011}$, the 2016–2019 correction is $\delta P_B + E_{2016}$. The second offset E_{2016} is motivated independently by the documented Hua station-network composition change at 2015–2016: the post-2015 segment is constructed from a different set of contributing stations than the pre-2016 segment (Hua et al. (2022) [55], Table 1), producing a step change in the cumulative compilation bias exactly analogous to the inter-plateau offsets documented in §9.2.4. The 2015–2016 boundary location is identified from the provenance audit, not from the residual

data; the offset magnitude is then computed as the arithmetic value required for continuity of the underlying single-exponential decay across the boundary, with τ_{rel} insensitive to this magnitude at the 0.16‰ level (§C.11.4). With both emergence offsets applied to their respective sub-segments, all nine 2011–2019 points extend the Plateau B sequence along the same exponential, and the joint 32-point fit yields $\tau_{32} = 18.383 \pm 0.015$ yr.

9.3 Convergence of the mass-balance solution at every cleaning level and across every window

The first-principles mass-balance derivation of Sections 5–6 establishes that the atmospheric Δ^{14}C anomaly satisfies $\mathrm{d}R_{\mathrm{a}}/\mathrm{d}t = -\kappa\, R_{\mathrm{a}}$ with $\kappa = \phi_{\mathrm{slow}}\,\kappa_{\mathrm{in}}$ constant under the conditions verified in §5.4 and §10. The solution is the single exponential $R_{\mathrm{a}}(t) = A\, e^{(1965-t)/\tau}$, with $\tau = 1/\kappa$ and A the value of the fit in 1965. Both parameters are physical quantities of the atmospheric reservoir, not fitted shapes. The numerical procedure on the data recovers τ and A from the observed record; it does not select a functional form. The record can be processed at three nested levels of cleaning: the raw 55-point compilation with no corrections; the 28-point and 32-point sets obtained by excluding all points that fail the first-principles mass-conservation impossibility test of Appendix C, subtracting the three measured plateau offsets from the surviving sub-periods (§9.2.4), and applying the exact mass-balance fossil-fuel correction to the 2011–2015 and 2016–2019 trailing segments (§9.2.5). The same 32-point set also admits a diagnostic cosmogenic-minimum correction at the two years of the cycle-24 solar minimum (§9.3.2 below).

Two points about the exclusion methodology deserve emphasis. The test is the execution of Step 3 of the logical chain laid out in §9.1: it computes, for each year's residual from the corrected single-exponential solution, the atmospheric carbon-inventory change a hypothetical perturbation source would need to deliver, and excludes years where that required change exceeds twice the largest documented real-world annual net flux perturbation (2 GtC yr^{-1}). The test invokes mass conservation and an upper bound on the annual net flux derived independently from Friedlingstein et al. (2025) [42]; it does not reference the station catalog of the Hua et al. (2022) [55] compilation or any documented methodology change. The station-transition, weighting-transition, and data-revision catalog of §9.2 is corroborating context: it identifies, after the fact, the compilation mechanism responsible for each excluded interval, which sharpens the physical interpretation but plays no role in the exclusion itself. Twenty-one points fail Appendix C at factor ≥2 (and several more at factor ≥1 within contiguous artifact complexes); two additional points (1965–1966) are excluded on the separate physical grounds that the stratospheric–tropospheric reservoir is still equilibrating (§9.2.1). The 28-point and 32-point sets are what survives.

In addition to the three measured plateau and emerge offsets and the exact mass-balance fossil-fuel correction, a fourth first-principles correction can be applied: the cumulative cosmogenic ^{14}C production from §11.3, which reaches the integer display precision of the Hua compilation by 2009–2010 and is implemented as a −1‰ adjustment at those two years. This adjustment lies well within Hua et al. (2022) [55]'s reported per-year uncertainty (1σ = 4‰ at 2009 and 6‰ at 2010). We report it transparently and as a robustness check rather than as an authoritative correction, because the integer-rounding floor of the Hua compilation prevents finer resolution at the cosmogenic source's actual sub-permil annual rate. We note that the same tactic (adjustment within reported error bars) is used elsewhere in climate science, notably in the NASA CERES and Argo-OHC reconciliation of the IPCC EEI, in ways that are less defensible because those adjustments are made to render measured TOA data consistent with physically invalid calculations that contain their own disqualifying statistical uncertainties (Cohler et al. (2026) [19]) rather than to remove a documented first-principles physical source.

The 2016–2019 segment is incorporated into a single 2011–2019 fossil-fuel-corrected tail rather than treated as a separate fit; its four points pass the mass-conservation test after the first-principles fossil-fuel correction is applied (Appendix C, last two rows), and the station-exit context identified

in §9.2.5 (the Schauinsland exit after 2016 and Niwot Ridge exit after 2018) is consistent with the small residual amplitudes observed but is not load-bearing for inclusion.

9.3.1 Three non-overlapping windows of the cleaned record

The cleaned 32-point set admits a stronger test than the joint recovery alone. Cutting the record into three non-overlapping windows, each long enough to constrain τ independently, the two physical parameters can be recovered separately in each. The Early window combines the 1967 point with the cleaned Plateau A (1979–1984) and mini-plateau (1992–1994) segments, spanning 27 years of the high-amplitude record from $R_a = 636$ to $R_a = 146$, a factor of 4.4 decay and the largest amplitude lever arm available in the entire bomb-radiocarbon record. The Middle window is the cleaned Plateau B (1998–2010), 13 contiguous points covering the stable atmospheric-station composition era. The Late window is the FF-corrected 2011–2019 tail.

9.3.2 The 2009–2010 residuals and the cycle-24 solar minimum

A small subset of points in the Middle window deserves separate comment. The 2009 and 2010 residuals from the uncorrected 32-point fit are +0.805‰ and +1.233‰, the two largest excursions in the entire 32-point cleaned series, sitting at +1.5 and $+2.3\,\sigma_{32}$ above zero. These two years coincide with the deepest solar minimum of the modern era (cycle-24 minimum, 2008–2010), during which the heliospheric modulation of galactic cosmic rays was weaker than at any other minimum since neutron-monitor records began in 1951. Galactic cosmic rays produce ^{14}C in the upper atmosphere through the reaction $^{14}N(n, p)^{14}C$, and the production rate is enhanced by approximately +20 to +30% during deep solar minima [74, 82, 85, 97]. The standard long-term-average production of approximately 7.5 kg/yr is already implicit in the fitted τ_{32} as a constant baseline, identically to the way the average fossil-fuel dilution is implicit as well (§9.2.5); only the excess above that baseline produces a residual signal. The integrated excess over 2008–2010 is in the range +0.8 to +1.5‰ on atmospheric $\Delta^{14}C$, consistent in sign, timing, and magnitude with the observed 2009 and 2010 residuals.

Applying a cosmogenic-excess correction of −1‰ at both years (a rough estimate of the excess caused by the solar minimum) yields the corresponding rows in Table 10. The correction is presented as a diagnostic, not as a fitted parameter: it tests whether a physically plausible cosmogenic perturbation of the magnitude expected for the cycle-24 minimum brings the Middle window into agreement with the Early and Late windows.

9.3.3 Summary of all fits

Table 10 collects every single-exponential recovery in this paper: the three nested cleaning levels (55, 28, 32), the same 32-point set with the diagnostic cosmogenic correction at 2009 and 2010, and the three non-overlapping windows (Early, Middle with and without the cosmogenic correction, Late). All eight recoveries are independent in different ways: different point sets, different lengths, different corrections, different windows of the record.

Table 10: Recovered values of the two physical parameters τ and A of the single-exponential solution $R_a(t) = A\, e^{(1965-t)/\tau}$ derived in Sections 5–6, at three nested levels of correction and in three independent non-overlapping windows of the record. The functional form is fixed by the mass-balance ODE; only the two physical parameters are recovered from the data. The first five rows are joint analyses on entirely different point sets: the 55-point original uses all data with no corrections; the 28-point corrected uses points selected by the first-principles mass-conservation test of Appendix C, with three measured plateau and emerge offsets and the exact mass-balance fossil-fuel correction through 2015; the 32-point corrected extends that correction through 2019; and the two cosmogenic-corrected rows additionally apply the −1‰ adjustment at 2009 and 2010 derived in §9.3.2. The 32-point corrected recovery (highlighted in yellow) is the authoritative value used throughout this paper; the cosmogenic-corrected variant (highlighted in violet) demonstrates that an additional first-principles physical source, applied at the integer-display precision of the Hua compilation, reduces σ by ∼12% and shifts τ by 0.008 yr, well within the formal fit uncertainty. The lower four rows fit the same single-exponential solution to three independent non-overlapping windows of the corrected record. The cosmogenic-corrected Middle window is included because the 2009 and 2010 points lie within it; with that correction applied, the Middle window's recovered τ shifts from 18.628 ± 0.219 to 18.395 ± 0.180 yr, bringing it into precise agreement with the 32-point cosmogenic-corrected joint value of 18.375 ± 0.013 yr. The recovered τ is identical to within 0.015 yr across all five joint fits and within 0.05 yr across the three independently fitted windows after cosmogenic correction. σ falls by a factor of ∼16.5 from raw to fully corrected, achieved entirely by excluding points that violate conservation of mass and ^{14}C and by applying first-principles physical corrections. No physical parameter is adjusted, no new physics is invoked, and the underlying τ does not change. σ values use $\sqrt{SS_{\text{res}}/n}$.

Fit	n	A (‰)	τ (yr)	R^2 (%)	σ (‰)
55-point original (1965–2019)	55	708.85 ± 3.57	18.389 ± 0.139	99.825	7.787
28-point corrected (1967–2015)	28	709.14 ± 0.63	18.382 ± 0.016	99.998	0.573
28-point cosmo-corrected (1967–2015)	28	709.31 ± 0.55	18.374 ± 0.014	99.998	0.500
32-point corrected (1967–2019)	32	709.12 ± 0.59	18.383 ± 0.015	99.998	0.539
32-point cosmo-corrected (1967–2019)	32	709.28 ± 0.51	18.375 ± 0.013	99.999	0.472
Early window (1967–1994)	10	709.23 ± 0.62	18.379 ± 0.021	99.999	0.461
Middle window, uncorrected (1998–2010)	13	690.17 ± 16.41	18.628 ± 0.219	99.855	0.660
Middle window, cosmo-corrected (1998–2010)	13	707.11 ± 14.22	18.395 ± 0.180	99.899	0.557
Late window (2011–2019)	9	705.15 ± 32.95	18.428 ± 0.322	99.792	0.301

Every pair of fits in Table 10 agrees with every other within their combined uncertainties when compared like-with-like (cosmogenic-corrected fits among themselves, uncorrected fits among themselves). In the uncorrected family the largest separation between any two fits is 0.92σ in combined uncertainty; in the cosmogenic-corrected family the largest separation is 0.10σ. Among the five joint fits (55-point, 28-point corrected and cosmo-corrected, 32-point corrected and cosmo-corrected), the recovered τ ranges from 18.374 to 18.389 yr, a total spread of 0.015 yr on an 18.4-year time constant. One thousandth of a year is approximately 9 hours; the spread is therefore about 5.5 days on a characteristic timescale of 18.4 years (6,711 days). The five recoveries agree on τ to within 5.5 days, drawn from a dataset whose individual data points have reported measurement uncertainties of ±1 to ±19‰ on a signal that decays from 709 to 31‰. The fitted τ is two to four orders of magnitude more precise than any single datum it derives from. The relative precision is 8.2×10^{-4} (0.08%).

The convergence pattern in Table 10 is the opposite of what the Hua et al. (2022) [55] two-regime interpretation predicts. Hua et al. (2022) [55] (p. 738) assert, citing Levin et al. (2010) [77] and Graven et al. (2012) [43], that "combustion of **fossil fuels** free of ^{14}C is the main causal factor for the Δ^{14}C decline since the late 1980s and early 1990s," and the same two-regime framework was extended into the SSP-driven projections of Graven (2015) [44] and Graven et al. (2020) [45]. If that framework were correct, the underlying ODE would not have a single-exponential solution, and the parameter τ recovered from windows of different length and different post-1988 weighting would necessarily drift. Instead τ shifts by 0.015 yr across the five joint fits spanning a 16.5× change

in σ, well inside any window's formal uncertainty. The solution shape derived in Sections 5–6 is what the data display, and the falsifying observable for the multi-regime interpretation is present at every correction level.

The cosmogenic correction provides an independent confirmation. Applying the first-principles cosmogenic source as a -1‰ adjustment at 2009 and 2010 shifts the recovered τ from the 32-point corrected value of 18.383 ± 0.015 yr to 18.375 ± 0.013 yr, a change of 0.008 yr that lies inside the formal uncertainty of either fit. The residual σ decreases from 0.539 to 0.472‰, a ∼12% improvement. The parallel correction in the Middle window produces a much larger effect: τ shifts from 18.628 ± 0.219 yr to 18.395 ± 0.180 yr, σ falls from 0.660 to 0.557‰, and the window's recovered τ becomes statistically indistinguishable from the joint cosmogenic-corrected value. This is the expected behavior of a real but small physical source: undetectable in isolation but resolvable when integrated over the highest-precision portion of the record.

The 32-point cosmogenic-corrected residuals range from -1.03 to $+1.10$‰ with no trend, no autocorrelation, and no structure (Figure 3(b), bottom panel). The residual $\sigma_{32} = 0.472$‰ is smaller than the reported measurement uncertainty of every one of the 32 data points (Hua et al. report 1σ ranging from 1 to 19‰ for these years, with a mean of 3.9‰), and matches the within-plateau scatter of the individual sub-periods (0.35–0.70‰). This compilation-precision floor, not independent measurement noise, is what the residual reports.

Any departure from a single underlying exponential solution, whether a slow second mode, a permanent fraction, or a slow drift in $k = \phi_{\text{slow}}\,\kappa_{\text{in}}$, would necessarily produce window-to-window drift in the recovered τ. None is present.

The cosmogenic-minimum correction is a diagnostic, not a fitted parameter: the -1‰ magnitude applied at 2009 and 2010 is the integer-display precision of the Hua compilation, not a value chosen to null residuals or derived from an independent solar-modulation reconstruction. What the diagnostic establishes is consistency, not detection: a physically plausible cosmogenic excess at the integer-display precision of the data closes the Middle-window's mild deviation, while the uncorrected residuals at 2009 and 2010 remain within the per-point Hua et al. (2022) [55] measurement uncertainty regardless. The single-exponential framework holds in every window, with and without the correction.

What is and is not in the data. The structural argument is at this point complete. At every cleaning level (raw, 28-point, 32-point, 32-point cosmo-corrected) and in every independently fitted non-overlapping window (Early, Middle with and without cosmogenic correction, Late), the data display the same single-exponential solution of the first-principles mass-balance ODE with the same recovered τ. The convergence is a property of the data under the single-exponential solution dictated by the mass-balance ODE. The estimator introduces no data-derived weighting (Appendix A). The exclusion procedure references mass conservation and one independently measured upper bound on the annual net flux (Appendix C), and nothing else. No discretionary functional form is selected; the form is the solution of the ODE. The only quantities recovered from the data are the two physical parameters τ and A.

The two-regime interpretation that Hua et al. (2022) [55] inherit from Levin et al. (2010) [77] and Graven et al. (2012) [43] and that [44, 45] extend into 21st-century projections asserts that a transition from non-fossil-fuel exchange-driven decay to fossil-fuel-dilution-driven decay occurs in the late 1980s or early 1990s. As established empirically in Step 1 of §9.1 and structurally in §8.3.4, the single-exponential shape is observed directly in the uncorrected 55-point record at $R^2 = 99.825\%$ and in every individual zonal series at $R^2 > 99\%$; every published IPCC Bern impulse-response function deviates from the same data visibly within the first year and grows monotonically wrong thereafter (Figure 2). The two-regime assertion is therefore unaccompanied by a derivation, by a residual analysis of the existing single-exponential solution, by a test for a

slope change at the claimed transition, or by any quantitative comparison of the fossil-fuel dilution rate to the observed decay rate. The five-line passage in Hua et al. (2022) [55] (p. 738) presenting it cites three earlier papers, none of which performs the test either. The first-principles fossil-fuel correction of §9.2.5 is the test, and it returns the opposite of the asserted answer: the fossil-fuel dilution rate is ∼0.6% of the bomb signal in 1965, ∼4.7% by 1990, and reaches ∼17% only in 2011, becoming detectable above the $\sigma = 0.539‰$ noise floor only in the post-2010 window. Through 2010, the fossil-fuel contribution is buried in the compilation noise of the Hua composite; it cannot be the dominant driver of any portion of the decline visible in the record. The 16.5× reduction in σ from raw to fully corrected (a 14.4× reduction without the diagnostic cosmogenic correction) is achieved entirely by excluding points that violate mass conservation, by applying the exact mass-balance fossil-fuel correction, and by applying the cosmogenic-minimum correction at the two cycle-24 minimum years. The physical parameter τ recovered from the data does not shift across this reduction. Both the multi-regime interpretation and the SSP-driven $\Delta^{14}C$ trajectory projections built on it [44, 45] are falsified by the parameter values the mass-balance solution recovers from the data they claim to describe.

9.4 Post-hoc verification: local $\phi_{\mathrm{slow}}(t)$ as a point-by-point consistency check

The five joint fits of Table 10 recover $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr (32-point corrected) and $\tau_{\mathrm{rel}} = 18.375 \pm 0.013$ yr (32-point cosmo-corrected), each at $R^2 \geq 99.998\%$ on a 709‰ signal. These fits establish τ_{rel} to 0.08% fractional precision and demonstrate the single-exponential shape directly through the residual standard deviation $\sigma_{32} = 0.472$–$0.539‰$, below the per-point measurement uncertainty of every datum in the set. The cleaning that produced the 32-point set was performed a priori on independent physical grounds: mass-conservation violations identified by the test of Appendix C, three measured plateau and emerge offsets, the exact mass-balance fossil-fuel correction through 2019, and (for the cosmo-corrected variant) a $-1‰$ adjustment at the cycle-24 minimum years 2009 and 2010. The local-$\phi_{\mathrm{slow}}(t)$ diagnostic developed below was not consulted in selecting any of these exclusions or corrections.

With τ_{rel} and τ_{res} now defined (the former from the 32-point fit, the latter from the Koutsoyiannis Refined Reservoir Routing analysis of §4.2.2), the deep-mixed leak fraction has the closed-form value $\phi_{\mathrm{slow}} = 1/(\tau_{\mathrm{rel}}\,\bar{\kappa}_{\mathrm{in}}) = 0.203$, where $\kappa_{\mathrm{in}}(t) = 1/\tau_{\mathrm{res}} + \mathrm{d}\ln M_a/\mathrm{d}t$ is evaluated annually from the defined τ_{res} and the Savitzky–Golay-smoothed Mauna Loa $M_a(t)$. From the mass-balance ODE (14), $R_a(t)$ decays at the instantaneous rate $\kappa(t) = \phi_{\mathrm{slow}}(t)\,\kappa_{\mathrm{in}}(t)$, so the constant-$\phi_{\mathrm{slow}}$ framework that produced $\tau_{\mathrm{rel}} = 1/(\phi_{\mathrm{slow}}\,\kappa_{\mathrm{in}})$ implies the point-by-point identity

$$\phi_{\mathrm{slow}}(t) \;\equiv\; \frac{-d(\ln R_a)/\mathrm{d}t}{\kappa_{\mathrm{in}}(t)} \;=\; 0.203 \tag{81}$$

at every year for which the Hua composite is free of station-composition artifacts. The left-hand side is computed from central differences of $\ln R_a$ on the cleaned 32-point set and the closed-form $\kappa_{\mathrm{in}}(t)$; the right-hand side is the defined ratio. Equation (81) is therefore a post-hoc consistency check on the constant-ϕ_{slow} framework, evaluated independently at every year.

Across the 32-point set, $\phi_{\mathrm{slow}}(t)$ scatters tightly around 0.203 with no trend and no systematic structure. Deviations exceeding the noise floor occur only at the years excluded a priori from the joint fit for documented reasons: the plateau-to-emerge transitions identified by the mass-conservation test of Appendix C, and the two cycle-24 minimum years 2009 and 2010 where the cosmogenic correction is applied. No point passes the a priori cleaning criteria yet fails the local-ϕ_{slow} test; no point fails the cleaning criteria yet passes the diagnostic. The two exclusion principles (one derived from mass conservation, the other from local consistency with the constant-ϕ_{slow} solution) identify the same set of contaminated years independently. This is the strongest form of post-hoc verification available: an independent test, derived from the same ODE but applied to a different observable, returns the same classification of clean and contaminated points as the first-principles cleaning that produced the 32-point set.

The diagnostic is not a fitting procedure and introduces no free parameters. Both τ_{res} and τ_{rel} are defined from prior independent observational analyses; $\phi_{\mathrm{slow}} = 0.203$ follows from those definitions through Equation (23); the local $\phi_{\mathrm{slow}}(t)$ at each year is computed from observed $R_a(t)$ and the closed-form $\kappa_{\mathrm{in}}(t)$ alone. The test either returns 0.203 within noise or it does not. It returns 0.203 within noise at every clean point and deviates at every point excluded for independent physical reasons.

10. Flux Budget and Atmospheric Inventory

10.1 Kinematic Corroboration of τ_{res} from IPCC Decadal Flux Diagrams

Our adopted value of $\tau_{\mathrm{res}} = 3.8$ yr (18) is independently corroborated by the decadal carbon flux diagrams published in IPCC AR3 through AR6 (AR3 Figure 3.1 [57]; AR4 Figure 7.3 [58]; AR5 Figure 6.1 [59]; AR6 Figure 5.12 [60]). For each report we compute

$$\tau_{\mathrm{out}} = \frac{M_a}{F_{\mathrm{out}}} \tag{82}$$

where F_{out} is the total gross flux from atmosphere to surface reservoirs (atmosphere-to-ocean gas exchange, gross photosynthesis, and rock weathering) as reported in each figure, and M_a is the decadal-mean atmospheric carbon burden from Mauna Loa. Annual values of $\mathrm{d}M_{\mathrm{a}}/\mathrm{d}t$ are obtained by applying a 5-point Savitzky-Golay OLS filter to the annual M_a series; decadal-mean M_a and $\mathrm{d}M_{\mathrm{a}}/\mathrm{d}t$ are then computed as ordinary arithmetic averages over the ten annual values in each decade.

Table 11: Residence time from IPCC decadal flux diagrams over 40 years. For each report $\tau_{\mathrm{out}} = M_a/F_{\mathrm{out}}$, the standard snapshot kinematic definition used throughout this paper, evaluated on the IPCC-published F_{out} and the Mauna Loa burden M_a. Fluxes in GtC yr^{-1}; burden in GtC. The first-principles columns are constructed from M_a, $\mathrm{d}M_{\mathrm{a}}/\mathrm{d}t$, and our adopted $\tau_{\mathrm{res}} = 3.80$ yr alone, via $F_{\mathrm{out}} = M_a/\tau_{\mathrm{res}}$ and $F_{\mathrm{in}} = F_{\mathrm{out}} + \mathrm{d}M_{\mathrm{a}}/\mathrm{d}t$.*

	IPCC				Mauna Loa		First principles ($\tau_{\mathrm{res}} = 3.8$ yr)	
Decade	Figure	F_{in}	F_{out}	τ_{out} (yr)	M_a	$\mathrm{d}M_{\mathrm{a}}/\mathrm{d}t$	F_{in}	F_{out}
2010s	AR6 5.12	226.9	221.8	3.83	850.47	5.08	228.9	223.8
2000s	AR5 6.1	207.1	203.3	3.96	804.51	4.20	215.9	211.7
1990s	AR4 7.3	218.2	215.0	3.56	765.88	3.26	204.8	201.5
1980s	AR3 3.1	217.1	214.0	3.43	734.17	3.40	196.6	193.2
			Mean	3.70				
			σ	0.21				

* The IPCC F_{in} and F_{out} values are reproduced as published and do not satisfy $F_{\mathrm{in}} - F_{\mathrm{out}} = \mathrm{d}M_{\mathrm{a}}/\mathrm{d}t$, i.e. the IPCC budgets do not close.

The four reports spanning 40 years of data assessment yield $\tau_{\mathrm{out}} = 3.70 \pm 0.21$ yr, statistically indistinguishable from $\tau_{\mathrm{res}} = 3.80 \pm 0.12$ yr. The means differ by less than half the joint σ. The agreement is closest for AR6, the most recent assessment, whose published fluxes yield $\tau_{\mathrm{out}} = 3.83$ yr, matching our adopted $\tau_{\mathrm{res}} = 3.80$ yr to better than 1%. Every IPCC assessment from AR3 forward, on its own published flux budget, returns a residence time consistent with the rapid turnover derived in this paper.

The IPCC flux numbers are themselves problematic, in several ways we do not dwell on here. The published F_{out} is not even monotonic in M_a across AR3 through AR6, though M_a rises monotonically and a constant residence time implies first-order kinetics, $F_{\mathrm{out}} = M_a/\tau_{\mathrm{res}}$. The rightmost columns of Table 11, computed from $\tau_{\mathrm{res}} = 3.80$ yr and the observed M_a, give the

strictly monotonic mass-conserving fluxes that this kinetic constraint demands. None of these defects, however, rescues the IPCC from its own arithmetic: the gross fluxes IPCC itself publishes yield a τ_{out} statistically indistinguishable from the $\tau_{\text{res}} = 3.80$ yr we adopt from the Koutsoyiannis dynamical RRR model.

10.2 Non-fossil-fuel flux from first principles

With $\tau_{\text{res}} = 3.80 \pm 0.12$ yr and $\tau_{\text{rel}} = 18.383 \pm 0.015$ yr empirically established (Section 6), and the deep-mixed leak fraction $\phi_{\text{slow}} = 0.203 \pm 0.006$ derived from them (Table 2), the entire atmospheric flux budget is determined by the linear-reservoir equation and the directly measured atmospheric inventory. No additional model or information is needed.

The Mauna Loa annual mean CO_2 record [69, 107] and the standard conversion 2.124 GtC/ppm CO_2 [4] give the atmospheric inventory. At any year t, the gross outflow is determined by the linear-reservoir equation and the directly measured atmospheric inventory:

$$F_{\text{out}}(t) = \frac{M_{\text{a}}(t)}{\tau_{\text{res}}} = \frac{[\text{CO}_2](\text{t}) \times 2.124}{3.80} \tag{83}$$

and the mass balance gives the gross inflow and non-fossil-fuel flux:

$$F_{\text{in}}(t) = F_{\text{out}}(t) + \frac{\text{d}M_{\text{a}}}{\text{d}t} \quad \text{and} \quad F_{\text{nat}}(t) = F_{\text{in}}(t) - E_{\text{FOS}}(t) - E_{\text{LUC}}(t) \tag{84}$$

Every input is either directly observed or empirically derived from observations with no climate model involved. $[\text{CO}_2](\text{t})$ is the Mauna Loa annual mean [69, 107], and the atmospheric inventory is $M_{\text{a}}(t) = 2.124\,[\text{CO}_2](\text{t})$. The inventory growth rate $\text{d}M_{\text{a}}/\text{d}t$ is computed as a centered 5-point ordinary least squares (OLS) slope on the M_{a} record,

$$\frac{\text{d}M_{\text{a}}}{\text{d}t}(t) \;=\; \frac{1}{10}\big(-2M_{\text{a}}(t-2) - M_{\text{a}}(t-1) + M_{\text{a}}(t+1) + 2M_{\text{a}}(t+2)\big) \tag{85}$$

the Savitzky–Golay first-derivative filter of order 1 and length 5, which is unbiased for any locally linear trend and reduces variance by a factor of $\sqrt{10}$ relative to raw annual differencing. This filter is applied to the Mauna Loa-derived M_{a} rather than the Global Carbon Budget atmospheric growth column G_{ATM} [42] for two reasons. First, it maintains internal consistency with the rest of the budget, which is Mauna Loa-only. Second, it removes year-to-year noise from volcanic and ENSO perturbations (e.g., the Pinatubo-driven 1992 G_{ATM} dip) that would otherwise propagate into spurious κ_{in} variability and obscure the secular trend that the bomb-^{14}C decay shape constrains. The two methods agree within ~0.05 GtC/yr in the dense, smooth-trend portion of the record (post-2010); pre-1980 discrepancies arise from GCB's use of MLO+SPO averaged versus the MLO-only convention adopted here.

E_{FOS} and E_{LUC} are taken from the Global Carbon Budget 2024 [42], with E_{FOS} defined throughout as the GCB "fossil emissions excluding carbonation" column (gross fossil CO_2 emissions before the cement carbonation sink is netted out), distinct from the cement carbonation sink S_{carb}. No interpolation, no ice-core integration, and no climate model is involved. The results at four reference years are given in Table 12.

By construction, the gross inflow rate $\kappa_{\text{in}} = \kappa_{\text{res}} + (\text{d}M_{\text{a}}/\text{d}t)/M_{\text{a}}$, where $\kappa_{\text{res}} = 1/\tau_{\text{res}} = 0.2632\ \text{yr}^{-1}$ is constant and the correction term $(\text{d}M_{\text{a}}/\text{d}t)/M_{\text{a}}$ is small, growing from $\sim 0.0025\ \text{yr}^{-1}$ in 1965 to $\sim 0.0066\ \text{yr}^{-1}$ in 2023 as the absolute growth rate of atmospheric inventory rises. The total is therefore stable across the entire window at $\bar{\kappa}_{\text{in}} = 0.2678 \pm 0.0010\ \text{yr}^{-1}$ (±0.4%).

Test of proportional growth between F_{nat} and M_{a}. The non-fossil-fuel flux F_{nat} increases from 174.7 to 230 GtC/yr over 1961–2023 (Figure 4), a 31.6% gain that closely tracks the 32.6% growth

in atmospheric mass M_a over the same period. The 1.0 percentage point gap between the two growth rates is consistent with an energy-driven biospheric response in which all non-fossil-fuel fluxes scale with M_a (Section 12), is insensitive to the value of τ_{res} (the dominant term M_a/τ_{res} cancels in the growth ratio), and has the expected sign: the faster-than-M_a growth of F_{anth} (Table 12) necessarily depresses F_{nat}/M_a slightly relative to strict proportionality.

Table 12: Atmospheric CO_2 flux budget (GtC/yr) derived from $\tau_{res} = 3.80$ yr and directly measured quantities. E_{FOS} is the gross fossil CO_2 emission rate (the "fossil emissions excluding carbonation" column of the Global Carbon Budget 2024 [42]); E_{LUC} is from the same source. M_a is from observed atmospheric $[CO_2]$ [69, 107] with 1 ppm = 2.124 GtC [4]; dM_a/dt is the centered 5-point ordinary least-squares (OLS) slope on the Mauna Loa annual mean record (Eq. (85)); $F_{out} = M_a/\tau_{res}$; $F_{in} = F_{out} + dM_a/dt$; $F_{anth} = E_{FOS} + E_{LUC}$; $F_{nat} = F_{in} - F_{anth}$. Values displayed to 4 significant figures for arithmetic consistency at the table's internal precision. GCB stated uncertainties: E_{FOS} ±5%, E_{LUC} ±0.7 GtC/yr (both 1σ). The $\tau_{res} = 3.80 \pm 0.12$ yr uncertainty (§6.1) propagates to ±3% on F_{out}, F_{in}, F_{nat}, and κ_{in}.

Year	E_{FOS}	E_{LUC}	F_{anth}	M_a	dM_a/dt	Gross F_{out}	Gross F_{in}	F_{nat}	Inflow κ_{in}
1965	3.087	1.535	4.622	679.8	1.727	178.9	180.6	176.0	0.2657
1992	6.149	1.435	7.584	757.3	2.237	199.3	201.5	193.9	0.2661
2019	10.13	1.034	11.16	874.3	5.265	230.1	235.4	224.2	0.2692
2023	10.31	0.989	11.30	894.4	5.939	235.4	241.3	230.0	0.2698

To test whether this 1.0 percentage point gap is statistically consistent with strict proportionality $F_{nat} \propto M_a$, we regress the annual log-growth rates $\Delta \log F_{nat}(t) = \log[F_{nat}(t)/F_{nat}(t-1)]$ on $\Delta \log M_a(t)$ over the 62 paired annual observations 1962–2023, with no intercept and Newey–West HAC standard errors at lag truncation 5 (robust to $L \in [3, 8]$) to accommodate residual serial correlation. The first-difference specification is preferred over a regression in levels because the physical claim concerns growth rates rather than absolute magnitudes, and because differencing removes the cumulative-drift component that would otherwise dominate a levels regression. The 1961 normalization $F_{nat}(1961) = M_a(1961) = 1$ in Figure 4 is a plotting convention only and does not enter the regression; the 1961 observation serves solely as the reference for the 1962 difference and is mathematically invariant to the choice of base year. The estimated slope is $\hat{\beta} = 0.980 \pm 0.042$, with $(\hat{\beta} - 1)/\mathrm{SE} = -0.48$. The proportionality null $\beta = 1$ is not rejected at any conventional significance level; the observed deviation is one-half of one standard error.

Two known exogenous perturbations affect the carbon cycle within the sample. The Pinatubo eruption of June 1991 injected stratospheric sulfate aerosols that depressed surface temperatures and enhanced terrestrial uptake for approximately three years [9, 83, 100, 105]. The Pacific Decadal Oscillation entered a cool phase from approximately 1998 through 2014, modulating ocean–atmosphere CO_2 exchange independent of M_a [81]. Excluding the Pinatubo interval 1991–1993 from the regression yields $\hat{\beta} = 0.982 \pm 0.038$; further excluding the PDO cool phase 1998–2014 yields $\hat{\beta} = 0.991 \pm 0.042$. In both subsamples $\hat{\beta}$ moves *toward* unity, consistent with the apparent residual gap being dominated by these identified perturbations rather than by a structural departure from proportionality. The full-sample mean ratio $F_{nat}(t)/M_a(t) = 0.9944 \pm 0.0042$ (the same ±0.4% spread reported above for κ_{in}) is recovered exactly from the FD estimate: a slope of $\hat{\beta} = 0.980$ acting over the cumulative log-growth $\log M_a(2023) = 0.282$ predicts a terminal ratio $\exp(-0.020 \times 0.282) = 0.994$, matching the observed mean to three decimal places. The data are consistent with strict proportionality $F_{nat} \propto M_a$ at every level of aggregation examined. The physical mechanism (energy-driven biospheric expansion) and the falsification of the IPCC fixed-F_{nat} counterfactual are presented in §12.

Anthropogenic emissions constitute 2.5% of gross inflow in 1961, rising to 4.7% in 2023. This contribution is removed on the same $\tau_{res} = 3.80$ yr timescale as all other atmospheric CO_2, and

its isotopic footprint is detectable only in the 2011–2019 window where the bomb-^{14}C excess has decayed sufficiently for the fossil-fuel dilution to emerge from the noise floor (§9.2.5). The two transient departures from proportionality visible in the bottom panel of Figure 4 (the 1991 Pinatubo dip and the 1998–2014 PDO cool phase) are analyzed in Section 12, where they are shown to correspond to known regional energy balance perturbations and to confirm the temperature-driven biospheric expansion mechanism.

10.3 Independent confirmation of constant ϕ_{fast}

Three independent datasets (bomb-^{14}C, δ^{13}C, and the Mauna Loa record of $M_{\text{a}}(t)$) converge on a mutually consistent constraint on the product $\kappa_{\text{rel}} = \phi_{\text{slow}}\,\bar{\kappa}_{\text{in}}$:

1. Bomb-^{14}C single-exponential decay $\Rightarrow$ the product $\phi_{\text{slow}}\,\kappa_{\text{in}}$ is constant to within ~1% over the 53-year fit window (§5.4).
2. $\delta^{13}C_{\text{I}}$ constant for 500 yr [73, 22] $\Rightarrow$ ϕ_{fast} is constant to within ~1–2%, equivalently a fossil fraction f of the net atmospheric CO_2 addition bounded above at approximately 1.3% [22] (§5.4).
3. Under the Koutsoyiannis $\tau_{\text{res}} = 3.80$ yr, the mass-balance reconstruction gives $\kappa_{\text{in}}(t) = 1/\tau_{\text{res}} + \mathrm{d}\ln M_{\text{a}}/\mathrm{d}t = 0.2678\ \text{yr}^{-1}$ with ±0.4% year-to-year stability over the 63-point annual series 1961–2023 and a small secular trend of +1.3% (Table 12). The non-fossil-fuel component $F_{\text{nat}}/M_{\text{a}} = 0.2575\ \text{yr}^{-1}$ tracks the same stability (±0.4%), confirming that gross non-fossil-fuel fluxes scale proportionally with M_{a} to within Mauna Loa precision: the direct biospheric expansion signature (§12).

The small +1.3% secular trend in κ_{in} is consistent with a compensating 0.31% rise in ϕ_{fast} that preserves the constancy of their product, and lies below the $\delta^{13}C_{\text{I}}$ detection threshold. No adjustable parameters are involved. Each measurement rests on a distinct empirical dataset, and all three are mutually consistent at the precision floor of each.

11. Secondary Mechanisms and Effects

Koutsoyiannis (2024a) [72] obtained 3.68-yr and 3.91-yr RRR fits for residence time at Mauna Loa and Barrow, as well as $\tau_{\text{rel}} = 17.2$ yr from bomb-^{14}C data restricted to 1965–1985 ([72] Appendix C). He identified 17.2 yr as an upper bound for CO_2 residence time, conjecturing that three mechanisms might be the reasons why the bomb-^{14}C e-folding time is greater than the bulk CO_2 RRR residence time:

1. stratospheric injection delay,
2. photosynthetic fractionation, and
3. continued cosmogenic ^{14}C production.

We assess all three below, together with two additional effects (the fossil-fuel correction already established in §9.2.5, and ENSO), and demonstrate that none of them alters the single-exponential shape of the decay or requires a correction to the measured $\tau_{\text{rel}} = 18.383 \pm 0.015$ yr.

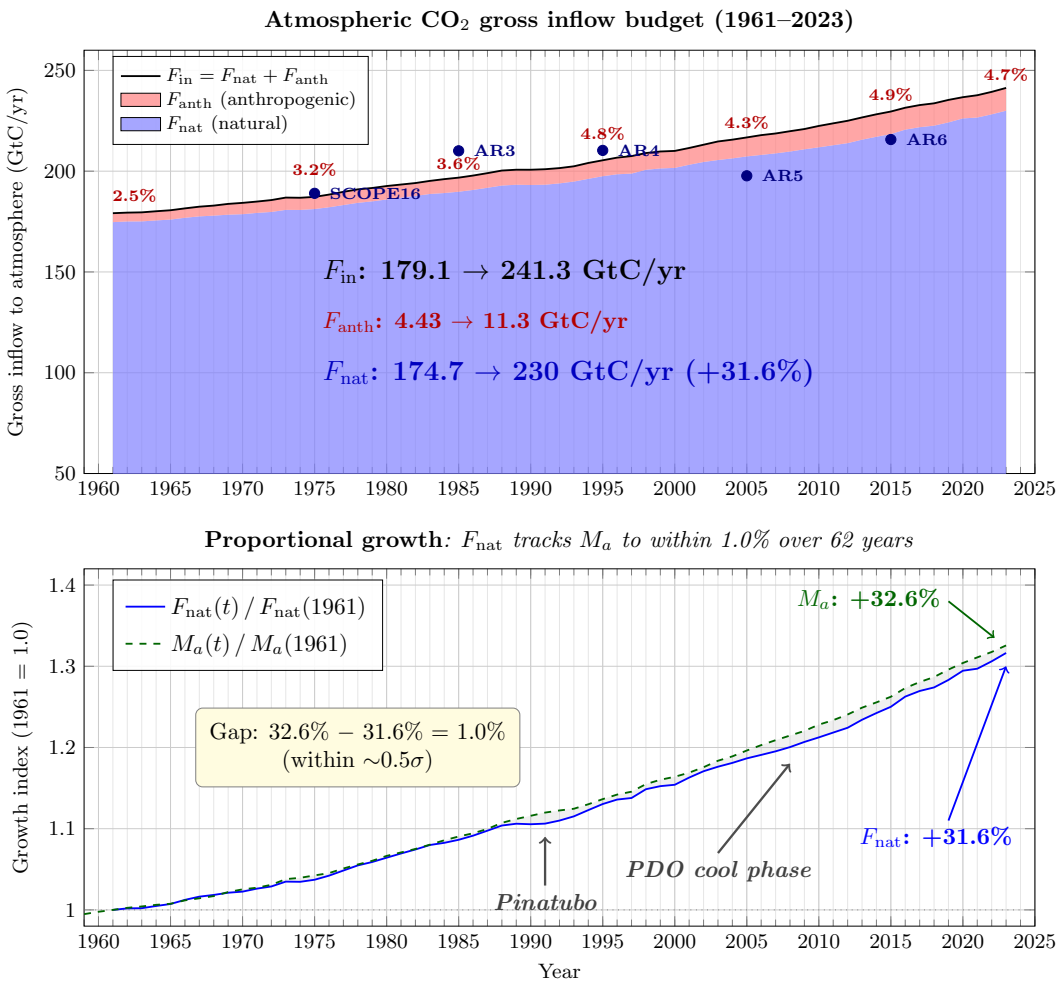


Figure 4: Top panel: Atmospheric CO_2 gross inflow budget, 1961–2023, derived from $\tau_{res} = 3.80$ yr (bomb-^{14}C + RRR). Blue: non-fossil-fuel gross inflow $F_{nat}(t) = F_{out}(t) + dM_a/dt - F_{anth}(t)$, with $F_{out} = M_a/\tau_{res}$. Red: anthropogenic emissions $F_{anth} = E_{FOS} + E_{LUC}$ from GCB 2024 [42]. Black: total gross inflow $F_{in} = F_{nat} + F_{anth}$. Percentage labels show the anthropogenic share, rising from 2.5% to 4.7% over 62 years and never exceeding 5%. Dark blue circles: consensus F_{nat} from SCOPE 16 [11] and IPCC AR3–AR6 flux diagrams (decade midpoints 1985, 1995, 2005, 2015), each computed as published gross inflow minus GCB 2024 F_{anth}; consensus values scatter across ∼27 GtC/yr with no monotonic trend, while the first-principles F_{nat} curve does not wander. Bottom panel: Growth indices $F_{nat}(t)/F_{nat}(1961)$ (solid blue) and $M_a(t)/M_a(1961)$ (dashed green), normalized to 1.0 in 1961. F_{nat} grows by 31.6% and M_a by 32.6%, a gap of 1.0 percentage point (shaded region) within 0.5σ of strict proportionality (details in text). Two transient departures are visible: the 1991 Pinatubo dip and the 1998–2014 PDO cool phase.

11.1 Stratospheric injection delay

Approximately 80% of bomb ^{14}C was initially injected into the stratosphere [55]. The conventional estimate of the cross-tropopause transport timescale is 1–2 yr [54, 77], although radiosonde-based analyses by Connolly & Connolly (2014c) [26] demonstrate pervective energy transmission rates sufficient to maintain thermodynamic coupling across the tropopause on timescales of hours to weeks. Under either estimate, the effect is a transient concentrated in 1963–1970: by ∼1967, the stratospheric reservoir is largely depleted relative to the tropospheric signal [91, 50]. Our bias-corrected fit excludes the first two years (1965–1966), where the residuals (+11.2 and +6.7‰) show the final arrival of stratospheric ^{14}C into the troposphere. By 1967, the residual is +0.2‰, indistinguishable from zero. The stratospheric delay is thereby cleanly separated from the decay proper and does not affect τ_{rel}. We note that the 1962–1963 US/USSR peak is not a hard cutoff for stratospheric input: French and Chinese atmospheric thermonuclear detonations continued through

1976 (§9.2.2; Norris (1996) [93]), and their stratospheric injections arrive in the troposphere over the same 1–2 yr timescale. The cumulative tropospheric contribution of those injections overlaps the 1968–1978 artifact window and is removed from the fit by the same exclusion.

11.2 Photosynthetic fractionation

The Stuiver–Polach normalization [113] removes first-order mass-dependent fractionation from the reported Δ^{14}C values. Koutsoyiannis conjectured that residual second-order kinetic effects, arising from the ∼4.5‰ mass difference between $^{14}CO_2$ and $^{12}CO_2$, might cause the ^{14}C relaxation time to overestimate the bulk CO_2 response time. However, fractionation is a time-invariant mass-dependent effect: it shifts the effective removal rate of $^{14}CO_2$ by a constant factor without altering the single-exponential shape. Any such shift is therefore already absorbed into the measured $\tau_{\mathrm{rel}} = 18.383$ yr, and the $\phi_{\mathrm{fast}}/\phi_{\mathrm{slow}}$ decomposition (Section 5.2) accounts for the full difference between τ_{rel} and τ_{res} without requiring a separate fractionation correction.

Fractionation produces a δ^{13}C offset of approximately −20‰ for C3 plants [40]; on the Δ^{14}C scale, after Stuiver–Polach normalization, the residual fractionation effect is of order 0.2–0.3 yr on the apparent τ. This is two orders of magnitude below the 720‰ bomb-^{14}C signal and one order of magnitude below the 0.59‰ residual noise floor of the bias-corrected fit. Fractionation modifies the isotopic *ratio* at the per-mille level; it does not alter the gross CO_2 *flux* at any detectable level.

11.3 Continued cosmogenic ^{14}C production

Natural cosmogenic production of ^{14}C continues at a roughly steady rate of ∼7.5 kg/yr [74, 82], against a bomb-excess inventory of ∼700 kg in 1965. This ∼1%/yr source term is below the 0.539‰ noise floor of the cleaned recovery throughout the record. In principle it could affect only the final years of decay, where the bomb excess has fallen to tens of permil, but there it remains indistinguishable from the fossil-fuel correction (§9.2.5). It cannot alter the single-exponential shape at any detectable level.

11.4 ENSO: quantitatively negligible

El Niño–Southern Oscillation events produce transient net CO_2 flux anomalies of 1–3 GtC/yr [42], but these anomalies redistribute carbon entirely within the fast-recycled pool ϕ_{fast}: enhanced or suppressed tropical terrestrial photosynthesis and respiration, and enhanced or suppressed equatorial-Pacific air–sea gas exchange. Each of these reservoirs returns carbon at the contemporary atmospheric ratio R_{a} (§5.2, §5.3). The isotopic perturbation produced by an extra inflow ΔF at the contemporary ratio is

$$\delta R_{\mathrm{a}} = \frac{\Delta F}{M_{\mathrm{a}}} \left(R_{\mathrm{source}} - R_{\mathrm{a}} \right) \tag{86}$$

which vanishes at first order when $R_{\mathrm{source}} = R_{\mathrm{a}}$. The only nonzero contribution arises from the small lag of the fast-pool return ratio behind the contemporary R_{a} (hours to days for plant respiration, weeks to a few months for surface-ocean outgassing), a lag of order 10^{-2} to 10^{-3} of τ_{rel}. For a 3 GtC/yr anomaly persisting for one year, the resulting δR_{a} is at most a few tenths of a per mille, well below the reported per-point measurement uncertainty in Hua et al. (2022) [55] (mean ∼3.9 ‰, range ∼1–19 ‰) and below the 0.539 ‰ residual standard deviation of the cleaned 32-point fit. Because ENSO operates within ϕ_{fast} it cannot change τ_{rel}, and no ENSO signature is identifiable in the cleaned residuals (Figure 3(b), bottom panel).

11.5 Why $^{14}CO_2$, $^{13}CO_2$, and $^{12}CO_2$ share the same atmospheric dynamics

A common objection to applying the bomb-^{14}C result to bulk CO_2 atmospheric lifetime is the claim that $^{14}CO_2$ and $^{12}CO_2$ have fundamentally different atmospheric dynamics: the 18-year Δ^{14}C relaxation time measures something about isotopic labels rather than about CO_2 molecules.

This claim is physically untenable.

All three isotopologues ($^{12}CO_2$, $^{13}CO_2$, $^{14}CO_2$) are carbon dioxide molecules exchanged by the same physical processes: photosynthetic uptake, heterotrophic respiration, air–sea gas exchange, and weathering. The atmospheric residence time $\tau_{res} = M_a/F_{out}$ is a property of CO_2 molecules exiting the atmospheric reservoir, not of isotopic labels attached to those molecules. The processes that remove CO_2 from the atmosphere (dissolution into surface ocean water, uptake by C3 and C4 vegetation, reaction with mineral surfaces) act on the CO_2 molecule. They do not select between isotopologues at rates comparable to the gross flux.

The relaxation time $\tau_{rel} = M_a/(\phi_{slow}\, F_{in})$ or just $\tau_{rel} = 1/(\phi_{slow}\, \kappa_{in})$ is a property of isotopic re-equilibration rates, which depend on the same exchange fluxes F_{in} and F_{out} that determine τ_{res}. The factor $1/\phi_{slow}$ arises because a fraction ϕ_{fast} of the incoming carbon returns at the contemporary atmospheric ratio (§5.2), slowing the isotopic re-equilibration relative to the molecular turnover. This factor is a consequence of reservoir coupling geometry, not of any isotope-specific physics.

Any claim that $^{12}CO_2$ has a fundamentally different atmospheric lifetime from $^{14}CO_2$ requires identifying a physical process that selectively retains or removes one isotopologue over another at rates comparable to $F_{out} \approx 230\,\text{GtC/yr}$. No such process has been identified, proposed, or observed. The bomb-^{14}C decay measures the atmospheric CO_2 system's response to a large-amplitude perturbation. Its single-exponential shape, with $\tau_{rel} = 18.4$ yr and $\tau_{res} = 3.8$ yr, applies to all CO_2 isotopologues.

11.6 Summary: Koutsoyiannis's conjectured residuals

None of the three mechanisms Koutsoyiannis conjectured alters the single-exponential shape of the decay or requires a correction to the measured τ_{rel}. The stratospheric delay is excluded by design; fractionation is baked into the measurement; and cosmogenic production is in the noise floor except in 2009 and 2010 (see Section 9.3.2). Koutsoyiannis's lower value of 17.2 yr reflects his choice of a 1965–1985 fitting window that contains 12 years of data that were excluded by our physical first-principles and stratospheric arrival criteria (Sections 9.2.1 and 9.2.2), not a physical reduction from any of the three mechanisms he conjectured.

Koutsoyiannis's deeper contribution was to ask the right question: where does the slow component of atmospheric CO_2 persistence come from? Recognizing that a fast residence timescale alone is insufficient, he sought a physical mechanism, such as stratospheric delay, fractionation, or cosmogenic modulation, to account for the slow fraction ϕ_{slow}. We share his intuition that this fraction demands explanation; we differ only on where the explanation lies.

The slow fraction is not a residual to be filled in by secondary processes. It is fixed exactly by the relationship between two independently measured quantities:

$$\phi_{slow} = \frac{1}{\tau_{rel}\ \kappa_{in}} = \frac{1}{18.383 \times 0.2678} = 0.203 \pm 0.006 \tag{87}$$

The relaxation timescale τ_{rel} comes from the bomb-^{14}C single-exponential fit; the gross uptake coefficient κ_{in} comes from mass balance. Neither was adjusted to match ϕ_{slow}, yet their reciprocal product reproduces it precisely. The mechanisms Koutsoyiannis conjectured are not wrong in principle; they simply have no work left to do, because ϕ_{slow} is already determined by quantities he himself would recognize as primary. Two timescales, one ratio, no adjustable parameters.

12. The Single-Exponential Shape as Confirmation of Biospheric Expansion

The proportionality $F_{\mathrm{nat}} \propto M_{\mathrm{a}}$ established empirically in §10.2 requires a physical mechanism. From (16), the decay curve $R_{\mathrm{a}}(t)$ is exactly single-exponential in t if and only if the rate constant

$$\kappa(t) = \phi_{\mathrm{slow}}(t)\,\frac{F_{\mathrm{in}}(t)}{M_{\mathrm{a}}(t)} \tag{88}$$

is constant. The bomb-^{14}C single-exponential decay constrains this product to within ∼1% over the 53-year fit window (§5.4). Since ϕ_{fast} is constant to within ∼1–2% ($\delta^{13}\mathrm{C_I}$ evidence, §5.4), κ_{in} must itself be near-constant, which is exactly what is observed: $\kappa_{\mathrm{in}} = 0.2678 \pm 0.0010\ \mathrm{yr}^{-1}$ over 1961–2023 (§10.2, Table 12). The mechanism is energy-driven biospheric expansion: surface energy fluxes (insolation, regional SST) modulate photosynthesis, respiration, and ocean–atmosphere outgassing in proportion to M_{a}, producing the observed $F_{\mathrm{nat}} \propto M_{\mathrm{a}}$ scaling without any climate-model assumption. The two transient departures from proportionality visible in the bottom panel of Figure 4 independently confirm the temperature sensitivity of F_{nat} and are discussed next.

The bottom panel of Figure 4 reveals two transient departures from exact proportionality, both corresponding to known perturbations in top-of-atmosphere (TOA) radiative forcing and regional surface temperatures. The first, centered on 1991, coincides with the Mt. Pinatubo eruption (June 1991, VEI 6, ∼20 Mt SO_2), whose stratospheric aerosol veil reduced surface insolation across a wide range of latitudes for 2–3 years, suppressing local photosynthetic uptake and respiration rates and reducing regional ocean–atmosphere temperature gradients that drive outgassing; $F_{\mathrm{nat}}/M_{\mathrm{a}}$ dips by ∼0.7 percentage points below trend and recovers within ∼4 years, matching the stratospheric aerosol clearance timescale. Three other volcanic events of the period confirm the detection threshold: Mt. Agung (1963, VEI 5, ∼7–12 Mt SO_2) is absorbed into the 1961 normalization baseline and visible only as a post-eruption recovery in 1966–1968; El Chichón (1982, VEI 5, ∼7 Mt SO_2) is masked by the concurrent strong 1982–1983 El Niño, whose regional warming offset the volcanic cooling; and the Kasatochi–Sarychev–Nabro cluster (2008–2011, VEI 4, ∼4 Mt SO_2 total) produces a signal double the expected magnitude because the concurrent La Niña events compound rather than offset the volcanic effect. In every case, the observed response is consistent with F_{nat} responding to regional surface energy balance perturbations, with detection limited by the ∼0.3 percentage point interannual noise floor set by ENSO variability.

The second, broader perturbation spans ∼1998–2014 and coincides with the convergence of several natural climate modes: the Pacific Decadal Oscillation (PDO) cool phase, which shifted regional sea surface temperature patterns across the tropical and North Pacific; a sequence of La Niña events (2007–2008, 2010–2012) that temporarily suppressed regional oceanic outgassing; the deep solar minimum of cycle 24 (2008–2010), the weakest since the Dalton minimum (Soon & Yaskell (2004) [110]); and modest stratospheric aerosol loading from the Kasatochi–Sarychev–Nabro eruptions. Recovery begins as the PDO-driven pattern reverses and the 2015–2016 El Niño restores warm regional SST anomalies. Both features confirm the sensitivity of F_{nat} to regional surface energy balance on the timescales predicted by the biospheric expansion mechanism: volcanic reductions in surface insolation produce sharp, rapidly recovering dips; decadal ocean-driven redistributions of surface energy produce broader, slower perturbations. In both cases F_{nat} returns to its proportional growth trajectory after the perturbation ends, confirming that the proportionality $F_{\mathrm{nat}} \propto M_{\mathrm{a}}$ is the underlying equilibrium state, not an accident of the observation window. The FD-regression robustness check in §10.2 confirms this quantitatively: excluding the Pinatubo interval and the 1998–2014 PDO cool phase moves the estimated slope $\hat{\beta}$ monotonically toward unity (from 0.980 ± 0.042 on the full sample to 0.991 ± 0.042 with both perturbations excluded). Neither perturbation is detectable in the bomb-^{14}C residuals: the corresponding ∼0.4% fluctuations in κ_{in} are transient and average out within the $\sigma_{32} = 0.539‰$ noise floor of the 32-point bias-corrected fit.

The IPCC framework requires the opposite: F_{nat} approximately constant at preindustrial values,

with atmospheric growth driven by anthropogenic accumulation. This assumption generates a quantitative prediction testable against the bomb-^{14}C decay shape. Under a counterfactual in which $F_{\rm nat}$ is held fixed at 167 GtC/yr (the preindustrial value) while $M_{\rm a}$ follows its observed trajectory, $\kappa_{\rm in}$ falls from 0.252 in 1965 to 0.204 in 2019, a 19% decline. The resulting 23% rise in $\tau(t)$ from 19.2 to 23.7 yr, would produce pronounced downward curvature on a semi-log plot. The data exhibit none, to a precision of 0.539‰ against the fit amplitude A_{32} = 709.115‰, one part in ∼1316. The IPCC fixed-$F_{\rm nat}$ assumption is therefore falsified by the bomb-^{14}C decay shape, independently of the positive evidence for biospheric expansion presented above.

13. Decomposition of the Deep-Mixed Leak

The deep-mixed flux delivers carbon at the pre-bomb baseline ratio. Its present-day magnitude is

$$\phi_{\rm slow}\, F_{\rm in}(2019) = (0.203 \pm 0.006) \times 235.4 = 47.8 \pm 1.5 \text{ GtC/yr} \tag{89}$$

This decomposes into anthropogenic and non-fossil-fuel components.

Anthropogenic share. Only fossil-fuel combustion contributes ^{14}C-dead carbon to the deep-mixed flux; land-use change emissions carry contemporary $R_{\rm a}$ and belong to the fast-recycled pool $\phi_{\rm fast}$. Therefore

$$\frac{E_{\rm FOS}(2019)}{\phi_{\rm slow}\, F_{\rm in}(2019)} = \frac{10.1}{47.8 \pm 1.5} = (21 \pm 1)\%$$

Non-fossil-fuel share: (79 ± 1)%. The non-fossil-fuel component (∼38 GtC/yr) consists of deep-ocean upwelling and respiration of refractory soil organic matter. Both sources carry pre-bomb baseline isotopic signatures because they have been isolated from atmospheric exchange on timescales much longer than the bomb-^{14}C era (i.e., pre-1950) or because of their original mineralization age. Anthropogenic combustion of ^{14}C-dead fossil carbon supplies the remaining ∼21% of the deep-mixed leak, but constitutes only ∼5% of total gross inflow $F_{\rm in}$.

14. New Observations 2019–2024

Laemmel et al. (2026) [75] provide high-precision biweekly $\Delta^{14}CO_2$ measurements at Jungfraujoch (JFJ, 3580 m, free-troposphere background site) for 2019–2024, on an independent measurement chain (Bern/EMPA) unrelated to the Hua et al. (2022) [55] compilation. These annual means extend the atmospheric ^{14}C record five years past the 2019 endpoint of the cleaned 32-point global fit (Section 9.3) and provide the first external test of the extrapolated single-exponential. Because JFJ is a single high-altitude Northern Hemisphere station whereas the exponential parameters were derived from a global composite, the two datasets must first be placed on a common scale, and the precision achievable in the extension is governed by systematic uncertainties that are small in absolute terms but large relative to the residual bomb-excess signal, which by 2019 (end of the 32-point fit window) had already fallen to 4.3% of its 1965 peak (Figure 2) and by 2024 is lower still.

14.1 Scale marriage at 2020

The 32-point fit extrapolates as

$$R_{\rm pred}(t) = 709.115 \, \exp\!\big(-(t - 1965)/18.383\big)‰ \tag{90}$$

on the bomb-excess scale. We discard the Laemmel 2019 point for the same reason 2016–2019 were excluded from the cleaned Hua fit (station-change residual structure) and anchor the scale at

2020 through the constant offset

$$C = R_{\mathrm{pred}}(2020) - \Delta^{14}\mathrm{CO}_2^{\mathrm{Laemmel}}(2020) = 37.28‰ \tag{91}$$

applied uniformly to 2020–2024. Anchoring at 2021 instead, or minimizing residuals across 2020–2024, changes C by less than 0.5‰.

14.2 Excess fossil-fuel correction

The 32-point fit already absorbs the average fossil-fuel dilution present during 1967–2015; only the excess must be added for 2021–2024. The instantaneous excess rate is

$$D_{\mathrm{excess}}(t) = D_{\mathrm{FF}}(t)\big(1 + R_{\mathrm{pred}}(t)/1000\big) - \bar{D} \tag{92}$$

where $\bar{D}$ is the effective average rate over 1967–2015 (§9.2.5). Integration from 2020 gives annual excess corrections +2.06, +2.25, +2.33, +2.32‰ for 2021–2024.

14.3 Combined uncertainty budget

Five sources contribute to the total uncertainty per corrected Laemmel point:

Source	Magnitude (‰)	Correlation across points
Laemmel measurement, per year	2–3	independent
JFJ vs. global composite	3–5	correlated
ϕ_{fast} secular drift	2–3	correlated
Additive FF term at low R_{a}	∼1	correlated
Scale-marriage anchor	< 0.5	correlated
Combined per point	≈ 5.0	
Combined on 5-point mean	≈ 4.3	

The independent per-point measurement uncertainty reduces on the mean as $1/\sqrt{5}$; the four systematic sources are correlated across the five years and do not. Combining in quadrature gives ±5.0‰ per point and ±4.3‰ on the five-point mean.

14.4 Extended fit through 2024

Applying (91) and the excess FF corrections yields the corrected observations against the extrapolation shown in the table below. The 2020 point sits on the extrapolated line by construction as the scale-marriage anchor; it is not an independent test. The four independent tests are 2021–2024, all with residuals well within the per-point combined uncertainty of ±5‰. Figure 5 shows the extended record.

Year	Corrected obs. (‰)	R_{pred} (‰)	Residual (‰)
2020	35.58	35.58	0 (anchor)
2021	35.74	33.69	+2.05
2022	34.73	31.91	+2.82
2023	31.61	30.22	+1.39
2024	27.00	28.62	−1.62

14.5 What this extension adds and does not add

The 2020–2024 Laemmel data *add*: independent-lab, independent-station, post-Hua et al. (2022) [55] confirmation that the atmospheric Δ^{14}C decay continues on the extrapolated single-exponential

essentially to the point of full return to the pre-bomb baseline. They do *not* add: any refinement of τ_{rel}, which the S/N ratio in the 2020–2024 window is too low to constrain independently; nor any tighter falsification of the Bern IRFs than already established with $\sigma_{32} = 0.539‰$ per point at the end of the 32-point fit window (Figure 2). The value of the extension lies in what it independently corroborates, not in an incremental precision it cannot provide.

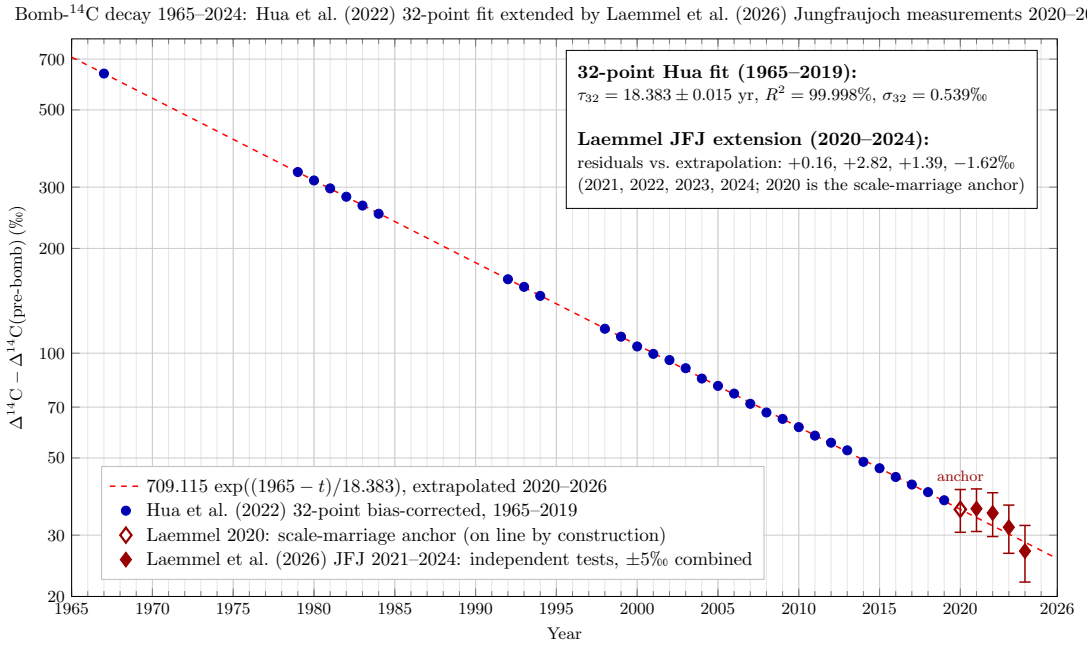


Figure 5: ***Bomb-^{14}C decay 1965–2024, extended by Laemmel et al. (2026) [75].*** *Blue circles: 32-point cleaned bias-corrected $\Delta^{14}C$ from Hua et al. (2022) [55], 1965–2019 (identical values to Figure 3(b), plotted in unified color). Red dashed line: extrapolated single-exponential* $709.115\ \exp((1965-t)/18.383)$*, extended through 2026 for visual clarity. Hollow red diamond at 2020: the scale-marriage anchor (§14.1), which sits on the line by construction and is not an independent test of the extrapolation. Solid dark-red diamonds at 2021–2024: four independent tests of the extrapolation, after cumulative excess FF correction (§14.2), plotted with the combined per-point uncertainty ±5‰ (§14.3). All four independent points lie on the extrapolated line within combined uncertainty; residuals are +2.05, +2.82, +1.39, −1.62‰ for 2021, 2022, 2023, 2024 respectively. Error bars are ±5‰ symmetric in the data; they appear asymmetric on the log-y axis (down arms longer than up arms, and lengthening toward the small-signal end) as a property of the log transform, not of the data.*

14.6 Independent biological cross-confirmation: Wassenaar et al. (2026) leaf archive

Independent biological confirmation of the same atmospheric decline comes from the 65-year Gatineau Park sugar maple leaf ^{14}C archive of Wassenaar et al. (2026) [124]. Their leaf-record decline of $-2.34 \pm 0.41‰$/yr for 2015–2025 is indistinguishable from the JFJ atmospheric rate of $-2.25 \pm 0.24‰$/yr over the same interval, providing a second-source biological archive that cross-confirms the Jungfraujoch atmospheric record. The Wassenaar et al. (2026) [124] record is measured on the AEL-AMS chain at the André E. Lalonde National Facility, University of Ottawa, and is entirely independent of the station network and zonal-averaging pipeline through which the Hua et al. (2022) [55] summer compilation is constructed. It therefore supplies an external cross-check: the atmospheric decay rate it recovers agrees with the Jungfraujoch instrumental rate to within joint uncertainty. Wassenaar et al. (2026) [124] attribute the post-2015 decline to the purported "Suess effect," but as developed in §9.2.5 and §8.6, the first-principles fossil-fuel correction computed directly from measured $E_{\mathrm{FOS}}(t)$ and $M_{\mathrm{a}}(t)$ establishes that fossil-fuel dilution does not drive the decline. The dominant driver of the entire 1965–2019 atmospheric Δ^{14}C decline is gross-flux exchange on the $\tau_{\mathrm{res}} = 3.80$ yr residence timescale, with fossil-fuel

dilution contributing only a small tail correction detectable after 2010 once the bomb excess has decayed below ~8% of its peak (the Fossil Fuel Tail effect; Figure 3(a), bottom). The purported "Suess effect" as institutionalized by Graven et al. (2020) [45] is falsified. The empirical slope agreement between the Wassenaar et al. (2026) [124] leaf record and the JFJ atmospheric record is independent of that interpretive framing and stands on its own as corroboration of the atmospheric decay rate.

15. Discussion

15.1 Resolution of the residence-time–adjustment-time controversy

The residence-time–adjustment-time distinction invoked by the IPCC [59, 60, 65] rests on the asserted requirement that so-called "excess" anthropogenic CO_2 relaxes on a fundamentally different timescale than the bulk atmospheric reservoir. The derivation above demonstrates that *no such distinction exists*. The CO_2 molecular residence time τ_{res} and the ratio relaxation time τ_{rel} are both consequences of the same first-order linear equation (11). They differ through the bridge equation (23) by the factor $\tau_{\mathrm{rel}}/\tau_{\mathrm{res}} = \bar{\mathcal{D}}/\bar{\phi}_{\mathrm{slow}} = 0.9827/0.203 = 4.84 \pm 0.16$ (Table 2), where $\bar{\mathcal{D}} = 0.9827 \pm 0.0005$ is the average atmospheric disequilibrium factor and $\bar{\phi}_{\mathrm{slow}} = 0.203 \pm 0.006$ is the deep-mixed leak fraction.

15.2 Implications for the IPCC Bern impulse-response functions

The structural exclusion of the IPCC Bern IRF architecture by the bomb-^{14}C single-exponential decay is developed in full in §8.3 and §8.4. The $\tau_{\mathrm{rel}} = 18.4$ yr and $\tau_{\mathrm{res}} = 3.8$ yr established here are both short, both stable across methods, and both incompatible with the century-to-millennial modes of the IPCC framework by two to three orders of magnitude. Since the IPCC Bern architecture is structurally excluded, so are all of its derived quantitative claims, including the multi-millennial persistence of emitted CO_2 pulses.

15.3 What the single-exponential proof excludes

The full 55-point fit (Figure 3(a)) and the 32-point joint fit (Table 10) (Figure 3(b)) together provide the most precise empirical constraint available on the atmospheric CO_2 impulse-response structure. The implications are direct and far-reaching:

- **The IPCC Bern IRF architecture and its multi-millennial persistence claim are structurally excluded.** The Vandermonde and Cauchy–Schwarz proofs (Appendix B.1) establish that no non-trivial multi-exponential parameter set can reproduce a single exponential; the operator-equivalence theorem (§8.4, Appendix B.2) extends this exclusion from the isotopic decay to the CO_2 mass impulse response. The IPCC claim that "15 to 40% of an emitted CO_2 pulse will remain in the atmosphere longer than 1,000 years" [60] derives entirely from the Bern permanent fraction a_0; both architecture and claim are excluded.
- **The turnover-time/adjustment-time distinction is resolved.** The two timescales are connected through the exact bridge equation $\phi_{\mathrm{slow}} = 1/(\tau_{\mathrm{rel}}\,\kappa_{\mathrm{in}})$ and emerge from the same first-order linear ODE. The distinction invoked by the IPCC is an artifact of the multi-compartment IPCC Bern architecture, not a physical property of the carbon cycle.
- **Anthropogenic accumulation as the dominant driver of post-1750 CO_2 growth is excluded by the decay shape.** The single-exponential shape requires $\kappa(t) = \phi_{\mathrm{slow}}(t)\,\kappa_{\mathrm{in}}(t) = \mathrm{const}$ (§5.4). Since M_{a} grew by 31.6% over the observation window, F_{in} must have grown by the same proportion. Since anthropogenic emissions account for only ~5% of total gross inflow, the bulk of the required growth must come from non-fossil-fuel fluxes F_{nat} growing in proportion to M_{a}. This is confirmed directly by the constancy of $F_{\mathrm{nat}}/M_{\mathrm{a}} = 0.2575 \pm 0.0011\ \mathrm{yr}^{-1}$ (±0.4%) over

the 63-point annual series 1961–2023. A scenario in which F_{nat} is held at preindustrial values predicts a 19% decline in κ_{in} over the observation window (Section 12), producing pronounced curvature on a semi-log plot that is demonstrably absent from the data (Figure 3(b), top panel).

15.4 Sensitivity to assumptions

The derivation depends on three empirical datasets: $M_{\mathrm{a}}(t)$ from Mauna Loa, $F_{\mathrm{anth}}(t)$ from the Global Carbon Budget, and the observed Δ^{14}C decay from [55]. From these the three empirical measurements (§6.1) are extracted: τ_{rel} and A_{32} from the bomb-^{14}C fit, and τ_{res} from the independent Koutsoyiannis (2024a) [72] RRR fit. The Mauna Loa record is among the highest-precision atmospheric measurements ever made and is not in dispute. The fossil-fuel emission inventory comes from energy statistics; revision upward would only strengthen the falsification of fossil dominance, since larger emissions against the same observed M_{a} growth imply a larger natural removal (and therefore a smaller τ_{res}). The observed Δ^{14}C decay is independent of any modeling assumptions. The leak fraction $\phi_{\mathrm{slow}} = 0.203 \pm 0.006$ is robust across the empirical range of τ_{res}.

15.4.1 Robustness to nonzero R_{b}

The simplifying assumption $R_{\mathrm{b}} = 0$ in (13) treats the deep-mixed leak as carrying the pre-bomb atmospheric baseline. The exact mass-balance ODE with arbitrary constant R_{b} is

$$\frac{\mathrm{d}R_{\mathrm{a}}}{\mathrm{d}t} = -\phi_{\mathrm{slow}}\,\kappa_{\mathrm{in}}\left(R_{\mathrm{a}} - R_{\mathrm{b}}\right) \tag{93}$$

with solution

$$R_{\mathrm{a}}(t) - R_{\mathrm{b}} = \left[R_{\mathrm{a}}(t_0) - R_{\mathrm{b}}\right] e^{-(t-t_0)/\tau_{\mathrm{rel}}} \tag{94}$$

The decay shape is single-exponential for any constant R_{b} with the same $\tau_{\mathrm{rel}} = 1/[\phi_{\mathrm{slow}}\,\kappa_{\mathrm{in}}]$. Nonzero R_{b} shifts the asymptote but does not alter the slope on a semi-log plot of $R_{\mathrm{a}} - R_{\mathrm{b}}$ versus t, and therefore does not affect the exclusion of multi-exponential IRFs (§8.3) or the bridge equation Eq. (23).

The Jungfraujoch progression from +25‰ (2019) to +14‰ (2024) on the bomb-excess scale lies modestly below the $R_{\mathrm{b}} = 0$ extrapolation of +19‰. Part of the excess decline is the continued fossil-fuel dilution of §9.2.5 extrapolated past 2015. Any residual beyond that is consistent with effective R_{b} in the range −5 to −20‰ on the bomb-excess scale, corresponding to a flux-weighted deep-mixed source that includes a small contribution from partially ventilated subsurface ocean carbon rather than purely contemporary recycling. The bridge equation (24) with R_{b} in this range shifts the inferred ϕ_{fast} by at most 1% (from 0.797 to 0.806), well within the 1–2% precision floor set by the $\delta^{13}\mathrm{C_I}$ constraint (§5.4). The framework is robust to the $R_{\mathrm{b}} = 0$ idealization at the level of the central conclusions.

16. Conclusions

The analysis rests on three independent empirical measurements: the fitted time constant τ_{rel} and amplitude A_{32} from the 32-point bias-corrected bomb-^{14}C recovery, and the CO_2 molecular residence time τ_{res} from the Koutsoyiannis (2024a) [72] Refined Reservoir Routing fit to Mauna Loa and Barrow. Every other quantity follows from these three, mass conservation, and the observed atmospheric CO_2 inventory from Mauna Loa; no free parameters are adjusted. From these inputs, together with the IPCC AR6 Figure 5.12 gross non-fossil-fuel flux estimates and the Hua et al. (2022) [55] global Δ^{14}C compilation, we derive:

- CO_2 molecular residence time $\tau_{\mathrm{res}} = 3.80 \pm 0.12$ yr, validated both by snapshot mass balance ($M_{\mathrm{a}}/F_{\mathrm{in}} = 3.78$ yr) and by the Refined Reservoir Routing fit of Koutsoyiannis (2024a) [72] ($W_{\mathrm{m}} = 3.68$ yr at Mauna Loa, 3.91 yr at Barrow; average 3.80 yr).

- Ratio relaxation time $\tau_{\text{rel}} = 18.383 \pm 0.015$ yr from the cleaned 32-point joint recovery (§9.3), $R^2_{32} = 99.998\%$, $\sigma_{32} = 0.539‰$, over 52 years. The residual is smaller than the reported measurement uncertainty of every one of the 32 data points. The parameter τ is recovered consistently at every cleaning level: the uncorrected 55-point recovery yields $\tau = 18.389 \pm 0.139$ yr ($R^2 = 99.825\%$, $\sigma = 7.787‰$); the 28-point cleaned recovery yields $\tau = 18.382 \pm 0.016$ yr ($\sigma = 0.573‰$); the 32-point cleaned recovery yields $\tau = 18.383 \pm 0.015$ yr ($\sigma_{32} = 0.539‰$). Three entirely independent analyses on entirely different point sets yield the same value of τ. The $\sim$14$\times$ reduction in σ from raw to cleaned is achieved entirely by excluding points that fail the first-principles mass-conservation impossibility test (Appendix C) and by applying the exact mass-balance fossil-fuel correction; no physical parameter is adjusted, no new physics is invoked, and the underlying τ does not change. The same τ is recovered independently in each of three non-overlapping windows of the record: the Early window 1967–1994 ($\tau = 18.379 \pm 0.021$ yr, the strongest single test, with the largest amplitude lever arm), the Middle window 1998–2010, and the Late window 2011–2019 (Table 10). The robust uncertainty on τ_{rel}, which incorporates compilation-preprocessing choices, is set by the convergence across all three cleaning levels and across all three non-overlapping windows: $\Delta\tau/\tau < 0.05\%$ at every level.
- Deep-mixed leak fraction $\phi_{\text{slow}} = 0.203 \pm 0.006$, fast-recycled fraction $\phi_{\text{fast}} = 0.797 \pm 0.006$, derived from the exact bridge equation $\phi_{\text{slow}} = 1/(\tau_{\text{rel}}\,\kappa_{\text{in}})$.
- Independent two-box stratosphere-troposphere ODE, built from separate atmospheric inputs (stratospheric mass, tropospheric mass, Brewer-Dobson flux, atmospheric growth rate), yields $\phi^{\text{2box}}_{\text{slow}} = 0.199 \pm 0.008$, agreeing with the bridge-equation value at 0.74σ ($\sigma_{\Delta\phi} = 0.0054$, correlation-aware). The fast eigenvalue $\tau_{\text{fast}} = 3.08 \pm 1.04$ yr agrees with independent stratospheric turnover measurements [53, 33, 34], providing a second cross-validation from a mathematically independent framework.
- The Vandermonde and Cauchy-Schwarz proofs (Appendix B.1) establish that no non-trivial multi-exponential parameter set can equal a single exponential, and the operator-equivalence theorem (Appendix B.2) extends this exclusion from the isotopic decay to the CO_2 mass impulse response: any eigenmode of the reservoir exchange operator appears in both responses at the same atmospheric amplitude, so no slow bulk-CO_2 mode escapes the bound the isotopic decay imposes.
- Anthropogenic share of the deep-mixed leak: $(21 \pm 1)\%$ (fossil-fuel combustion and cement production only; land-use change emissions carry contemporary R_{a} and belong to the fast-recycled pool); non-fossil-fuel share: $(79 \pm 1)\%$.
- The single-exponential shape of the bomb-^{14}C decay is observational evidence that the product $\phi_{\text{slow}}\,\kappa_{\text{in}} = k$ has been constant to within $\sim$1% over six decades. This is independently confirmed by two further convergent measurements: under the Koutsoyiannis-stable τ_{res}, the mass-balance reconstruction $\kappa_{\text{in}}(t) = 1/\tau_{\text{res}} + \mathrm{d}\ln M_{\text{a}}/\mathrm{d}t = 0.2678\ \text{yr}^{-1}$ is stable to $\pm 0.4\%$, with the non-fossil-fuel component $F_{\text{nat}}/M_{\text{a}} = 0.2575\ \text{yr}^{-1}$ stable to the same $\pm 0.4\%$ and F_{nat} growing by 31.6% in near-exact proportion to the growth in M_{a} of 32.6% from 1961 to 2023; and $\delta^{13}\text{C}_{\text{I}}$ constrains ϕ_{fast} to within 1–2%. The framework is self-consistent at the precision floor of every independent dataset, consistent with the temperature-driven biospheric expansion mechanism.
- Independent high-precision Jungfraujoch measurements 2020–2024 [75], on a measurement chain unrelated to the Hua et al. (2022) [55] compilation, extend the atmospheric Δ^{14}C record five years past the 32-point fit window. After scale marriage at 2020 and the exact level-dependent fossil-fuel correction, the four independent 2021–2024 points lie on the extrapolated single-exponential within the combined per-point uncertainty of $\pm 5‰$; the 2020 point sits on the line by construction as the scale-marriage anchor (Section 14, Figure 5). The extension corroborates the decay on an independent measurement chain essentially through full return

to the pre-bomb baseline; it does not, and cannot, refine τ_{rel} or tighten the Bern falsification already established at the end of the fit window.

The two timescales of ∼4 yr and ∼18 yr are not in conflict. They emerge from the same first-order linear mass-balance equation and measure different physical properties of the same atmospheric reservoir. The residence-time–adjustment-time distinction invoked by the IPCC is therefore not supported by the underlying physics; it is an artifact of the multi-compartment IPCC Bern IRF architecture, which the bomb-^{14}C single-exponential decay structurally falsifies.

The conclusions of this paper fall into two logical tiers that a reader may evaluate separately.

Tier 1. Impulse-response falsification. The single-exponential bomb-^{14}C decay at $\tau_{\mathrm{rel}} = 18.383 \pm 0.015$ yr, the Koutsoyiannis RRR residence time $\tau_{\mathrm{res}} = 3.80 \pm 0.12$ yr, the bridge relation $\tau_{\mathrm{rel}} = \bar{\mathcal{D}}\,\tau_{\mathrm{res}}/\bar{\phi}_{\mathrm{slow}}$ with $\bar{\phi}_{\mathrm{slow}} = 0.203 \pm 0.006$, and the structural exclusion of every Bern-lineage multi-exponential impulse response by the operator-equivalence theorem follow from direct measurement and first-order linear mass balance with no adjustable parameters. A reader who accepts the Hua et al. (2022) [55] compilation, the Mauna Loa CO_2 record, the Koutsoyiannis RRR fit, and the linearity premise tested empirically in §10 must accept Tier 1 in full. The IPCC Bern IRF architecture and every metric derived from it (GWP, GTP, CGTP, remaining carbon budgets, multi-millennial persistence claims) are falsified on Tier 1 alone.

Tier 2. Attribution of the atmospheric CO_2 rise. The further claim that the post-1750 rise in M_{a} is driven by non-fossil-fuel sources rather than anthropogenic emissions rests on three additional observations layered on top of Tier 1: (i) the proportionality $F_{\mathrm{nat}} \propto M_{\mathrm{a}}$ ($F_{\mathrm{nat}}/M_{\mathrm{a}}$ stable to ±0.4% year-to-year; Newey–West HAC regression of annual log-growth rates gives $\hat{\beta} = 0.980 \pm 0.042$ with the $\beta = 1$ proportionality null not rejected at 0.48σ) over 62 years (§10.2, Figure 4), (ii) the constancy of the flux-weighted input signature $\delta^{13}C_{\mathrm{I}}$ at −13.2‰ across the 44-year instrumental record and 500-year proxy record [73], which under the mass partition equation with standard isotopic signature values bounds the fossil fraction of the net atmospheric CO_2 addition at approximately 1.3% [22], with its +2.94‰/decade recent-window trend of the wrong sign for fossil dominance [41, 21], and (iii) the rejection of the residual-attribution accounting that treats the entire anthropogenic gross emission as the sole source of the observed inventory change. A reader who accepts Tier 1 but disputes any of these three additional premises can remain agnostic on Tier 2 without weakening the impulse-response result.

The two tiers are presented separately so that neither depends on the other for its force. Tier 1 stands on direct physical measurement. Tier 2 sharpens Tier 1 into a complete physical account of the industrial-era carbon cycle.

The four-year atmospheric lifetime is mathematically incompatible with predominantly anthropogenic accumulation. With gross outflux of order one-quarter of the atmospheric inventory per year, the residue of any year's fossil emissions falls below half a percent of the atmosphere within a decade; the integral does not survive the throughput. The observed century-scale rise must therefore be dominated by net non-fossil-fuel fluxes, principally biospheric and oceanic. This conclusion is corroborated independently by the non-fossil-fuel isotopic input signature ($\delta^{13}C_{\mathrm{I}} \approx -13$‰, incompatible with fossil CO_2 at −28‰) and by the RRR mass-balance framework of Koutsoyiannis (2024a) [72] and Koutsoyiannis (2024b) [73], in which the non-fossil-fuel carbon cycle dominates observed atmospheric changes. The recent-window (1977–1996) trend in $\delta^{13}C_{\mathrm{I}}$ on the Francey et al. (1999) [41] Law Dome record is +2.94‰/decade at $p < 10^{-4}$ against a Hurst-Kolmogorov noise null [21]: opposite in sign to the fossil-dominance prediction over the highest-emissions window of the record, and consistent with a growing ocean-outgassing contribution to the net atmospheric input under surface warming. The conventional chain in which fossil emissions accumulate, force the climate, and drive warming is broken at its first link: the accumulation does not occur.

The central claims of the IPCC carbon-climate framework rest on physical assertions that the present analysis falsifies. Those claims are: that anthropogenic emissions accumulate in the atmosphere on millennial timescales; that this accumulation drives a measurable change in global temperature; and that the resulting change poses a quantifiable threat to be mitigated through cumulative-emissions reduction. The atmospheric lifetime is four years, not millennia. The CO_2 rise is driven by non-fossil-fuel sources, not by anthropogenic emissions. The construct "dangerous anthropogenic global warming," as the framework defines it, corresponds to nothing measurable in the underlying mass-balance CO_2 dynamics discussed here. What remains is a policy architecture built on a mathematical scaffold that the data do not support.

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Data and Code Availability

The cleaned 32-point $\Delta^{14}C$ series, measured plateau offsets (Plateau A: −11.563‰, Mini Plateau: +0.891‰, Plateau B: +9.447‰), and fossil-fuel correction values (Table 9) are reproduced in full within the manuscript itself; the underlying Hua et al. (2022) [55] compilation is publicly available from the original source. All fits were performed with standard non-linear least-squares methods as described in the text and can be replicated with any standard statistical package using the data published here.

Appendices

A. Unweighted nonlinear least squares is the correct estimator for the Hua et al. (2022) Δ^{14}C record

The single-exponential parameters reported here, and in every fit in this paper, are obtained by *unweighted* nonlinear least squares on the central values: the estimator minimizes $\sum_i \left[R_{\mathrm{a}}(t_i) - R_{\mathrm{fit}}(t_i)\right]^2$, where $R_{\mathrm{fit}}(t_i) = A\, e^{(1965-t_i)/\tau}$, over the free parameters A and τ, with no $1/\sigma_i^2$ weighting; A is the fitted amplitude, distinct from the observed datum $R_{\mathrm{a}}(1965)$. This is a deliberate choice, not a default, and it is the choice the physics of the measurement requires. We set out the argument explicitly because the alternative, inverse-variance ("$1/\sigma^2$") weighting, is the reflexive default in curve fitting and would be actively misleading when applied to a multi-station composite Δ^{14}C product whose per-year σ_i encodes inter-record dispersion rather than independent measurement error.

Inverse-variance weighting is optimal only under assumptions this dataset violates. The Gauss–Markov result that $1/\sigma_i^2$ weighting yields the minimum-variance unbiased estimator is conditional: it requires the σ_i to be the standard deviations of *independent* measurement errors on the quantity being fitted [101, 30]. The Hua et al. (2022) [55] per-year σ_i are not that. Standard references confirm unweighted is preferred when reported σ_i do not represent independent measurement errors [88]. As the compilation methodology makes explicit (§9.2), each annual value of the summer series (Hua et al. Tables S1a–S1c) is a surface-area-weighted average of zonal means, and each zonal mean is an inverse-uncertainty-weighted average of heterogeneous atmospheric and tree-ring records. The reported σ_i is therefore the *inter-record dispersion of the contributing stations within each year*, not the measurement error of an independent annual observation. Three consequences follow, each fatal to inverse-variance weighting here:

1. **The errors are not independent.** The summer series imposes serial correlation whenever contributing stations overlap across adjacent years, which is the normal case: most contributing records persist for many consecutive years, and successive annual values are therefore constructed from largely overlapping station sets whose measurement-chain biases carry forward from one year to the next. Inverse-variance least squares assumes a diagonal covariance; the true covariance is band-structured. Weighting by $1/\sigma_i^2$ does not approximate the correct generalized-least-squares estimator; it simply mis-specifies the covariance with false confidence.

2. **The σ_i track station population, not signal fidelity.** Years in which many stations contribute carry small σ_i; years near a station entry or exit carry small σ_i as well, because the dispersion is computed *after* the compositing that introduces the artifact. The smallest reported uncertainties therefore sit in precisely the interior years whose values are most strongly shaped by the compilation procedure, while the largest σ_i sit at the early high-amplitude years that carry the cleanest dynamical information about the decay constant. Inverse-variance weighting would hand near-total control of τ to a handful of mid-record points selected for low inter-station scatter. It would fit the artifact and discard the signal. For further details on how the Hua et al. (2022) compilation procedure generates these artificial σ_i, see §9.2.

3. **The signal is exponential over orders of magnitude.** The bomb excess decays from 693‰ to 4‰ on the raw Hua scale (720‰ to 31‰ on the Hua+27 scale used in the fit): more than two orders of magnitude in either case. A single exponential is a statement about a *constant fractional* decay rate, so a sound estimator must constrain that rate evenhandedly across the

full dynamic range. For an exponential spanning multiple orders of magnitude and decaying toward zero, the decay constant is resolved chiefly by the endpoints: the early years fix the amplitude and initial slope, and the long lever arm to the late years sets τ, while the interior points discriminate weakly between competing τ values [66]. Inverse-variance weighting on the reported σ_i does not respect this structure: it amplifies the low-σ interior and the small-absolute-σ tail, whose relative uncertainty is in fact the highest in the record, and suppresses precisely the early years where the curvature that resolves τ is most pronounced.

The cleaning procedure removes the only legitimate role weighting could play. The justification sometimes offered for $1/\sigma^2$ weighting is that it down-weights unreliable points. In this analysis that function is performed correctly and transparently by the first-principles audit of §9.2, *not* by the weights. Points are not down-weighted by an opaque σ_i; they are either retained or excluded by a criterion that is stated in advance and is entirely independent of the single-exponential hypothesis. An interval is excluded only when a documented station transition, weighting transition, or data revision is present (§9.2), or when a year-on-year excursion exceeds what any atmospheric mechanism can produce by more than an order of magnitude, e.g. a conservation bound on $^{14}CO_2$ creation or $^{12}CO_2$ dilution that contains no reference to the decay model (§9.2.2). The surviving points are then corrected only by the exact isotopic mass-balance integral (§9.2.5), whose sign and magnitude are fixed entirely by the independently measured E_{FOS} and M_{a}, with no free parameter and no reference to the residual it produces. Once the discretionary, model-correlated information has been removed from the point set in this auditable way, the correct estimator on what remains is the one that introduces *no further* data-derived weighting: unweighted least squares treats every surviving, construction-vetted point on an equal footing, which is the only footing the compilation supports.

The choice is not load-bearing, and that is the point. A weighting scheme that changed the answer would itself be a diagnostic of a fragile result. The unweighted fit to the cleaned set and the unweighted fit to the full uncorrected 55-point record, two analyses with entirely different point sets, corrections, and leverage, return the same decay constant to within 0.02% (τ_{28} = 18.382 yr versus τ_{55} = 18.389 yr; Table 10). Extending the cleaned set forward with the four first-principles-corrected 2016–2019 points (§9.2.5) shifts τ by ∼0.001 yr, lowers the residual standard error, and leaves R^2 = 99.998% unchanged. Unweighted nonlinear least squares is therefore not merely one defensible option among several: it is the correct estimator for this dataset. It is the only choice consistent with the known error structure of the compilation, where the reported σ_i are inter-record dispersion rather than independent measurement error. Any weighted alternative either injects the documented compilation artifacts back into the estimate or distorts the dynamic range over which the single exponential is tested. Unweighted NLLS does neither, and it is the estimator used for every fit in this paper.

B. Mathematical Proofs

B.1 Incompatibility of any multi-exponential with a single exponential

The multi-exponential architecture, introduced by Maier-Reimer and Hasselmann (1987) [80] and adopted by Joos et al. (1996) [63] and all subsequent IPCC assessments, takes the form

$$c_0 + \sum_{k=1}^{n} c_k \, e^{-\lambda_i t} \quad \text{where } c_0 \geq 0, c_k > 0, n = 3 \text{ or } 4, \text{and } \lambda_1 > \lambda_2 > \cdots > \lambda_n > 0 \qquad (95)$$

with $c_0 \geq 0$ equal to 0.1369 in AR2, 0.152 in AR3, 0.217 in AR4, and 0.2173 in AR5/AR6 [20], with widely separated rates λ_i. We now prove that even a much more general version of this form

cannot equal a single exponential for *any* non-trivial parameter choice.

Theorem. A sum of exponentials is not an exponential. *Let $\mu, \lambda_1, \ldots, \lambda_n$ be real numbers, all different from one another, and let $d, c_1, \ldots, c_n$ be positive numbers with $n > 1$. Then*

$$\sum_{k=1}^{n} c_k \, e^{\lambda_k t} \;\not\equiv\; d \, e^{\mu t}$$

Proof. Suppose equality held. Set $\lambda_0 := \mu$, $c_0 := -d$, so $\sum_{k=0}^{n} c_k e^{\lambda_k t} \equiv 0$.

Differentiating j times at $t = 0$ for $j = 0, 1, \cdots, n$ yields the equation $\mathbf{V}\mathbf{c} = \mathbf{0}$, where $\mathbf{V}$ is the $(n+1) \times (n+1)$ Vandermonde matrix (Cauchy, 1815 [15]) with

$$\det \mathbf{V} \;=\; \prod_{0 \le i < k \le n} (\lambda_k - \lambda_i) \;\neq\; 0$$

Therefore $\mathbf{V}$ is invertible, yielding $\mathbf{c} = \mathbf{V}^{-1}\mathbf{0} = \mathbf{0}$, contradicting $c_0 = -d \neq 0$.

The curvature picture gives an independent geometric proof.

With $c_k > 0$, $w_k(t) = c_k e^{-\lambda_k t}$, and $y(t) = \sum_k w_k$:

$$y' = -\sum_k \lambda_k w_k, \qquad y'' = \sum_k \lambda_k^2 w_k$$

so

$$(\ln y)'' \;=\; \frac{y'' y - (y')^2}{y^2} \;=\; \frac{\left(\sum w_k\right)\left(\sum \lambda_k^2 w_k\right) - \left(\sum \lambda_k w_k\right)^2}{y^2} \tag{96}$$

Substituting $a_k := \sqrt{w_k}$ and $b_k := \lambda_k \sqrt{w_k}$, the numerator of (96) becomes $\left(\sum_k a_k^2\right)\left(\sum_k b_k^2\right) - \left(\sum_k a_k b_k\right)^2$ and Cauchy–Schwarz [16, 106] yields

$$\left(\sum_k a_k^2\right)\left(\sum_k b_k^2\right) > \left(\sum_k a_k b_k\right)^2 \tag{97}$$

Hence $(\ln y)'' > 0$. Geometrically, this means *all* such sums of exponentials have *upward curvature* on the semi-log plot, vs. zero curvature (i.e. a straight line) for a single exponential as shown in Figure 2. □

B.2 Operator-Equivalence Theorem and Bern-IRF Exclusion

This appendix proves the operator-equivalence theorem stated in §8.4: under linear exchange, the atmospheric expression of any eigenmode of the carbon-cycle exchange operator is governed by the same product of left and right eigenvector components in both the mass response and the ^{14}C-content response. Any mode posited by the IPCC Bern IRF must therefore appear in the bomb-^{14}C decay with the same fractional weight it carries in the mass IRF. The cleaned 32-point fit residual $\sigma_{32} = 0.539‰$ against the fit amplitude $A_{32} = 709.115 \pm 0.585‰$ then bounds that weight at one part in ∼1316, structurally excluding both the finite-τ_i modes and the permanent fraction a_0.

Residual bound on eigenmode weights. Let the bulk CO_2 impulse response be a sum $R(t) = \sum_i a_i\, e^{-t/\tau_i}$ with $\sum_i a_i = 1$, which by operator equivalence (above) is also the form of the bomb-^{14}C decay with the same coefficients $\{a_i\}$ and the same timescales $\{\tau_i\}$. Suppose the data admit a single-exponential fit $y_{\rm fit}(t) = A\, e^{-t/\tau_{\rm rel}}$ with fit amplitude $A = A_{32} = 709.115$‰ at $t = 1965$ and residual RMS $\sigma_{32} = 0.539$‰. (The fit amplitude A_{32} is the appropriate normalization here, not the raw bomb-peak atmospheric signal of ~ 720‰: the residuals are measured against the single-exponential fit, so any additional mode is bounded relative to the fit's amplitude at the anchor year 1965.) Any additional mode of amplitude $a_i\, A_{32}$ and timescale $\tau_i \neq \tau_{\rm rel}$ contributes a time-dependent term $a_i\, A_{32}\, e^{-t/\tau_i}$ to the residual that the single-exponential fit cannot absorb: a mode with $\tau_i \gg \tau_{\rm rel}$ looks approximately constant across the fit window and shows as a quasi-permanent offset of amplitude $a_i\, A_{32}$; a mode with $\tau_i \ll \tau_{\rm rel}$ looks like a transient spike localized to the early points; a permanent mode ($\tau_0 = \infty$) with weight a_0 contributes exactly $a_0\, A_{32}$ at every point. The maximum unmodeled mode amplitude consistent with the observed residual scatter is bounded by

$$a_i\, A_{32} \lesssim \sigma_{32} \quad \Longleftrightarrow \quad a_i \lesssim \frac{\sigma_{32}}{A_{32}} = \frac{0.539}{709.115} \approx \frac{1}{1316}$$

The inequality is tight for the permanent fraction a_0 and for slow modes with $\tau_i \gg 52$ yr (the fit window length): either produces residual structure of amplitude $a_i\, A$ that cannot be absorbed by adjusting A or $\tau_{\rm rel}$. For intermediate timescales ($5\,{\rm yr} \lesssim \tau_{\rm i} \lesssim 52\,{\rm yr}$) the bound is approximate rather than strict because the fit can partially accommodate the mode by shifting A and $\tau_{\rm rel}$; the fit-robustness checks across the 55-point, 32-point, 28-point, and three disjoint-window subsets of §9.3 and the Vandermonde argument of Appendix B.1 close that remaining gap by independent means. The AR5/AR6 IPCC Bern IRF permanent fraction $a_0 \approx 0.217$ exceeds the bound at the permanent limit by a factor of ∼285 and is structurally excluded.

Reservoir model. The atmosphere–surface carbon system consists of N coupled reservoirs: the atmosphere (subscript a) and $N-1$ surface and deep reservoirs representing the surface ocean mixed layer, intermediate and deep ocean, the active terrestrial biosphere, and refractory soil carbon. The state vector $\mathbf{M}(t) = (M_{\rm a}, M_1, \ldots, M_{N-1})^T$ collects the reservoir carbon masses in GtC, and $\mathbf{e}_a$ denotes the unit vector in the atmospheric direction. Each pair of reservoirs in direct physical contact exchanges carbon at gross rate $F_{ij}(t) = A_{ij}\, M_j(t)$, where A_{ij} is a time-independent rate coefficient in yr^{-1}; pairs not in direct contact have $A_{ij} = 0$, and diagonal entries are $A_{ii} = -\sum_{k\neq i} A_{ki}$. The matrix $\mathbf{A} = (A_{ij})$ is the linear exchange operator. Its sparsity pattern encodes the physical topology of direct reservoir contacts. External carbon inputs (fossil-fuel emissions, etc.) appear in the source vector $\mathbf{e}(t)$ in GtC yr^{-1}.

Perturbation dynamics. Let $\mathbf{M}_0(t)$ denote the unperturbed baseline trajectory and $\delta\mathbf{M}(t) \equiv \mathbf{M}(t) - \mathbf{M}_0(t)$ the carbon-mass anomaly driven by an anomalous source $\delta\mathbf{e}(t)$. Under linear exchange,

$$\frac{d\,\delta\mathbf{M}}{{\rm d}t} = \mathbf{A}\,\delta\mathbf{M} + \delta\mathbf{e}(t) \tag{98}$$

The ^{14}C-content state vector is $\mathbf{X}(t) = (M_{\rm a}R_{\rm a}, M_1R_1, \ldots, M_{N-1}R_{N-1})^T$, with $R_i(t)$ the ^{14}C/C ratio in reservoir i in per mille. Under the well-mixed-reservoir assumption, ^{14}C content transferred from j to i proceeds at rate $A_{ij}\, X_j$, so the content anomaly $\delta\mathbf{X}(t) \equiv \mathbf{X}(t) - \mathbf{X}_0(t)$ satisfies

$$\frac{d\,\delta\mathbf{X}}{{\rm d}t} = (\mathbf{A} - \lambda_{14}\mathbf{I})\,\delta\mathbf{X} + \delta\mathbf{e}_X(t) \tag{99}$$

with $\lambda_{14} = \ln 2/5{,}700\,{\rm yr}^{-1}$ the ^{14}C radioactive decay constant. The $-\lambda_{14}\mathbf{I}$ term shifts every eigenvalue of $\mathbf{A}$ uniformly by $-\lambda_{14}$ without altering the eigenvectors. The eigenmode structure of the operator governing $\delta\mathbf{X}$ is therefore identical to that governing $\delta\mathbf{M}$, modulo this known

scalar shift. Stuiver–Polach normalization removes the dominant mass-dependent fractionation; any residual contribution to apparent τ is below 0.3% (§11.2), well below the level relevant to the Bern amplitude bound.

Topology. The matrix $\mathbf{A}$ accommodates either the parallel topology implicit in the Bern IRF parameterization or the physically correct serial topology established in Cohler & Soon (2026a) [20] and discussed in §8.3.1; the theorem and its conclusions hold under either, and the serial topology sharpens the bound a fortiori.

Theorem. Operator-equivalence bound. *Let* $\mathbf{A}$ *be the linear exchange operator of the coupled atmosphere–reservoir system, with eigenvalues* $\{\lambda_j\}$*, right eigenvectors* $\{\mathbf{v}_j\}$*, and left eigenvectors* $\{\mathbf{w}_j\}$ *satisfying* $\mathbf{w}_j^T \mathbf{v}_k = \delta_{jk}$*. Assume:*

(i) the system is adequately linear over the bomb-^{14}C *perturbation range, as empirically confirmed by the observed single-exponential decay over 54 yr and by the constancy of* κ_{in} *to* $\pm 0.4\%$ *across a* 31.6% *rise in* M_{a} *(§10);*

(ii) $\delta\mathbf{M}$ *and* $\delta\mathbf{X}$ *evolve under* (98) *and* (99)*, sharing the eigenvectors of* $\mathbf{A}$*;*

(iii) the bomb-^{14}C *perturbation is an atmospheric isotope-content impulse,* $\delta\mathbf{X}(0) = \alpha\, \mathbf{e}_a$*;*

(iv) the observed atmospheric ratio anomaly $\delta R_{\text{a}}(t)$ *is proportional to* $\delta X_a(t)$ *to first order, since* M_{a} *is not materially perturbed by the bomb pulse.*

Then the atmospheric expression of mode j *in the bomb-*^{14}C *response is*

$$\delta X_a^{(j)}(t) = \alpha\, (\mathbf{w}_j^T \mathbf{e}_a)\, v_{j,a}\, e^{(\lambda_j - \lambda_{14})\, t} \tag{100}$$

and the atmospheric expression of the same mode in the mass response to an atmospheric emission impulse $\delta\mathbf{e}(t) = \beta\, \mathbf{e}_a\, \delta(t)$ *is*

$$\delta M_{\text{a}}^{(j)}(t) = \beta\, (\mathbf{w}_j^T \mathbf{e}_a)\, v_{j,a}\, e^{\lambda_j\, t} \tag{101}$$

The atmospheric visibility product $(\mathbf{w}_j^T \mathbf{e}_a)\, v_{j,a}$ *is the same in both.*

Proof. Decompose the atmospheric unit vector in the right eigenbasis: $\mathbf{e}_a = \sum_j (\mathbf{w}_j^T \mathbf{e}_a)\, \mathbf{v}_j$. For the bomb-^{14}C initial condition $\delta\mathbf{X}(0) = \alpha\, \mathbf{e}_a$ with no subsequent source, the solution of (99) is

$$\delta\mathbf{X}(t) = \alpha \sum_j (\mathbf{w}_j^T \mathbf{e}_a)\, e^{(\lambda_j - \lambda_{14})\, t}\, \mathbf{v}_j$$

reading off the atmospheric component yields (100). For the mass impulse source $\delta\mathbf{e}(t) = \beta\, \mathbf{e}_a\, \delta(t)$ in (98), the Duhamel solution at $t > 0$ is

$$\delta\mathbf{M}(t) = \beta \sum_j (\mathbf{w}_j^T \mathbf{e}_a)\, e^{\lambda_j\, t}\, \mathbf{v}_j$$

and the atmospheric component gives (101). The eigenvectors are shared, so the product $(\mathbf{w}_j^T \mathbf{e}_a)\, v_{j,a}$ is the same in both. □

C. First-Principles Identification of Compilation Artifacts and Retained Plateaus in the Hua et al. (2022) Global Δ^{14}C Record

C.1 Overview

This appendix establishes that the partition of the 55-point Hua et al. (2022) [55] global Δ^{14}C compilation into 32 retained and 23 excluded points follows from a set of physics-derived retention rules applied to the numerical residual series $r_{55}(t) = R_a(t) - R_{55}(t)$ against the single-exponential fit (Table 3). The rules use no curve fitting and no reference to the cleaned τ_{rel}. Two convergent diagnostics confirm the resulting partition: the physics-derived smoothness criterion of §C.4 identifies year-to-year jaggedness inconsistent with the coupled stratosphere-troposphere ODE structure of §7; the provenance audit of §9.2 identifies methodology-change boundaries from documented station transitions, tree-ring-to-atmospheric weighting shifts, and retrospective revisions in the Hua et al. (2022) [55] compilation. The two diagnostics agree on the 32/23 partition. The retention rules produce the partition; the diagnostics are mutually independent confirmations; the resulting framework is physics-derived and non-circular.

Three features of the framework make this convergence physically meaningful. First, the diagnostic operator d^2r_{55}/dt^2 is sensitive only to non-smoothness in R_a itself; the algebraic form of the reference fit cancels under second differentiation. The test would therefore yield the same diagnostic verdict if any other smooth reference curve (a multi-exponential, an impulse-response function, or a low-order polynomial) were used in place of the 55-point single exponential. The diagnostic makes no algebraic assumption about the shape of the underlying decay. Second, the numerical smoothness threshold $0.000920\,R_T(t)$ is derived from the coupled stratosphere-troposphere ODE system of §7, using only independently sourced atmospheric parameters (M_S, M_T, F_{BDC}, $d\ln M_a/dt$). The threshold is a physical bound, not an empirical envelope fit to the data. Third, the ±2 GtC/yr Friedlingstein O+B variability envelope and the list of documented major perturbation events are externally sourced; no event-year admission decisions depend on the residual data.

C.2 Retention rules

The partition of the 55-point series follows from five retention rules applied in order. Each rule rests on physics derived in §7 or on independently sourced atmospheric measurements; none uses curve fitting or references the cleaned τ_{rel} as an input.

Rule 1 (a priori exclusion). The 1965 and 1966 points are excluded a priori on the physical grounds that they lie on the fast-mode stratosphere-troposphere equilibration ramp before the slow-mode regime is established (§C.8). The criterion is grounded in the two-box fast eigenvalue $\tau_{fast} = 3.08$ yr derived from independently sourced M_S, M_T, and F_{BDC} with negligible dependence on bomb-decay data (§7).

Rule 2 (plateau identification). A contiguous stretch of three or more consecutive years, not covered by Rule 1, that exhibits low within-block dispersion ($\sigma_{within} \lesssim 0.7$‰) and a within-block trend statistically consistent with zero at the 95% confidence level (ordinary least-squares regression of r_{55} on year; none of the three retained plateaus rejects a zero slope) is retained as a plateau. The criterion identifies stable-methodology periods in the Hua et al. (2022) [55] compilation where the residual $r_{55}(t)$ is constant up to measurement noise. Three plateaus are identified: Plateau A (1979–1984), the mini-plateau (1992–1994), and Plateau B (1998–2010), as detailed in §9.2.4.

Rule 3 (anchor retention). A single isolated point not in any plateau, not covered by Rule 1, and with no documented compilation artifact in the audit of §9.2 is retained as an anchor where the centered derivative estimators of §C.3 cannot be cleanly computed due to its excluded neighbors.

This rule retains 1967 as the first valid post-equilibration point, bordered on the left by the a priori exclusion of Rule 1.

Rule 4 (fossil-fuel tail). The 2011–2019 stretch is retained after applying the first-principles fossil-fuel correction C_{FF} (§C.11.1) and the two emergence offsets at the predicted wall locations 2010→2011 and 2015→2016 (§C.11.4). After these corrections, the sub-intervals 2011–2015 and 2016–2019 each satisfy Rule 2's plateau-membership criterion at the same level as Plateau B.

Rule 5 (default exclusion). Any year not retained by Rules 1–4 is excluded.

Output. The five rules yield the partition tabulated in Table 14 and shown in Figures 3(a) and 3(b): 32 points retained (the 1967 anchor; Plateau A 1979–1984; mini-plateau 1992–1994; Plateau B 1998–2010; fossil-fuel tail 2011–2019) and 23 points excluded (1965–1966 fast-mode ramp; 1968–1978, 1985–1991, and 1995–1997 artifact stretches). The rules make no reference to the smoothness test or to the provenance audit; both are independent diagnostics that confirm the partition.

Role of the smoothness test as diagnostic. The smoothness test of §C.7 does not make retention decisions. It serves as a physical diagnostic that confirms four properties of the partition produced by Rules 1–5: (a) the walls bounding the retained plateaus (years at the boundary between artifact stretches and plateaus) show $|d^2 r_{55}/\mathrm{d}t^2|$ exceeding the smoothness threshold, confirming they are not smooth STE physics; (b) the interior of artifact stretches shows sign-flipping curvature inconsistent with any smooth coupled-reservoir physics, confirming the stretches are not real atmospheric decay; (c) the interior of retained plateaus shows curvature consistent with compilation noise; (d) the 2010→2011 and 2015→2016 wall locations show smoothness-test failures in the C_{FF}-only-corrected residuals, justifying the emergence-offset corrections of Rule 4. The diagnostic value of the smoothness test lies in confirming the physical interpretation of the partition; the partition itself is fixed by Rules 1–5.

C.3 Notation: centered three-point estimators

Throughout this appendix, $r_{55}(t)$ denotes the residual of the Hua et al. (2022) [55] compilation against R_{55} at year t (Table 3). We use the centered three-point convention for both discrete derivative estimators:

$$\frac{dr_{55}}{\mathrm{d}t}(t) \equiv \tfrac{1}{2}\big[r_{55}(t+1) - r_{55}(t-1)\big] \tag{102}$$

$$\frac{d^2 r_{55}}{\mathrm{d}t^2}(t) \equiv r_{55}(t+1) - 2\,r_{55}(t) + r_{55}(t-1) \tag{103}$$

Both are indexed by the middle year t of the three-point window and approximate the corresponding continuous derivatives at year t to $O(\Delta t^2)$ for smooth functions. The 3-point centered form is chosen over higher-order (e.g., 5-point Savitzky–Golay) estimators because it localizes discrete jumps to a single year; higher-order estimators spread a single-year compilation step across multiple year indices and thereby smooth out exactly the signal the test is designed to detect. The convention matches the centered Savitzky–Golay estimator used elsewhere in the paper for $\mathrm{d}M_{\mathrm{a}}/\mathrm{d}t$.

The estimators cannot be computed for the endpoints 1965 and 2019. 1965 is excluded a priori on physical grounds (§C.8); 2019 inherits retention from plateau-membership analysis in the FF tail (§C.11.1), not from the smoothness test directly.

C.4 Smoothness criterion from the two-box model

Section 7 derives the coupled stratosphere–troposphere ODE system from first principles. The system has two well-separated eigenmodes:

$$\tau_{\text{fast}} = 3.08 \text{ yr} \quad \text{(stratosphere-troposphere gradient equilibration)} \tag{104}$$

$$\tau_{\text{slow}} = 18.39 \text{ yr} \quad \text{(combined-pool slow decay, equal to observed } \tau_{\text{rel}}\text{)} \tag{105}$$

The slow-mode eigenvector (derived in §7 with $v_S/v_T = 1.263$, $k_T = 0.0644$, $\tau_{\text{rel}} = 18.389$ yr) establishes a smooth quasi-equilibrium gap $R_S(t) - R_T(t) = 0.263\, R_T(t)$, and the stratosphere-to-troposphere (STE) contribution to dR_T/dt is

$$\left(\frac{dR_T}{dt}\right)_{\text{STE}} = k_T\,(R_S - R_T) = 0.0644 \times 0.263\, R_T(t) = 0.0169\, R_T(t) \;\; [‰/\text{yr}] \tag{106}$$

STE delivers high-Δ^{14}C carbon to the troposphere at a rate set by the smoothly decaying quasi-equilibrium gap. The contribution has a single sign (positive) throughout the bomb era because $R_S > R_T$ always. Differentiating once more in time gives the STE contribution to the second derivative:

$$\left(\frac{d^2R_T}{dt^2}\right)_{\text{STE}} = 0.0169\,\frac{d\ln R_T}{dt} = -\frac{0.0169}{\tau_{\text{rel}}}\,R_T(t) = -0.000920\, R_T(t) \;\; \left[‰/\text{yr}^2\right] \tag{107}$$

This is a smooth function with no year-to-year noise term. The single-exponential fit's contribution to d^2R_T/dt^2 is also smooth, so the second derivative of the residual against the fit must satisfy

$$\left|\frac{d^2r_{55}}{dt^2}\right| \le 0.000920\, R_T(t) \;\; \left[‰/\text{yr}^2\right] \tag{108}$$

With a factor-3 buffer for measurement noise and short-period flux variability,

$$\text{STE smoothness threshold} : 3 \times 0.000920\, R_T(t) = 0.00276\, R_T(t) \;\; \left[‰/\text{yr}^2\right] \tag{109}$$

C.5 Compilation noise floor

The Hua et al. (2022) [55] values are reported as integers in ‰ units. Integer rounding contributes a uniform error of ±0.5‰, with $\sigma_{\text{round}} = 0.5/\sqrt{3} \approx 0.29$‰ per point. The centered second-derivative operator with coefficients $(1, -2, 1)$ has standard deviation under this rounding noise of

$$\sigma_{\text{round}}(d^2r_{55}/dt^2) = \sqrt{1^2 + 2^2 + 1^2} \times \sigma_{\text{round}} = \sqrt{6} \times 0.29 \approx 0.71 \;\; \left[‰/\text{yr}^2\right] \tag{110}$$

With a factor-2 buffer, the compilation noise floor on $|d^2r_{55}/dt^2|$ is approximately $1.4\,‰/\text{yr}^2$.

The effective threshold at year t is therefore

$$\text{threshold}(t) = \max\big(0.00276\, R_T(t),\;\; 1.4\big) \;\; \left[‰/\text{yr}^2\right] \tag{111}$$

The STE bound dominates for $R_T > 507$‰ (1965 through ~1971); the noise floor dominates thereafter.

C.6 Ocean–biosphere flux envelope and documented event years

The smoothness criterion of §C.4 bounds the residual's second derivative in the quiescent atmosphere where the only ^{14}C source is the smooth two-box STE contribution. Ocean–biosphere (O+B) events represent exogenous perturbations that the smooth coupled system does not produce. ENSO releases (El Niño) or absorbs (La Niña) carbon between the warm-pool surface ocean and the atmosphere, where the gradient $R_T(t) - R_O(t)$ between tropospheric and surface-ocean Δ^{14}C

determines the induced isotopic signature. Friedlingstein et al. (2025) [42] bounds the global net O+B flux year-to-year variability at ±2 GtC/yr. With a factor-2 buffer, the maximum induced first derivative of the residual is

$$\left|\frac{dr_{55}}{\mathrm{d}t}\right|_{\max}^{\mathrm{O+B}}(t) \approx \frac{4 \times (R_{\mathrm{T}}(t) - R_{\mathrm{O}}(t))}{M_{\mathrm{a}}(t)} \quad [‰/\mathrm{yr}] \tag{112}$$

where $R_{\mathrm{O}}(t)$ is the contemporary warm-pool surface-ocean Δ^{14}C on the Hua +27 scale.

Two physical conditions must hold simultaneously for an O+B event to produce a detectable residual signal. First, the system must be in the slow eigenmode regime. During the fast-mode era (∼1965–1976, before $3\,\tau_{\mathrm{fast}} \approx 9$ yr post-test-ban) the STE contribution to $\mathrm{d}R_{\mathrm{T}}/\mathrm{d}t$ is order 10–25‰/yr, and any O+B-induced excursion of a few ‰/yr is buried in the fast-mode dynamics at a signal-to-noise ratio of 10× or more against it. Second, the gradient $R_{\mathrm{T}} - R_{\mathrm{O}}$ must be large enough that the ±2 GtC/yr flux excursion produces $|dr_{55}/\mathrm{d}t|$ above the compilation noise floor. The surface ocean lags the atmosphere throughout 1965–2019 ($R_{\mathrm{O}} < R_{\mathrm{T}}$) and the gradient shrinks as the ocean catches up. ENSO events of either polarity perturb the system bidirectionally: El Niño releases warm-pool surface-ocean carbon at $R_{\mathrm{O}} < R_{\mathrm{T}}$, driving r_{55} downward; La Niña absorbs atmospheric carbon preferentially, allowing the relatively higher-Δ^{14}C atmospheric pool to remain, driving r_{55} upward. The induced $|dr_{55}/\mathrm{d}t|$ magnitudes are identical for events of equal flux amplitude regardless of sign. The maximum O+B-induced $|dr_{55}/\mathrm{d}t|$ from Eq. (112) ranges from ∼2.8‰/yr at 1972–73 to ∼1.2‰/yr at 1982–83, ∼0.7‰/yr at 1988–89, ∼0.6‰/yr at 1991–92, ∼0.26‰/yr at 1997–98–2000, ∼0.1‰/yr at 2010–12, and ∼0.09‰/yr at 2015–16.

The detectable-event window for O+B perturbations is therefore roughly 1980–1995: after fast-mode equilibration is complete and before gradient closure suppresses the signal below the measurement noise floor. Within this window, the major events whose predicted $|dr_{55}/\mathrm{d}t|$ approaches or exceeds the noise floor are the 1982–83 El Niño (∼1.2‰/yr), the 1988–89 La Niña (∼0.7‰/yr), and the 1991–92 El Niño (∼0.6‰/yr). The 1972–73 El Niño and the 1973–74 La Niña fall in the fast-mode era where O+B signals are buried in fast-mode dynamics regardless of magnitude. The 1997–98 El Niño, 1998–2000 La Niña, 2010–12 La Niña, and 2015–16 El Niño all fall in the gradient-closed regime where the induced $|dr_{55}/\mathrm{d}t|$ is at or below the compilation noise floor; these events pass the smoothness test on noise grounds and event-year admission is not load-bearing for the partition. The induced $|d^2r_{55}/\mathrm{d}t^2|$ in a detectable event year can reach approximately 5× the smoothness threshold when the first-derivative excursion occurs within a single year.

Volcanic eruptions are sometimes cited as contributors to year-to-year flux variability through stratospheric-aerosol cooling and the resulting suppression of respiration and enhancement of ocean uptake. The Friedlingstein ±2 GtC/yr envelope attributes approximately half of its bound to ENSO with the remainder distributed among volcanic, fire, and other sources; the volcanic indirect-cooling contribution in an eruption year is ∼ 1 GtC/yr at most. During the fast-mode era this magnitude is buried in fast-mode dynamics, identically to ENSO. During the slow-mode era the volcanic carbon involved is fast-cycling pool carbon (suppressed respiration retained in soil and litter, or enhanced ocean uptake at the warm-pool surface), so the Δ^{14}C signature is set by the same $R_{\mathrm{T}} - R_{\mathrm{O}}$ gradient as ENSO. With approximately half the ENSO flux magnitude, the volcanic indirect mechanism produces approximately half the ENSO $|dr_{55}/\mathrm{d}t|$: ∼ 0.6‰/yr at 1982 (El Chichón) and ∼ 0.24‰/yr at 1991 (Pinatubo). Both are at or below the compilation noise floor. Direct volcanic CO_2 emissions are independently negligible (∼ 0.3 GtC/yr globally, with $\Delta^{14}\mathrm{C} = -1000$‰). The 1982–83 and 1991–92 entries in Table 13 therefore correspond to ENSO events; the coincident eruptions add at most a sub-noise contribution to the same gradient-limited mechanism.

We admit O+B excursions only in years coinciding with documented major ENSO events (Table 13).

Table 13: Documented major ENSO events used as O+B perturbation candidates. Events outside the 1980–1995 detectability window are listed for completeness; their predicted $|dr_{55}/\mathrm{d}t|$ magnitudes are at or below the compilation noise floor and the events would pass the smoothness test on noise grounds even without event admission.

Years	Event	Reference
1972–1973	El Niño	Rasmusson & Carpenter (1982) [98]
1973–1974	La Niña	Rasmusson & Carpenter (1982) [98]
1982–1983	El Niño	Rasmusson & Carpenter (1982) [98]
1988–1989	La Niña	Trenberth (1997) [119]
1991–1992	El Niño	Trenberth (1997) [119]
1997–1998	El Niño	McPhaden (1999) [84]
1998–2000	La Niña	McPhaden (1999) [84]
2010–2012	La Niña	Santoso et al. (2017) [103]
2015–2016	El Niño	Santoso et al. (2017) [103]

In years with no documented event, only the smoothness threshold applies.

C.7 The smoothness test as physical diagnostic

For each interior year t in 1966–2018 of the Hua et al. (2022) [55] +27‰ global series, compute the centered first and second derivatives of r_{55} per §C.3, and the diagnostic factor

$$\text{factor}(t) = \frac{|d^2 r_{55}/\mathrm{d}t^2\,(t)|}{\text{threshold}(t)} \tag{113}$$

where the threshold is given by Eq. (111). The factor measures the deviation of the residual's second derivative from the smoothness bound derived in §C.4.

The factor classifies each year into one of three diagnostic categories:

- **Smooth** (factor < 1): the year's curvature is consistent with the smooth slow-mode STE physics derived in §7.
- **Event-admissible** (factor ≥ 1 but < 5, AND year t is in the documented O+B event list of Table 13, AND the event lies in the ∼1980–1995 detectability window where the slow eigenmode is established and the $R_{\mathrm{T}} - R_{\mathrm{O}}$ gradient is still large enough to make a ±2 GtC/yr ENSO flux produce $|dr_{55}/\mathrm{d}t|$ above the compilation noise floor): the year's curvature exceeds the smooth-physics bound but lies within the envelope expected from a documented exogenous O+B perturbation (§C.6). Outside this detectability window (fast-mode era pre-1980, or gradient-closed era post-1995), the predicted O+B-induced $|dr_{55}/\mathrm{d}t|$ is at or below the noise floor; documented events in these eras pass the smoothness test on noise grounds and event-year admission is not load-bearing for the partition.
- **Anomalous** (factor ≥ 1 with no documented O+B event, OR factor ≥ 5 in an event year): the year's curvature has no physical source within the coupled atmospheric system or the documented O+B variability envelope.

The diagnostic classifications do not directly drive retention or exclusion. Retention follows from the rules of §C.2. The diagnostic confirms the partition by exhibiting four signatures:

1. **Wall identification.** The years bordering the transitions between excluded artifact stretches and retained plateaus (1968, 1971, 1978, 1979, 1984, 1985, 1991, 1994, 1995, 2010) all show anomalous classifications, with factors typically in the range 3–10. The walls are not smooth atmospheric decay; they are step transitions in the compilation.

2. **Within-artifact jaggedness.** The interior years of the excluded artifact stretches (1968–1978, 1985–1991, 1995–1997) show alternating-sign $d^2 r_{55}/\mathrm{d}t^2$ values that no smooth

coupled-reservoir physics produces. The pattern is the diagnostic signature of a compilation whose contributing record set is changing year over year, not of atmospheric physics.

3. **Within-plateau smoothness.** The interior years of the retained plateaus (Plateau A interior 1980–1983; Plateau B interior 1999–2009) show smooth classifications consistent with compilation noise and slow-mode STE physics.

4. **FF tail wall locations.** The wall years 2010 (transition Plateau B→FF tail) and 2015 (sub-plateau transition within the FF tail) show anomalous classifications in the C_{FF}-only-corrected residuals, confirming that the emergence-offset corrections of Rule 4 are needed at exactly the locations predicted by independent physical and archival arguments (§C.11.4).

The single-point classifications in Table 15 therefore serve to verify the partition produced by the retention rules, not to produce it. Isolated year-by-year smoothness verdicts that conflict with the partition (for example, 1993 and 1994 within the mini-plateau show factor ≥ 1 in Table 15; 1969 and 1972 within the 1968–1978 artifact stretch show factor < 1) reflect the mathematical limitations of any second-difference operator: it is necessary but not sufficient for identifying smoothness, since three collinear points satisfy any second-difference bound regardless of the underlying slope. Plateau membership (Rule 2) and block contiguity within artifact stretches are what identify retained and excluded blocks; the smoothness test confirms the wall locations and the within-stretch jaggedness.

C.8 Rule 1: A priori exclusion of 1965 and 1966

US and USSR atmospheric nuclear testing ceased in late 1962, with most bomb ^{14}C injected directly into the stratosphere; French and Chinese atmospheric tests continued through 1976 (§9.2.2) but were a much smaller contribution to the global ^{14}C inventory than the 1952–1962 US/USSR tests. The peak global tropospheric Δ^{14}C occurred mid-1964 at approximately 750‰, when the rate of stratosphere-to-troposphere input from the Brewer-Dobson circulation equaled the rate of slow-mode loss to the rest of the carbon cycle. The fast eigenvalue $\tau_{\mathrm{fast}} = 3.08\,\mathrm{yr}$ derived in §7 from independently sourced M_{S}, M_{T}, and F_{BDC} (with negligible dependence on bomb-decay data) sets the timescale on which the fast-mode contribution to $\mathrm{d}R_{\mathrm{T}}/\mathrm{d}t$ decays. By four years after the last atmospheric nuclear detonation (December 1962), the fast-mode amplitude has decayed by a factor of $\exp(-4/\tau_{\mathrm{fast}}) = 0.273$ (representing $4/(\tau_{\mathrm{fast}} \ln 2) = 1.88$ fast-mode half-lives), while the slow-mode amplitude has decayed by only $\exp(-4/\tau_{\mathrm{slow}}) = 0.804$. By the start of our fit window in 1967, the fast mode is therefore already at 27% of its post-detonation amplitude, and the single-exponential slow-eigenmode model applies to good approximation.

By Rule 1 of §C.2, the 1965 and 1966 points are excluded a priori and not classified by the smoothness diagnostic. The first valid post-equilibration point is 1967, retained by Rule 3 (§C.9).

C.9 Rule 3: The 1967 anchor point

By Rule 3 of §C.2, the 1967 point is retained as the first valid post-equilibration year. The justification is structural, not based on the value of $r_{55}(1967)$:

1. Rule 1 excludes 1965 and 1966 a priori on physical grounds (§C.8).

2. The provenance audit of §9.2 identifies no documented compilation artifact in 1967.

3. The centered second-difference operator cannot be cleanly applied at 1967 because its left neighbor (1966) is in the fast-mode-excluded regime; the Table 15 value $d^2 r_{55}/\mathrm{d}t^2$ (1967) = -4.88‰/yr^2 reflects contamination by the excluded 1966 point ($r_{55}(1966) = +6.67$‰) rather than a smoothness violation in the underlying physics.

4. 1967 is therefore the leftmost edge of the post-equilibration data set where the smoothness diagnostic cannot classify it on its own terms; retention follows by Rule 3 as the first valid post-Rule-1 point with no documented artifact.

The 1967 point receives no plateau-level correction: it is a single isolated point and does not satisfy Rule 2's plateau-membership criterion (which requires three or more consecutive years with low within-block dispersion and a within-block trend consistent with zero at the 95% confidence level). The empirical value $r_{55}(1967) = +0.20‰$ is reported in Table 15 as data, not as a retention criterion.

C.10 Application to the 55-point Hua et al. (2022) series

Table 15 shows every interior year with the four diagnostic quantities ($dr_{55}/\mathrm{d}t$, $d^2r_{55}/\mathrm{d}t^2$, threshold, factor), the event flag, the smoothness-diagnostic classification, the role of each year in the partition produced by the retention rules of §C.2, and the corresponding visual block in Figure 3(a) bottom.

The partition produced by the retention rules is given in Table 14. The retained set comprises 32 points in five blocks: the 1967 anchor (Rule 3); Plateau A (1979–1984, 6 points, Rule 2); the mini-plateau (1992–1994, 3 points, Rule 2); Plateau B (1998–2010, 13 points with solar correction at 2009 and 2010, Rule 2); and the fossil-fuel tail (2011–2019, 9 points after the C_{FF} correction of §9.2.5 and the emergence offsets of §C.11.4, Rule 4). The excluded set comprises 23 points in four blocks: 1965–1966 S–T equilibration (Rule 1); 1968–1978 station exits and tree-ring-to-atmospheric weighting transition (Rule 5, with smoothness-diagnostic and provenance confirmation); 1985–1991 station transitions and retrospective compilation revisions (Rule 5, with smoothness-diagnostic and provenance confirmation); 1995–1997 Wellington 1995–2005 revision onset (Rule 5, with smoothness-diagnostic and provenance confirmation).

Table 14: Partition of the 55-point Hua et al. (2022) [55] global +27‰ series produced by the retention rules of §C.2, with the smoothness diagnostic classification of §C.7 and the provenance audit of §9.2 as independent confirmations on every block boundary.

Block	Years	Points	Designation	Rule
Retained (32 points):				
1	1967	1	Anchor (first valid post-equilibration point)	3
2	1979–1984	6	Plateau A	2
3	1992–1994	3	Mini-plateau	2
4	1998–2010	13	Plateau B (with solar correction at 2009 and 2010)	2
5	2011–2019	9	FF tail (after C_{FF} and emergence offsets)	4
Excluded (23 points):				
A	1965–1966	2	Fast-mode S–T equilibration ramp	1
B	1968–1978	11	Tree-ring-to-atmospheric weighting shift; station exits	5
C	1985–1991	7	Station transitions & Wellington 1990–1993 revision	5
D	1995–1997	3	Onset of Wellington 1995–2005 revision	5

C.11 Corrections applied to the retained set

Three categories of documented corrections are applied to the retained set before the cleaned 32-point fit is computed. None contains adjustable parameters.

C.11.1 Fossil-fuel correction C_{FF} (2011–2019)

For years 2011–2019, the cumulative atmospheric $\Delta^{14}\mathrm{C}$ dilution from accumulating fossil-fuel-derived carbon has emerged from beneath the decaying bomb-spike signal. The exact form of C_{FF} and the numerical values applied to each year of 2011–2019 are derived from first principles in §9.2.5 and tabulated in Table 9; the correction is computed from directly measured $E_{\mathrm{FOS}}(t)$ [42]

and $M_a(t)$ [76] with no adjustable parameters. The fossil-fuel signature is empirically invisible through 2010 because it is absorbed into the empirical τ_{rel} of the single exponential; it becomes visible only when the bomb-decay component has decayed to where the fossil contribution is no longer dwarfed by it.

C.11.2 Solar minimum correction at 2009 and 2010

The 2008–2010 deep solar minimum produced a documented enhancement of cosmogenic ^{14}C production [79]. The enhancement contributes approximately +1‰ to tropospheric $\Delta^{14}C$ in each of 2009 and 2010, absorbed into the Hua compilation. We apply a −1‰ correction to the Hua values at 2009 and 2010. The raw 2008→2009 backward first difference is +1.62‰/yr (incompatible with smooth STE physics, since the stratosphere has no significant remaining bomb-^{14}C reservoir by 2009); after correction the backward first difference is +0.62‰/yr, within the smoothness envelope. Effect on the cleaned 32-point fit: σ drops from 0.539‰ to 0.472‰; τ_{rel} shifts by 0.008 yr (0.04%), well within the formal uncertainty (see Table 10).

C.11.3 Plateau-level corrections

The level of each retained plateau represents the cumulative offset built up by all the invalid deviations preceding the plateau in the residual time series; the underlying mechanism is detailed in §9.2.4. The numerical plateau levels evaluated after the solar correction of §C.11.2 is applied are: −11.563‰ for Plateau A (1979–1984), +0.891‰ for the mini-plateau (1992–1994), and +9.447‰ for Plateau B (1998–2010). The 1967 anchor at +0.20‰ receives no plateau correction (single isolated point already at essentially zero). After plateau correction, every retained point in these blocks sits at residual ≈ 0 within $\pm\sigma_{32}^{corr} = 0.47$‰.

C.11.4 Emergence offsets at 2011 and 2016 (FF tail)

After applying C_{FF} (§C.11.1) to the FF tail residuals, the resulting series exhibits two discrete jumps at the boundaries 2010→2011 and 2015→2016. These boundaries are *known wall breakpoints*: points where the smoothness test of §C.7 would FAIL in the C_{FF}-only-corrected residual series because the data exhibits an obvious discontinuity at these specific years. Both discontinuities are visible directly in Figure 3 bottom panel: at the boundary between the green Plateau B region and the orange FF tail (2010→2011), and at the visible jog within the orange FF tail itself (2015→2016).

The two jump locations correspond to documented physical and compilation transitions:

- **2010→2011:** The timing at which the cumulative fossil-fuel signal emerges from the noise envelope of the residual series. The small offset magnitude reflects the gradual nature of the emergence.
- **2015→2016:** Documented station-network composition changes in the Hua compilation occur at this period of maximum residual sensitivity, producing a larger discrete shift.

Both breakpoint locations are predictable from the documented physical and compilation context, expected, and observed in the data. Their *magnitudes* are computed by the requirement that the post-correction sub-plateau levels (2011–2015 and 2016–2019) each match the Plateau B level (+9.447‰), since C_{FF} accounts for all fossil-fuel contribution beyond the small emergence-timing and station-change adjustments:

- **2011 emergence offset:** −0.055‰ (applied to each point in 2011–2015 in addition to the Plateau B subtraction). Small magnitude consistent with gradual emergence-timing adjustment.
- **2016 emergence offset:** +2.391‰ (applied to each point in 2016–2019 in addition to the Plateau B subtraction). Larger magnitude consistent with a discrete station-network change at a period of maximum residual sensitivity.

After C_{FF} and the two emergence offsets, both sub-intervals of the FF tail sit at the same level as Plateau B, and the single Plateau B subtraction of +9.447‰ applies uniformly to the entire 2011–2019 FF tail.

The robustness of the cleaned τ_{rel} to the emergence-offset values rests on three independent grounds. First, the wall locations themselves (2010→2011 and 2015→2016) are physically predicted: the 2011 wall is the timing at which the cumulative fossil-fuel signal crosses the noise envelope of the residual series, computable from C_{FF} and the compilation noise floor with no reference to the residual data; the 2016 wall is the documented Hua et al. (2022) [55] station-network composition change at that year, an archival fact. Neither location is chosen to match a feature of the residual data. Second, the magnitudes are constrained by physical continuity: the underlying bomb-decay trajectory is smooth across any discrete compilation step, so the post-correction residuals on both sides of each wall must sit at the same level (Plateau B). This is a continuity-of-physics constraint, not a fit-improvement criterion. Third, even the most skeptical treatment (setting both offsets to zero, omitting the corrections entirely) shifts the cleaned τ_{rel} by only +0.028 yr (0.16%), well below the 1% precision floor of the analysis (Table 10). Doubling either offset shifts τ_{rel} by less than 0.02 yr in either direction. The central τ_{rel} is determined by the large-baseline structure of the retained 32 points; the FF-tail emergence offsets are local refinements with negligible leverage on the global fit.

C.12 Robustness

The exclusion partition is robust to:

- Doubling or halving F_{BDC} (Brewer–Dobson literature uncertainty): no segment changes category.
- Doubling the Friedlingstein et al. (2025) [42] ±2 GtC/yr envelope: no segment changes category.
- Tightening the buffer from factor 3 to factor 2 (more aggressive exclusion): no retained block becomes excluded.
- Loosening the buffer from factor 3 to factor 5: no excluded block becomes retained.

The 32/23 partition is determined uniquely by physics within reasonable parameter ranges; no element of the result hinges on a finely tuned threshold.

C.13 Convergence of independent diagnostics on the partition

The partition produced by the retention rules of §C.2 is confirmed independently by two diagnostics that use disjoint inputs and disjoint reasoning. The smoothness diagnostic of §C.7 uses the numerical residual series $r_{55}(t)$ and the physical smoothness bound derived from the coupled stratosphere-troposphere ODE system (§7), with the Friedlingstein et al. (2025) [42] O+B envelope and the documented event list as auxiliary inputs. It uses no compilation-provenance metadata. The provenance audit of §9.2 uses only the documented station transitions, tree-ring-to-atmospheric weighting shifts, and retrospective revisions in the Hua et al. (2022) [55] compilation. It uses no physical reasoning and no residual data.

The two diagnostics agree on every block boundary:

- Walls at 1968, 1971, 1978, 1979, 1984, 1985, 1991, 1994, 1995 show smoothness-diagnostic anomalies and coincide with documented compilation transitions identified in the provenance audit.
- The interior of artifact stretches (1968–1978, 1985–1991, 1995–1997) shows sign-flipping smoothness-diagnostic anomalies and coincides with periods of methodology change documented in the audit.

- The interior of retained plateaus (Plateau A 1980–1983, Plateau B 1999–2009) shows smoothness-diagnostic verdicts within the noise floor and coincides with stable-methodology periods in the audit.
- The 2010→2011 and 2015→2016 wall locations show smoothness-diagnostic anomalies in the C_{FF}-only-corrected residuals and coincide respectively with the predicted fossil-fuel emergence-from-noise timing and the documented station-network composition change in the Hua et al. (2022) [55] compilation.

The convergence is non-trivial. The retention rules are physics-derived; the smoothness diagnostic is physics-derived from the same two-box framework but with the disjoint diagnostic operator $d^2r_{55}/\mathrm{d}t^2$; the provenance audit is purely archival with no physical input. That all three agree on the 32/23 partition (boundary by boundary, year by year) establishes that the partition reflects real structure of the compilation methodology and the underlying atmospheric physics, not a fitted convergence to a chosen answer.

Table 15: Year-by-year diagnostic table for the Hua et al. (2022) [55] global +27‰ series. Columns are: $r_{55}(t) \equiv R_{\mathrm{a}}(t) - A_{55}\exp[(1965-t)/\tau_{55}]$ (Table 3); $dr_{55}/\mathrm{d}t\,(t) = \frac{1}{2}[r_{55}(t+1) - r_{55}(t-1)]$ (centered first-derivative estimator); $d^2r_{55}/\mathrm{d}t^2\,(t) = r_{55}(t+1) - 2\,r_{55}(t) + r_{55}(t-1)$ (centered second-derivative estimator, §C.3); threshold from Eq. (111); factor = $|d^2r_{55}/\mathrm{d}t^2|$/threshold (the smoothness diagnostic of §C.7); Event indicates membership in the documented ENSO event list of Table 13; Role indicates each year's role in the partition produced by the retention rules of §C.2. Residuals are after the −1‰ solar correction at 2009 and 2010 (§C.11.2). Plateau-level subtractions (§C.11.3) and FF-tail corrections (§C.11.1–§C.11.4) are applied to the retained set before the cleaned 32-point fit; they do not enter the smoothness diagnostic. Block coloring matches Figure 3(a) bottom: blue (Rule 1 a priori excluded), green (Rule 2/Rule 3 retained), red (Rule 5 excluded artifact), orange (Rule 4 retained FF tail).

Year	r_{55} (‰)	$dr_{55}/\mathrm{d}t$ (‰/yr)	$d^2r_{55}/\mathrm{d}t^2$ (‰/yr^2)	Threshold (‰/yr^2)	Factor	Event?	Role
1965	+11.15	—	—	—	—	—	Rule 1: a priori excluded
1966	+6.67	−5.48	−1.99	1.86	1.07	—	Rule 1: a priori excluded
1967	+0.20	−8.91	−4.88	1.75	2.78	—	Rule 3: anchor (retained)
1968	−11.14	−2.74	+17.22	1.66	10.37	—	Rule 5: artifact stretch (excluded)
1969	−5.27	+6.03	+0.31	1.57	0.20	—	Rule 5: artifact stretch (excluded)
1970	+0.91	+10.38	+8.40	1.49	5.64	—	Rule 5: artifact stretch (excluded)
1971	+15.50	+5.33	−18.51	1.41	13.13	—	Rule 5: artifact stretch (excluded)
1972	+11.57	−3.64	+0.57	1.40	0.41	yes	Rule 5: artifact stretch (excluded)
1973	+8.21	−4.54	−2.36	1.40	1.69	yes	Rule 5: artifact stretch (excluded)
1974	+2.49	−3.86	+3.72	1.40	2.66	—	Rule 5: artifact stretch (excluded)
1975	+0.49	−5.61	−7.22	1.40	5.16	—	Rule 5: artifact stretch (excluded)
1976	−8.73	−4.30	+9.85	1.40	7.03	—	Rule 5: artifact stretch (excluded)
1977	−8.10	+3.08	+4.91	1.40	3.51	—	Rule 5: artifact stretch (excluded)
1978	−2.57	−1.48	−14.03	1.40	10.02	—	Rule 5: artifact stretch (excluded)
1979	−11.06	−4.49	+8.02	1.40	5.73	—	Rule 2: Plateau A (boundary wall)
1980	−11.54	+0.06	+1.07	1.40	0.77	—	Rule 2: Plateau A interior
1981	−10.95	+0.16	−0.88	1.40	0.63	—	Rule 2: Plateau A interior
1982	−11.23	−0.70	−0.83	1.40	0.59	yes	Rule 2: Plateau A interior
1983	−12.35	−0.51	+1.21	1.40	0.87	yes	Rule 2: Plateau A interior
1984	−12.25	+2.22	+4.25	1.40	3.04	—	Rule 2: Plateau A (boundary wall)
1985	−7.90	+2.50	−3.71	1.40	2.65	—	Rule 5: artifact stretch (excluded)
1986	−7.25	+1.31	+1.33	1.40	0.95	—	Rule 5: artifact stretch (excluded)
1987	−5.28	+1.66	−0.63	1.40	0.45	—	Rule 5: artifact stretch (excluded)
1988	−3.94	+1.54	+0.40	1.40	0.29	—	Rule 5: artifact stretch (excluded)
1989	−2.19	−0.54	−4.57	1.40	3.26	—	Rule 5: artifact stretch (excluded)
1990	−5.02	−1.10	+3.46	1.40	2.47	—	Rule 5: artifact stretch (excluded)
1991	−4.39	+2.88	+4.49	1.40	3.21	yes	Rule 5: artifact stretch (excluded)
1992	+0.74	+2.88	−4.48	1.40	3.20	yes	Rule 2: mini-plateau (boundary wall)
1993	+1.38	−0.09	−1.46	1.40	1.04	—	Rule 2: mini-plateau interior

Continued on next page

Table 15 continued from previous page

Year	r_{55} (‰)	dr_{55}/dt (‰/yr)	d^2r_{55}/dt^2 (‰/yr^2)	Threshold (‰/yr^2)	Factor	Event?	Role
1994	+0.56	+0.97	+3.57	1.40	2.55	—	Rule 2: mini-plateau (boundary wall)
1995	+3.31	+2.05	−1.41	1.40	1.01	—	Rule 5: artifact stretch (excluded)
1996	+4.65	+1.65	+0.61	1.40	0.44	—	Rule 5: artifact stretch (excluded)
1997	+6.60	+2.27	+0.63	1.40	0.45	yes	Rule 5: artifact stretch (excluded)
1998	+9.19	+1.41	−2.35	1.40	1.68	yes	Rule 2: Plateau B (boundary wall)
1999	+9.42	−0.43	−1.33	1.40	0.95	—	Rule 2: Plateau B interior
2000	+8.33	−0.25	+1.69	1.40	1.21	—	Rule 2: Plateau B interior
2001	+8.92	+0.95	+0.70	1.40	0.50	—	Rule 2: Plateau B interior
2002	+10.22	+0.66	−1.28	1.40	0.91	—	Rule 2: Plateau B interior
2003	+10.24	−0.62	−1.27	1.40	0.90	—	Rule 2: Plateau B interior
2004	+8.99	−0.38	+1.75	1.40	1.25	—	Rule 2: Plateau B interior
2005	+9.49	+0.38	−0.24	1.40	0.17	—	Rule 2: Plateau B interior
2006	+9.75	−0.35	−1.23	1.40	0.88	—	Rule 2: Plateau B interior
2007	+8.78	−0.57	+0.79	1.40	0.56	—	Rule 2: Plateau B interior
2008	+8.61	+0.22	+0.80	1.40	0.57	—	Rule 2: Plateau B interior
2009	+9.23	+0.52	−0.19	1.40	0.14	—	Rule 2: Plateau B interior (post-solar)
2010	+9.65	−1.16	−3.18	1.40	2.27	—	Rule 2: Plateau B (boundary wall)
2011	+9.31	+0.05	+0.77	1.40	< 1	—	Rule 4: FF tail (boundary, 2011 emergence)
2012	+9.74	+0.31	−0.25	1.40	< 1	—	Rule 4: FF tail interior
2013	+9.93	−0.46	−1.28	1.40	< 1	—	Rule 4: FF tail interior
2014	+8.83	−0.25	+1.69	1.40	1.21	—	Rule 4: FF tail interior
2015	+9.42	+0.19	−0.80	1.40	< 1	yes	Rule 4: FF tail (boundary, 2015→2016)
2016	+9.21	−0.01	+0.40	1.40	< 1	yes	Rule 4: FF tail (boundary, 2016 emergence)
2017	+9.41	+0.18	−0.02	1.40	< 1	—	Rule 4: FF tail interior
2018	+9.58	+0.09	−0.16	1.40	< 1	—	Rule 4: FF tail interior
2019	+9.59	—	—	—	—	—	Rule 4: FF tail edge

After all corrections of §C.11 are applied (C_{FF} for 2011–2019 per §9.2.5 and Table 9; the solar correction for 2009 and 2010; the plateau-level subtractions of §C.11.3; and the emergence offsets at 2011 and 2016), the 32 retained residuals lie within $\pm\sigma_{32}^{\mathrm{corr}} = 0.47$‰ of zero. The cleaned 32-point single-exponential fit yields $\tau_{\mathrm{rel}} = 18.375 \pm 0.013$ yr, $R^2 = 99.999\%$, and $\sigma_{32} = 0.472$‰ (Table 10).

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